

Supplementary Information for *Dendritic morphology as a dictionary for local credit assignment*

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Scope of the Supplementary Information

The supplement explains how the models compute synaptic credit, how spatially restricted learning signals approximate it, and how those restrictions affect learning. Here *credit* means how a small synaptic change would affect task error. A *route dictionary* is a set of spatial profiles through which learning signals can be distributed. These signals are mathematical quantities used by the learning rules; the biological datasets do not directly measure them.

Sections S1–S2 derive exact gradients and define the available feedback profiles. S3–S4 address what those profiles can represent, how noisy updates affect progress, and how task interactions constrain a tree. S5 gives the artificial learning experiments; S6–S7 describe reconstruction and anatomical dictionaries; S8 covers inhibitory contacts; S9 tests focal changes in modeled transport. Measured responses, imposed alignment and external-data interpretation follow in S10–S12. Statistics and source records are consolidated in S13. The supplementary cohort index indexes experiments, and Table S34 summarizes what each comparison can establish.

The three uses of a tree should be kept distinct. In a directed artificial neuron, activity travels from terminal compartments toward the soma. An abstract feedback tree can instead specify which coordinates receive a shared learning signal, without adding forward compartments. Reconstructed arbors use reciprocal cable coupling, which allows current flow in both directions. Physical depth D_p , feedback resolution D_r , topological depth and electrotonic distance therefore describe different properties. D1–D4 denote stage counts within a stated cohort; their branching vectors are supplied locally.

The independent replicate is a paired training seed, reconstructed cell, postsynaptic target or external animal, depending on the comparison. Synapses, intervention sites, channel draws, scans, data splits and Monte Carlo streams are repeated observations within those units. The initial and disjoint minnie65 cohorts come from one mouse; Pinky is a second mouse. Neither provides population-random cell sampling. The six-animal external reanalysis concerns neuronal coordinates rather than within-arbor routing. Structural capacity, modeled transport and visual-response alignment are separate outcomes. We call exact transport, best-fit projection coefficients and supplied reliability moments *oracles* when they provide information unavailable to the restricted rule; their use does not establish a biological mechanism for estimating that information.

Symbols are defined locally because the derivations use several coordinate systems. Bold lower-case symbols generally denote vectors; matrix dimensions are stated when an operator is introduced. Unless a subsection specifies otherwise, the principal indices are u (modeled neuron), n (compartment), i (synapse) and k (feedback route). H_c denotes task-credit hierarchy in the operator experiment, H_p latent hierarchy depth in the physical-depth tasks and K the number of communicated feedback channels. Table S13 consolidates the notation, while each derivation introduces the symbols needed to follow it.

S1 Exact gradients in a directed conductance tree

To determine which factors a local learning rule can compute, we first derive the exact conductance gradient, following the factorization and LocalCA rule introduced in our predecessor preprint [1]. This requires the forward steady state, the local synaptic sensitivity and transport of the somatic loss derivative through the tree.

1.1 Steady-state compartment voltage

Consider a rooted dendritic tree with soma node 0. Let V_n and C_n be the voltage and capacitance of compartment n , and let $\mathcal{C}_n$ be its set of children. A non-somatic compartment n receives excitatory contacts $\mathcal{S}_n^E$ and inhibitory contacts $\mathcal{S}_n^I$, with $\mathcal{S}_n = \mathcal{S}_n^E \cup \mathcal{S}_n^I$. A contact of type $X \in \{E, I\}$ has nonnegative maximal conductance g_i^X , nonnegative presynaptic activity x_i^X , open conductance $g_i^X x_i^X$, and reversal E_X . We use $E_i^{\text{rev}} = E_E$ on $\mathcal{S}_n^E$ and $E_i^{\text{rev}} = E_I$ on $\mathcal{S}_n^I$; when contact type is not at issue, we write the generic g_i^{syn} , x_i , and E_i^{rev} . Define the aggregate open conductances

$$G_n^E = \sum_{i \in \mathcal{S}_n^E} g_i^E x_i^E, \quad G_n^I = \sum_{j \in \mathcal{S}_n^I} g_j^I x_j^I.$$

129 The compartment also receives child-to-parent coupling conductances $g_{c \rightarrow n}^{\text{den}}$, and has leak g_n^{L} with reversal E_{L} . A
 130 child transmits $a_c = f_c(V_c)$, where f_c is the branch output nonlinearity (named “reactivation” in the implementation)
 131 and is the identity when that mechanism is disabled. Before taking the steady state, the directed compartmental
 132 dynamics are

$$C_n \dot{V}_n = g_n^{\text{L}}(E_{\text{L}} - V_n) + G_n^{\text{E}}(E_{\text{E}} - V_n) + G_n^{\text{I}}(E_{\text{I}} - V_n) + \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}}(a_c - V_n). \quad (1)$$

The total child coupling is $G_n^{\text{child}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}}$ and its voltage-weighted contribution is $J_n^{\text{child}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c$.
 Setting $\dot{V}_n = 0$ and collecting the coefficients of V_n then gives

$$g_n^{\text{tot}} = g_n^{\text{L}} + G_n^{\text{E}} + G_n^{\text{I}} + G_n^{\text{child}}, \quad (2)$$

$$V_n = R_n^{\text{tot}} (g_n^{\text{L}} E_{\text{L}} + G_n^{\text{E}} E_{\text{E}} + G_n^{\text{I}} E_{\text{I}} + J_n^{\text{child}}), \quad R_n^{\text{tot}} = (g_n^{\text{tot}})^{-1}. \quad (3)$$

133 Here g_n^{tot} and R_n^{tot} are the total conductance and its reciprocal, respectively. The soma obeys the same current
 134 balance with $n = 0$, including its child couplings and any explicitly modeled somatic inputs, but has no parent
 135 term. The controlled regular-tree architectures used here have no external somatic input synapses unless a control
 136 is explicitly described otherwise. For nonnegative conductances and activities, Eq. 3 is a conductance-weighted
 137 average of leak, synaptic reversal potentials and child activities. This boundedness is useful for numerical stability,
 138 but is not required for the chain-rule factorization below. Both E and I increase g_n^{tot} ; they differ through the driving
 139 forces $E_{\text{E}} - V_n$ and $E_{\text{I}} - V_n$, not through a signed conductance. When $E_{\text{I}} < V_n < E_{\text{E}}$, the former is depolarizing
 140 and the latter hyperpolarizing. At $V_n = E_{\text{I}}$, an inhibitory conductance has zero net current but still reduces R_n^{tot} ,
 141 producing a pure shunt at that operating point.

142 The directed artificial-tree simulations use the dimensionless convention $g_n^{\text{L}} = 1$, $E_{\text{L}} = E_{\text{I}} = 0$, and $E_{\text{E}} = 1$, so
 143 inhibition is denominator-only. The reconstructed focal-cable analysis uses $E_{\text{E}} = 1$ and $E_{\text{I}} = -0.2$, so its inhibitory
 144 perturbation is both hyperpolarizing and shunting. Neither convention is a fit in millivolts.

145 1.2 Local sensitivities and synaptic eligibility

Writing the numerator of Eq. 3 as $b_n = g_n^{\text{tot}} V_n$, quotient differentiation gives

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = \frac{x_i E_i^{\text{rev}} g_n^{\text{tot}} - x_i b_n}{(g_n^{\text{tot}})^2} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n), \quad (4)$$

$$\frac{\partial V_n}{\partial V_c} = f'_c(V_c) g_{c \rightarrow n}^{\text{den}} R_n^{\text{tot}}, \quad (5)$$

$$\frac{\partial V_n}{\partial g_{c \rightarrow n}^{\text{den}}} = R_n^{\text{tot}} (a_c - V_n). \quad (6)$$

146 For a differentiable task loss $\mathcal{L}$, the exact synaptic conductance gradient therefore factors as

$$\frac{\partial \mathcal{L}}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\frac{\partial \mathcal{L}}{\partial V_n}}_{\text{compartment error}}. \quad (7)$$

147 The eligibility contains quantities available at the synapse or its host compartment. All remaining task dependence
 148 lies in the compartment error. In particular, excitatory and inhibitory contacts use $x_i^{\text{E}} R_n^{\text{tot}} (E_{\text{E}} - V_n)$ and $x_j^{\text{I}} R_n^{\text{tot}} (E_{\text{I}} -$
 149 $V_n)$, respectively. Near $V_n = E_{\text{I}}$, the inhibitory contact’s own gradient can be small even though its conductance still
 150 reduces R_n^{tot} and changes the sensitivity of other inputs.

151 1.3 Transport of the somatic error

152 Suppose the loss depends on the tree through its somatic output. Let $p(n)$ be the unique parent of non-somatic node
 153 n , and define the somatic voltage error $\delta_0^V = \partial \mathcal{L} / \partial V_0$. The chain rule gives

$$\frac{\partial \mathcal{L}}{\partial V_n} = \frac{\partial \mathcal{L}}{\partial V_{p(n)}} f'_n(V_n) R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}. \quad (8)$$

Because the graph is a tree, iterating Eq. 8 along the unique path from n to the soma yields

$$\tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial V_n} = \tilde{\alpha}_n \delta_0^V. \quad (10)$$

The effective path gain $\tilde{\alpha}_n$ includes every local transfer derivative. With identity reactivation, it reduces to the conductance-stage gain α_n^{cond} obtained by removing the f'_i factors. Equation 10 is an exact derivative, not a separately fitted model parameter.

1.4 Raw-parameter transforms

Trainable raw parameters θ are transformed to nonnegative conductances by a monotone map $g = \psi(\theta)$. The optimizer receives

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial g} \psi'(\theta). \quad (11)$$

Where $\psi'(\theta) > 0$, this transform preserves the sign of the conductance gradient. Active synapse masks are applied after forming the conductance-space gradient. The exact-gradient diagnostics compare gradients in the same parameter space and include the transform derivative.

1.5 General implicit-adjoint form

The preceding path product is a corollary of a more general steady-state identity. Let $\mathbf{V}$, $\mathbf{g}$ and $\mathbf{x}$ collect compartment voltages, conductances and inputs. Suppose their steady state is defined by the residual equations $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = \mathbf{0}$, with nonsingular voltage Jacobian $J_V = \partial \mathbf{F} / \partial \mathbf{V}$. Define the adjoint $\mathbf{q}$ by

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}. \quad (12)$$

The implicit-function theorem gives

$$\frac{\partial \mathcal{L}}{\partial g_i} = -\mathbf{q}^T \frac{\partial \mathbf{F}}{\partial g_i}. \quad (13)$$

For a passive residual $\mathbf{F} = G\mathbf{V} - \mathbf{b}$, let G be the conductance matrix, $\mathbf{b}$ the vector of imposed current contributions and $\mathbf{e}_n$ the coordinate vector for compartment n . A conductance synapse at n has $\partial \mathbf{F} / \partial g_i = x_i(V_n - E_i^{\text{rev}})\mathbf{e}_n$, and therefore

$$\frac{\partial \mathcal{L}}{\partial g_i} = x_i(E_i^{\text{rev}} - V_n)q_n. \quad (14)$$

Both contact classes add positive diagonal loading to G , while their different reversals enter $\mathbf{b}$ and the local driving force in Eq. 14. Equations 12–14 recover the directed path product when J_V is triangular. For the reciprocal passive cable, $J_V = G$ is symmetric positive definite, so $G^{-T} = G^{-1}$. A general nonsymmetric or recurrent operator instead requires the inverse transpose. Active steady states enter through their Jacobian whenever the required differentiability and nonsingularity conditions hold; no passive symmetry is assumed by the general identity.

1.6 Conductance-specific control of path gain

Holding coupling conductances and the forward state fixed, add effective inhibitory conductance $\Delta G_k^I \geq 0$ at compartment k . This is an activity-weighted aggregate conductance, not an individual maximal synaptic conductance. For a compartment n whose path crosses k ,

$$\frac{\alpha_n^{\text{cond}}(\Delta G^I)}{\alpha_n^{\text{cond}}(0)} = \prod_{k \in \mathcal{A}(n)} \frac{g_k^{\text{tot}}}{g_k^{\text{tot}} + \Delta G_k^I}, \quad \frac{\partial \log \alpha_n^{\text{cond}}}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathcal{A}(n), \\ 0, & k \notin \mathcal{A}(n), \end{cases} \quad (15)$$

where $\mathcal{A}(n) = \{k : (i \rightarrow k) \in \text{path}(n \rightarrow 0)\}$ contains the parent endpoints of the path edges: the strict ancestors of n , including the soma if its conductance is varied. Equation 15 proves path-selective attenuation. It does not prove that attenuation improves optimization. No reversal potential appears in Eq. 15 because this calculation retains only the total-conductance denominator while holding the forward state fixed. Any positive conductance has the same direct denominator effect. It is specifically a shunt when E_I is near the local voltage, so that its accompanying current is small; the full forward and gradient effects retain E_I .

Let $z_n = \log \alpha_n^{\text{cond}}$ and let β_n be the accumulated log-gain attenuation along path n . The perturbed log gain is $z'_n = z_n - \beta_n$. Across any specified ensemble of paths,

$$\text{Var}(z') = \text{Var}(z) + \text{Var}(\beta) - 2 \text{Cov}(z, \beta). \quad (16)$$

Shunting narrows this distribution only if $\text{Cov}(z, \beta) > \text{Var}(\beta)/2$. Path-gain compression is therefore an operating-point-dependent empirical outcome rather than a theorem.

S2 Local rules and feedback objects

The exact compartment error derived above is generally unavailable to a local synapse. We therefore retain the exact local eligibility and ask what restricted teaching field can replace the transported error.

2.1 Three-factor update

The three-factor rule retains the exact local eligibility and substitutes an available voltage-space teaching field $\hat{\delta}_n$ for the exact compartment error:

$$\hat{\nabla}_{g_i^{\text{syn}}} \mathcal{L} = \left\langle x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \hat{\delta}_n \right\rangle_B, \quad \hat{\nabla}_{g_{c \rightarrow n}^{\text{den}}} \mathcal{L} = \left\langle R_n^{\text{tot}} (a_c - V_n) \hat{\delta}_n \right\rangle_B, \quad (17)$$

where B is the number of examples in a minibatch and $\langle \cdot \rangle_B$ denotes their average. Excitatory and inhibitory contacts use the same expression; their reversal potentials set the sign and magnitude of the driving force.

The four- and five-factor rules multiply Eq. 17 by slowly varying, bounded stage-level scalar reliability estimates. For stage ℓ , mini-batch t , and example b , the implementation averages the recorded layer voltage and corresponding somatic population activity over their feature axes, giving $\bar{V}_{\ell,b}^{(t)}$ and $\bar{V}_{0,b}^{(t)}$. The runs reported here used an uncentered dot product,

$$s_{\ell,t} = \frac{1}{B} \sum_{b=1}^B \bar{V}_{\ell,b}^{(t)} \bar{V}_{0,b}^{(t)}, \quad (18)$$

$$m_{\ell,1} = s_{\ell,1}, \quad m_{\ell,t} = 0.9m_{\ell,t-1} + 0.1s_{\ell,t} \quad (t > 1), \quad (19)$$

$$r_{\ell,t}^{4F} = \text{clip}_{[0.1, 2.0]}(m_{\ell,t}). \quad (20)$$

Thus $r_{\ell,t}^{4F}$ is one scalar for a recorded stage, not a separate branch-wise signal, and is not a centered covariance or a Pearson correlation. Recorded voltages are detached from automatic differentiation, and the same minibatch supplies $s_{\ell,t}$ and the local update. The positive clamp prevents a negative or very small dot product from reversing or eliminating the update.

The five-factor rule retains this multiplier and adds a conditional variance-ratio factor. Let $Y_\ell^{(t)}$ be the recorded stage voltage and $X_\ell^{(t)}$ its source-resolved parent proxy, each centered across mini-batch t . Let $\mathcal{E}_{0.1}[\cdot]$ denote an exponential moving average (EMA) with update rate 0.1. Define

$$v_{Y,\ell,t} = \mathcal{E}_{0.1}[\langle (Y_\ell^{(t)})^2 \rangle], \quad v_{X,\ell,t} = \mathcal{E}_{0.1}[\langle (X_\ell^{(t)})^2 \rangle], \quad c_{YX,\ell,t} = \mathcal{E}_{0.1}[\langle Y_\ell^{(t)} X_\ell^{(t)} \rangle], \quad (21)$$

$$\beta_{\ell,t} = \frac{c_{YX,\ell,t}}{v_{X,\ell,t} + 10^{-3}}, \quad v_{\text{res},\ell,t} = \max\{v_{Y,\ell,t} - \beta_{\ell,t} c_{YX,\ell,t}, 10^{-12}\}, \quad (22)$$

$$\phi_{\ell,t} = \text{clip}_{[0.25, 4.0]} \left(\frac{v_{Y,\ell,t}}{v_{\text{res},\ell,t}} \right), \quad \hat{\nabla}^{5F} = r_{\ell,t}^{4F} \phi_{\ell,t} \hat{\nabla}^{3F}. \quad (23)$$

202 If $v_{X,\ell,t} \leq 10^{-12}$, the implementation sets $\phi_{\ell,t} = 1$. These factors are empirical preconditioners. They are neither
 203 needed for the exact factorization nor extra task-error channels. Under exact transported error, the three-factor
 204 rule recovers the exact dendritic parameter gradient in the stated steady-state model. This separates feedback
 205 approximation from the validity of the local eligibility term; agreement of trained endpoints with backpropagation
 206 is tested empirically rather than assumed from the identity.

207 2.2 Feedback fields

208 For neuron u with N_u routed dendritic compartments, indexed $n = 1, \dots, N_u$ while the soma retains index $n = 0$, let
 209 $\boldsymbol{\delta}_u^V = (\delta_{1,u}^V, \dots, \delta_{N_u,u}^V)^\top$ collect the exact compartment errors and let $\widehat{\boldsymbol{\delta}}_u^V$ collect the values available to the local rule.
 210 Let $\mathbf{c}_u = (c_{u,1}, \dots, c_{u,K})^\top$ contain the signed values carried by K feedback channels. Let $A_u \in \mathbb{R}^{N_u \times K}$ be their fixed
 211 compartmental address matrix: column k is the spatial field produced by unit amplitude on channel k , and $(A_u)_{nk}$ is
 212 its delivery weight at compartment n . Linear superposition of the channel fields gives

$$\widehat{\boldsymbol{\delta}}_u^V = A_u \mathbf{c}_u, \quad \widehat{\delta}_{n,u}^V = \sum_{k=1}^K (A_u)_{nk} c_{u,k}. \quad (24)$$

213 Equation 24 specifies the allowed feedback routes; it is a modeling assumption separate from the membrane current
 214 balance. The address matrix specifies where signals can be delivered; the coefficient vector specifies the values
 215 delivered on the current example. The model is exact for $A_u = I$ and $\mathbf{c}_u = \boldsymbol{\delta}_u^V$. It becomes a neuron-wide broadcast
 216 for $K = 1$ and $A_u = \mathbf{1}_{N_u}$. A binary subtree column is one on descendants of its branch point and zero elsewhere.
 217 The equation leaves coefficient estimation open: projection analyses use best-fitting oracle coefficients to measure
 218 representational capacity, whereas learning experiments use the feedback rule specified for that task.

219 Here K counts fixed spatial delivery profiles, not an intrinsic number of independent external errors. Conditional
 220 on the forward state, a directed single-output tree has the scalar somatic error multiplied by its state-dependent path
 221 profile (Eq. 10). Exact path transport uses that scalar together with the sample-specific derivatives and is tested
 222 as a separate transport resource. A context gate can likewise reside in the local eligibility. The dimension of a
 223 field ensemble therefore measures the demand on a fixed dictionary across states; it does not exclude scalar error
 224 delivery combined with additional local transport or gating.

225 Table S14 defines the feedback objects without conflating bandwidth with routing. The strict scalar field used for
 226 the plotted MNIST ladder applies the scalar reduction defined below at every local-rule stage. The matched-width
 227 scalar-fallback field reuses a somatic vector only at a stage of the same width; at wider stages it uses the same
 228 robust scalar reduction as the strict scalar condition. For an N -dimensional somatic error vector $\boldsymbol{\delta}_b$ in example b ,
 229 the code first computes $\mu_b = N^{-1} \sum_u \delta_{b,u}$. With $\varepsilon = 10^{-6}$,

$$s_b = \begin{cases} \mu_b, & |\mu_b| > \varepsilon, \\ \delta_{b,u^*}, & u^* = \arg \max_u |\delta_{b,u}|, \quad |\mu_b| \leq \varepsilon. \end{cases} \quad (25)$$

230 For a one-dimensional error, the coordinate is retained. Selecting the largest-magnitude coordinate when the mean
 231 nearly cancels prevents the reduction from producing a near-zero teaching signal. The replacement count was not
 232 recorded in the initial runs and was therefore measured directly in the same-seed computational rerun. The matched-
 233 width field does not repeat each neuron’s coordinate over that neuron’s descendants. The neuron-specific feedback
 234 field does exactly that, using one communicated coordinate per neuron and the tree as a fixed coordinate-to-arbor
 235 map. Exact path transport supplies the entire path derivative and is an information upper bound.

236 In the regular [3, 3] networks, the matched-width scalar-fallback field retained the 128-dimensional somatic
 237 vector only at the 128-wide stage. It reduced the field to one scalar at the 384- and 1,152-wide stages, so all
 238 2,408,448 candidate feedforward synaptic parameters (92,160 active contacts) received strict scalar feedback.
 239 The exception comprised 384 final branch-to-soma couplings and 256 somatic gain and threshold parameters:
 240 640 of 2,413,312 core parameters (0.027%). We therefore trained a separate 15-seed strict-scalar condition in
 241 each architecture and use it, rather than matched-width scalar-fallback feedback, for the scalar rung in the main

figure. Despite its small parameter count, this near-somatic exception affected learning. Strict scalar reduced mean accuracy relative to matched-width scalar-fallback feedback by 4.859 percentage points in shunting networks (95% paired-seed interval, 3.994–5.669; fallback higher in 15/15 seeds) and by 2.274 points in raw-additive networks (1.796–2.699; fallback higher in 14/15 seeds, one tie). Both intervals lay well outside the prespecified ± 0.5 -point practical-equivalence margin. These results show that a small near-somatic parameter group can exert disproportionate control over learning; they do not imply that the remaining neurons collapse under strict scalar feedback, because their local eligibility traces remain distinct. Table S16 reports this paired comparison.

2.3 Conditional gradient alignment

For one tree and one example, suppose the correct somatic error is available but path gain is omitted. Write local eligibility as e_i and let $n(i)$ be the host compartment. Then

$$g_i^{\text{exact}} = e_i \tilde{\alpha}_{n(i)} \delta_0^V, \quad g_i^{\text{shared}} = e_i \delta_0^V. \quad (26)$$

Let g^{exact} and g^{shared} collect these synaptic components, and let CV_w denote the weighted coefficient of variation defined below. For nonnegative path gains and nonzero gradient vectors,

$$\cos(g^{\text{shared}}, g^{\text{exact}}) = \frac{1}{\sqrt{1 + \text{CV}_w(\tilde{\alpha})^2}}, \quad w_i = (e_i \delta_0^V)^2. \quad (27)$$

Here

$$\bar{\alpha}_w = \frac{\sum_i w_i \tilde{\alpha}_{n(i)}}{\sum_i w_i}, \quad \text{CV}_w(\tilde{\alpha})^2 = \frac{\sum_i w_i [\tilde{\alpha}_{n(i)} - \bar{\alpha}_w]^2}{\bar{\alpha}_w^2 \sum_i w_i}, \quad (28)$$

so CV_w is the eligibility-energy-weighted coefficient of variation of the path gain. This identity explains why path-gain dispersion can matter after the correct somatic coordinate has been supplied. It does not solve the separate problem of supplying the correct coordinate across network layers.

S3 Topology-matched credit capacity

A restricted field is useful only if its spatial supports approximate the required compartment-level credit. We therefore ask how much of that field can be represented by K ancestry-defined routes, and how this compares with a point-neuron partition at the same bandwidth.

3.1 Ancestry dictionary

Let A_T be the ancestry matrix of tree T , with one row per synaptic coordinate and one column per candidate route. Let $\boldsymbol{\beta}$ contain the state-dependent gains of these routes. A low-bandwidth teaching coefficient vector $\mathbf{c} \in \mathbb{R}^K$ generates a synaptic field

$$\hat{\mathbf{g}} = \Phi_{T,K} \mathbf{c}, \quad \Phi_{T,K} = D_e A_T \text{diag}(\boldsymbol{\beta}) S_K, \quad (29)$$

where the diagonal matrix D_e holds local eligibility or anatomical weights and S_K assigns the K communicated channels to route sites. In this dictionary model, topology fixes binary support and $\boldsymbol{\beta}$ rescales route columns. Nonzero column rescaling preserves their span at a fixed selected route set, although it can affect selection, conditioning and coefficient dynamics. The focal cable interventions studied below are different: a physical shunt changes spatial transfer and local eligibility, and need not be a diagonal gain on these fixed columns. The compartment-level matrix A_u in Eq. 24 is the general address map. Here A_T is its anatomy-specific lift to synapse coordinates, and $\Phi_{T,K}$ is the resulting synaptic-update dictionary after eligibility, route gain and channel selection are included.

3.2 Address dimension and the point-neuron resource comparison

Separate the synapse-local eligibility from the task-dependent target credit field and let $\mathbf{t}_u \in \mathbb{R}^{N_u}$ denote the latter for neuron u . For this comparison, Φ acts on the chosen task-credit coordinates; eligibility-weighted update dictionaries use the same projection construction in their own coordinates. Let W be a positive diagonal coordinate-weight matrix, and let $\dagger$ denote the Moore–Penrose pseudoinverse. Define

$$\|\mathbf{v}\|_W^2 = \mathbf{v}^\top W \mathbf{v}, \quad P_{\Phi, W} = \Phi(\Phi^\top W \Phi)^\dagger \Phi^\top W. \quad (30)$$

This is the W -orthogonal projector onto $\text{col}(\Phi)$. Equivalently, with $\bar{\Phi} = W^{1/2} \Phi$, $\bar{P}_\Phi = \bar{\Phi} \bar{\Phi}^\dagger = W^{1/2} P_{\Phi, W} W^{-1/2}$ is the ordinary Euclidean projector in the transformed coordinates. For a prespecified nested family of dictionaries $\Phi_{u, K}$, a residual tolerance $0 \leq \varepsilon < 1$ and positive expected target energy, define

$$d_{\varepsilon, u} = \min \left\{ K : \frac{\mathbb{E} \|P_{\Phi_{u, K}, W} \mathbf{t}_u\|_W^2}{\mathbb{E} \|\mathbf{t}_u\|_W^2} \geq 1 - \varepsilon \right\}. \quad (31)$$

We set $d_{\varepsilon, u} = +\infty$ if no tested dictionary attains the threshold. This definition concerns the task-dependent coefficient after eligibility has been factored out; heterogeneous presynaptic activity alone is therefore not counted as an extra teaching coordinate. It is conditional on the model, state ensemble, coordinate metric and dictionary family. It specifies a resolution at a stated residual tolerance, rather than an intrinsic task complexity or a unique optimal morphology.

For any fixed $\Phi \in \mathbb{R}^{N_u \times K}$, all fields generated as $\Phi \mathbf{c}$ lie in $\text{col}(\Phi)$ and hence occupy at most $\text{rank}(\Phi) \leq K$ dimensions. Conversely, the weighted least-squares field in that space is $P_{\Phi, W} \mathbf{t}_u$, and its expected weighted residual energy is

$$\mathbb{E} \|(I - P_{\Phi, W}) \mathbf{t}_u\|_W^2 = \text{tr}[(I - \bar{P}_\Phi) \bar{\Sigma}_t], \quad \bar{\Sigma}_t = \mathbb{E}[W^{1/2} \mathbf{t}_u \mathbf{t}_u^\top W^{1/2}]. \quad (32)$$

Equations 31–32 prove the fixed-dictionary dimension statement used in the main text; $\bar{\Sigma}_t$ is an uncentered second moment. The ordinary point-neuron local rule has $\Phi = \mathbf{1}$ and therefore one task-dependent coefficient per neuron. A K -route dendritic dictionary can use up to K structured fields inside that neuron. A point-neuron network is not mathematically forbidden from matching them, but doing so requires K separately routed teaching coefficients, K gated parameter groups, or an equivalent expansion into additional units. Accordingly, the comparison is about where address resources are represented, not about an absolute expressive separation between all possible dendritic and point-neuron networks.

3.3 Field capture

We now measure how much of a target field lies in a dictionary’s span. Let $\mathbf{g}$ denote that field in the coordinates under consideration, and use the positive weighting from Eq. 30. Set $\bar{\mathbf{g}} = W^{1/2} \mathbf{g}$, $\bar{\Phi}_{T, K} = W^{1/2} \Phi_{T, K}$, and let $\bar{P}_\Phi$ be the ordinary Euclidean orthogonal projector onto the columns of $\bar{\Phi}_{T, K}$. For a nonzero target field, capture is

$$\mathcal{C}_K(\mathbf{g}; \Phi, W) = \frac{\|\bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2} = 1 - \frac{\|\bar{\mathbf{g}} - \bar{P}_\Phi \bar{\mathbf{g}}\|_2^2}{\|\bar{\mathbf{g}}\|_2^2}. \quad (33)$$

The unweighted case has $W = I$. The MICrONS structural analysis uses summed mapped excitatory synaptic area on the diagonal of W . For the transformed uncentered second moment $\bar{\Sigma}_g = \mathbb{E}[\bar{\mathbf{g}} \bar{\mathbf{g}}^\top]$, aggregate energy capture for a fixed projector is

$$\mathcal{C}_K(\bar{\Sigma}_g; \Phi, W) = \frac{\text{tr}(\bar{P}_\Phi \bar{\Sigma}_g)}{\text{tr}(\bar{\Sigma}_g)}. \quad (34)$$

This is the ratio of expected projected energy to expected total energy. It is generally different from the mean per-field fraction $\mathbb{E}[\mathcal{C}_K(\mathbf{g}; \Phi, W)]$, which weights nonzero fields equally. The two coincide for a fixed field norm, or when the second moment is formed after normalizing each field. The MNIST dictionary gallery averages per-field

fractions; the cable-response operator analysis aggregates energy over its columns. The second moment equals a centered covariance only for mean-zero fields. Equation 34 states the topology–task matching condition: a sparse anatomical basis is useful only to the extent that task credit lies in its span.

Proposition 1 (Ky–Fan credit-capacity bound). *Let $\bar{\Sigma}_g \in \mathbb{R}^{p \times p}$ be positive semidefinite, with eigenvalues satisfying $\lambda_1 \geq \dots \geq \lambda_p \geq 0$ and $\text{tr}(\bar{\Sigma}_g) > 0$. For every rank- K orthogonal projector P ,*

$$\frac{\text{tr}(P\bar{\Sigma}_g)}{\text{tr}(\bar{\Sigma}_g)} \leq \frac{\sum_{j=1}^K \lambda_j}{\sum_{j=1}^p \lambda_j}. \quad (35)$$

Equality is attained by any rank- K projector whose range is a Ky–Fan-maximizing invariant subspace: it contains every eigenspace with eigenvalue strictly greater than λ_K and may choose the remaining directions within the eigenspace at λ_K . The difference between the right side and anatomical capture is the rank-matched spectral regret.

Proof. Ky–Fan’s maximum principle [2] gives $\max_{\substack{P=P^T=P^2 \\ \text{rank}(P)=K}} \text{tr}(P\bar{\Sigma}_g) = \sum_{j=1}^K \lambda_j$. Division by the fixed total spectral energy proves Eq. 35. $\square$

The spectral screen interpolates between two task-field second moments C_0 and C_1 with equal trace, using $C(\rho) = (1 - \rho)C_0 + \rho C_1$ for $0 \leq \rho \leq 1$. Thus, for two fixed projectors P and Q , the normalized capture difference has the sign of

$$\Delta(\rho) = \text{tr}[(P - Q)C_0] + \rho \text{tr}[(P - Q)(C_1 - C_0)], \quad (36)$$

so an interior crossover is determined exactly by its two endpoints. This affine result distinguishes alignment from rank: increasing K raises the oracle bound, whereas increasing ρ determines whether the anatomical subspace approaches it.

3.4 One-step descent bound

When the target field is a true loss gradient, capture also determines a one-step descent guarantee in the corresponding metric. Let $\mathbf{w}$ be the parameter vector and $\mathbf{g} = \nabla_{\mathbf{w}} \mathcal{L}$. To express the weighted calculation as Euclidean descent, define

$$\bar{\mathbf{w}} = W^{-1/2} \mathbf{w}, \quad \mathbf{w} = W^{1/2} \bar{\mathbf{w}}, \quad (37)$$

and define $\bar{\mathcal{L}}(\bar{\mathbf{w}}) = \mathcal{L}(W^{1/2} \bar{\mathbf{w}})$. Then $\nabla_{\bar{\mathbf{w}}} \bar{\mathcal{L}} = W^{1/2} \mathbf{g} = \bar{\mathbf{g}}$. If $\bar{\mathcal{L}}$ has L_W -Lipschitz gradients and η is the step size, the descent lemma gives

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_{\Phi} \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \bar{\mathbf{g}}^T \bar{P}_{\Phi} \bar{\mathbf{g}} + \frac{L_W \eta^2}{2} \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2. \quad (38)$$

Orthogonal projection implies $\bar{\mathbf{g}}^T \bar{P}_{\Phi} \bar{\mathbf{g}} = \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2$, hence

$$\bar{\mathcal{L}}(\bar{\mathbf{w}} - \eta \bar{P}_{\Phi} \bar{\mathbf{g}}) \leq \bar{\mathcal{L}}(\bar{\mathbf{w}}) - \eta \left(1 - \frac{L_W \eta}{2}\right) \|\bar{P}_{\Phi} \bar{\mathbf{g}}\|_2^2. \quad (39)$$

At $\eta = 1/L_W$, the ratio between the projected- and full-gradient guarantees is Eq. 33. In the original coordinates, the projected step is

$$\mathbf{w}^+ = \mathbf{w} - \eta W^{1/2} \bar{P}_{\Phi} W^{1/2} \mathbf{g}, \quad (40)$$

which is a particular preconditioned update rather than an ordinary Euclidean gradient step in $\mathbf{w}$. The implementation was checked on 10,000 random positive-definite quadratic losses: there were no bound violations, minimum numerical slack was 2.67×10^{-7} , and the capture identity agreed to 1.89×10^{-15} . This numerical test checks the Euclidean projector algebra, which also applies after an explicit coordinate transformation; it does not validate synaptic-area weighting as a physical parameter metric. The result is a one-step guarantee only when the tested field is an exact gradient in the stated coordinates. It is not a convergence or generalization theorem and does not show that a biological circuit can infer the oracle coefficients.

S4 Signal–noise theory of restricted credit routing

Capture alone favors increasing route resolution, but a broader operator also admits more stochastic error. We next formalize this signal–noise trade-off and ask when restricting the communicated credit improves expected progress.

4.1 Expected descent from routed-update moments

Let $\mathcal{L} : \mathbb{R}^p \rightarrow \mathbb{R}$ have L -Lipschitz gradient, with $L > 0$, on a neighborhood containing every update segment under consideration. At a fixed parameter state, write $\mathbf{g} = \nabla \mathcal{L}(\mathbf{w})$ and let $\mathbf{d}$ be a routed update estimator with finite second moment. Define $\mathbf{m} = \mathbb{E}[\mathbf{d}]$ and $V = \mathbb{E}[(\mathbf{d} - \mathbf{m})(\mathbf{d} - \mathbf{m})^\top]$. The expectation is over the stated sampling or feedback noise, conditional on $\mathbf{w}$. The update is

$$\mathbf{w}^+ = \mathbf{w} - \eta \mathbf{d}. \quad (41)$$

Proposition 2 (Routed-update moment bound). *For Eq. 41,*

$$\mathbb{E}\mathcal{L}(\mathbf{w}^+) \leq \mathcal{L}(\mathbf{w}) - \eta \mathbf{g}^\top \mathbf{m} + \frac{L\eta^2}{2} [\|\mathbf{m}\|_2^2 + \text{tr}(V)]. \quad (42)$$

Write $a = \mathbf{g}^\top \mathbf{m}$ and $b = \|\mathbf{m}\|_2^2 + \text{tr}(V)$. For $b > 0$, minimizing the bound over nonnegative steps gives $\eta^* = [a]_+ / (Lb)$.

Provided the stated smoothness condition holds along the update segments at this step, the certified decrease is

$$U = \frac{[\mathbf{g}^\top \mathbf{m}]_+^2}{2L\{\|\mathbf{m}\|_2^2 + \text{tr}(V)\}}, \quad [a]_+ = \max(a, 0). \quad (43)$$

For $b = 0$, $\mathbf{d} = 0$ almost surely and we set $U = 0$. A zero score means that this bound certifies no positive one-step decrease at the evaluated state; it does not establish failure of every later update or learning rule.

Proof. Apply the descent lemma [3] to $-\eta \mathbf{d}$, take expectations and use $\mathbb{E}\|\mathbf{d}\|_2^2 = \|\mathbf{m}\|_2^2 + \text{tr}(V)$. Minimizing the resulting scalar quadratic over $\eta \geq 0$ gives the claim. $\square$

A fixed linear routing operator gives a useful special case. If $\mathbf{d} = M(\mathbf{g} + \boldsymbol{\xi})$ with deterministic $M \in \mathbb{R}^{p \times p}$, $\mathbb{E}\boldsymbol{\xi} = 0$ and $\text{Cov}(\boldsymbol{\xi}) = \Sigma$, then $\mathbf{m} = M\mathbf{g}$ and $V = M\Sigma M^\top$. Equations 42–43 then recover the credit-operator expressions, denoted $U(M)$ below. The identity, orthogonal projectors and fixed block-gain screens use this specialization.

The more general moment form is needed for sample-dependent eligibilities and gates. In the context-gated task, changing an unselected distractor leaves the exact parameter gradient unchanged but changes a shared-feedback update through that distractor’s eligibility. A fixed matrix of the exact parameter gradient therefore cannot represent every routed update. The output-error-to-update map remains linear conditional on the complete forward state; that conditional map is a different object. The factorial reanalysis measures routed means and second moments directly and does not require a fixed map from exact parameter gradients to routed updates.

At fixed infinitesimal update norm, $-\mathbf{g}$ is the steepest Euclidean descent direction by Cauchy–Schwarz. Consequently, no routed local operator provides a uniform first-order improvement over exact full-batch backpropagation on the same parameters and objective. Equation 43 instead compares restricted or noisy estimators and exposes when rejecting stochastic energy compensates for discarded signal.

Corollary 1 (Projected stochastic crossover). *Consider the identity-curvature quadratic $\mathcal{L}(\mathbf{w}) = \frac{1}{2}\|\mathbf{w} - \mathbf{w}^*\|_2^2$, initialize at $\mathbf{w} = 0$, set $\mathbf{g} = -\mathbf{w}^*$, use unit step size, and let P be an orthogonal projector. The expected post-update losses of the unprojected and projected unbiased gradients are*

$$\mathbb{E}\mathcal{L}[-(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} \text{tr}(\Sigma), \quad (44)$$

$$\mathbb{E}\mathcal{L}[-P(\mathbf{g} + \boldsymbol{\xi})] = \frac{1}{2} [\|(I - P)\mathbf{g}\|_2^2 + \text{tr}(P\Sigma P)]. \quad (45)$$

368 *The projected estimator is better precisely when*

$$\text{tr}[(I - P)\Sigma] > \|(I - P)\mathbf{g}\|_2^2, \quad (46)$$

369 *that is, rejected noise exceeds discarded signal. A local estimator using the same P with additional independent,*
 370 *zero-mean coefficient noise cannot achieve a lower expected loss than backpropagation followed by P .*

371 For completeness, at a common step $0 < \eta < 2$ the unprojected-minus-projected expected loss is

$$\frac{\eta}{2} \{ \eta \text{tr}[(I - P)\Sigma] - (2 - \eta) \|(I - P)\mathbf{g}\|_2^2 \}. \quad (47)$$

372 Hence projection is better exactly when $\text{tr}[(I - P)\Sigma] > [(2 - \eta)/\eta] \|(I - P)\mathbf{g}\|_2^2$; the unit-step result above is the
 373 special case $\eta = 1$. Smaller common steps therefore require proportionally more rejected noise to compensate for
 374 the discarded signal.

375 4.2 Interior optimum along the nested route ladder

376 The scalar, neuron-specific and correctly routed ancestry fields in the controlled hierarchy analyses span nested
 377 subspaces. Their orthogonal projectors form the ladder $P_{\text{scalar}} \subset P_{\text{neuron}} \subset P_{\text{ancestry}, K} \subset I$. Along such a ladder
 378 the capture of Eq. 33 is monotone by construction, but the guaranteed utility is not. For an orthogonal projector
 379 with retained signal energy $S = \mathbf{g}^\top P \mathbf{g} = \|P\mathbf{g}\|_2^2$ and admitted noise $N = \text{tr}(P\Sigma P)$, the optimal-step utility of the
 380 credit-operator bound is

$$U(S, N) = \frac{S^2}{2L(S + N)}. \quad (48)$$

381 Let one rung up the ladder add marginal signal $\Delta S \geq 0$ and marginal noise $\Delta N \geq 0$. Substituting into Eq. 48 and
 382 clearing denominators gives the exact marginal condition

$$U(S + \Delta S, N + \Delta N) > U(S, N) \iff \Delta S [(S + N)(2S + \Delta S) - S^2] > S^2 \Delta N. \quad (49)$$

383 The bracket is positive, so for small marginal additions the condition reduces to a threshold on the marginal
 384 signal-to-noise ratio, $\Delta S/\Delta N > S/(S + 2N)$: the utility keeps rising while each added route scale contributes
 385 proportionally more signal than the running threshold, and falls once the added scales admit mostly noise. When
 386 the signal spectrum decays faster than the admitted noise across ladder scales, U can attain an interior maximum.
 387 On a finite tested ladder, the marginal condition must become unfavorable before the last rung; spectral decay alone
 388 does not guarantee that crossing or a unique optimum. The displayed marginal condition assumes $S > 0$; if $S = 0$,
 389 a rung with positive retained signal has positive utility whenever its total second moment is finite and nonzero.
 390 Equations 48 and 49, together with the interior-maximum construction, are verified numerically in the released
 391 test suite (`tests/test_laminar_optimality.py`). Random, deranged and independently gained route families
 392 elsewhere in the paper need not lie on this same nested ladder.

393 Two prespecified experiments locate this optimum empirically (Supplementary Fig. S16; analysis code is
 394 included in the release). In the hierarchy-depth routing experiment the initialization utility of the aligned tree,
 395 computed per seed and task depth, placed its argmax at the trained loss minimum in 135 of 200 seed–task pairs,
 396 and the group-mean argmax matched the trained optimum exactly at $H_c = 1, 2, 3$; at the full-rank boundary $H_c = 4$
 397 the one-step bound was conservative and favoured $D_r = 2$, reflecting its single-step horizon.

398 The trained subtree factorial provides a second, retrospective test. Its full-batch moment score uses the
 399 global logistic curvature bound and the same four-control comparator as the accuracy contrast: deranged, depth-
 400 interleaved, random-sparse and random-rank routes. The mean ancestry-minus-best-control utility was positive at
 401 $K = 4$, negative at $K = 1, 2$ and zero at $K = 8$. Its contrast-maximizing budget was $K = 4$ in all 20 seeds, matching
 402 the observed accuracy-contrast maximum in 13 of 20. Mean accuracy-contrast selection regret was 1.94 percentage
 403 points. Adding the hypothetical minibatch variance made the mean $K = 4$ utility contrast negative, reduced
 404 optimum agreement to 8 of 20 and gave mean regret 2.75 points. Thus the score matched to the deterministic

training recipe recovers the mean intermediate-bandwidth ancestry peak, while its seed-level selection remains imperfect. This retrospective result is distinct from the raw within-family performance optimum at $K = 8$ and from prospective morphology selection. The comparator definitions, scores and prediction errors are supplied in `source_data/review_evidence_reanalysis/`.

4.3 Laminar route families and hierarchical task second moments

The distinction between a point neuron and a dendritic tree can be stated as a constraint on the feasible credit operators at fixed rank. A point neuron addresses its synapses with the single shared column **1**: a rank-one operator with uniform support. An ancestry dictionary uses structured columns whose supports form a *laminar* family: any two supports are nested or disjoint. An unconstrained rank- K feedback matrix is the comparison class. For the balanced binary tree used in the factorial, the span of the K -route ancestry dictionary equals the span of the K coarsest vectors of the orthonormal tree-Haar basis $U = [\mathbf{h}_1, \dots, \mathbf{h}_n]$ (constant, then sibling-subtree differences, coarse to fine) for every admissible budget $K \in \{1, 2, 4, \dots, n\}$. Here n is the number of leaf coordinates. Let $P_{A,K}$ denote the orthogonal projector onto the selected ancestry span.

Proposition 3 (Laminar capture of a hierarchical second moment). *Let the task-credit second moment be $C = U\Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ along the coarse-to-fine Haar order, with $\text{tr}(C) > 0$. Then for every admissible budget K the ancestry dictionary attains the Ky–Fan bound,*

$$\frac{\text{tr}(P_{A,K}C)}{\text{tr}(C)} = \frac{\sum_{i \leq K} \lambda_i}{\text{tr}(C)} = \max_{\substack{P=P^\top=P^2 \\ \text{rank}(P)=K}} \frac{\text{tr}(PC)}{\text{tr}(C)}. \quad (50)$$

For a general second moment C with eigenpairs $(\lambda_i, \mathbf{v}_i)$ ordered by decreasing eigenvalue, the shortfall from the Ky–Fan bound decomposes exactly as

$$\sum_{i \leq K} \lambda_i - \text{tr}(P_{A,K}C) = \sum_{i \leq K} \lambda_i \|(I - P_{A,K})\mathbf{v}_i\|_2^2 - \sum_{j > K} \lambda_j \|P_{A,K}\mathbf{v}_j\|_2^2, \quad (51)$$

the leading-eigenspace energy outside the laminar span minus the trailing energy that the span retains. It is therefore bounded above by the former term. The shortfall is zero exactly when $P_{A,K}$ is Ky–Fan maximizing; when $K < n$ and $\lambda_K > \lambda_{K+1}$, this is equivalent to the ancestry span being the unique leading- K eigenspace.

Proof. The ancestry span equals the span of the K coarsest Haar vectors, which under the stated ordering are the top- K eigenvectors of C ; a rank- K orthogonal projector maximizes $\text{tr}(PC)$ exactly on a top- K eigenspace (Ky–Fan). For general C , expanding $\text{tr}(P_{A,K}C) = \sum_j \lambda_j \|P_{A,K}\mathbf{v}_j\|_2^2$ and $\|P_{A,K}\mathbf{v}_i\|_2^2 = 1 - \|(I - P_{A,K})\mathbf{v}_i\|_2^2$ for $i \leq K$ gives the stated decomposition. Ky–Fan’s equality condition, including arbitrary choices inside a degenerate cutoff eigenspace, gives the last statement. $\square$

This ordered hierarchical second moment is one sufficient class for which ancestry attains the rank- K spectral optimum. At a fixed K , the equality condition is the leading-eigenspace condition in the proposition; it does not require the entire second moment to have a Haar eigenbasis and does not identify a unique physical tree. Hierarchical organization without the stated coarse-to-fine spectral ordering is not sufficient. The spectral-alignment experiment realizes both regimes: at full alignment and $K = 4$ ancestry attained the rank-matched Ky–Fan bound to numerical precision, whereas at random alignment its advantage over sampled random routes vanished (main text). The attainment, the shortfall identity and the failure case with a fine-scale-dominant second moment are verified numerically in `tests/test_laminar_optimality.py`.

4.4 Point-neuron partition residual and route-coefficient error

Let t_i be the exact task-dependent target remaining after local eligibility has been factored out, let $w_i > 0$ be coordinate weights, and let $\pi(i)$ assign synapse i to a group that must share one teaching coefficient. The best allowed field minimizes $\sum_i w_i [t_i - c_{\pi(i)}]^2$.

443 **Proposition 4** (Shared-coefficient residual). *For every group G , the optimum is its weighted mean*

$$c_G^* = \frac{\sum_{i \in G} w_i t_i}{\sum_{i \in G} w_i}, \quad (52)$$

444 *and the irreducible residual is*

$$\mathcal{R}(\pi) = \sum_G \sum_{i \in G} w_i (t_i - c_G^*)^2. \quad (53)$$

445 *Refining a partition cannot increase $\mathcal{R}$; it decreases the residual only when an old group contains coefficient*
 446 *variation resolved by the new groups.*

447 *Proof.* Differentiate the separable weighted least-squares objective with respect to each c_G . The refinement
 448 statement follows because the old solution remains feasible when two or more refined coefficients are constrained
 449 to be equal. $\square$

450 The ordinary point-neuron rule has one group per neuron. Dendritic ancestry offers a particular nested sequence
 451 of refinements, but is helpful only if the task coefficient varies across those subtrees. An explicitly grouped point
 452 model can implement the same partition, which is why the resource-matched point controls in the main article
 453 coincide with the dendritic implementation.

454 For a general dictionary Φ and positive diagonal W , let $\mathbf{c}^* = \arg \min_{\mathbf{c}} \|\mathbf{t} - \Phi \mathbf{c}\|_W^2$ and let $\hat{\mathbf{c}}$ be the coefficients
 455 actually supplied by a local circuit. Weighted projection orthogonality yields the exact decomposition

$$\|\mathbf{t} - \Phi \hat{\mathbf{c}}\|_W^2 = \underbrace{\|\mathbf{t} - \Phi \mathbf{c}^*\|_W^2}_{\text{address / representation error}} + \underbrace{\|\Phi(\hat{\mathbf{c}} - \mathbf{c}^*)\|_W^2}_{\text{coefficient-estimation error}}. \quad (54)$$

456 If the coefficient error is unbiased with covariance Ω_c , the expected second term is $\text{tr}(\Phi^T W \Phi \Omega_c)$. The Gram
 457 matrix $G_\Phi = \Phi^T W \Phi$ therefore controls how coefficient error is expressed. Estimating $\mathbf{c}^*$ is sensitive to noise
 458 along directions with small positive Gram eigenvalues. Correlated route columns can likewise make individual
 459 coefficients poorly identifiable even when their combined field is stable. Address rank alone is not a complete
 460 learning metric.

461 4.5 Static route gains preserve span but can change conditioning

462 **Proposition 5** (Route-wise static-gain invariance). *Let D_β be an invertible diagonal matrix of fixed nonzero route*
 463 *gains. Then*

$$\text{col}(\Phi D_\beta) = \text{col}(\Phi), \quad P_{\Phi D_\beta, W} = P_{\Phi, W}. \quad (55)$$

464 *Thus route-wise static scalar gains cannot change address rank, projection residual or spectral capture. They can*
 465 *change the Gram matrix from $\Phi^T W \Phi$ to $D_\beta^T \Phi^T W \Phi D_\beta$ and hence alter conditioning and coefficient dynamics.*

466 This proposition concerns nonzero scalar rescaling of existing route columns. Spatially varying row gains,
 467 state-dependent gating and gains that reach zero can change the effective span or active support; those cases are
 468 not covered by Eq. 55. Physical shunts also change the cable Green’s function. For a somatic adjoint source on
 469 a passive tree, that change has an exact diagonal-gain representation in the baseline-weighted ancestry partition
 470 associated with the shunt site (Corollary 2). Its partition indicators are differences of nested subtree indicators. The
 471 diagonal form does not extend to an arbitrary selected overlapping dictionary, and the local synaptic driving force
 472 can still vary within each block.

473 4.6 Finite-time coefficient learning has a bias–variance crossover

474 Equal span need not give equal learning trajectories. Let $\mathbf{f}_t = \Phi \mathbf{c}_t$ be the field produced by coefficients $\mathbf{c}_t$ after
 475 update t , and let $\lambda_{\max} > 0$ be the largest eigenvalue of $\Phi \Phi^T$. Normalizing ordinary coefficient descent by this
 476 eigenvalue gives $H_\Phi = \Phi \Phi^T / \lambda_{\max}$. For normalized step size η and noisy rate-based target estimates $\mathbf{g} + \boldsymbol{\xi}_t$, with

477 $\mathbb{E}\xi_t = 0$ and $\text{Cov}(\xi_t) = \sigma^2 I/n$, where n is the effective sample size and σ^2 the single-observation noise scale, the
 478 field iteration is

$$\mathbf{f}_{t+1} = \mathbf{f}_t - \eta H_\Phi(\mathbf{f}_t - \mathbf{g} - \xi_t). \quad (56)$$

479 **Proposition 6** (Exact finite-time coefficient risk). Assume $\mathbf{f}_0 = 0$, $\mathbf{g} \in \text{col}(\Phi)$ and let $(\mu_j, \mathbf{u}_j)$ be the positive
 480 eigenpairs of H_Φ , with $\theta_j = \mathbf{u}_j^\top \mathbf{g}$. For independent isotropic noise and a stable step satisfying $|1 - \eta\mu_j| < 1$ for
 481 every positive mode, the expected population loss after T updates is

$$\mathbb{E} \frac{\|\mathbf{f}_T - \mathbf{g}\|_2^2}{2} = \frac{1}{2} \sum_j \left[(1 - \eta\mu_j)^{2T} \theta_j^2 + \frac{\sigma^2 (\eta\mu_j)^2 [1 - (1 - \eta\mu_j)^{2T}]}{n (1 - (1 - \eta\mu_j)^2)} \right]. \quad (57)$$

482 For Gram-preconditioned descent, $H_\Phi = P_\Phi$ and all positive $\mu_j = 1$; therefore every dictionary with the same
 483 projector has the same field trajectory under common observations.

484 *Proof.* Project Eq. 56 onto each eigenvector. The deterministic error contracts by $1 - \eta\mu_j$ per step. Independent
 485 innovations have variance $(\eta\mu_j)^2 \sigma^2/n$ and accumulate as a finite geometric series. Summing half the modal second
 486 moments gives Eq. 57. Gram preconditioning replaces $\Phi^\top \Phi$ by its pseudoinverse on the positive subspace, so the
 487 induced field operator is P_Φ . $\square$

488 Equation 57 makes the low-data boundary explicit. Small μ_j retain more finite-time bias but admit less estimator
 489 variance. For two operators with the same span, let ΔB be the difference in the bias term and ΔV the difference in the
 490 coefficient of the noise term. Their risk difference is $\Delta B + (\sigma^2/n)\Delta V$. If $\Delta B > 0$ and $\Delta V < 0$, the ordering reverses
 491 at $n^* = -\sigma^2 \Delta V / \Delta B$: slow coordinates can help below n^* by implicit noise filtering, whereas well-conditioned
 492 coordinates help above it by reducing bias. This is a finite-horizon regularization effect, not new address capacity
 493 and not an unconditional advantage of ill-conditioning.

494 4.7 Branch reliability shrinkage

495 Partition the parameter coordinates into disjoint branch blocks b . Let $\mathbf{g}_b$ be the clean gradient in block b , Σ_b its
 496 noise covariance, and I_b the identity on that block. Define signal and noise energies $S_b = \|\mathbf{g}_b\|_2^2$ and $N_b = \text{tr}(\Sigma_b)$.
 497 With attenuation a_b in each block, $M = \text{blockdiag}(a_b I_b)$ and a fixed step $\eta = c/L$ for $c > 0$, Eq. 42 guarantees a
 498 loss decrease of

$$\mathcal{G}_c(\mathbf{a}) = \frac{c}{L} \sum_b \left[a_b S_b - \frac{c a_b^2}{2} (S_b + N_b) \right]. \quad (58)$$

499 **Proposition 7** (Fixed-step reliability-optimal attenuation). For each block with $S_b + N_b > 0$, Eq. 58 is maximized
 500 independently at

$$a_b^* = \min \left\{ 1, \frac{S_b}{c(S_b + N_b)} \right\}, \quad (59)$$

501 for a physical attenuator constrained to $0 \leq a_b \leq 1$. A block with $S_b = N_b = 0$ contributes zero for every gain
 502 and can be omitted from the ratios below. If at least one block has positive energy, the best common gain is
 503 $a_g^* = \min\{1, \sum_b S_b / [c \sum_b (S_b + N_b)]\}$, and the branch-specific guarantee is no smaller because its feasible set
 504 contains every common gain. At $c = 1$, Eq. 59 reduces to the reliability factor $S_b / (S_b + N_b)$; in that case
 505 $\mathcal{G}_{\text{branch}} = (2L)^{-1} \sum_b S_b^2 / (S_b + N_b)$ and $\mathcal{G}_{\text{global}} = (2L)^{-1} (\sum_b S_b)^2 / \sum_b (S_b + N_b)$, with equality when reliability is
 506 constant across positive-energy blocks. If all blocks have zero energy, every common gain is optimal and both
 507 guarantees are zero.

508 *Proof.* Differentiate each concave quadratic in Eq. 58 and project its unconstrained maximizer onto $[0, 1]$. Restricting
 509 all coordinates to a common value gives a_g^* . The branchwise optimum cannot be worse than this restricted solution.
 510 For $c = 1$, the closed-form comparison also follows from Cauchy–Schwarz applied to $S_b / \sqrt{S_b + N_b}$ and $\sqrt{S_b + N_b}$.
 511 $\square$

512 This optimum assumes access to S_b and N_b . A point implementation provided with the same branch partition
 513 and multiplicative factors produces the same update. Its biological relevance depends on whether local conductance
 514 can estimate or realize those factors.

4.8 Passive focal shunts attenuate a positive adjoint

Proposition 8 (No sign reversal in a passive grounded tree). *Let G be a symmetric positive-definite nonsingular M -matrix, with nonpositive off-diagonal entries. For a nonnegative adjoint source $\mathbf{s}$, define $\mathbf{q} = G^{-1}\mathbf{s}$. Adding a nonnegative shunt κ at compartment k gives $G' = G + \kappa \mathbf{e}_k \mathbf{e}_k^T$ and*

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa q_k}{1 + \kappa(G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (60)$$

Both G^{-1} and $(G')^{-1}$ are elementwise nonnegative, hence $0 \leq \mathbf{q}' \leq \mathbf{q}$ componentwise. A focal passive shunt attenuates but cannot reverse a positive adjoint field.

Proof. Equation 60 is the Sherman–Morrison identity. A nonsingular M -matrix has a nonnegative inverse; adding a nonnegative diagonal term preserves the M -matrix property [2]. The subtracted vector is nonnegative, and $\mathbf{q}' = (G')^{-1}\mathbf{s} \geq 0$. $\square$

This ordering does not by itself guarantee descendant selectivity relative to a matched current injection; that spatial comparison depends on the Green’s function, focal site and electrotonic regime and is evaluated numerically in the focal experiments.

Corollary 2 (A focal shunt applies gains to its ancestry partition). *Assume additionally that the nonzero off-diagonal entries of G form a connected tree rooted at the soma s , and that the adjoint source is $\ell \mathbf{e}_s$ for a fixed scalar somatic loss derivative ℓ . Write $R = G^{-1}$ and $\mathbf{h} = R \mathbf{e}_s > 0$. For a shunt at k , let $\mathcal{A}_k$ be the vertices on the path from s to k , including both endpoints. For each $a \in \mathcal{A}_k$, define the disjoint block $P_a = \{i : \text{LCA}(i, k) = a\}$, where LCA is the lowest common ancestor. Then*

$$q'_i = \eta_a q_i \quad (i \in P_a), \quad \eta_a = 1 - \frac{\kappa R_{ks} R_{ak}}{(1 + \kappa R_{kk}) R_{as}}, \quad \eta_k = \frac{1}{1 + \kappa R_{kk}}. \quad (61)$$

For finite $\kappa \geq 0$, every gain satisfies $0 < \eta_a \leq 1$. Let B_k have columns $\mathbf{1}_{P_a}$ and define the baseline-weighted partition dictionary $D_k = \text{diag}(\mathbf{h}) B_k$. With $\Gamma_k = \text{diag}(\eta_a)$,

$$\mathbf{q} = \ell D_k \mathbf{1}, \quad \mathbf{q}' = \ell D_k \Gamma_k \mathbf{1}. \quad (62)$$

Thus the adjoint effect is exactly a diagonal gain in this partition basis.

Proof. Removing a vertex a separates a tree into components. If a lies on the path from i to j , solving the source-free equations in the component containing i , with boundary voltage at a , gives $R_{ij} = R_{ia} R_{aj} / R_{aa}$. For $a = \text{LCA}(i, k)$, apply this identity to (i, k) and (i, s) ; it also holds when an endpoint equals a . Hence $R_{ik} / R_{is} = R_{ak} / R_{as}$. Substitution into Eq. 60 gives Eq. 61. Positivity of both tree inverse matrices gives positive ratios, and the preceding proposition bounds them by one. The formula also holds when $\ell = 0$, when both adjoints vanish. The disjoint blocks partition all vertices, yielding Eq. 62. $\square$

The block P_k is the whole subtree below the shunt. For an ancestor a above k , P_a contains a and its sister subtrees off the path to k . If c is the next vertex toward k , its indicator is $\mathbf{1}_{P_a} = \mathbf{1}_{\text{Desc}(a)} - \mathbf{1}_{\text{Desc}(c)}$, whereas $\mathbf{1}_{P_k} = \mathbf{1}_{\text{Desc}(k)}$. The gain field therefore lies in the span of nested ancestry indicators. The diagonal form requires the disjoint partition basis; an arbitrary selected set of overlapping routes need not have diagonal gains. Even for the ancestry chain, a change from partition to nested-subtree coordinates generally makes the gain matrix non-diagonal, while preserving that full span.

For the synaptic gradient, the local driving force remains a separate factor: $\gamma'_i = \eta_a q_i (E_E - V'_i)$ for $i \in P_a$. Its within-block variation is therefore not constrained to a single gain relative to $\gamma_i = q_i (E_E - V_i)$. The corollary describes somatic-source adjoint transport on a passive tree, rather than the complete synaptic update, a distributed-error source or a nonlinear finite-dose response. It connects the morphology-defined dictionary to a conductance gain without implying a learning advantage.

552 **Corollary 3** (Spectral size of a focal passive perturbation). *Under the preceding assumptions, let $\mathcal{R} = G^{-1}$ and*
553 *$\mathcal{R}' = (G + \kappa \mathbf{e}_k \mathbf{e}_k^\top)^{-1}$. The inverse-operator change has rank at most one, with rank one for $\kappa > 0$, and*

$$\|\mathcal{R}' - \mathcal{R}\|_2 = \frac{\kappa \|\mathcal{R} \mathbf{e}_k\|_2^2}{1 + \kappa \mathcal{R}_{kk}} \leq \frac{\kappa \|\mathcal{R}\|_2^2}{1 + \kappa / \lambda_{\max}(G)}. \quad (63)$$

554 *Consequently, for any adjoint source $\mathbf{s}$, $\|(\mathcal{R}' - \mathcal{R})\mathbf{s}\|_2 \leq \|\mathcal{R}' - \mathcal{R}\|_2 \|\mathbf{s}\|_2$.*

555 *Proof.* Sherman–Morrison gives $\mathcal{R}' - \mathcal{R} = -\kappa(\mathcal{R} \mathbf{e}_k)(\mathcal{R} \mathbf{e}_k)^\top / (1 + \kappa \mathcal{R}_{kk})$. The norm of this symmetric outer-
556 product matrix is its scalar coefficient times $\|\mathcal{R} \mathbf{e}_k\|_2^2$. Use $\|\mathcal{R} \mathbf{e}_k\|_2 \leq \|\mathcal{R}\|_2$ and $\mathcal{R}_{kk} \geq \lambda_{\min}(\mathcal{R}) = 1/\lambda_{\max}(G)$ for
557 the bound. $\square$

558 Equation 63 provides only a global magnitude ceiling on the total adjoint change. It does not predict the
559 localization contrast or the observed crossover with membrane conductance. Those spatial effects depend on the
560 Green’s-function profile and electrotonic length constant of the cable.

561 4.9 Why depth can help or hurt

562 Let $P_1 \preceq P_2 \preceq \dots \preceq P_{D_r}$ be nested tree-route projectors. Increasing depth reduces the address bias $\|(I - P_{D_r})\mathbf{g}\|^2$ but
563 increases admitted noise $\text{tr}(P_{D_r} \Sigma P_{D_r})$ whenever the new modes carry stochastic energy. These competing terms
564 can therefore produce an interior optimum when task signal is concentrated above one hierarchy level and noise
565 continues into finer levels; its location depends on their magnitudes, as specified by Eq. 46. This is the controlled
566 role isolated by the hierarchy-depth routing experiment. It is distinct from nonlinear forward composition, which
567 can make deeper dendritic subunits representationally useful even when the backward address span is unchanged.

568 Depth can also impose a transport cost. In an idealized ensemble with equal local eligibility, let the route gain
569 be $A = \prod_{\ell=1}^{D_p} G_\ell$ and assume independent Gaussian log gains, $\log G_\ell \sim \mathcal{N}(\mu_\ell, \sigma_\ell^2)$. The cosine between a shared
570 coefficient and the exact gain-weighted field in the large-route limit is

$$\frac{\mathbb{E}A}{\sqrt{\mathbb{E}A^2}} = \exp\left(-\frac{1}{2} \sum_{\ell=1}^{D_p} \sigma_\ell^2\right). \quad (64)$$

571 Additional levels can therefore add address resolution while simultaneously making uncompensated shared feedback
572 less aligned. The balance depends on task hierarchy, gain variability, noise and the available coefficient estimator;
573 “more depth” has no unconditional sign.

574 4.10 Controlled signal–noise experiment

575 The signal–noise experiment used 16 Euclidean coordinates represented in the orthonormal Haar basis of a balanced
576 binary tree, ordered from the global mode (level 0) to individual sibling contrasts (level 4). The configuration,
577 50 analysis seeds 7000–7049, two preliminary validation seeds and numerical tolerances were specified before
578 the analysis outcomes were inspected. The preliminary runs verified basis orthogonality, projector idempotence,
579 static-gain span invariance and exact agreement with the explicit point-gain control. All planned observations were
580 completed: 3,750 spectral, 1,800 depth, 3,750 projection and 1,500 reliability rows.

581 In the spectral screen, the task-gradient second moment assigned total energy (0.15, 0.35, 0.30, 0.20, 0) to tree
582 levels 0–4. An alignment coefficient $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ mixed this second moment with an independently
583 rotated endpoint having the same eigenvalues. Intermediate mixtures do not preserve the spectrum and jointly
584 vary alignment and concentration. Ancestry, random rank-matched and dense principal-eigenspace projectors
585 were compared at $K \in \{1, 2, 4, 8, 16\}$. The dense value was also calculated as the Ky–Fan sum of the leading
586 K eigenvalues divided by total second-moment energy. Because the endpoint second moments had equal trace,
587 ancestry-minus-random capture varied affinely with alignment. Endpoint differences therefore gave the exact
588 per-seed crossover; seeds already favoring ancestry at zero alignment were assigned threshold zero. In these
589 unweighted coordinates, let $\bar{\Sigma}_g$ denote the uncentered task-gradient second moment; energy-weighted capture was
590 $\text{tr}(P \bar{\Sigma}_g) / \text{tr}(\bar{\Sigma}_g)$.

The same-span diagnostic compared raw nested indicators, their tree-Haar basis and indicators multiplied by fixed nonzero lognormal column gains. Rank, coherence, positive-spectrum Gram condition number and capture were calculated directly.

In the depth screen, a unit-norm target was sampled from modes at levels 0 through H_c , with $H_c \in \{1, 2, 3, 4\}$. The gradient-noise standard deviation at level ℓ was $0.65[1 + 1.5 \max(\ell - H_c, 0)]$. Starting from zero, 80 common-noise updates of size 0.12 used the projector onto levels 0 through $D_r \in \{1, 2, 3, 4\}$. A fixed leaf permutation provided the rewired control, and the identity provided stochastic backpropagation.

The projection screen independently set retained signal r and retained noise n to $\{0.25, 0.5, 0.75, 0.9, 1\}$. It compared an unprojected stochastic gradient, backpropagation followed by the same projection, and a routed estimator with an added total coefficient-noise fraction of 0.05. For a unit step on the identity-curvature quadratic, expected losses were calculated analytically and stochastic losses were recorded from paired draws.

In the reliability screen, eight disjoint blocks had signal and noise energies proportional to $\exp(hx_b)$ and $\exp(-hx_b)$, respectively, with equally spaced $x_b \in [-1, 1]$ and heterogeneity $h \in \{0, 0.5, 1, 1.5, 2\}$. We compared no attenuation, the best common gain, $S_b/(S_b + N_b)$, seed-shuffled gains, reversed gains and a point-gain control receiving the same branch factors. Intervals used 20,000 resamples of the 50 paired seeds, and paired tests were two-sided Wilcoxon signed-rank tests.

The resolved protocol is `configs/credit_phase_theory/confirmatory.json`. Source tables are in `source_data/credit_phase_theory/`. All full-rank projectors agreed to 4.44×10^{-16} , route-wise static gain rescaling preserved the projector to 1.33×10^{-15} , and the explicit point-gain control matched reliability-aligned attenuation exactly. This fixed-state quadratic experiment validates the signal-noise prediction but does not join oracle reliability to learned positive conductances.

4.11 Initialization score in the trained subtree factorial

The existing factorial used full-batch gradient descent. Its secondary stochastic diagnostic reconstructed the frozen training data, initialization and routes, then estimated routed-update variance from 64 hypothetical minibatches of 64 examples per seed. The full-data routed mean and that variance supplied Eq. 43. This variance was not noise experienced by the full-batch optimizer and cannot by itself explain the trained contrasts. The original released score used the largest logistic Hessian eigenvalue at initialization as its curvature scale. The corrected analysis uses the global logistic bound

$$L_{\text{glob}} = \max_c \lambda_{\text{max}} \left(\frac{8}{4N} X_c^\top X_c \right),$$

where rows of X_c contain the selected-stream features for examples in context c , N is the total training-set size, and eight is the gradient context-rescaling factor. This bound is 1.00083–1.02125 times the initialization Hessian maximum across seeds. We scored the deterministic full-batch update with $V = 0$, the hypothetical minibatch estimator with its estimated V , and the deterministic bound at the actual training step $\eta = 0.3$.

Across 540 dendritic-tree fits, the deterministic optimized-step score had Spearman correlations 0.9509 with norm-matched one-step progress and 0.9242 with final accuracy; the hypothetical-minibatch score gave 0.9373 and 0.9157, respectively (Table S3). The deterministic bound at the training step gave 0.9686 and 0.9409. Correlation intervals use 5,000 whole-seed bootstrap draws over 20 seeds. These are associations in the reanalyzed factorial, not validation on independent task families.

To characterize the task-coordinate spectrum, the analysis used the uncentered second moment of $\delta_i \mathbf{e}_{c_i}$, where δ_i is the output error on example i and $\mathbf{e}_{c_i}$ is the coordinate vector for its context c_i . Eight balanced contexts and logits near zero make this spectrum nearly isotropic, giving the reported participation rank near eight. This value characterizes the chosen coefficient representation at initialization; it is not an independent estimate of task complexity or a morphology law. The mean within-family utility and accuracy maxima occur at $K = 8$ for all five nondegenerate route families. Across their 25 implementation-family cells, there are nine distinct nonconstant mean utility curves: four implementations coincide, whereas leaf rewiring changes four of the five family curves; the principal component analysis (PCA) oracle curve is unchanged. In the dendritic-tree implementation, deterministic utility selected an accuracy-maximizing bandwidth in 95 of 100 seed-family curves, the same success count as

always choosing the largest K ; the hypothetical-minibatch score matched in 97 of 100. The 20 derangement curves had zero optimized-step utility throughout and were reported as degenerate. Repeated equivalent parameterizations do not provide independent tests of endpoint ordering. The distinct interior-optimum and ancestry-minus-control analyses, including their prediction failures, are reported in Section 4.2. The corrected scores, per-family selection errors and regret are provided in `source_data/review_evidence_reanalysis/`.

4.12 Prospective costed selection in tree-constrained linear learning

The prospective experiment separated a valid local descent bound from its use to choose a model for later learning. Five explicit 15-node trees supplied eight labeled leaves: four balanced hierarchies with distinct leaf orders and one contiguous comb. Each supplied orthonormal subtree-cut dictionaries at $K = 1, 2, 4, 8$, giving 20 candidates per arm. Every candidate retained 64 trainable leaf-input weights. In the feedback-only arm, all candidates used identity forward transfer $H_T = I_8$. In the joint arm, H_T was the leaf block of $(I_{15} + \kappa L_T)^{-1}$, where L_T was the full-tree weighted Laplacian, $\kappa = 1$, unit leak was present at every node and edge conductance was inverse embedded length. Internal nodes were retained. For input $\mathbf{x} \in \mathbb{R}^8$, trainable weights $W \in \mathbb{R}^{8 \times 8}$ and an imposed context decoder $\mathbf{a} \in \mathbb{R}^8$, the scalar prediction was $\mathbf{a}^\top H_T W \mathbf{x}$. Here W denotes model weights, rather than the coordinate-weight matrix used in the earlier projection theory. This is an explicit passive-tree linear abstraction; the readout is not a natural somatic voltage and the model is not the nonlinear DendriNet used in the physical-depth experiments.

A random orthogonal teacher mapped isotropic Gaussian inputs to labels. Context decoders were sampled uniformly from the first r columns of an orthogonal basis Q , with $r \in \{1, 2, 4, 8\}$ specifying the reference credit rank. Observation-noise standard deviation (SD) was $\sigma_y = 0$ or 0.75. At zero initialization, the exact reference decoder-space credit second moment was

$$S_0 = (1 + \sigma_y^2) Q \text{diag}(\underbrace{1/r, \dots, 1/r}_r, 0, \dots, 0) Q^\top.$$

Orthogonal rotation changed orientation while preserving this spectrum exactly. The joint arm’s transfer could change transported-credit and parameter-gradient spectra; invariance is claimed only for the stated reference moment. Angle and noise comparisons shared underlying random draws within seed and rank. Five development seeds (6100–6104) used angles $0, \pi/4, \pi/2$; 20 separate confirmation seeds (7200–7219) used held-out angles $\pi/8, 3\pi/8$. The 16 rank–noise–angle conditions within each confirmation seed were paired, giving 320 tasks per arm.

For dictionary projector P , the actual synaptic descent direction was $\mathbf{d}_i = P H_T^\top \mathbf{a}_i \delta_i \mathbf{x}_i^\top$, where δ_i was the scalar prediction error on example i . For inner products, norms and covariance calculations, the 8×8 update was identified with its vector of 64 coordinates. Its coefficients used the known decoder and transfer, an explicitly prescribed information source. They were not learned by a biological local encoder. Each candidate had 64 fixed task-decoder coefficients, K channels, $8K$ possible context-to-route encoder coefficients, eight nonzero delivery entries, 15 electrical nodes and 14 edges. Let ℓ_T be the total cable length in the fixed two-dimensional embedding and $\ell_{\max}$ the largest candidate length. The normalized geometric cable cost was $\ell_T / \ell_{\max}$; the resource cost was $C_T = K/8 + 0.1 \ell_T / \ell_{\max}$. It is a declared proxy, not a measured biological wiring or energy cost. Some candidates have identical feedback projectors but distinct cable costs; these equivalence classes are retained in the candidate manifest and are not independent replicates.

Independent calibration, training and test sets contained 1,024, 2,048 and 4,096 examples. Calibration evaluated the actual routed update mean and empirical single-example covariance, giving $\mathbb{E} \|\mathbf{d}_{\text{batch}}\|^2 = \|\mathbb{E} \mathbf{d}_i\|^2 + \text{tr Cov}(\mathbf{d}_i)/32$ for minibatches of 32 independent and identically distributed (iid) draws. The initialization score was the bound at the actual step $\eta = 0.35$,

$$U_T = \eta \mathbf{g}^\top \mathbb{E} \mathbf{d}_i - \frac{\eta^2 L_T^{\text{cal}}}{2} \left(\|\mathbb{E} \mathbf{d}_i\|^2 + \frac{\text{tr Cov}(\mathbf{d}_i)}{32} \right).$$

Here L_T^{cal} is the largest Hessian eigenvalue of the finite calibration loss, defined as half the mean squared error (MSE), and is global in W . This is a bound on that calibration loss, not on independent-test or population loss. All

weights started at zero. Paired candidates received the same iid minibatch stream for 256 updates, with checkpoints at 16, 64 and 256. Only the final checkpoint defined the primary endpoint, test half-MSE plus $0.08C_T$.

Development comprised 4,800 candidate fits. Within-task centered development data fitted one nonnegative utility-to-final-loss scale s per arm, clipped to $[0, 256]$. The moment policy selected the candidate minimizing $\mathcal{L}_{\text{cal}}(0) - sU_T + 0.08C_T$. Development also selected the fixed candidate, the tree for a maximum-budget candidate, and a separate candidate for each known rank. Sampled random choice and its exact uniform expectation were references. All 3,200 held-out policy choices were recorded with cryptographic hashes at 16:11:16 UTC on 5 September 2026, before any confirmation training. Every confirmation shard checked the protocol, source, calibration and selection hashes. No policy was retuned after confirmation. The design was prospectively frozen for this computational experiment, not publicly preregistered. Confirmation contained 12,800 fits and 38,400 checkpoint rows; no outcome was excluded.

Regret was final costed loss minus the retrospectively best candidate’s costed loss on the same task. That oracle was unavailable to selection. All intervals used 10,000 bootstrap resamples of 20 whole-seed averages. Moment-policy regret was 0.08055 (95% interval 0.07262–0.08781) for feedback-only selection and 0.10404 (0.09975–0.10791) for joint selection. Fixed maximum-budget regret was 0.04495 and 0.04269; rank-only regret was 0.00831 and 0.00821. The moment selector lost to both in every seed average in both arms. Rank-only improved over fixed by 0.03664 (0.03630–0.03697) and 0.03448 (0.03427–0.03468). The retrospectively best budget equaled the imposed rank on every confirmation task, while the contiguous balanced tree won 308/320 feedback-only and 303/320 joint tasks. This supports a restricted rank–capacity relation in the constructed task class; it supplies little evidence for fine tree selection.

We then examined why the selector failed, without changing the primary analysis or its records. Exact expected calibration one-step decrease, using the full Hessian and actual minibatch covariance, satisfied the local bound in all 12,800 conditions (minimum slack 3.48×10^{-5}). Reconstructing each first training update gave mean within-task rank correlations of initialization utility with independent- test decrease of 0.98369 (0.98069–0.98638) and 0.97391 (0.96951–0.97755). Thus local prediction can succeed while the calibrated long-horizon selector fails. Laplacian and Schur identities, zero-coupling equivalence, exact spectrum invariance, minibatch covariance and finite-difference checks passed; 120 independently replayed endpoints agreed to 5.55×10^{-17} . Frozen configurations, full matrices, choices, outcomes and audits are in `source_data/prospective_morphology_selection/`.

4.13 Interaction structure at fixed input-sensitivity rank

The preceding linear experiment allowed a dense trainable input map, which can compensate for generic route rotations. Let Φ contain K orthonormal route columns, write $W = \Phi V$ with $V \in \mathbb{R}^{K \times 8}$, and let Q_r contain the r active context columns of Q . The context-visible transfer is then $Q_r^T H_T \Phi$. When this matrix has row rank r , a suitable V reproduces every noiseless context. Finite-time conditioning and cost can still differ, but detailed shape then need not restrict representation. We therefore conducted a separate mechanism study with explicit local input access. This analysis followed the original prospective outcomes and is an exhaustive population diagnostic, not another prospective test of the original selector.

To impose local input access, we placed eight independent Rademacher inputs (each equally likely to be -1 or $+1$) at named leaves of a binary tree. For internal node v , let u_L and u_R be its child outputs and let a_v, b_v, c_v, d_v be its four trainable coefficients. Each of the seven internal nodes computes

$$u_v = a_v + b_v u_L + c_v u_R + d_v u_L u_R. \quad (65)$$

Only the root is read out. Every candidate has 28 free coefficients and 14 edges, with no dense input mixing, intermediate readout or access to inputs outside a node’s subtree. The fixed pool combines three shapes—a balanced tree, a comb and a tree with a $3 + 5$ root split—with the same four predetermined leaf permutations, giving twelve candidates. Equal edge and coefficient counts do not imply equal cable length, latency or biological energy cost. These signed algebraic nodes are not conductance compartments.

723 The primary target census contains all 105 perfect matchings M and all 35 unordered four-plus-four partitions
 724 A of the eight inputs:

$$f_M(\mathbf{x}) = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j, \quad f_A(\mathbf{x}) = \frac{1}{2} \left(\prod_{i \in A} x_i + \prod_{i \notin A} x_i \right). \quad (66)$$

725 The targets are enumerated independently of the candidate trees. Evaluating all $2^8 = 256$ equally weighted inputs
 726 gives target variances 1 and $1/2$, respectively. For the continuous derivative of each target's unique multilinear
 727 extension,

$$\mathbb{E}[\nabla_{\mathbf{x}} f \nabla_{\mathbf{x}} f^T] = I_8/4. \quad (67)$$

728 Thus both algebraic and participation-ratio effective rank are eight, and the complete input-sensitivity spectrum
 729 agrees. This is a property of the target, not equality of changing model parameter-gradient or synaptic-credit
 730 moments. Normalized MSE (NMSE) divides population squared error by the stated target variance.

731 Interaction-tensor descriptions have previously been used to analyze structured neural function classes [4]. The
 732 centered constant-feature criterion below is derived for the specific local model in Eq. 65.

733 4.14 A necessary and sufficient centered-cut condition

734 For each subset U of input indices, define the Walsh function $\chi_U(\mathbf{x}) = \prod_{i \in U} x_i$, with $\chi_\emptyset = 1$. These functions form
 735 an orthonormal basis under the uniform input distribution. Writing c_U for the coefficient of χ_U gives $f = \sum_U c_U \chi_U$.
 736 If $\mathbf{c}_f$ and $\mathbf{c}_g$ collect the coefficients of two functions f and g , Parseval's identity gives $\mathbb{E}[(f - g)^2] = \|\mathbf{c}_f - \mathbf{c}_g\|^2$.
 737 For the leaf set S of an actual subtree, reshape the target coefficients into C_S , with row index a subset of S and
 738 column index a subset of its complement $\bar{S}$. Let C_S° denote this matrix after removing the row for the empty subset
 739 of S .

740 **Proposition 9** (Exact representation by a scalar multi-affine tree). *For a fixed binary tree with one named*
 741 *Rademacher input at each leaf, a target is represented exactly by Eq. 65 and a root-only output if and only*
 742 *if $\text{rank}(C_S^\circ) \leq 1$ at every proper subtree cut.*

743 *Proof.* Every ancestor is affine in a particular child's output, and sibling inputs are disjoint. Consequently a
 744 represented function can be written $g = A(\mathbf{x}_{\bar{S}}) + h_S(\mathbf{x}_S)B(\mathbf{x}_{\bar{S}})$. Its full cut matrix has rank at most two; removing
 745 the empty- S row leaves rank at most one. This proves necessity.

746 For sufficiency, call a cut active when $C_S^\circ \neq 0$. At each active cut, choose a centered function h_S whose Fourier
 747 coefficients span the column space of C_S° , and define $V_S = \text{span}\{1, h_S\}$. If $C_S^\circ = 0$, take $h_S = 0$ and $V_S = \text{span}\{1\}$.
 748 At a leaf use $V_i = \text{span}\{1, x_i\}$. The condition implies $f \in V_S \otimes \mathcal{H}_{\bar{S}}$, where $\mathcal{H}$ is the complete finite-dimensional
 749 function space on the indicated inputs.

750 For sibling leaf sets L and R , the orthogonal projections onto V_L and V_R commute because they act on disjoint
 751 input factors. Hence $f \in V_L \otimes V_R \otimes \mathcal{H}_{L \cup R}$. For a non-root parent $P = L \cup R$ with nonzero centered component, write
 752 $f = 1_P \otimes A + h_P \otimes B$, with $B \neq 0$. Contracting the outside factor against a linear functional that maps B to one gives
 753 $h_P + c1_P \in V_L \otimes V_R$, where c is a scalar contributed by A . Since the constant function also belongs to this product
 754 space, so does h_P . It is therefore a linear combination of $1, h_L, h_R, h_L h_R$, exactly the four allowed features. Inactive
 755 parent features are zero and cause no exception. At the root, the complete target itself belongs to the product of its
 756 children's spaces. Induction supplies all local coefficients and proves sufficiency. $\square$

757 Retaining the constant feature in every subtree makes this criterion stronger than an uncentered rank-two
 758 constraint. Let $\mathcal{T}_{\text{proper}}$ be the leaf sets of the tree's proper subtrees, and let $\sigma_j(C_S^\circ)$ be the singular values in
 759 decreasing order. The optimal low-rank approximation identity and Parseval then give

$$\text{MSE}(f, g) \geq \max_{S \in \mathcal{T}_{\text{proper}}} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (68)$$

760 Taking the maximum preserves a lower bound. Summing tails from overlapping cuts instead gives a heuristic score,
 761 without the same guarantee. A positive maximum-cut bound need not equal the true optimum. The condition applies

to actual subtree cuts; the complement of a subtree need not itself be a subtree. For the matching (12)(34)(56)(78), an even-odd root split has centered tail 0.75, whereas the full rank-two tail is 0.5. For the two-quartet target with supports 1234 and 5678, a five-leaf subtree 12345 has centered tail 0.25 despite a zero full rank-two tail. Supplementary Fig. S36 compares both bounds with fitted population error.

The proof requires ancestors to be affine in each child’s output. A general nonlinear scalar ancestor can generate higher cut rank from one scalar feature. Neither the necessity condition nor Eq. 68 therefore applies automatically to a tanh tree or a conductance model.

4.15 Population fitting, complete-tree construction and depth

All twelve candidates were fitted to every primary target by 32 alternating least-squares (ALS) sweeps with four fixed random restarts. Conditional on the other nodes, the design for one node is $q(\mathbf{x})[1, u_L, u_R, u_L u_R]$, where q is the derivative of the root with respect to that node’s output. The offset is the current root output minus qu_v . Sweeps alternate postorder and preorder. Hidden mean/scale gauges are normalized with exact parent compensation. Every restart and checkpoints at 0, 2, 8 and 32 sweeps are retained: 6,720 fits and 26,880 checkpoint rows. The endpoint is the smallest achieved NMSE among the four restarts. The difference between this endpoint and Eq. 68 can reflect both incomplete optimization and a loose bound, so it is not a certified optimization gap. Full-domain fitting has no held-out generalization interpretation.

For score-based selection among the twelve fixed candidates, values within 10^{-10} are treated as ties and resolved by lexicographic candidate identifier. A separately retained summary applies this tolerance to the raw floating-point minima without changing any fit. The two-sweep pilot fits every candidate; the best-fixed baseline chooses a single candidate per family in hindsight. The uniform-random baseline is its exact candidate expectation. Family averages and paired shape/assignment counts are descriptive censuses, with error comparisons at tolerance 10^{-7} and no sampling confidence intervals over the enumerated tasks. The lower bound is attained within this tolerance in 1,407/1,680 endpoints. Although some restarts develop large cancelling coefficients, every case selected by a deterministic policy has an existing equivalent restart within 10^{-7} NMSE whose coefficient norm is at most 2.645752. All original fits, including the remaining poorly conditioned case, remain included.

To search beyond the fixed pool, we used an exploratory dynamic program over all input subsets. Let $c(S)$ be the normalized cut tail from Eq. 68, with zero cost at the root and leaves. Let $b(S)$ be the smallest achievable maximum cut cost for a subtree on leaf set S , with base case $b(\{i\}) = 0$. Then

$$b(S) = \max \left\{ c(S), \min_{L \cup R = S} \max[b(L), b(R)] \right\}. \quad (69)$$

Every binary tree decomposes into one root split and two independent children, so this recurrence covers all trees. A second pass restricts every cut to the globally minimal threshold, then minimizes total tail, depth and deterministic split ties. Costs are rounded to 12 decimals for numerical ties. This corrected two-pass version is retained separately from the first exploratory construction and agrees with exhaustive enumeration of all 945 six-input trees on four random cost tables. At a zero threshold, the theorem makes the minimum-depth solution the shallowest exact tree in this class; for a positive threshold only the cut score, not fitted error, is optimized.

Dominant left singular vectors of the centered cut matrices define local scalar features, and four-feature least squares recovers each node’s coefficients. All 105 matching and 35 quartic targets reconstruct at minimum depth three. A separate positive control uses 24 unique seeded permutations π of the nested target

$$f_\pi(\mathbf{x}) = \frac{1}{2} \sum_{k \in \{2,4,6,8\}} \prod_{j=1}^k x_{\pi(j)}. \quad (70)$$

Its rank is eight but its spectrum $(1/4, 1/4, 1/2, 1/2, 3/4, 3/4, 1, 1)$ differs from the primary families; it is constant only within this control family. Every nested target requires depth four. An independent depth-constrained recurrence gives minimum centered-cut NMSE bound 0.146446609407 at depth at most three and zero at depth four. All 164 constructions have NMSE below 7×10^{-30} and maximum absolute coefficient one up to rounding

(Supplementary Fig. S37). The algorithm uses the complete target Fourier tensor: these are exact-target capacity certificates, not predictions from limited calibration data.

4.16 Finite, noisy calibration of interaction-based tree selection

The exact-target construction leaves open whether a useful tree can be chosen from a limited sample. We therefore tested selection from finite noisy observations without access to the target’s exact Fourier coefficients. The forward model was the eight-input algebraic tree used in the preceding representability analysis. Each of its seven internal nodes used the four-coefficient local computation in Eq. 65, giving 28 fitted coefficients, 14 edges and a root-only output. Leaves received individual named inputs; no global input mixer, additional decoder or bypass was available. This experiment concerns that constrained multilinear model, rather than conductance dynamics or biologically local plasticity.

Tasks were drawn from four equally weighted families: sums of four disjoint pair products, sums of two complementary quartic products, sums of nested products of degrees 2, 4, 6, 8, and a random-interaction control. The control selected four distinct nonconstant monomials uniformly from all degree-2–8 masks, independently of the candidate trees. Each task received a random input permutation where applicable, independent random coefficient signs and magnitudes sampled uniformly between 0.5 and 1.5. Coefficients were normalized to unit squared norm. Under independent Rademacher inputs, each target therefore had mean zero and population variance one. These random coefficients generated new functions; they did not preserve an identical input-gradient second moment across tasks. The earlier exact capacity experiment provides the separate isospectral comparison.

The primary candidate pool retained the same twelve trees: three shapes (balanced, comb and 3+5) under four fixed input assignments. An ℓ_1 -penalized linear estimator (Lasso) received input/output observations and the dictionary of all 255 nonconstant Walsh monomials. It received no family label, planted interaction support, true coefficient, teacher derivative or full-domain truth table. Its prior information was the known eight-variable Rademacher domain and a sparse multilinear dictionary. The intercept was fitted separately and excluded from the interaction score, because a global constant does not constrain subtree compatibility.

Calibration comprised 64, 256 or 1,024 independent input draws, with independent Gaussian label noise of standard deviation zero or 0.5. The primary condition was 256 queries with noise standard deviation 0.5. The first 75% of observations were available for fitting the Lasso and short-pilot trees; the remaining 25% formed a held-out calibration subset used to choose among the pilot fits (the calibration gate). Thus the primary estimator used 192 observations, while the shared total calibration-query budget was 256. Let n_{fit} be the number of fitting observations and $\text{sd}(y_{\text{fit}})$ their label standard deviation. The Lasso penalty was $c \text{sd}(y_{\text{fit}}) \sqrt{2 \log(256)/n_{\text{fit}}}$. Fitting used cyclic coordinate descent, tolerance 10^{-8} and at most 20,000 iterations. The scale c was selected from 0.15, 0.35, 0.65 using five development seed blocks. The frozen value was 0.15. The same development outcomes fixed the best constant candidate and a lookup rule assigning a candidate to the estimated input-gradient rank needed to explain 99.5% of its energy. Both reduced to `balanced_p0`; bins with fewer than two development tasks used the constant fallback.

For each candidate subtree, we matricized the estimated coefficients across the subtree/complement split, removed the row corresponding to the constant function inside the subtree, and computed the squared singular-value tail beyond its first component. The selector minimized the maximum normalized tail. Scores within 10^{-10} were treated as ties; ties were resolved by the sum of overlapping cut tails and then by lexicographic candidate name, using the same tolerance. The maximum is a population-MSE lower bound when computed from the true coefficients. Under noisy estimation it is a selection score, not a certified lower bound on the true task. The summed tails are a tie-breaking heuristic and are not a bound.

Baselines comprised fixed representatives of each shape, the development-best fixed candidate, the development-trained estimated-rank-only rule, uniform random choice averaged exactly over the candidate pool, and a two-sweep fitting pilot. The pilot independently fitted every candidate on the same calibration-fit observations from one fixed initialization and selected the smallest MSE on the calibration gate. Pilot parameters were discarded before final training, so the selected candidate received no additional training from this procedure. A secondary selector applied the previously audited subset dynamic program to the estimated coefficients at the primary calibration condition

only. It could select a tree outside the twelve-candidate pool while retaining the same coefficient and edge budget.

All model definitions, estimator choices, baselines, training settings, primary comparisons and effect-size margin were frozen after development. Twenty new seed blocks generated one new function from each family, giving 80 confirmatory tasks. Each task was evaluated under all six calibration conditions. Cryptographic hashes recorded every calibration score, pilot choice, adaptive-tree choice and calibration dataset before any confirmatory candidate completed final training or exposed its final test endpoint. Two implementation errors stopped workers before fitting: lexicographic JSON ordering of numerical node IDs, and a launcher importing a historical module with the same name. The failure logs were retained. Separately hashed amendments corrected only node decoding and module dispatch. The original frozen selector files and all sealed choices remained unchanged.

For each task, final candidate training used a separate fixed stream of 1,024 input draws with Gaussian label noise of standard deviation 0.3. Every candidate used 32 alternating least-squares sweeps and two paired random initializations. Coordinate updates solved the four-parameter local least-squares subproblem, with output-preserving gauge normalization. A separate 512-observation noisy validation stream selected between the two final restarts. No test observation was used for fitting, restart choice or calibration. The short pilot and final fits used separate data and began independently from their declared initializations.

Final evaluation used 4,096 independent test inputs. The primary endpoint was MSE against the clean target, divided by its known population variance of one. MSE against independently noisy test labels and exact MSE over all 256 input patterns were secondary evaluation-only endpoints. Because the input domain is finite, independent streams can contain the same input pattern. They represent independent observations with fresh noise and distinct sample identifiers; they are not partitions of a reused noisy-label cache. The task generator knows the target coefficients to draw observations, but selectors receive only their stated calibration inputs. Complete coefficients and full-domain values are reserved for post-seal target-informed references and evaluation.

The twelve fixed candidates, the estimated adaptive tree and a privileged true-coefficient adaptive reference were each fitted twice per task, giving 2,240 confirmatory fits. The same final candidate outcomes were reused across calibration conditions. Primary regret was the selected candidate's independent-test normalized MSE minus the retrospective minimum among the twelve fixed candidates, each represented by its validation-selected restart. The adaptive selector's regret against this fixed pool can be negative; such values were retained and describe an advantage from searching a larger tree space. The true-coefficient cut selector, true-coefficient adaptive tree and retrospective best trained candidate in the pool were explicitly privileged references, not deployable baselines.

Uncertainty used 20 seed blocks. The four family-specific paired differences were averaged within each seed before bootstrapping 10,000 seed samples. The two primary comparisons were the estimated-cut selector against the development-best fixed candidate and against the two-sweep pilot, at 256 queries/noise 0.5. Descriptive intervals used 95% coverage. Bonferroni-adjusted two-sided 97.5% intervals were used for the two primary comparisons. Meaningful superiority required both a positive adjusted lower bound and mean improvement of at least 0.01 normalized MSE. Other calibration sizes, the noise-free conditions, family-specific results and adaptive construction were prespecified secondary analyses. Failed fits and convergence warnings were retained; nonfinite or linear-algebra failures received the predeclared loss of $1e6$ and were counted explicitly.

The prespecified task mixture defines the scope of inference. Confirmation tests fresh functions and observations within that mixture; it does not establish transfer to unknown task families, a scalable compiler at larger input counts, or optimal biological dendritic morphology. The fixed Walsh dictionary grows exponentially with the number of inputs, and the tree fitter is an alternating least-squares procedure rather than a synapse-local plasticity rule.

Supplementary Fig. S41 reports the complete calibration sensitivity, family-specific effects, primary adjusted contrasts and implementation cost. Section 4.17 tests subsequent gradient learning on estimated architectures. The separate study in Section 4.18 uses oracle-compatible trees to examine task, assignment and feedback-resolution effects.

4.17 Direct confirmation from noisy morphology estimates to gradient learning

The calibration study evaluated selected trees with an alternating least-squares fitter. A separate cohort tested whether its estimates also supported ordinary gradient learning, completing the sequence from noisy calibration to tree construction and parameter updates (main Fig. 6, Supplementary Fig. S41G,H). Twenty new seed blocks (930910–930929) were disjoint from the preceding calibration and credit cohorts. Each generated one fresh matching, quartic, nested-prefix and random-interaction target using the same prespecified mixture of signs, coefficient magnitudes and input assignments. Every target had population variance one; the input-gradient second moment was not held identical across this mixture.

We retained the 255-monomial Lasso with penalty scale 0.15, the same intercept treatment and the same subset dynamic program. The selector received 192 noisy observations from a total calibration budget of 256, with label-noise SD 0.5. The remaining 64 observations were reserved and were not used to select or tune this follow-up. Neither family labels nor true support or coefficients were supplied to the estimator. All eighty estimated trees were recorded together with cryptographic hashes before parameter learning. The fixed balanced tree was the earlier development-best baseline. A third tree was constructed from true coefficients after sealing as a privileged structural reference. No construction weights or other target-derived initialization were used.

Each tree was trained with either exact path credit or unit root broadcast, giving six conditions per task and 480 fits. Root sensitivity remained one; broadcast replaced the six other internal sensitivities by one. No oracle projection coefficients or proximal-zone definition were needed. Each forward model had seven multi-affine internal units, 28 coefficients, fourteen edges and root-only output; trees with a leaf directly below the root were permitted.

Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. Learning rates were copied without retuning from the previous credit-development record: 0.01 for exact credit and 0.003 for broadcast. The comparison therefore tests frozen optimizer recipes at different learning rates. All six conditions shared a label-independent Gaussian initialization with SD 0.5, zero biases, and the same minibatch indices. Initial coefficients and subsequent updates were clipped to $[-2, 2]$; each condition's gradient norm was clipped at ten. Learning differentiated half-MSE divided by the known target variance, using 1,024 updates of 64 observations sampled with replacement from a fixed 2,048-observation training dataset. Training label-noise SD was 0.15. Separate 4,096-observation test data had independent noise of the same SD. Calibration, training, test, initialization and minibatch streams each used a separate deterministic random-seed assignment. Repeated Rademacher input patterns were allowed; observation identities and noise draws were independent across splits. There was no validation selection, early stopping, new rate search or continuation from calibration fits.

The primary endpoint was noisy-test NMSE after 1,024 updates, with expected noise floor 0.0225 for every family. Clean independent-test and exact 256-pattern population errors were secondary. Two primary paired contrasts compared fixed minus estimated tree error under exact credit, and broadcast minus exact error on estimated trees. Four family differences were averaged within each seed before 10,000 whole-seed bootstrap resamples. The two primary contrasts used Bonferroni-adjusted two-sided 97.5% intervals and the prespecified 0.01-NMSE meaningful-improvement margin. Family-specific and true-coefficient-reference comparisons used descriptive pointwise 95% intervals. The primary pooled improvements were 0.45913 NMSE [0.38489, 0.53227] and 0.60750 [0.56916, 0.64610], respectively.

Matching, quartic and nested targets approached the noise floor under the estimated-tree exact-credit recipe, while random interactions retained substantial error. Broadcast performed slightly better on matching tasks; exact credit did not dominate every family. The tree constructed from the full target is a privileged structural reference, not an oracle for the final endpoint: a better interaction score need not produce a better finite-training outcome. These results test the declared finite-data construction and learning pipeline, with its specified function class, optimizer and credit assumptions, rather than a general morphology law or a biophysical coefficient mechanism.

A pre-training interface mismatch in an oracle helper stopped the initial array before initialization or fitting. Retained failure records and a hashed runtime adapter document the repair; scientific settings and sealed estimates were unchanged. An excluded development smoke (seed 721000) subsequently exercised all six conditions through the complete training horizon. Its six fits are excluded from the fresh sample. All 480 fresh trajectories and final weights were retained. Independent replay reproduced all eighty Lasso-to-tree choices from sealed observations and

all reported test, clean-test and population errors from saved final weights, with maximum metric difference zero. Initialization and source-role hashes matched, and every parameter remained in the declared box. The complete protocol, runtime adapter, excluded implementation check, selection record and validation results are supplied under `source_data/morphology_calibration/end_to_end`.

4.18 Learning compatible trees with exact and restricted credit

To separate credit assignment from uncertainty in structure estimation, this experiment supplied a tree compatible with the full target. Three development seeds and twenty fresh seeds generated independent signed and permuted matching, quartet and nested-interaction targets. The complete target Fourier tensor supplied a compatible tree through the preceding dynamic program; no target-derived coefficients initialize learning. An independent permutation of leaf labels preserved the exact shape, 28 coefficients and 14 edges. Chance compatible permutations were retained. The primary matching and quartet families retained identical input-sensitivity spectra, while the nested family had the previously stated anisotropic spectrum.

All seven internal units initialized their non-bias coefficients independently from $\mathcal{N}(0, 0.5^2)$, with zero biases. The same initial coefficients, training cache and minibatch draws were used across paired assignments, credit rules and optimizers. Local features were $(1, u_L, u_R, u_L u_R)$ and the exact path derivative $q_v = \partial u_{\text{root}} / \partial u_v$ supplied credit. Exact gradients agreed with central finite differences for every coefficient in all three families. Root sensitivity was one under every rule. The six nonroot internal units received either the exact field, a unit root broadcast, the orthogonal projection onto a single constant profile, or the projection onto two proximal-subtree indicator profiles. A size-matched permutation of the latter profiles served as a control. All projected coefficients were computed from the current exact field and are therefore oracle coefficients; the experiment did not learn a circuit estimator for them.

Training used 2,048 independently sampled Rademacher examples with output noise SD 0.15 and 1,024 minibatch updates of size 64. The gradient corresponded to half-MSE divided by the known clean target variance. Stochastic gradient descent (SGD) and Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$) each tested learning rates 0.003, 0.01 and 0.03. Update gradients were clipped at norm 10 and parameters projected to $[-2, 2]$, which contains all exact constructions. Every clipping/projection count was recorded. These are constrained finite-horizon training results, not estimates of unrestricted optimization optima.

A separate learning rate for each optimizer and rule minimized mean development population NMSE over all families and both assignments. A common rate per optimizer was also selected on development. All choices were recorded with cryptographic hashes before any fresh fit; all three rates remain in the released outcomes. Every SGD rule selected the upper tested rate, so the grid does not establish its best attainable optimizer setting. The independent test set contained 4,096 examples with fresh SD 0.15 output noise. Test NMSE, normalized by the clean target variance, was primary; noiseless population NMSE on all 256 inputs was secondary. Evaluation at steps 0, 1, 16, 64, 256 and 1,024 retained 21,600 checkpoint rows from 3,600 fresh fits. The endpoints at the development-selected rates contained 1,200 fits. No run was excluded. Ten thousand bootstrap samples of twenty whole paired seed blocks were drawn for pointwise 95% intervals, averaging families within seed for pooled contrasts. Multiple subgroup intervals are descriptive, not separately multiplicity-adjusted discoveries.

Exact credit strongly improves learning of compatible quartet and nested targets under both frozen optimizer recipes. Broadcast is already effective for matching targets and outperforms exact credit there under SGD. Restricted two-subtree projections help in some Adam conditions but do not consistently beat shuffled profiles, and have essentially no pooled advantage over broadcast under SGD. For compatible trees trained with exact credit and Adam, mean test NMSE was 0.0329; permuting the input labels increased it to 0.6080. At the common Adam rate 0.01, the compatible exact/broadcast averages remain 0.0329/0.5973. Supplementary Fig. S40 reports the full family, optimizer, profile, assignment and rate controls. Numerical outcomes and source/selection hashes are in `source_data/morphology_credit/`.

4.19 Input grouping and restricted credit in directed conductance trees

We used a minimal directed shunting model to test input grouping while holding the branching shape and parameter counts fixed. Four terminal compartments fed two proximal compartments and a soma. Every model had seven compartments, six child couplings, four excitatory contacts, six inhibitory contacts and sixteen trainable conductances. Leak conductance was one; excitatory and inhibitory reversal potentials were one and zero. All compartment activations were the identity, and the soma had no direct synaptic inputs or additional decoder. Let $i \in \{0, 1, 2, 3\}$ index terminal compartments and $j \in \{1, 2\}$ proximal compartments. Write T_i , P_j and S for terminal, proximal and somatic voltages. The pair $\mathcal{S}_j$ contains the terminals assigned to proximal compartment j . Terminal i has synaptic conductances g_i^E and g_i^I , and b_{ji} couples it to proximal compartment j . The maximal proximal shunt conductance is h_j , and a_j couples that compartment to the soma. For terminal activities x_i^E, x_i^I and proximal shunt input z_j ,

$$T_i = \frac{g_i^E x_i^E}{1 + g_i^E x_i^E + g_i^I x_i^I}, \quad (71)$$

$$P_j = \frac{\sum_{i \in \mathcal{S}_j} b_{ji} T_i}{1 + \sum_{i \in \mathcal{S}_j} b_{ji} + h_j z_j}, \quad (72)$$

$$S = \frac{a_1 P_1 + a_2 P_2}{1 + a_1 + a_2}. \quad (73)$$

The positive local input z_j controls shunt conductance; it carries no teaching signal. Conductances were parameterized as $g = \exp(\theta)$, with $\theta \in [-5, 5]$. The matched resources were compartment, contact, coupling and parameter counts and the admissible conductance range; total learned conductance was not constrained to be equal. No fresh run reached a parameter bound. The NumPy implementation was checked compartment by compartment against the repository’s production `forward_branch_dynamics` function in its identity-activation, zero-numerical-offset configuration. All sixteen log-conductance derivatives and ten input derivatives agreed with autograd to the numerical tolerances reported in the accompanying tests. This is a directed steady-state model, not a reciprocal cable or spiking model.

The three groupings were (01 | 23), (02 | 13) and (03 | 12). A planted teacher used the same conductance equations, with independent Gaussian perturbations of standard deviation 0.25 in log conductance around $g^E = 2$, $g^I = 0.7$, $h = 3$, $b = 3$ and $a = 4$. Independent student perturbations had standard deviation 0.4 around these centers and were shared across paired candidate groupings, credit rules and optimizer settings. Student initialization did not use teacher coefficients. Teacher tasks permuted terminal excitatory/inhibitory input pairs together; the proximal inputs retained their identities. Paired task inputs were inversely permuted so that the realized target values were identical across the three tasks. Thus each task had the same input distribution and the same spectrum of the input-gradient second moment. The latter was verified from 2,048 paired probes per seed; its matrix entries could differ by the corresponding coordinate permutation. Compatibility meant that the student’s input grouping matched the teacher’s grouping. These tasks are planted mechanistic positive controls within one model class, rather than tests of transfer to an unseen class of biological computations.

Each input was $\exp(\sigma Z - \sigma^2/2)$ for independent standard-normal Z . We used $\sigma = 0.7$ for terminal excitatory and inhibitory inputs and $\sigma = 1.2$ for proximal shunt inputs. All activities and conductances were strictly positive, all voltages lay between the reversal potentials, and there was no label noise. Each task used 1,024 training, 1,024 validation, 4,096 test and 256 calibration examples from independent random streams. Within a task, every candidate and credit rule received the same examples and minibatch indices. Training minimized half-MSE divided by the training target variance. Reported NMSE is MSE divided by the target variance of the evaluation split; it is dimensionless and is not a percentage.

Four feedback conditions shared the forward model. Let V_n be the voltage of compartment n and L the training loss. Write $\rho_n = \partial S / \partial V_n$ for its current exact path sensitivity and $\delta_0 = \partial L / \partial S$ for the somatic loss derivative. Exact transport supplied $\delta_0 \rho_n$. A fixed calibrated broadcast supplied $\delta_0 \gamma_n$, where γ_n was the mean initial path sensitivity of that student’s own forward states on the label-free calibration inputs. The profile was frozen during training.

A one-profile projection used the same γ and supplied its best Euclidean fit to the current nonsomatic field. The two-subtree condition restricted γ to each proximal compartment and its two terminal descendants, giving disjoint columns $\mathbf{A}_j$. Its coefficients were

$$c_j = \frac{\mathbf{A}_j^\top \boldsymbol{\rho}}{\mathbf{A}_j^\top \mathbf{A}_j}. \quad (74)$$

The one-profile projection used the analogous coefficient with $\mathbf{A} = \gamma$. These per-trial coefficients had access to the current exact student path field; both projection rules are oracle diagnostics of restricted feedback, not learned coefficient estimators. Soma error remained exact in every condition. Each conductance update multiplied the delivered compartment error by its local conductance eligibility, including the chain derivative of the exponential parameterization. For a child coupling, the local conductance derivative was $(V_{\text{child}} - V_{\text{parent}})/G_{\text{parent}}^{\text{tot}}$, where $G_{\text{parent}}^{\text{tot}}$ is the parent's total open conductance. Credit geometry was evaluated at the corresponding exact-learning model's common checkpoint state; the reported path-capture normalization excluded the soma.

We used 1,000 minibatch updates of size 128, with checkpoints at 0, 10, 100, 300 and 1,000. Stochastic gradient descent (SGD) and Adam were evaluated separately; Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. One learning rate per optimizer was chosen by averaging final validation NMSE over five development seeds (15300–15304), all task/student groupings and all four credit rules. The original SGD grid was 0.001, 0.003, 0.01; because its optimum was at the upper boundary, an additional development-only bracket tested 0.03, 0.1, 0.3, 1. Every result, including unstable high-rate trajectories, was retained. Adam tested 0.003, 0.01, 0.03. The selected rates were 0.3 for SGD and 0.01 for Adam. These settings were frozen before the twenty fresh seeds (15400–15419), producing $20 \times 3 \times 3 \times 4 \times 2 = 1,440$ fits and 7,200 checkpoint outcomes. No learning rate, horizon or credit rule was selected using fresh training outcomes. The two primary contrasts were the Adam exact-gradient grouping effect and the Adam calibrated-broadcast-minus-exact effect within compatible groupings. Other contrasts are descriptive optimizer and feedback-resolution sensitivities.

Inference used the seed as the independent unit. Each seed's compatible mean averaged its three teacher-task permutations; its incompatible mean additionally averaged the two mismatched student groupings per task. Confidence intervals used 10,000 bootstrap resamples of these paired seed summaries. Intervals are pointwise rather than simultaneous across all reported subgroup comparisons. All sources, rates, states, physical-domain checks and numerical tests are retained under `source_data/morphology_conductance`; execution code is under `scripts/morphology_conductance`.

An interaction lower bound for this conductance model. Let R_i denote a teacher terminal response, $A_j = (1 + \sum_i b_{ji} + h_j z_j)^{-1}$ its proximal attenuation, and $\beta_j = a_j/(1 + a_1 + a_2)$. The sum in A_j runs over the teacher terminals assigned to proximal compartment j . The teacher output is a sum of terms $\beta_j b_{ji} R_i A_j$. Let $p_{\text{teacher}}(i)$ and $p_{\text{student}}(i)$ denote the proximal parent assigned to terminal i in the teacher and student, respectively, and write $p(i) = p_{\text{teacher}}(i)$. For terminal i , set $j = p(i)$. Because terminal inputs and proximal shunt inputs are independent, a term whose terminal and teacher shunt belong to different *student* proximal input blocks has the centered interaction

$$H_i = \beta_j b_{ji} (R_i - \mathbb{E}R_i)(A_j - \mathbb{E}A_j). \quad (75)$$

Every admissible student output is additive across its two proximal input blocks. Each such H_i is orthogonal to all functions additive across those blocks, and the omitted H_i are mutually orthogonal. Therefore, for every choice of the student's conductances,

$$\mathbb{E}[(S_{\text{teacher}} - S_{\text{student}})^2] \geq \sum_{i: p_{\text{student}}(i) \neq p_{\text{teacher}}(i)} (\beta_{p(i)} b_{p(i)i})^2 \text{Var}(R_i) \text{Var}(A_{p(i)}). \quad (76)$$

The bound is zero for compatible groupings. It is a lower bound for the larger class of arbitrary functions additive across the student's proximal blocks, not a claim that the constrained rational student attains it. Its validity relies on the specified independent inputs, identity somatic activation and absence of direct somatic synaptic inputs; it does

not extend automatically to arbitrary conductance architectures or nonlinear readouts. An equivalent obstruction is the absence of a mixed derivative between a terminal input and a shunt input assigned to the other student subtree.

We specified this diagnostic before the fresh training outcomes and retained the learner and rate-selection procedure unchanged. Population lognormal moments were evaluated with 32-, 64- and 128-node Gauss–Hermite quadrature. The primary bound used 128 nodes; the maximum change from 64 to 128 nodes was below 8×10^{-17} in normalized-MSE units. This is a converged numerical evaluation of an analytic bound, not a formally certified quadrature error. Across the fresh teacher draws, the mean incompatible population-NMSE lower bound was 0.013222, with a minimum of 0.005367 across task/candidate pairs. These are population lower bounds, distinct from held-out sample NMSE.

4.20 Finite-horizon forecasts for the contextual linear learner

To diagnose the original scalar extrapolation independently of structural capacity, we froze a new finite-horizon predictor before fresh calibration or training. The candidate pool, learning rate $\eta = 0.35$, minibatch size $B = 32$, 256-update horizon and objective cost remained unchanged. Five original development seeds were diagnostic. Twenty new seeds (9300–9319) crossed four ranks, two noise levels, two held-out angles, both transfer arms and both training regimes. Every candidate was trained, yielding 25,600 new fits and 76,800 checkpoint outcomes at updates 16, 64 and 256. The new non-pilot scores were hash-sealed before each seed’s outcomes. No predictor hyperparameter was fitted to the new endpoints. This separates mode-dependent learning dynamics from a scalar initial slope, as motivated by analyses of deep linear learning and task alignment with kernel eigenfunctions [5, 6].

To forecast learning, we express the weights in route coordinates. For orthonormal routes Φ , the learner stays in $W = \Phi V$. For context j with decoder $\mathbf{a}_j$, define the effective route vector $\mathbf{c}_j = \Phi^\top H_T^\top \mathbf{a}_j$. The scalar prediction is $\mathbf{c}_j^\top V \mathbf{x}$ and the actual update is minibatch descent in V . Let $\mathbf{v} = \text{vec}(V)$ stack the route-coordinate weights. For calibration example i with input $\mathbf{x}_i$, label y_i and effective route vector $\mathbf{c}_i$, define $\mathbf{z}_i = \text{vec}(\mathbf{c}_i \mathbf{x}_i^\top)$. With $\mathbb{E}_i$ denoting an average over the calibration-fit examples, set $H_{\text{emp}} = \mathbb{E}_i[\mathbf{z}_i \mathbf{z}_i^\top]$ and $\mathbf{b}_{\text{emp}} = \mathbb{E}_i[y_i \mathbf{z}_i]$. Write $H_{\text{emp}} = U \text{diag}(\lambda) U^\top$, with orthonormal eigenvectors in U and eigenvalues in λ . After T updates, full-batch descent from zero gives

$$\mathbf{v}_T = U \text{diag}\left(\frac{1 - (1 - \eta\lambda)^T}{\lambda}\right) U^\top \mathbf{b}_{\text{emp}}, \quad (77)$$

The diagonal expression is evaluated separately at each eigenvalue, with continuous value ηT when that eigenvalue is zero. The empirical selector fits the first 512 calibration examples and evaluates on the other 512; Eq. 77 is exact for that full-batch surrogate, not for the original cached-data SGD process.

For fresh inputs $\mathbf{x} \sim \mathcal{N}(0, I_d)$, where d is the input dimension, a smaller recurrence describes stochastic gradient descent (SGD). Let p_j be the context probabilities and $\mathbf{t}_j \in \mathbb{R}^d$ the target coefficient vector in context j . Suppose $y = \mathbf{t}_j^\top \mathbf{x} + \varepsilon$, with independent zero-mean noise of variance σ_j^2 . The exact recurrence closes on the mean weights $M = \mathbb{E}[V] \in \mathbb{R}^{K \times d}$ and the route-space variance matrix $\Gamma = \mathbb{E}[(V - M)(V - M)^\top]$. The latter sums weight covariance over the input coordinates; it is not the full vectorized parameter covariance. Define

$$C = \sum_j p_j \mathbf{c}_j \mathbf{c}_j^\top, \quad D = \sum_j p_j \mathbf{c}_j \mathbf{t}_j^\top, \quad \bar{G} = CM - D, \quad (78)$$

$$R_j = \|M^\top \mathbf{c}_j - \mathbf{t}_j\|^2 + \mathbf{c}_j^\top \Gamma \mathbf{c}_j, \quad A_\eta = I - \eta C, \quad (79)$$

$$J = \sum_j p_j \mathbf{c}_j \mathbf{c}_j^\top [(d+2)R_j + d\sigma_j^2], \quad K_G = C^\top C + \bar{G} \bar{G}^\top. \quad (80)$$

Here $\bar{G}$ is the mean gradient, R_j is the expected squared residual excluding observation noise, J is the second moment of the single-example gradient Gram matrix, and K_G is the corresponding second moment of the conditional-mean gradient. For a fixed residual coefficient vector $\mathbf{e}$, Gaussian fourth moments yield $\mathbb{E}[(\mathbf{e}^\top \mathbf{x} - \varepsilon)^2 \|\mathbf{x}\|^2] = (d+2)\|\mathbf{e}\|^2 + d\sigma_j^2$. Conditioning first on the current weights, the Gram of a minibatch-mean gradient is its

conditional-mean Gram plus $1/B$ times its conditional variance. Total covariance therefore gives

$$M_{t+1} = M_t - \eta \bar{G}_t, \quad (81)$$

$$\Gamma_{t+1} = A_\eta \Gamma_t A_\eta^\top + \frac{\eta^2}{B} (J_t - K_{G,t}), \quad (82)$$

$$\mathbb{E}[\mathcal{L}_T] = \frac{1}{2} \sum_j p_j [R_j(T) + \sigma_j^2]. \quad (83)$$

Thus the forecast updates gradient noise as the residual and weight variance change during learning. Gaussian input fourth moments and fresh independent minibatches are essential. Noise need only have the stated first two moments. This closure is not asserted for nonlinear targets, anisotropic inputs, adaptive optimizers or correlated samples.

The primary plug-in predictor estimates context frequencies, conditional coefficient vectors and residual noise variances using 1,024 independent calibration examples. Coefficients use ordinary least squares without an intercept, and noise variances use residual degrees of freedom. Its recurrence is exact for the fitted Gaussian population and is a plug-in approximation for the generating population. Population-oracle versions instead receive true teacher coefficients and noise moments and are labeled privileged. The 16-step pilot performs real candidate updates and evaluates on calibration observations; its computation is included. An observed-context-count baseline chooses the smallest permitted K covering the number of distinct observed context vectors, followed by the cheapest tree. This baseline and a conditioning baseline restricted to the same observed count were added after inspecting development results, then frozen separately before the new outcomes. Context count equals generating rank in this constructed family; it is not a general estimator of credit rank.

We ran both fresh-example SGD and minibatches resampled from a fixed 2,048-example cache, retaining a common 4,096-example test set. Reusing a cache makes later weights dependent on those examples and violates the fresh recurrence’s independence assumption. Population-oracle prediction minus exact population loss of the realized weights quantifies that discrepancy. All task conditions are averaged within each independent seed before 10,000 whole-seed bootstrap draws. Intervals are pointwise; the many secondary selector, rank and regime comparisons are not interpreted as separately confirmed discoveries. No new multiplicity-adjusted significance claim is made from these interval plots. Undefined correlations in numerically constant cases are retained with explicit defined-task counts rather than used to exclude outcomes.

Supplementary Fig. S38 shows endpoint regret, strong baselines and implementation time; Supplementary Fig. S39 shows prediction accuracy and cache sensitivity. The primary forecast reaches mean fixed-cache regret 0.000064 and 0.000042 in the feedback and joint arms, while the cheap observed-count baseline reaches 0.000134 and 0.000197. Thus the large improvement over the original scalar score does not establish a large advantage for fine morphology selection. Focused tests compare singular spectral formulas with direct descent, stochastic moments with exact Gaussian quadrature, and enforce that predictors cannot request training or test observations. Checks of source hashes and pre-outcome selection records passed. Numerical data, protocols, complete outcomes and exported source hashes are provided in `source_data/morphology_structure/` and `source_data/morphology_finite_horizon/`.

4.21 Boolean interactions, scalar tree structure and conditional credit

We used four-input Boolean functions to make the structural and credit questions explicit. The complete domain comprises sixteen equally weighted patterns. Writing $b_i = (x_i + 1)/2$ for Rademacher inputs $x_i \in \{-1, 1\}$, the raw labels $y \in \{0, 1\}$ are defined below. The symbols $\wedge$, $\vee$ and $\oplus$ denote AND, OR and exclusive OR (XOR),

respectively:

$$y_{\text{AND}} = b_1 \wedge b_2 \wedge b_3 \wedge b_4, \quad y_{\text{OR}} = b_1 \vee b_2 \vee b_3 \vee b_4, \quad (84)$$

$$y_{\text{parity}} = b_1 \oplus b_2 \oplus b_3 \oplus b_4, \quad (85)$$

$$y_{\text{OR(AND)}} = (b_1 \wedge b_2) \vee (b_3 \wedge b_4), \quad (86)$$

$$y_{\text{XOR(AND)}} = (b_1 \wedge b_2) \oplus (b_3 \wedge b_4), \quad (87)$$

$$y_{\text{AND(XOR)}} = (b_1 \oplus b_2) \wedge (b_3 \oplus b_4), \quad (88)$$

$$y_{\text{nested}} = b_1 \wedge [b_2 \vee (b_3 \wedge b_4)]. \quad (89)$$

1119 These are specified mechanistic examples rather than a representative sample of natural tasks. The associative AND,
1120 OR and parity families control for claims that greater depth or a particular input grouping is inherently necessary.
1121 Task permutations preserve a family's truth-table distribution; comparisons between families need not preserve its
1122 input-gradient spectrum.

1123 Every candidate has three scalar internal units, twelve trainable coefficients and six edges. For child outputs ℓ, r
1124 and trainable coefficients a, b, c, d , each unit computes $u = a + b\ell + cr + d\ell r$. Only the root supplies the output.
1125 All local coefficients can change during learning. There is no additional dense decoder or access to inputs outside
1126 a unit's descendant subtree. This is the multi-affine algebraic class of Proposition 9, not a positive-conductance
1127 neuron. An independent enumeration of all fifteen unordered, rooted, full binary trees on four labeled inputs
1128 supplies the structural analysis: three balanced trees have depth two, and twelve comb trees have depth three.
1129 An actual subtree cut is evaluated by removing the Fourier row for the constant function inside that subtree and
1130 retaining the singular-value tail beyond rank one. The maximum cut tail is a lower bound on regression MSE for
1131 the entire tree; zero tails on every actual cut are necessary and sufficient for exact representation in this class. These
1132 capacity statements do not by themselves establish convergence of a particular optimizer or attainability under an
1133 additional parameter box.

1134 We distinguish exact truth-value regression from threshold classification. Let $\mu_y = \mathbb{E}[y]$ and $\sigma_y = \sqrt{\text{Var}(y)}$
1135 be the mean and standard deviation over all sixteen patterns. Training targets are $t = (y - \mu_y)/\sigma_y$. Mean squared
1136 error on t therefore equals raw-output MSE divided by $\text{Var}(y)$. A predicted normalized output $\hat{t}$ corresponds
1137 to the raw Boolean decision $\hat{t} \geq (0.5 - \mu_y)/\sigma_y$. High classification accuracy does not imply small regression
1138 error, and a positive regression lower bound does not establish a classification impossibility. AND and OR are
1139 imbalanced: a constant majority prediction has accuracy 15/16. Balanced accuracy averages the true-positive rate
1140 (TPR) and true-negative rate (TNR), $(\text{TPR} + \text{TNR})/2$, and distinguishes this trivial result from accurate decisions
1141 on both classes. All sixteen truth patterns contribute to each reported population endpoint; they are not independent
1142 biological replicates.

1143 **Proposition 10** (An explicit Boolean grouping obstruction). *Let $f = (b_1 \wedge b_2) \text{ XOR } (b_3 \wedge b_4)$ on uniform inde-*
1144 *pendent bits, and let $z = (f - 3/8)/\sqrt{15/64}$. A scalar multi-affine balanced tree grouped as $(b_1, b_3) \mid (b_2, b_4)$ or*
1145 *$(b_1, b_4) \mid (b_2, b_3)$ has population NMSE at least 8/15 for every choice of its coefficients. The aligned grouping*
1146 *$(b_1, b_2) \mid (b_3, b_4)$ represents z exactly with coefficients in $[-2, 2]$.*

1147 *Proof.* For the crossed cut $b_1 b_3 \mid b_2 b_4$, the raw-target centered Walsh matrix, with nonconstant inside rows ordered
1148 as $x_1, x_3, x_1 x_3$, is

$$C = \frac{1}{8} \begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{pmatrix}.$$

1149 The columns correspond to $1, x_2, x_4, x_2 x_4$. Its three orthogonal rows each have squared norm 1/16, so the rank-one
1150 tail is exactly 1/8. Dividing by $\text{Var}(f) = 15/64$ gives 8/15; the other crossed pairing is symmetric. For the aligned
1151 tree, set

$$u = 2b_1 b_2 - 1 = \frac{-1 + x_1 + x_2 + x_1 x_2}{2}, \quad v = 2b_3 b_4 - 1 = \frac{-1 + x_3 + x_4 + x_3 x_4}{2}, \quad z = \frac{1 - 4uv}{\sqrt{15}}.$$

1152 The largest absolute coefficient is $4/\sqrt{15} < 2$. □

1153 The independent audit computed raw Walsh coefficients and centered ranks using exact rational arithmetic,
 1154 followed by numerical singular-value tails of the exact matrices. It retained all 105 family–tree pairs and constructed
 1155 an exact model for each of the 49 zero-bound cases. AND, OR and parity admit all fifteen trees; each mixed-gate
 1156 target admits exactly one. The independent constructions used nonconstant signed Boolean slices and exact
 1157 four-corner interpolation. Every raw truth table and normalized coefficient bound was checked, giving a maximum
 1158 absolute coefficient of $4/\sqrt{15}$ across these constructions. Thus compatible solutions exist within the experimental
 1159 parameter box, without assuming convergence to them.

1160 The OR-of-AND target and the nested target each use two AND gates and one OR gate, yet their minimum
 1161 exact depths are two and three. For the nested function, the smallest lower bound among all three balanced trees
 1162 is $(22 - 2\sqrt{57})/55 \simeq 0.12546$ NMSE; the stated comb tree is exact. All seven functions have input-gradient
 1163 rank four, but their full spectra generally differ. This is a rank-versus-interaction example, not an isospectral
 1164 comparison between the mixed-gate families. The maximum over cuts remains a lower bound; we do not assume
 1165 every incompatible whole tree attains it.

Gate coordinates illustrate conditional derivatives. The elementary gates can also illustrate how another branch affects a local derivative. In raw Boolean branch-output coordinates $u, v \in \{0, 1\}$, their multilinear extensions are

$$F_{\text{AND}}(u, v) = uv, \quad \partial_u F_{\text{AND}} = v, \quad (90)$$

$$F_{\text{OR}}(u, v) = u + v - uv, \quad \partial_u F_{\text{OR}} = 1 - v, \quad (91)$$

$$F_{\text{XOR}}(u, v) = u + v - 2uv, \quad \partial_u F_{\text{XOR}} = 1 - 2v. \quad (92)$$

1166 The other branch thus gates a derivative’s magnitude or sign in these canonical coordinates. The XOR-of-AND
 1167 diagram illustrates this dependence and a specific input-grouping constraint. Its gate labels are explanatory; the
 1168 learned units are not constrained to remain Boolean gates or to preserve that representation. A broadcast learner can
 1169 change internal coordinates, so the derivative diagram is not a proof that broadcast cannot learn the task. Positive
 1170 affine normalization of the raw target changes derivative scale but not these signs. Actual learned update fields and
 1171 finite-training outcomes require the separate experimental comparisons below.

1172 **Frozen learning comparisons.** Four candidate trees were tested: the three balanced pairings and the comb
 1173 $b_1 \mid [b_2 \mid (b_3, b_4)]$. These are four of the fifteen trees in the independent capacity analysis. Each seed applied one
 1174 common physical input permutation to both the semantic task variables and candidate leaves. Per-family streams
 1175 supplied initialization, a 256-observation training cache, 1,024 independent test observations and minibatch indices.
 1176 All paired tree, optimizer, credit and rate conditions shared these streams within family. Data were drawn uniformly
 1177 with replacement from the sixteen patterns; training and test labels received independent Gaussian noise of SD 0.15
 1178 in normalized-target units. Independent draws could repeat the same input pattern.

1179 Every coefficient began from an independent mean-zero Gaussian with SD 0.5, except zero biases; no target
 1180 gates, coefficients or exact-construction weights were used. Initial weights and every subsequent update were
 1181 clipped to $[-2, 2]$. Training used 2,048 updates, minibatches of 32, and gradient norm clipping at ten. Exact credit
 1182 multiplied each local eligibility by the current complete path derivative. Broadcast set the two nonroot internal
 1183 sensitivities to one; root sensitivity remained one under both rules. All twelve coefficients were trainable. Stochastic
 1184 gradient descent (SGD) and Adam were tested separately; Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$.

1185 Five development seeds (2026110–2026114) selected a rate separately for each optimizer and rule from 0.003,
 1186 0.01 and 0.03, minimizing mean final clean population NMSE over all seven families and four trees. Ties within
 1187 10^{-12} selected the smaller rate. Adam selected 0.003 for exact credit and 0.01 for broadcast; SGD selected
 1188 0.03 and 0.01, respectively. These choices were recorded with cryptographic hashes before twenty new seeds
 1189 (2026210–2026229), which retained all three rates. The resulting 6,720 fresh fits supplied 53,760 records at updates
 1190 0, 1, 16, 64, 256, 512, 1,024 and 2,048. An implementation check using seed 2026000 exercised all 336 conditions

and was excluded from inference. These seeds vary initialization, permutation and sampled noisy data for seven fixed logical templates; they do not sample new Boolean function families.

The primary endpoint was clean population NMSE evaluated on all sixteen patterns. No sampled test variance or observed class-frequency denominator entered this endpoint. Noisy-test NMSE, raw threshold accuracy and balanced accuracy were secondary. Gradient cosines compare the delivered and exact aggregate clean-population gradients at each learner’s own trained state, before clipping or optimizer transformation; they are not mean per-example cosines or common-state comparisons between separately trained rules.

Two primary Adam contrasts concerned XOR-of-AND: the mean of the two crossed balanced-tree errors minus aligned-tree exact-credit error, and broadcast minus exact error on the aligned tree. The latter compares the prespecified development-selected recipes, not identical learning rates. Ten thousand resamples of twenty paired seed blocks supplied two-sided Bonferroni-adjusted 97.5% intervals for these two contrasts. A positive adjusted lower limit and mean improvement of at least 0.01 NMSE were both required. Other task, optimizer and common-rate contrasts have descriptive pointwise 95% intervals.

The grouping improvement was 0.56317 NMSE [0.54914, 0.58275], satisfying both primary criteria. The credit difference was 0.002507 [0.001780, 0.003318]: its adjusted interval excluded zero, but it did not meet the prespecified 0.01-NMSE mean-effect margin. Both recipes learned the aligned XOR-of-AND task well and classified all sixteen patterns correctly in every fresh seed. All three same-rate Adam credit differences were also below 0.003 NMSE. The low-rate SGD contrast had a wide interval and is retained. The larger parity credit differences are descriptive outcomes of a prespecified control task, not primary discoveries. Neither the gate illustration nor these observations establish an impossibility theorem for broadcast learning, or show that a particular change of internal coordinates caused its success. Full-rate outcomes, constant-mean and random-candidate references, clipping records, weights and source/selection hashes are retained under `source_data/boolean_morphology`.

Post hoc diagnostic of local target information. After examining the fresh learning outcomes, we asked how much target information was available within each proper input subset S . In this diagnostic, f denotes a raw Boolean target, $z = (f - \mathbb{E}[f]) / \sqrt{\text{Var}(f)}$ its normalized version, and $\hat{f}_U$ its Walsh coefficient on input subset U . The squared L^2 norm below averages over the uniform input distribution. We defined

$$P_S = \|\mathbb{E}[z \mid x_S]\|_{L^2}^2 = \frac{1}{\text{Var}(f)} \sum_{\emptyset \neq U \subseteq S} \hat{f}_U^2.$$

We cross-checked all ninety-eight task–subset values by exact rational evaluation of conditional means on the full truth table. Parity4 has $P_S = 0$ for every proper subset. XOR-of-ANDs instead has $P_S = 1/5$ for every two-input subset, including compatible and crossed pairs. Thus this diagnostic quantifies accessible lower-order target information rather than grouping compatibility.

Let F denote the model’s normalized prediction. At fixed parameters, a local eligibility $e_S(x_S)$ at a nonroot subtree is a function only of that subtree’s inputs. Conditional expectation gives $\mathbb{E}[ze_S] = \mathbb{E}[\mathbb{E}[z \mid x_S]e_S]$; when $P_S = 0$, the direct target contribution to the population broadcast update $\mathbb{E}[(F - z)e_S]$ vanishes. This identity does not imply an inability to learn: label-dependent changes in root coefficients can subsequently affect F , and finite training samples and stochastic updates need not exhibit exact population cancellation. The diagnostic was added after observing the learning results and did not change the training protocol or its predeclared criteria.

4.22 Matched fixed-profile learning and credit-field geometry

This experiment separates forward compatibility, spatial profile calibration and learning. The cut construction specializes hierarchical tensor representations to scalar multi-affine messages [7, 8]. Each algebraic tree has seven binary multi-affine units, four coefficients per unit, and only a root readout. The six nonroot units receive the manipulated credit field. The root derivative is one in every condition. Pairwise and quartic targets share a compatible balanced tree and a seed-specific permutation of eight independent Rademacher inputs. Their coefficients have magnitude one-half and independently sampled signs. Their input-sensitivity second moments

both equal $I_8/4$, but their clean output variances are one and one-half. A nested target combines products over prefixes of sizes two, four, six and eight on its separately compatible tree; it is not part of the input-isospectral comparison.

Let $q_n(x) = \partial u_0 / \partial u_n$ be the exact path derivative for unit n , and let $\delta(x)$ be the scalar derivative of normalized half-squared error with respect to root output. Local coefficient eligibilities are $1, u_L, u_R, u_L u_R$. Exact credit multiplies these by δq_n . Unit broadcast replaces q_n by one. An initially calibrated profile uses $a_n = 256^{-1} \sum_x q_n(x)$ over 256 unlabeled initialization examples, then keeps a_n fixed. The sign control retains only $\text{sign}(a_n)$. Profiles require no target labels. Learning-rate choices are separate from the profile calculation. Fixed positive coordinate scaling largely cancels under Adam, so calibrated and sign-only Adam trajectories are expected to be nearly identical; they are not independent mechanistic confirmations.

The protocol and code were frozen before development. Three disjoint development seeds tested the same three rates for every rule and optimizer. The selected rate minimized final validation NMSE averaged across the three task families, separately for each model, optimizer and rule. A common rate per model and optimizer was also selected. Both choices were sealed before twenty fresh paired seed blocks. Matching and quartic conditions share initial weights, calibration inputs, example streams, additive label noise and minibatch order. Both SGD and Adam were retained; the primary contrast was predefined for Adam as the quartic-minus-pairwise difference in calibrated-minus-exact held-out NMSE. The experiment contains 432 development and 1,200 fresh fits, with the selected and common-rate union retained without endpoint exclusions. Forty-eight short benchmark fits were excluded from selection and inference.

Algebraic training, validation and test sets contain 2,048, 1,024 and 4,096 examples, with minibatches of 64 and gradient-norm clipping at ten. Initial nonbias coefficients are independent $\mathcal{N}(0, 0.5^2)$ draws and biases are zero. Adam uses $\beta = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. Both algebraic optimizers test rates 0.003, 0.01 and 0.03. The conductance teacher uses 1,024 training, 1,024 validation and 4,096 test examples without label noise, constraining log-conductances to $[-5, 5]$. Its SGD rates are 0.001, 0.003 and 0.01, and Adam rates are 0.003, 0.01 and 0.03.

Algebraic fits stop after 1,024 steps, retaining coefficient bounds $[-2, 2]$ and all recorded bound contacts. Saved checkpoints occur at steps 0, 1, 16, 64, 256, 512 and 1,024. Additive label noise has standard deviation 0.15, giving expected noise-only NMSE of 0.0225 for matching and nested targets and 0.045 for quartic targets. The gradient divides the residual by each family’s own clean variance. Clean full-domain population error is a secondary endpoint. All twenty endpoint seeds, including poorly fitted exact quartic cases, remain in the figures and means. Bootstrap intervals use 10,000 whole-seed resamples and are descriptive 95% intervals; the common-rate comparison tests sensitivity to tuning, not a new independent cohort.

For each saved state, let Q have one row per evaluation example and six columns containing q_n . For a dictionary A , energy capture is $\|QP_A\|_F^2 / \|Q\|_F^2$, where P_A is its orthogonal projector. The best fixed rank-one profile is the leading right singular vector of Q , with capture $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$. Its profile and per-example amplitudes are oracle fits. The uncentered second moment is $C = Q^T Q / N$, and participation-ratio effective rank is $(\text{tr} C)^2 / \text{tr}(C^2)$. We additionally evaluated unit broadcast, the fixed initial profile, ancestry spans, full-rank capture, and loss-error-weighted fields $\text{diag}(\delta)Q$. The latter has its own optimal profile; applying the path-optimal profile after error weighting answers a different question.

Eligibility-weighted reconstruction multiplies exact and projected path fields by their local coefficient features before comparing parameter updates. Site-normalized spectra divide each path coordinate by its root-mean-square magnitude, exposing sensitivity to fixed coordinate scaling. Raw capture energies are not invariant to arbitrary hidden-unit rescaling. At identical weights and inputs, Q is independent of the target; a scalar residual also cancels from normalized per-example projection capture whenever it is nonzero. Eighty common-state relabeling controls verify that identity. These diagnostics characterize learned-state geometry, rather than infer a task-only credit rank at initialization.

An independent analysis recomputed 5,040 saved states in the matched cohort, checked 20,160 metrics to maximum absolute discrepancy 8.7×10^{-15} , and verified 120 matching–quartic pairing identities. The earlier 600 selected compatible-tree fits had metric records but no saved parameter states. Faithful replay reconstructed 3,600 checkpoints; all 28,800 archived deterministic metrics matched to 8.9×10^{-16} . Those reconstructions provide

1283 replication and trajectory comparisons, not previously stored checkpoints. A separate independent learning audit
 1284 recomputed held-out NMSE for 138 task files with maximum discrepancy 2.22×10^{-16} .

1285 The positive-conductance comparison uses the prior compatible teacher across three paired input permutations
 1286 and the same credit definitions. It does not implement the algebraic quartic target. All four rules learn it accurately
 1287 with Adam; exact credit has mean NMSE 9.39×10^{-6} , compared with 1.35×10^{-6} for the fixed-profile controls.
 1288 Positive path derivatives make sign and unit broadcasting identical. Both models' local updates were independently
 1289 checked by finite differences. Protocols, selections, checkpoints, full trajectories, numerical audits and scripts
 1290 are supplied in `source_data/credit_rule_bridge/`, `source_data/credit_resolution_bridge/` and their
 1291 corresponding script directories.

1292 4.23 State-matched positive-conductance reliability experiment

1293 We next tested whether positive conductances could realize the attenuation prescribed by Eq. 59 in a separate trained
 1294 branch model (Table S4). Here b indexes branches, i examples and d inputs within a branch. Every input rate x_{bid}
 1295 was nonnegative, and each excitatory conductance was parameterized by $g_{bd} = \text{softplus}(q_{bd})$, with trainable raw
 1296 parameter q_{bd} . With leak conductance g^L , define total unshunted conductance and voltage on example i

$$G_{bi} = g^L + \sum_d x_{bid} g_{bd}, \quad V_{bi} = \frac{\sum_d x_{bid} g_{bd}}{G_{bi}}. \quad (93)$$

1297 A fixed shunt $\kappa_b \geq 0$ had reversal zero. To isolate its effect on input resistance from its effect on the forward feature,
 1298 an oracle current $\kappa_b V_{bi}$ was added on each example. The perturbed state is therefore

$$V'_{bi} = \frac{\sum_d x_{bid} g_{bd} + \kappa_b V_{bi}}{G_{bi} + \kappa_b} = V_{bi}, \quad \frac{\partial V'_{bi}}{\partial q_{bd}} = \frac{x_{bid}(1 - V_{bi})}{G_{bi} + \kappa_b} \text{sigmoid}(q_{bd}). \quad (94)$$

1299 Relative to the unshunted eligibility, the example-specific physical input-resistance factor is $G_{bi}/(G_{bi} + \kappa_b)$. The
 1300 compensating current was recomputed from the pre-shunt state and treated as a clamp, not differentiated as an
 1301 independently learned pathway. Consequently this manipulation deliberately removes forward feature suppression
 1302 while retaining conductance attenuation in the local update.

1303 The binary task used eight branches, four inputs per branch, 512 training examples and 2,048 held-out examples.
 1304 Input rates were generated from a positive baseline plus a class-dependent branch pattern and clipped at zero. Initial
 1305 full-data gradients supplied $S_b = \|\mathbf{g}_b\|_2^2$. Coefficient-noise variance was then scaled so that

$$N_b = S_b \exp(-2hx_b), \quad x_b \in [-1, 1], \quad (95)$$

1306 where x_b is the evenly spaced branch-order coordinate and $h \in \{0, 0.5, 1, 1.5, 2\}$ controls reliability heterogeneity.
 1307 The statistical reliability was $r_b = S_b/(S_b + N_b)$, lower-bounded at 0.03 for numerical stability. Because the common
 1308 step was $\eta = c/L$ with $c = 0.5$, the corrected fixed-step target was $a_b^* = \min\{1, r_b/c\}$ and the initial shunt was
 1309 $\kappa_b = \overline{G}_b[(a_b^*)^{-1} - 1]$, where $\overline{G}_b$ is the training-example mean of G_{bi} at initialization. Shunts remained fixed during
 1310 the 40 updates; the excitatory conductance parameters were trained, and the same coefficient-noise realizations
 1311 were used across matched methods within seed.

1312 Nine conditions separated the relevant alternatives: exact clean backpropagation, noisy credit without a shunt,
 1313 a state-matched unattenuated-eligibility control, the step-consistent best single global shunt, reliability-aligned,
 1314 seed-shuffled and anti-aligned shunts, an explicit point-gain control receiving the identical branch factors, and
 1315 aligned shunting with clean credit. The point-gain control is a resource control, not a biological assertion: because
 1316 it receives the same branch partition and factors, its update must equal Eq. 94. The clean-backpropagation condition
 1317 retains the unattenuated exact gradient and is the optimization reference.

1318 An exploratory implementation used r_b despite taking $c = 0.5$ and was excluded from inference. After correcting
 1319 the target factor, we fixed the analysis plan, evaluated two implementation-check seeds (also excluded), and then
 1320 used 50 paired analysis seeds (8100–8149). All 2,250 expected seed-condition rows completed. The minimum
 1321 input rate was zero, maximum state mismatch was 2.22×10^{-16} , and independently computed point-gain and

unattenuated-eligibility trajectory differences were 4.00×10^{-15} and 1.31×10^{-14} . A finite-difference validation held the compensating current fixed while perturbing an excitatory parameter; its maximum relative error was 3.60×10^{-8} . All prespecified numerical criteria were met.

At $h = 2$, reliability-aligned minus best-global one-step test-loss reduction was 0.001817 (95% paired seed-bootstrap interval 0.001618–0.002020), positive in 49/50 seeds (two-sided Wilcoxon $P = 3.55 \times 10^{-15}$). Aligned minus unshunted reduction averaged 0.001339 (0.000594–0.002096), positive in 33/50 seeds ($P = 0.0030$). After 40 updates, aligned-minus-global test loss was -0.020404 (-0.022220 to -0.018705), favoring aligned in 50/50 seeds ($P = 1.78 \times 10^{-15}$). Aligned shunting also had lower final loss than shuffled and anti-aligned shunts by 0.009204 and 0.013781, respectively. However, aligned-minus-unshunted final loss was 0.000425 (-0.002456 –0.003071; $P = 0.154$), so the immediate improvement did not establish a sustained loss difference from no shunt. Exact clean backpropagation retained the lowest mean final loss. These results validate a positive-conductance approximation and ordering of fixed-step oracle reliability factors under a state clamp; they do not establish autonomous shunt learning, dendrite-exclusive computation or general superiority to unshunted learning.

4.24 Adaptive local reliability estimation

The preceding experiment supplied the reliability moments. We next asked whether they could be estimated from noisy local observations, specifying the adaptive-gain experiment before examining its outcomes (Table S5). At the current parameter state, branch b received two conditionally independent noisy local gradient observations, $\hat{\mathbf{g}}_{1b}$ and $\hat{\mathbf{g}}_{2b}$, on the same examples. It formed

$$\hat{\mathbf{S}}_b^{\text{obs}} = \langle \hat{\mathbf{g}}_{1b}, \hat{\mathbf{g}}_{2b} \rangle, \quad \hat{N}_b^{\text{obs}} = \frac{1}{2} \|\hat{\mathbf{g}}_{1b} - \hat{\mathbf{g}}_{2b}\|_2^2. \quad (96)$$

For independent zero-mean credit noise, these estimate the squared clean branch gradient and the noise energy of one observation. Eight paired observations initialized the moments; exponential averages with decay 0.9 were then updated before each of 40 training steps. Negative finite-sample signal estimates and moments below 10^{-12} were clipped only when the gain was constructed. The step-consistent gain followed Eq. 59, was bounded to $[0.03, 1]$, and was converted to a shunt using the branch’s current mean total conductance.

The task and state-matched positive-conductance model were unchanged. Three heterogeneity levels $h \in \{0, 1, 2\}$ crossed exact clean backpropagation, noisy no-shunt learning, the fixed initial-oracle shunt, adaptive local, adaptive global and fixed-permutation adaptive shunts, and an independently computed adaptive point gain. Main update noise and estimator-probe noise were paired across methods. Two implementation-check seeds were excluded from inference; the analysis then used 50 paired seeds (8200–8249), producing all 1,050 planned seed-condition outcomes. Maximum state error was 2.22×10^{-16} , maximum physical-shunt finite-difference error was 4.01×10^{-8} , and the adaptive physical-shunt/point-gain trajectories agreed within 4.00×10^{-15} .

At $h = 2$, adaptive-local minus adaptive-global one-step loss decrease was 0.001420 (95% paired-seed bootstrap interval 0.001158–0.001684; 45/50 positive; two-sided Wilcoxon $P = 4.49 \times 10^{-13}$). Adaptive local reduced final loss relative to adaptive global by 0.016217 (0.014501–0.017913; 50/50 seeds; $P = 1.78 \times 10^{-15}$) and relative to shuffled adaptive placement by 0.005528 (0.003766–0.007418; 41/50; $P = 7.80 \times 10^{-8}$). The one-step local-minus-shuffled mean was positive, but only 26/50 signs favored local and therefore did not meet the predefined directional criterion.

Adaptive estimation nevertheless performed worse than the no-shunt and fixed-oracle references. Adaptive-local final loss exceeded no shunt by 0.004250 (0.002499–0.005975; 39/50; $P = 1.78 \times 10^{-5}$) and the fixed initial oracle by 0.003449 (0.002577–0.004305; 43/50; $P = 4.44 \times 10^{-9}$). The adaptive final gain ordering had mean within-seed Spearman $\rho = 0.756$ with the initial oracle and mean absolute gain error 0.205. Thus local estimates identified useful relative gain placements, but did not provide a sustained advantage over unshunted learning. Paired gradient probes, known optimizer step fraction and the exact state clamp remain supplied; this is not a kinetic or fully autonomous model of inhibitory plasticity.

1365 4.25 Same-span noisy coefficient-learning experiment

1366 To isolate coefficient learning from address capacity, we compared three dictionaries with the same span in a
1367 rate-based factorial (Table S6). The 16-leaf balanced-tree Haar basis supplied the common rank-eight span through
1368 depth three. The tested dictionaries were the eight orthonormal Haar modes, all 15 nested indicators through depth
1369 three, and the same nested indicators scaled by positive gains evenly spaced on the log scale from $\exp(-3)$ to
1370 $\exp(3)$. The dictionaries’ projectors agreed to 9.44×10^{-16} and their positive-spectrum Gram condition numbers
1371 were 1, 15 and 388.52.

1372 Each independent seed drew a unit-norm target in the common span. At every update the estimator observed
1373 the target plus isotropic Gaussian noise with standard deviation $2/\sqrt{n_{\text{eff}}}$, where $n_{\text{eff}} \in \{4, 16, 64, 256\}$. The
1374 same 80 observation vectors were used for every matched parameterization. Vanilla coefficient descent used
1375 $0.2/\lambda_{\max}(\Phi^T \Phi)$; the oracle coordinate control used $0.2(\Phi^T \Phi)^+$. Population field loss was recorded after 1, 5, 20
1376 and 80 updates. The target address residual was below 1.45×10^{-30} in every condition, so all reported loss is
1377 within-span coefficient-estimation error.

1378 The predictions were specified before two implementation-check seeds, which were excluded from inference,
1379 and 50 paired seeds (9100–9149). The results contradicted the prediction that Haar coordinates would always
1380 reduce finite-time error. At $n_{\text{eff}} = 4$, raw-minus-Haar final loss was -0.2774 (95% paired interval -0.3260 to
1381 -0.2313) and scaled-minus-Haar was -0.1906 (-0.2500 to -0.1335): slower modes filtered high estimator
1382 noise. At $n_{\text{eff}} = 256$, the effects reversed to 0.02555 (0.02171 – 0.02952) and 0.17091 (0.14576 – 0.19686), positive
1383 in 50/50 seeds for both contrasts. Equation 57, derived after these outcomes to explain the reversal, predicted
1384 median crossovers at effective sample sizes 38.53 and 8.25. The Gram-preconditioned field trajectories agreed
1385 across dictionaries to 1.15×10^{-15} . Thus static parameterization can alter finite-data learning without altering the
1386 attainable credit field.

1387 S5 Artificial-tree experiments

1388 The exact conductance gradient separates local eligibility from transported error. We used controlled artificial trees
1389 to ask how much of that error must be resolved: neuron identity, within-arbor path identity, or both. We began with
1390 regular directed morphologies and increased feedback resolution while holding the forward task fixed within each
1391 comparison.

1392 5.1 Architecture and input sparsity

1393 MNIST images were converted to tensors, flattened to 784 inputs and not normalized or augmented. The 60,000-
1394 image training set was split by seed into 48,000 training and 12,000 validation images; the official 10,000-image
1395 test set was evaluated only after training. The core contained 128 modeled excitatory neurons and no separate
1396 inhibitory-neuron population. Each neuron had morphology [3,3]: three proximal compartments, three distal
1397 children per proximal compartment and one soma, for 13 compartments per neuron. There were no somatic input
1398 synapses.

1399 At each of the 12 dendritic compartments, separate sparse parameter arrays (TopK) selected 40 excitatory and
1400 20 inhibitory contacts from the same 784-pixel input. The model therefore contained 92,160 active input contacts
1401 among 2,408,448 candidate synaptic parameters. Inactive local weights were not updated. Child-to-parent couplings
1402 and all active conductances were learned after a softplus transform. A linear 128×10 decoder produced the class
1403 scores. The archived model-resource record contained 2,414,602 trainable raw parameters in total, including
1404 candidate synaptic arrays, couplings, reactivation parameters and decoder.

1405 5.2 Raw-additive and normalized-additive controls

The raw-additive control used the configuration alias `dendritic_additive`, with shunting and additive normalization both disabled. For compartment n , the implemented forward computation was

$$J_n^{\text{E,add}} = \sum_{i \in \mathcal{I}_n^{\text{E}}} g_i^{\text{E}} x_i^{\text{E}}, \quad J_n^{\text{I,add}} = \sum_{j \in \mathcal{I}_n^{\text{I}}} g_j^{\text{I}} x_j^{\text{I}}, \quad (97)$$

$$J_n^{\text{child,add}} = \sum_{c \in \mathcal{C}_n} g_{c \rightarrow n}^{\text{den}} a_c, \quad V_n^{\text{add}} = J_n^{\text{E,add}} - J_n^{\text{I,add}} + J_n^{\text{child,add}}, \quad a_n = f_n(V_n^{\text{add}}). \quad (98)$$

1406 Consequently,

$$\frac{\partial V_n^{\text{add}}}{\partial g_i^{\text{E}}} = x_i^{\text{E}}, \quad \frac{\partial V_n^{\text{add}}}{\partial g_j^{\text{I}}} = -x_j^{\text{I}}, \quad \frac{\partial V_n^{\text{add}}}{\partial g_{c \rightarrow n}^{\text{den}}} = a_c, \quad \delta_c^V = \delta_n^V g_{c \rightarrow n}^{\text{den}} f'_c(V_c). \quad (99)$$

1407 This control retained excitatory currents and subtractive inhibitory currents on the same branching graph, with
 1408 the same reactivation function. It omitted the conductance denominator, reversal-potential driving force and
 1409 parent-compartment resistance. Its inhibitory term is a negative current contribution, not a negative conductance.
 1410 Raw-parameter gradients in both architectures also included the softplus derivative.

1411 The main regular-tree experiments use this raw-additive model. Supplementary Figs. S1–S3 and S4A,B instead
 1412 use a stage-normalized variant that standardizes the raw voltage separately within each example s across the output
 1413 coordinates n of every stage. Writing $Z_{sn} = J_{sn}^{\text{E,add}} - J_{sn}^{\text{I,add}} + J_{sn}^{\text{child,add}}$, the normalized model is

$$V_{sn}^{\text{norm}} = \frac{Z_{sn} - \langle Z_s \rangle_n}{\text{sd}_n(Z_s) + \epsilon}, \quad a_{sn} = \frac{\tanh(0.1 V_{sn}^{\text{norm}}) + 1}{2}. \quad (100)$$

1414 Its reactivation slope and midpoint were fixed at $m = 0.1$ and $b = 0$, respectively. We call this the *normalized-*
 1415 *additive model*. Because normalization changes the forward computation, its absolute accuracies are not treated as
 1416 repeated measurements of the raw-additive model.

1417 We verified the normalized-additive implementation by reproducing its backpropagation endpoints. Across
 1418 five seeds, normalized-additive and shunting networks reached 48.300% and 49.542% on flattened CIFAR-10,
 1419 compared with reference values of 48.300% and 49.516%. We then selected a raw-additive backpropagation (BP)
 1420 configuration using validation accuracy only. Adaptive initialization, empirical reactivation calibration and the
 1421 same optimizer settings reached 49.960% validation accuracy and 49.723% test accuracy across three calibration
 1422 seeds. This procedure defines the operating point for the independent learning-rule comparison; it does not separate
 1423 initialization from reactivation calibration and is not itself a comparison among learning rules. A provisional
 1424 raw-additive configuration with a matched-BP mean of 38.98% was excluded because it did not provide an adequate
 1425 operating point.

1426 In the MNIST, Fashion-MNIST and depth experiments, every compartment used a learned reactivation function
 1427 with slope m_n and midpoint b_n :

$$f_n(V) = \frac{\tanh[m_n(V - b_n)] + 1}{2}, \quad f'_n(V) = \frac{m_n}{2} \{1 - \tanh^2[m_n(V - b_n)]\}. \quad (101)$$

1428 The shunting architecture used its analytical initialization policy. Because the raw-additive voltage scale differs,
 1429 the raw-additive architecture used its predefined occupancy-quantile policy: the empirical 0.1 and 0.9 voltage
 1430 quantiles were mapped to reactivation outputs 0.1 and 0.9, with minimum quantile width 10^{-3} , maximum slope 50
 1431 and reversion on an invalid fit. These were architecture-specific, prespecified reactivation-initialization policies
 1432 rather than identical initial values. The independent raw-additive CIFAR-10 confirmation instead used the coupled
 1433 empirical-calibration procedure selected by the validation-only screen above. All runs met the predefined complete-
 1434 ness, calibration, convergence and backpropagation-adequacy criteria. The prespecified neuron-specific bandwidth
 1435 criterion was met, whereas the prespecified exact-path superiority criterion was not (Supplementary Fig. S4D).

5.3 Training protocols and feedback cohorts

The primary feedback-definition experiment used the three-factor rule, a locally updated decoder, mini-batches of 256 and 180 epochs. Adam used $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ and no weight decay. Raw TopK parameters, child couplings, reactivation parameters and decoder used learning rates 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} and 1.5×10^{-3} , respectively. Local gradients were clipped elementwise at 5. The best validation checkpoint was restored; early stopping and mixed precision were disabled. Runs used deterministic PyTorch settings, disabled TensorFlow-32 (TF32) and cuDNN benchmarking, and used the highest float32 matrix-multiplication precision.

The main feedback ladder used 15 seeds (42–56) for each of three feedback fields—strict scalar feedback, neuron-specific feedback and exact path transport—in each architecture (90 runs). A matched-width scalar-fallback condition at the same seeds and architectures added 30 runs. The decision to replace an earlier mixed-source set with these 120 consistently executed runs was made before their outcomes were examined. Versioned configurations, checkpoints and source identifiers are retained for all runs; the superseded results are not used in the article.

The Fashion-MNIST replication retained these shunting and raw-additive architectures, contact pattern, three-factor local rule, local decoder, optimizer, 180-epoch limit and architecture-specific initialization without retuning. Images were flattened to 784 inputs without normalization or augmentation. For each paired seed 10200–10209, 48,000 of the 60,000 training images were used for fitting and 12,000 for validation; the official 10,000-image test set was used only after checkpoint selection. Sixty fits crossed shunting and raw-additive trees with matched-width scalar-fallback feedback, neuron-specific feedback and exact path transport.

Mean accuracies for matched-width scalar-fallback, neuron-specific and exact path fields were 0.8393, 0.8839 and 0.8829 in shunting trees and 0.8442, 0.8820 and 0.8833 in raw-additive trees. Neuron-specific feedback improved over matched-width scalar-fallback feedback by 4.46 percentage points (95% paired-seed interval 4.09–4.88) in shunting trees and 3.78 points (3.43–4.14) in raw-additive trees; every seed difference was positive in both architectures (exact two-sided sign-flip $P = 0.001953$). Exact path minus neuron-specific feedback was -0.11 points (-0.21 to -0.01 ; 2/10 positive; $P = 0.0957$) and 0.13 points (-0.08 to 0.35 ; 8/10 positive; $P = 0.314$), respectively. The benefit of preserving neuron identity therefore replicated, whereas within-tree transport added no reliable benefit on this dataset (Supplementary Fig. S4E).

The prespecified replication criterion required at least 8/10 positive neuron-specific-minus-scalar-fallback differences and an interval above zero in each architecture. All 60 runs produced finite endpoints and complete checkpoints at the specified seeds, without activating a fallback pathway. Seed-level outcomes, contrasts and versioned source identifiers are provided in `source_data/fashion_feedback_ladder/`.

5.4 Raw-additive CIFAR-10 feedback ladder

The CIFAR-10 analysis used flattened inputs without normalization or augmentation and a 90:10 training–validation split. The model contained one 20-soma dendritic population with branch factors [3, 3, 3, 3], 25 direct contacts of each input sign per branch and a 20–32–16–10 decoder. With `input_mode=1`, the stored inhibitory-population fields are inactive unless a separate build flag is enabled. That flag was absent, so the realized network had direct excitatory and inhibitory input streams but no separate inhibitory-cell population. This architecture was verified from the source before the accuracies were examined.

A validation-only backpropagation screen selected adaptive network initialization together with empirical per-layer reactivation calibration from three training batches. Because these choices were selected jointly, the experiment does not attribute recovery to either component alone. All four feedback conditions used the resulting raw-additive forward model, Adam at 10^{-3} , batch size 256, at most 200 epochs, early-stopping patience 40 and restoration of the validation-best checkpoint. Adam’s built-in weight decay was zero. The separate configuration field equal to 0.01 controlled sparse active-weight maintenance rather than L2 regularization. Twenty paired analysis seeds (10800–10819) crossed strict scalar, neuron-specific, exact-path and matched-backpropagation feedback, for 80 fits.

All 80 runs produced finite endpoints, complete checkpoints and empirical calibration records while preserving the specified seed pairing; no convergence flags occurred. Mean test accuracies were 0.34472 for strict scalar

feedback, 0.50865 for neuron-specific feedback, 0.50006 for exact path transport and 0.50122 for matched backpropagation. Neuron-specific sharing minus strict scalar feedback was 0.16393 (95% paired- t interval, 0.15878–0.16908; 20/20 positive; Holm-adjusted directional $P = 5.57 \times 10^{-24}$; exact sign-flip $P = 9.54 \times 10^{-7}$). Exact path minus neuron-specific was -0.00859 (-0.01171 to -0.00547 ; 2/20 positive), so every prespecified exact-path superiority criterion failed. Exact path minus backpropagation was -0.00115 (95% interval, -0.00457 to 0.00226) and passed a formal equivalence assessment using two one-sided tests (TOST) within ± 0.01 (90% interval, -0.00398 to 0.00167 ; $P_{\text{TOST}} = 1.58 \times 10^{-5}$). Thus this harder-data cohort confirms a feedback-bandwidth benefit at the neuron-identity level but not an advantage of exact within-arbor transport. It is a compact transfer assay, not a competitive CIFAR-10 benchmark.

The exact-transport factorial was a separate five-seed protocol (seeds 42–46): it used 200 epochs, a rectified linear unit (ReLU) core output, and crossed three- versus five-factor eligibility with local versus backpropagated decoder updates. TopK, coupling, reactivation and decoder learning rates were 1.5×10^{-3} , 7×10^{-4} , 5×10^{-4} and 1.5×10^{-3} . Its same-architecture backpropagation reference used the same seeds, 200-epoch budget, ReLU output, and validation-based checkpoint selection, with method-specific learning rates of 10^{-3} for TopK and decoder and 5×10^{-4} for coupling and reactivation. Test data were not used for checkpoint selection. Tables S10, S11 and S12 provide the complete specifications and provenance; Table S15 reports the principal outcomes.

Across the two tested architectures, preserving neuron-specific feedback recovered most of the scalar-feedback deficit. This comparison does not imply that shunting universally improves learning. The separate exact-transport experiment establishes that local three-factor eligibility is sufficient when the remaining compartment error is supplied accurately.

We compared the analytic gradient with automatic differentiation in 100 diagnostic runs. Median relative error was 7.43×10^{-8} and the maximum was 1.88×10^{-7} . Parameter tensors, rather than scalar tensor elements, were the diagnostic units.

5.5 Depth-by-feedback experiment and fixed-checkpoint diagnostic

Table S20 accounts for all planned, excluded and included units in this and the related artificial-tree cohorts. This experiment crossed two tasks, raw-additive and shunting neuron models, one to four dendritic stages, and three local teaching fields at ten paired seeds. A matched backpropagation condition was included for each task–model–depth–seed combination, giving 480 local-learning and 160 backpropagation runs. The local conditions all used the theorem-derived three-factor eligibility. One modeled population of 128 neurons was used so that equal neuron indices in different feedforward populations could not share a teaching coordinate. The design, configurations, seeds, inclusion criteria and analysis code were specified before the outcomes were examined.

Runs were included if they contained the specified configuration and seed, a checkpoint, complete learning curves, all expected dendritic stages, nondegenerate learned couplings and finite held-out metrics, without a transport-shape fallback. All 640 runs met these completion criteria. A later, outcome-independent conductance-validity criterion excluded the 160 synthetic-noise shunting runs because their signed transfer output is not a valid positive-conductance drive. The 480 retained runs contain both neuron models on MNIST and the raw-additive model on the noise task. Nominal depth changed compartment and parameter count and is therefore interpreted as a depth stress rather than a capacity-matched comparison. Table S17 summarizes the retained feedback contrasts.

To separate feedback dimension from assignment, an additional prespecified cohort compared correct neuron-specific feedback with a fixed derangement at depths two and four. The derangement preserves the number and per-example distribution of teaching coordinates, the forward network, and the training path; only the coordinate-to-tree address changes. Applying the same validity rule removed the 40 synthetic-noise shunting runs. Table S19 reports the six retained paired contrasts from 120 runs.

A second prespecified cohort tested inhibitory dose on the noise-resilience task. It crossed two neuron models, inhibitory-contact counts of 0, 5, 10, 20 and 40 per branch, backpropagation or three local feedback fields, and ten paired seeds, for 400 runs. Its primary endpoint was the change from dose 0 to dose 40 in the shunting-minus-raw-additive accuracy difference. Because all 200 shunting cells used the invalid signed-input convention, the full family is excluded from figures and inference. Its configurations and exclusion status are retained in Source Data.

5.6 Exact transport, spatial connectivity and depth controls

We repeated the exact-path-transport versus backpropagation comparison with valid conductance inputs. The factorial design crossed the two learning methods with MNIST or the three-class noisy-line task (defined below), raw-additive or positive-conductance shunting cores, depths one through four and seeds 42–51. The analysis therefore contained 320 training runs and 160 method pairs. All included runs used source commit 74792ca91724066148f36c758191744a315d62b3; the nested synthetic-data repository used commit 4f3612a52756c3691e5bd269c6465c02f240c1e8. The raw-additive synthetic model retained its signed transfer output. The positive-conductance shunting model used an explicit ReLU output transfer for both learning methods before conductances entered the denominator. This rerun therefore uses valid nonnegative shunting inputs and is not a direct re-execution of the excluded signed-input conditions.

Runs were included if they contained the specified configuration and seed, a complete learning summary, finite final metrics and a finite stage-complete checkpoint, without a nonfinite training event or transport-shape fallback. If device memory was insufficient only to compute a metric on the full dataset at once, the same final metric was recomputed in streaming batches and still had to be finite. Initial attempts that lacked the required nested data repository, had insufficient memory or reached the signed-shunting input check produced no endpoint and were excluded independently of outcome. Missing configurations were re-executed without changing their scientific settings. Inference paired exact path transport and backpropagation by seed within task, core and depth. The four plotted task–core effects first averaged the four depths within seed and then used a 20,000-draw paired-seed bootstrap interval; two-sided Wilcoxon tests are supplied as descriptive method-agreement checks rather than equivalence tests. All 320 runs met the inclusion criteria. Forty-two used streaming recomputation for at least one terminal metric; every included final metric was finite. Table S18 reports the four depth-averaged comparisons. Across all 160 depth-specific method pairs, mean exact-minus-backpropagation accuracy was -0.0545 percentage points and the largest absolute individual pair difference was 5.90 points. The latter precludes interpreting the analysis as pointwise equality even though each depth-averaged task–core interval included zero.

To test the role of input organization, a separate cohort compared fixed spatial and random sparse input maps at depth four. The 320 runs crossed two tasks, two neuron models, backpropagation or three local feedback fields, two maps, and ten paired seeds. Each distal branch had 21 contacts. Spatial branches recursively partitioned the 28×28 feature plane into 16 disjoint regions per neuron, whereas random branches sampled independently from all 784 coordinates. Supplementary Fig. S7 reports the 240 conductance-valid learning contrasts together with the corresponding coverage analysis. MNIST retains both cores; the noise task retains the raw-additive core. Because the effect persists under backpropagation and on the randomly projected task, it is treated as a forward sparse-connectivity prior rather than credit-routing evidence.

A separate prespecified cohort isolated depth from terminal-branch count by holding 16 terminal branches and an approximately fixed input contact budget. The four depth conditions contained 960, 960, 968 and 960 active contacts per soma, respectively. It crossed two neuron models, backpropagation or three local feedback fields, four depths and ten paired seeds, giving 320 runs. The conductance-valid synthetic comparison retains the 160 raw-additive runs. This control matches terminal branches and contacts, not trainable parameters: deeper networks still contain more compartments and more coupling and reactivation parameters. Supplementary Fig. S8 reports their complete depth profiles and the prespecified depth-four-minus-depth-one contrasts.

We then evaluated matched-width scalar-fallback, neuron-specific and exact teaching fields on one fixed held-out batch, at each of the 120 conductance-valid backpropagation checkpoints. Only nonsomatic stages and their trainable branch parameters entered the field and gradient vectors. For each candidate field we computed scaled error-field capture, eligibility-weighted gradient capture and cosine, and the loss change after matching its update norm to the exact gradient. The primary relative step was 10^{-5} ; 10^{-6} , 10^{-4} and 10^{-3} are retained as sensitivity values. The association interval resampled checkpoints while keeping paired feedback fields together. The diagnostic did not update or select checkpoints. Table S22 reports the primary operating point. Supplementary Fig. S9 displays the complete diagnostic.

1578 5.7 Context-gated Fashion-MNIST protocol

1579 The context-gated task used Fashion-MNIST classes 0 and 6. Inputs were average-pooled to 7×7 , standardized
 1580 using training data and augmented with a constant coordinate. Each seed used 8,192 training and 2,000 test draws
 1581 with replacement from the official binary splits, balanced across context and label. Branch counts were $B = 2, 4, 8$;
 1582 conflict doses were 0, 0.25, 0.5, 4/7, 0.6, 2/3, 0.75, 1. Thresholding one underlying uniform mask made conflict
 1583 nested across doses. Correct, shared, cyclically deranged, BP and gated-point rules reused inputs and initialization.
 1584 Full-batch descent used 250 updates at step 0.04 and an initial weight standard deviation (SD) of 0.01. Three
 1585 exploratory seeds (100–102) selected a stable setting; an additional implementation-check seed was also excluded.
 1586 Analysis seeds 11000–11019 were fixed before outcomes, with no outcome-dependent stopping.

1587 The primary estimand was each seed’s least-squares slope of correct-minus-shared accuracy on conflict,
 1588 separately for each B . Prespecified criteria required positive slopes in at least 16/20 seeds, at least ten percentage
 1589 points of separation at full conflict, less than two points at zero conflict, correct routing above derangement and
 1590 exact BP/gated-point agreement. All criteria passed. The maximum analytic-versus- autodifferentiation discrepancy
 1591 was 3.43×10^{-7} ; equivalent-rule accuracy differences were zero. Intervals used 20,000 paired-seed bootstrap draws;
 1592 tests were two-sided Wilcoxon tests. Mean-field assumptions and empirical chance crossings are distinguished
 1593 below.

1594 The original 20 branch-conflict seeds were reexecuted to measure gradient geometry throughout learning, using
 1595 the same task configuration and inputs. The original shared filesystem cache was unavailable before data loading; the
 1596 same official Fashion-MNIST files were copied to a readable cache and verified against their published MD5 values.
 1597 Only the cache location changed. The input SHA manifest is retained in `analysis/completion_20260905/`
 1598 `fashion_cache_manifest.json`. Held-out diagnostic batches were evaluated at epochs 0, 1, 10, 50, 100 and
 1599 250. Candidate fields were compared both at each method’s own state and at the common exact-rule state, yielding
 1600 28,800 diagnostic rows. Maximum drift from archived endpoints was 4.99×10^{-11} in accuracy and 4.98×10^{-9} in
 1601 loss. Supplementary Fig. S29D–F displays common-state gradient cosine at epochs 0, 50 and 250; all states and
 1602 checkpoints remain in `source_data/review_branch_trajectories/`. At conflict dose 0.5, shared-gradient
 1603 cosine fell from 0.987 to -0.519 for four branches and from 0.952 to -0.701 for eight branches by epoch 250.
 1604 These changes show why the centered mean-field boundary derived at initialization need not describe trained
 1605 states. We retained that boundary without refitting it and treated gradient direction at a checkpoint and the accuracy
 1606 crossing after training as separate endpoints.

1607 5.8 Mean-field boundary for context-gated branch selection

1608 The Fashion-MNIST conflict experiment selects one branch c uniformly from B branches. Only $\mathbf{w}_c^\top \mathbf{x}_c$ enters the
 1609 logit, so the exact derivative of every unselected branch is zero. Shared feedback omits that selector and updates
 1610 every active eligibility. For the centered mean-field prediction, assume balanced class signs $s \in \{-1, 1\}$, class
 1611 means $\mathbb{E}[\mathbf{x} | s] = s\mathbf{v}$, and zero logits, so the logistic error is $\delta = -s/2$. An unselected input copies the selected
 1612 image with probability $1 - \chi$ and is independently drawn from the opposite class with probability χ . Its mean class
 1613 evidence relative to the selected view is therefore $1 - 2\chi$. The resulting compatibility matrix is

$$C_\chi = (1 - 2\chi)\mathbf{1}\mathbf{1}^\top + 2\chi I.$$

1614 Its shared-mode eigenvalue is $B - 2\chi(B - 1)$, which vanishes at $\chi_c = B/[2(B - 1)]$. Under these population and
 1615 initialization assumptions, the expected shared update loses positive alignment with the exact update at that
 1616 boundary. Logistic derivatives and class-conditional moments need not preserve this relation as weights change.
 1617 The empirical trained chance crossing is consequently a separate descriptive endpoint, not a theorem about the
 1618 entire optimization trajectory. The missing branch selector may be supplied in feedback or in local eligibility; it is
 1619 not evidence that the unselected branches require opposite exact errors.

5.9 Two-branch subtree-address experiment

We first tested the prespecified within-neuron support contrast in a minimal two-sibling model. The subsequent architecture-by-budget factorial varied the number of feedback routes K . One pilot seed (4099), excluded from inference, checked numerical integrity and nondegeneracy: neuron-shared capture was 0.506, correct-route capture was 1.000, correct-route held-out accuracy was 0.814, and analytic exact path transport agreed identically with analytic backpropagation. The same configuration was then used for ten paired analysis seeds (4100–4109).

For seed-specific unit vectors $\mathbf{u}_A$ and $\mathbf{u}_B$, binary label sign $y \in \{-1, 1\}$ and context $c \in \{0, 1\}$, the two input means were

$$\mathbb{E}[\mathbf{x}_A | y, c] = y [\mathbb{K}(c = 0) - 0.75\mathbb{K}(c = 1)]\mathbf{u}_A, \quad (102)$$

$$\mathbb{E}[\mathbf{x}_B | y, c] = y [\mathbb{K}(c = 1) - 0.75\mathbb{K}(c = 0)]\mathbf{u}_B. \quad (103)$$

Each stream had 12 coordinates and independent Gaussian noise with standard deviation 1.15. The 2,048-example train and test splits contained exactly 512 examples in each context–label cell. The somatic logit selected branch A when $c = 0$ and branch B when $c = 1$. Both inputs nevertheless remained active, preventing the one-coordinate rule from acquiring branch specificity through a zero eligibility on the distractor.

For somatic logit z and logistic sigmoid σ , let $\delta = \sigma(z) - \mathbb{K}(y = 1)$ and $\mathbf{r}(c) = (\mathbb{K}[c = 0], \mathbb{K}[c = 1])$. Exact and correct-subtree coefficients were $\delta\mathbf{r}(c)$; within-neuron derangement reversed the two entries, giving $\delta\mathbf{r}(c)_{::-1}$; the shared neuronal and same-depth controls used $\delta(1, 1)$; and the random rank-two control used $\delta\mathbf{r}(c)R_s^T$ for a seed-fixed orthogonal rotation R_s . The gated point emulation used one somatic coefficient together with the context-gated features $(\mathbb{K}[c = 0]\mathbf{x}_A, \mathbb{K}[c = 1]\mathbf{x}_B)$, making its alternative address resource explicit. All conditions had 24 trainable weights and the same paired initialization and batch order. The random dense control had four routing nonzeros; the sparse two-route conditions had two.

Training used mini-batch stochastic gradient descent (SGD), learning rate 0.3, batch size 64 and 60 epochs. The switch endpoint applied another 12 epochs to 1,024 context-1 examples with balanced labels and measured the change in context-0 accuracy. For gradient geometry, exact and candidate per-example gradients were concatenated over both branches. Scaled capture was squared cosine after optimal scalar rescaling. The one-step candidate was normalized to the exact batch-gradient norm and evaluated at step 0.25. Table S1 reports every tested condition; no condition was removed on the basis of its outcome.

Table S1: Two-branch within-neuron routing and context-switch results. Accuracy and forgetting are fractions; initial $\cos^2$ measures directional agreement after optimal rescaling, as defined below the table. Values are means across ten independent paired analysis seeds; intervals for the principal differences are shown in Supplementary Fig. S19A and reported in Source Data.

Condition	Channels	Test accuracy	Initial $\cos^2$	Switch forgetting	Accuracies and
Neuron-shared	1	0.7624	0.5095	0.5566	
Same-depth neuron-shared	1	0.7624	0.5095	0.5566	
Correct sibling subtrees	2	0.8063	1.0000	0.0000	
Within-neuron derangement	2	0.1963	0.0000	0.0006	
Random dense rank two	2	0.5847	0.2791	−0.0014	
Exact path transport	2	0.8063	1.0000	0.0000	
Analytic backpropagation	2	0.8063	1.0000	0.0000	
Gated point emulation	1 + gated eligibility	0.8063	1.0000	0.0000	

forgetting are fractions. Initial $\cos^2$ is the squared cosine between concatenated per-example exact and delivered updates, equivalently capture after optimally rescaling that single update direction; it is not capacity of the full route span and does not retain orientation. Signed cosines are supplied in Source Data. Switch forgetting is the loss of retained-context accuracy after context-specific retraining; small values in the initially below-chance deranged condition reflect its poor pre-switch solution. Gated eligibility supplies branch selection while communicating one scalar error.

The very poor deranged loss and the variable random-map loss were retained as prespecified negative controls. The equality of correct subtree routing, exact transport, backpropagation and gated point emulation is equally

important: in this shallow model the exact path coefficient is only the binary branch gate. The result therefore isolates address alignment but does not establish a dendrite-specific advantage over a point system supplied with equivalent gated parameter groups.

5.10 Subtree-address bandwidth and architecture factorial

To vary feedback bandwidth, the complete factorial extended the two-sibling task to eight context-selected streams arranged in a balanced binary hierarchy. Context and binary label were exactly balanced in train and test sets, every stream remained active on every example, and the sign of the task signal reversed outside the selected context. Thus a shared coefficient could not recover the required address simply because off-route eligibilities were zero. The prespecified feedback budgets were $K \in \{1, 2, 4, 8\}$ nested routes. Twenty paired analysis seeds (4200–4219) crossed five representations and 27 method–budget combinations, giving 135 fits per seed and 2,700 fits in total.

The dendritic tree, explicitly gated point model, flat-compartment model and grouped-point model were implementation-equivalent controls: each contained 64 trainable weights, 64 active input contacts and one output nonlinearity, and each received the same routed coefficient field. The fifth representation was a degree- and depth-matched rewired tree. Its resource counts were unchanged, but its ancestry groups were permuted relative to the task hierarchy. Feedback families comprised correct ancestry, fixed within-neuron derangement, depth-interleaved groups, random sparse groups, a seed-fixed random rank- K map, and a task-fitted rank- K upper bound. Neuron-shared, exact-transport and analytic-backpropagation references completed the design. Route fields were unit-normalized per example. Data, initialization and every stochastic control map were keyed to seed, method family and K , so matched representations saw identical stochastic realizations.

Initial pilot checks verified exact context balance, zero context–label correlation, nonzero activity in all streams, neuron-shared gradient capture below 0.2, exact capture of one, numerical equality of analytic exact path transport and backpropagation, and zero fallback counts. After the first large run, we found that stochastic control maps had not been paired across representation labels. Those outcomes were excluded before scientific analysis. We corrected the random streams, added cross-representation implementation equivalence to the pilot checks and repeated the complete 2,700-fit cohort. Only the corrected cohort is reported.

Table S2 gives the primary bandwidth contrasts. Correct nested-subtree routing strongly exceeded route derangement at $K = 2$ and $K = 4$, but its comparison with the best matched non-anatomical control was deliberately more stringent: ancestry lost at $K = 1$ and $K = 2$, won by 0.0127 accuracy at $K = 4$ (95% paired seed-bootstrap interval 0.00593–0.01946; 15/20 seeds; paired Wilcoxon signed-rank Holm-adjusted $P = 0.0101$ across the four budgets; Table S2 retains the paired mean-sign-flip sensitivity test), and tied at full rank. Correct routing on the task-matched tree exceeded the rewired tree at $K = 2$ and $K = 4$ in all 20 seeds. The four matched representations were exactly equal in every condition (maximum held-out-accuracy difference zero). The experiment therefore supports an intermediate-bandwidth advantage for a task-aligned nested address, while showing that an explicitly routed point implementation can reproduce it.

5.11 Trained partition-residual reconstruction

Using the saved configurations and outputs, we quantified irreducible address error in a post hoc diagnostic of the complete factorial. For example b , let c_b denote context, δ_b the somatic error, $e_{c_b,i}$ the indicator selecting leaf i , and $\mathbf{x}_{bi}$ its input vector. We defined the exact coefficient field $c_{bi} = \delta_b e_{c_b,i}$, coordinate weight $w_{bi} = \|\mathbf{x}_{bi}\|_2^2$, and active route field ϕ_{bi} . Allowing the best single coefficient for that route on each example gives

$$z_b^* = \frac{\sum_i w_{bi} \phi_{bi} c_{bi}}{\sum_i w_{bi} \phi_{bi}^2}, \quad \mathcal{C}_{\text{part}} = 1 - \frac{\sum_{b,i} w_{bi} (c_{bi} - z_b^* \phi_{bi})^2}{\sum_{b,i} w_{bi} c_{bi}^2}. \quad (104)$$

This is the normalized complement of the shared-coefficient residual in Eq. 53. It grants an optimal route coefficient and therefore isolates address/support error from the coefficient scale actually used during training.

All 20 seeds and 135 conditions per seed were reconstructed at initialization and after 80 training epochs. Reconstructed losses agreed with the recorded values within 4.97×10^{-9} and accuracies within 5.01×10^{-11} ,

the precision of the saved decimal tables; the address-plus-coefficient Pythagorean decomposition closed within 7.28×10^{-12} . On the dendritic-tree representation, trained correct-ancestry capture was 0.162, 0.315, 0.547 and 1.000 at $K = 1, 2, 4, 8$. Fixed coordinate-to-arbor derangement had the common scalar value at $K = 1$ and zero capture at $K = 2, 4, 8$, because its active support excluded the target leaf (Supplementary Fig. S23A).

For each seed we averaged implementation-equivalent representations within feedback family and budget, then correlated trained capture with held-out accuracy across conditions. The mean within-seed Spearman correlation was 0.824 (95% seed-bootstrap interval 0.810–0.838; 20/20 positive; two-sided Wilcoxon $P = 1.91 \times 10^{-6}$). Restricting the analysis to the six route families and $K < 8$ removed the exact full-rank boundary and gave 0.639 (0.610–0.665; 20/20 positive; $P = 1.91 \times 10^{-6}$; Supplementary Fig. S23B,C). This association is descriptive and post hoc. In particular, ancestry, depth-interleaved and random sparse partitions had similar address capture at matched K , so the residual does not explain the small trained ancestry advantage at $K = 4$.

5.12 Learning coefficients from an explicit local context cue

To test whether coefficients could be learned from a supplied cue, we used a four-route dictionary on fresh versions of the balanced eight-context task. We retained its 64 forward weights, paired initialization, 2,048-example train and test sets, and 80 full-batch task-training updates. The dictionary was fixed before task training: $\Phi_{bk} = \mathbf{1}[k = \lfloor b/2 \rfloor] / \sqrt{2}$ for zero-indexed blocks $b = 0, \dots, 7$ and routes $k = 0, \dots, 3$. For trial i with context c_i , the supplied local cue was $\mathbf{h}_i = \mathbf{e}_{c_i} + \sigma_h \boldsymbol{\epsilon}_i$, with $\boldsymbol{\epsilon}_i \sim \mathcal{N}(0, I_8)$. Here $\mathbf{e}_{c_i}$ is the one-hot context vector and σ_h is cue-noise SD. The cue is an additional information resource; the experiment does not infer a biological signal.

Separate balanced calibration trials supplied the target route-activation vector $\mathbf{e}_{\lfloor c_i/2 \rfloor}$. A four-output softmax encoder with an intercept learned this mapping by minibatch delta-rule updates. Weights started at zero; calibration used 30 epochs, batch size 16, step 0.2 and weight decay 0.001 on non-bias weights. Calibration sets of 16, 64 and 256 trials were nested within each cue-noise condition. Because epoch count was fixed, increasing sample count also increased the number of encoder updates. This axis therefore changes both calibration data and computation. No task labels, exact task gradients or held-out task outcomes entered encoder training. Its parameters were frozen before the classification weights were trained.

At cue delays $d = 0, 1, 4$ trials, the encoder received the cue array circularly shifted by d . Predicted coefficients $\mathbf{c}_i$ produced the normalized field $\Phi \mathbf{c}_i / \|\Phi \mathbf{c}_i\|$. Delay is a mismatch between the current eligibility and an earlier independent context; it is not a calibrated physiological time constant. Controls used exact one-hot route selection, a uniform frozen profile, or a cyclic permutation of the learned coefficient coordinates. Exact and frozen controls repeated across nuisance conditions are algebraic repetitions rather than independent evidence. Task updates used the original factor-eight context rescaling and step 0.3.

Twenty analysis seeds (52000–52019) followed the excluded implementation-check seed 51999. Crossing three calibration sizes, cue-noise SDs 0, 0.5, 1, delays and four rules produced 2,160 task fits and 10,800 checkpoint rows. At task epochs 0, 1, 10, 40 and 80, we recorded held-out accuracy/loss and delivered-versus-exact gradient cosine at each method’s own state. Held-out cue classification, coefficient mean squared error (MSE) and route-norm errors supplied separate encoder diagnostics. The protocol, fixed before new outcomes but not publicly preregistered, is `configs/review_completion/coefficient_encoder.json`.

The primary condition used 256 calibration trials, cue-noise SD 0.5 and no delay. Learned coefficients improved accuracy over the frozen profile by 7.16 percentage points (95% paired-seed bootstrap interval 6.26–8.07; 20/20 positive) and over mismatched coefficients by 7.10 points (6.19–7.99; 20/20). They remained 54.47 points below exact route selection (53.54–55.45; 20/20 lower). Two-sided Wilcoxon tests restored the exact 1/2048 accuracy-count grid before ranking and used Holm correction across these three primary contrasts. Adjusted P values were 5.72×10^{-6} versus the frozen rule and 1.77×10^{-4} versus each other control. Intervals used 20,000 paired-seed draws; other grid contrasts are descriptive sensitivities. All fits were finite. Learning from the supplied cue improved performance relative to fixed or misassigned coefficients, but left a large gap to the oracle. The experiment does not establish autonomous discovery of credit coordinates in a biological neuron. The complete outputs are in `source_data/review_coefficient_encoder/`.

An exploratory follow-up was specified after observing the soft-encoder outcomes. The same trained encoder,

original paired seeds and task data were retained, but its maximum-probability route was delivered as a one-hot vector. No encoder parameter was refitted and no hyperparameter was selected from the task outcomes. This hard readout is a paired sensitivity, not a new independent replication or a replacement for the primary soft rule (Supplementary Fig. S32). At the primary noise SD 0.5, mean accuracy was 25.825% for soft coefficients, 64.031% for hard selection, 80.291% for exact selection and 18.669% for the frozen profile. Hard-minus-oracle accuracy remained -16.26 percentage points (95% paired-seed interval -17.83 to -14.83), while hard-minus- frozen was 45.36 points (43.69 – 46.92). At zero noise and 256 calibration trials, hard selection equaled the oracle numerically, whereas the soft rule reached 79.978%. Both learned readouts failed when cues were delayed across independent contexts. Thus accurate cue identification does not guarantee that graded coefficients are suitable for the learning rule: readout and timing are separate constraints. All exploratory fits and unchanged encoder settings are in `source_data/review_coefficient_hard_readout/`.

5.13 Nonlinear physical depth in the DendriNet model

Changing the resolution of a linear route projector does not change the physical depth of a forward computation. We therefore tested physical depth separately in the nonlinear DendriNet population model. The `hierarchical_gain_load` generator emits 64 strictly positive excitatory rates followed by 64 strictly positive inhibitory rates. Three independent zero-mean Gaussian node fields define fine, coarse and global lognormal gain factors, giving physical task hierarchy $H_p = 3$ (H3). Independent Gaussian rate noise is added to each baseline before clipping the rate to a floor of 10^{-4} ; gains are applied after this clipping. The factor product multiplies signal-bearing excitatory coordinates in the distal block; the three inhibitory blocks separately measure the corresponding factors. Lognormal means are corrected by $-\sigma^2/2$ at every factor, so changing hierarchy or sensor alignment does not silently change the means of the multiplicative factors. The additive noise before clipping is zero mean; the resulting rate perturbation is not. This distinction matters in middle and proximal excitatory blocks, whose class-independent baseline is 0.02 and noise standard deviation is 0.08.

The matched condition uses the same latent node in excitatory signal and its inhibitory sensor. In the zero-alignment condition, each sensor instead uses an independent node field with the same scale and grouping. In the shuffled condition, a nonzero cyclic trial shift is applied to the complete inhibitory field after class balancing. The tree-placement condition leaves every data tensor unchanged and reverses the fine and global feature ranges across the fine/coarse/proximal inventory tiers. These controls remove task–tree alignment in different ways: latent correspondence, example-wise access and physical placement, respectively. The final placement control reverses the fine and global ranges and preserves their 21/22/21 widths. An initial three-cycle placement changed the number of candidate mask slots and therefore violated the predefined resource-matching criterion. It was excluded before inference and is reported only as an excluded design check.

In DendriNet, branch factors are ordered from soma to distal dendrite and D_p denotes the number of physical dendritic stages. For architecture $[b_1, \dots, b_{D_p}]$, the number of nonsomatic units is $\sum_{j=1}^{D_p} \prod_{r=1}^j b_r$. Thus D1 [8] has eight distal units per soma; D2 [2, 3] has two proximal and six distal units; and D3 [2, 1, 2] has two proximal, two middle and four distal units. The unary middle expansion is deliberate: all depths have eight nonsomatic units. The structured masks assign all 4/2/2 tiers to the single D1 stage, fine+coarse tiers to the D2 distal stage and the global tier to its proximal stage, and one fine, coarse or global tier to each D3 stage. The architectures were matched for trainable parameters (66,178), active synapses (14,336), candidate slots (21,760) and persistent state scalars per sample (2,944). The physical branch law was the same positive-conductance division used elsewhere in the paper, with identity reactivation and learned softplus-positive synaptic weights.

The exploratory calibration used signal contrast 0.24, training gain SD 0.25 and an unseen test gain SD 1.2. Deeper networks fit training better, but BP and both local credit assignment (LocalCA) transports remained at chance on the shifted test distribution (Supplementary Fig. S15A). This condition did not meet the predefined accessibility criterion and is reported as a negative calibration result. A two-seed test-gain sweep showed monotone erosion under increasing shift. A coupling sweep at matched train/test gain identified a seed-consistent interaction: from child conductance 1 to 64, the D3-minus-D1 accuracy contrast increased by 0.1236 on average, while D1 changed minimally. A signal-contrast sweep at child conductance 16 then located the non-ceiling signal-contrast

1784 boundary. Contrast 0.72 narrowly missed the predefined D1 accessibility threshold (mean 0.597), whereas contrast
1785 0.80 met it, with D1, D2 and D3 means 0.6086, 0.6737 and 0.9227. These exploratory seeds 10100–10141 were
1786 excluded from inference.

1787 Before observing any outcome for seeds 10200–10209, we prespecified a 270-fit analysis at signal contrast
1788 0.80, per-factor train/test log-gain SD 0.25 and child conductance 16. The nine feedback–placement conditions
1789 were aligned shunting BP; zero-alignment, shuffled-sensor and reversed-placement shunting BP; aligned raw-
1790 additive BP; and aligned or reversed-placement shunting LocalCA with shared-soma and exact-path feedback.
1791 Each condition crossed D1, D2 and D3 over the same ten paired seeds 10200–10209. We specified the primary
1792 contrasts, resource- and accessibility-matching criteria, bootstrap procedure and limits on interpretation in advance.
1793 In particular, aligned D3–D1 alone tests forward capacity. Task–tree specificity requires an aligned-minus-control
1794 depth interaction, and a local-credit claim requires the corresponding LocalCA contrast. The raw-additive tree is a
1795 forward-mechanism control; a point model with sufficient flexible capacity can in principle emulate the hierarchy, a
1796 boundary tested directly in the same-task controls below.

1797 The accessibility criteria required at least nine of ten aligned-BP seeds above 0.55 at each depth and every
1798 condition mean below 0.98. All 270 fits met the completeness, finite-metric, no-fallback and exact-resource
1799 criteria. With aligned shunting BP, mean accuracy was 0.6096, 0.6752 and 0.9182 at D1–D3, giving a paired
1800 D3-minus-D1 effect of 30.86 percentage points (95% seed-bootstrap interval 30.45–31.30; 10/10 positive; exact
1801 sign-flip $P = 0.001953$). The raw depth effects were 0.02 points for independent sensors, -0.01 for shuffled trials
1802 and -0.43 for reversed placement. Their aligned-minus-control depth interactions were 30.84, 30.87 and 31.28
1803 points, respectively, all positive in 10/10 paired seeds. Aligned raw-additive BP instead fell from 0.5733 to 0.5190.
1804 Under LocalCA, aligned D3-minus-D1 was 16.95 points (16.37–17.44) for shared-soma LocalCA and 27.90 points
1805 (27.24–28.53) for exact-path LocalCA. The corresponding aligned-minus-reversed interactions were 17.72 points
1806 (17.27–18.18) and 28.68 points (28.07–29.27), both positive in 10/10 seeds. These interactions, rather than the
1807 aligned depth effects alone, support task–tree specificity.

1808 In a post hoc mechanism analysis, we evaluated one fixed held-out batch at each of the 30 aligned-shunting BP
1809 checkpoints. Exact-path LocalCA reproduced the selected conductance-gradient direction computed by automatic
1810 differentiation (autograd) at every checkpoint (minimum cosine 0.9999999999994; no missing gradient tensors).
1811 Mean shared-soma cosine was 0.9457 at D1, 0.7990 at D2 and 0.7674 at D3. Its paired D3-minus-D1 change was
1812 -0.1784 (95% seed-bootstrap interval -0.2082 to -0.1490 ; 0/10 positive pairs). This analysis verifies the path
1813 recursion and shows that a common neuronal error becomes a poorer directional proxy as additional physical stages
1814 differentiate branch states. It was not prespecified as a performance endpoint.

1815 In Supplementary Fig. S31D, an aligned row maximum is outlined only when its 95% bootstrap interval clears
1816 the runner-up's; reversed rows are unoutlined. Panel E outlines descriptive maxima only when the within-row
1817 spread exceeds one percentage point; blank cells were untested.

1818 **5.14 Same-task point, nonserial and credit-coordinate controls**

1819 We added three controls without changing the calibrated task, data, seeds, initialization policy or stopping rule.
1820 First, an all-active grouped-star model used exactly the same branch layers, structured masks and parameters as the
1821 serial tree, but each nonsomatic layer read the external input independently; the child-aggregation parameters were
1822 preserved for resource matching but summed only once at the soma. Stored model states and resource counts were
1823 identical at initialization. D1 logits were numerically identical, as required. At D1 the serial-minus-star contrast was
1824 -0.005 percentage points (-0.067 to 0.054 ; 6/10 positive; $P = 0.844$). At D2 and D3 it was 6.54 (6.20–6.88) and
1825 30.81 points (30.41–31.26), respectively, both positive in 10/10 seeds ($P = 0.001953$). Under reversed placement
1826 the D3 serial-minus-star contrast was -0.43 points (-0.68 to -0.16 ; 1/10 positive; $P = 0.0137$), and the paired
1827 alignment interaction was 31.24 points (30.65–31.77; 10/10 positive; $P = 0.001953$). Serial divisive composition,
1828 rather than the inventory of branch units, is therefore required for the aligned effect in this model.

1829 Second, point multilayer perceptron (MLP) widths were chosen before fitting to match either 15,055 active
1830 parameters or, to the nearest realizable integer width, 66,200 total parameters (the tree has 66,178). Their mean
1831 test accuracies were 0.9861 and 0.9902. They exceeded serial D3 by 6.79 points (6.19–7.34) and 7.20 points

1832 (6.75–7.68), respectively, in 10/10 paired seeds ($P = 0.001953$ for each). Neither point model was given the
1833 hierarchy. These results rule out an absolute expressivity advantage for the tree on this task, while retaining the
1834 narrower interpretation from the controls with matched branch resources.

1835 Third, soma-broadcast autograd used the standard synaptic derivatives but replaced each nonsomatic loss
1836 derivative by the loss derivative at its owning soma. Forward values and logits were identical to full BP before
1837 training. With the standard BP optimizer, full BP minus soma broadcast at D3 was 0.64 points (0.15–1.17; 8/10
1838 positive; $P = 0.0332$). After this 140-fit analysis was complete, we found that the LocalCA reference used a
1839 different, split optimizer. We therefore specified 60 additional soma-broadcast fits using the LocalCA optimizer
1840 before viewing their outcomes. This optimizer-matched broadcast reached 0.7771 at aligned D3, compared with
1841 0.7778 for shared-soma LocalCA: soma-broadcast autograd minus shared-soma LocalCA was -0.06 points (-0.23
1842 to 0.09 ; 5/10 positive; $P = 0.492$). Changing only the broadcast optimizer cost 13.47 points (12.89–14.05; 10/10
1843 positive). Within LocalCA, exact-path LocalCA exceeded shared-soma LocalCA by 10.96 points (10.34–11.58;
1844 10/10 positive), while full BP remained 3.09 points above exact-path LocalCA (2.48–3.79; 10/10 positive). After
1845 matching the teaching coordinate and optimizer, we detected no performance penalty from the tested eligibility
1846 expression on this task. The 10.96-point path-specificity contrast is conditional on the split LocalCA optimizer; it is
1847 not an optimizer-independent requirement of three serial stages. The corresponding full-BP versus soma-broadcast
1848 contrast under the standard optimizer was only 0.64 points. The optimizer-matched comparison is a post hoc
1849 diagnostic rather than an independent confirmation. Automatic differentiation serves as an implementation control;
1850 it does not specify a biological mechanism.

1851 Across these controls, all 200 fits had finite outcomes, contained no fallback warning and used exactly seeds
1852 10200–10209. Serial and star models had 66,178 trainable parameters, 14,336 active contacts, 21,760 candidate
1853 slots and 2,944 persistent state scalars. Seed-level outcomes, paired contrasts, model configurations and machine-
1854 readable validation records are provided in `source_data/point_dendrite_credit_controls/`.

1855 **5.15 Longer training budget for the serial credit comparison**

1856 The original physical-depth comparisons used a maximum of 180 epochs and validation-loss early stopping with
1857 patience thirty. An audit of one hundred original D1/D3 trajectories verified their configurations and final metrics
1858 against the archived files. All fifty D3 fits across the five relevant credit/optimization conditions reached 180 epochs,
1859 with validation loss still decreasing. These endpoints therefore cannot establish a converged advantage.

1860 The follow-up retained all ten original seeds in five aligned D3 conditions: exact autograd and soma-broadcast
1861 autograd under the BP recipe, soma-broadcast autograd under the LocalCA recipe, and exact-path and shared-soma
1862 LocalCA. Ten exact-BP D1 fits supplied the depth reference, giving sixty fits. Every model had 66,178 trainable
1863 parameters and 14,336 active synapses. Both recipes used Adam; the BP recipe used learning rate 0.003 without split
1864 parameter groups, while the LocalCA recipe retained its original split groups and nominal rate 0.0008. Complete
1865 group-specific rates, weight-boosting, decay, clipping and update options are retained in the configurations. Only
1866 the maximum epoch count and output names changed. The original patience and validation-based state selection
1867 were preserved, without tuning against the extended outcomes.

1868 Historical final-model files did not contain optimizer and random-generator states. The extension therefore
1869 restarted the original seeds using the pinned implementation a99c3a7, rather than claiming exact continuation
1870 from epoch 180. Four seed blocks used GPUs and six used CPUs, with the device class held fixed across all six
1871 conditions within a seed. At epochs 180, 360 and 600, and on earlier termination, additional records retained
1872 current and best weights, optimizer state and Python, NumPy and Torch random states. A separate audit confirmed
1873 that every final model exactly equaled its saved validation-selected best state, and that all configuration changes
1874 were limited to the epoch cap and output naming.

1875 For each restart, we evaluated the best validation-selected state available within the first 180 epochs and within
1876 the complete run. If a run stopped before 180, both windows used its final selected state. Test accuracy and
1877 cross-entropy were reporting metrics, never selection criteria. Learning curves show best validation loss; stopped
1878 runs are carried forward so every displayed mean includes the same ten seeds. The compact Source Data retain all
1879 34,159 observed epochs, every recorded epoch metric, 120 original/extension configurations, sixty run records and

the complete underlying raw-file hash inventory. Carried plotting rows are separately identified.

At 600 epochs, the exact-BP D3-minus-D1 accuracy difference was 38.17 percentage points (95% paired interval, 37.98–38.39). The LocalCA exact-path-minus-shared-soma difference changed from 10.86 points (10.15–11.62) at 180 epochs to -1.52 (-1.83 to -1.23) at 600. The BP-recipe exact-minus-broadcast difference at 600 was 0.48 points (0.40–0.55), and the broadcast BP-minus-LocalCA-recipe difference was 1.54 points (1.39–1.70). Secondary loss contrasts retain the metric dependence: exact-path LocalCA had mean test cross-entropy 0.2544 versus 0.2758 for shared-soma LocalCA, despite lower accuracy. All contrasts use the same ten paired seeds and 10,000 whole-seed bootstrap draws; intervals are descriptive, without a new confirmatory multiplicity claim.

All fifty D3 fits and two D1 fits reached the 600-epoch cap. Every D3 condition still had decreasing validation loss over its final thirty epochs. Relative to the historical first-180 trajectories, the largest absolute validation-loss difference was 0.005885. The median absolute discrepancy in validation-selected test accuracy was 0.070 percentage points and the largest was 1.830 points. These differences are retained per run; budget contrasts use only the two windows of the same new trajectories. The extension establishes budget and metric dependence, not convergence or fresh-seed replication.

5.16 Alignment-dose interpolation of the physical-depth effect

We interpolated sensor alignment α after observing the $\alpha = 0$ and 1 endpoint results. Before viewing any intermediate outcome, we fixed $\alpha \in \{0.25, 0.50, 0.75\}$ and left the generator, architecture, optimizer, stopping and seed settings unchanged. The 90 fits crossed three physical depths with ten paired seeds. Combined with the endpoints, paired D3-minus-D1 effects were 0.019, 0.207, 0.820, 3.126 and 30.855 percentage points at $\alpha = 0, 0.25, 0.50, 0.75, 1$, respectively. The corresponding positive-pair counts were 6, 8, 10, 10 and 10 of ten. Exact two-sided sign-flip tests gave $P = 0.844$ at $\alpha = 0$, 0.0703 at 0.25 and 0.00195 at every higher level.

The predefined summary fit an ordinary least-squares line to the five paired depth effects within each seed. Its mean slope was 25.84 percentage points per unit α (95% paired-seed bootstrap interval 25.43–26.26; 10/10 positive; exact sign-flip $P = 0.001953$). The fitted slope summarizes the increase but does not establish a linear response. The change from $\alpha = 0.25$ to 0.75 was only 2.92 points (2.65–3.17), whereas the final 0.75 to 1 step contributed 27.73 points (27.35–28.11). The interpolation therefore supports a steep boundary near complete alignment in this calibrated task. It does not supply a second task family, hierarchy depth or biological replication.

The 90-fit dataset contains results obtained immediately before and after an unrelated source-code edit. Direct comparison showed that the edit introduced only an optional rule override that none of these model configurations used; condition-level deduplication yielded one result per condition. All 90 unique intermediate fits met the finite-metric, no-fallback, seed and equal-resource criteria. Software-version records, machine-readable outcomes and paired contrasts are in `source_data/physical_alignment_dose/`.

5.17 Literal grouped-point control and independent hierarchy depth

The all-active grouped star retained the serial model’s child-aggregation modules in an arrangement of independent branches. To remove that residual architectural feature, we implemented a literal grouped-point control: each nonsomatic level received the raw external excitatory and inhibitory streams independently and projected directly to its owning soma. Every local branch module, structured mask and shunting equation is preserved, but no level inherits another branch state. Each direct projection has the same scalar count as the serial child aggregator it replaces and copies its initialized raw conductances. We verified the copied projection parameters, finite gradients and exact D1 logits against the serial hierarchy.

This control was designed after the preceding $H_p = 3$ endpoints were known. Its 60 literal grouped-point fits and comparisons were specified before their outcomes were viewed; they crossed aligned or reversed placement, D1–D3 and the same paired seeds 10200–10209. The previously reported serial outcomes served as the fixed reference. Under alignment, grouped-point means were 0.6096, 0.6100 and 0.6095 at D1–D3. Serial-minus-grouped-point was 30.87 percentage points at D3 (95% paired-seed bootstrap interval 30.48–31.31; 10/10 positive; exact sign-flip $P = 0.001953$), but -0.44 points after placement reversal (-0.69 to -0.16). The resulting alignment interaction

was 31.31 points (30.73–31.84; 10/10 positive). Grouped star minus the literal grouped point was 0.06 points at aligned D3 (0.01–0.13; 7/10 positive; $P = 0.0840$), which did not meet the predefined directional criterion.

We prospectively tested a second hierarchy depth, fixing its design before observing outcomes. The $H_p = 2$ (H2) generator used two 32-feature sensor tiers, fixed inventory counts 4/4 and exact-resource D1 [8] and D2 [4, 1] morphologies. All remaining task, signal, gain, axial-coupling, optimizer and stopping settings were inherited without retuning. Aligned and reversed placement were crossed with serial BP, literal grouped-point BP and serial LocalCA with shared-soma or exact-path transport over ten paired seeds (10300–10309), yielding 160 fits.

Aligned serial BP rose from 0.6588 at D1 to 0.9672 at D2. The 30.84-point depth effect (30.46–31.23; 10/10 positive; $P = 0.001953$) became -1.56 points under reversed placement, giving a 32.40-point interaction (32.10–32.70). Grouped-point depth effects were 0.02 points under alignment (-0.02 to 0.06) and -0.02 under reversal (-0.06 to 0.02); their alignment interaction was 0.04 points (-0.03 to 0.11). Serial-minus-grouped at aligned D2 was 30.82 points (30.41–31.23), and its architecture-by- placement interaction was 32.36 points (32.02–32.71). Shared-soma LocalCA showed a 21.20-point aligned depth effect (19.34–23.00) and a 22.93-point placement interaction (21.02–24.77). Exact-path LocalCA showed 31.38 points (30.95–31.83) and a 33.36-point interaction (33.00–33.75). Every positive interaction was ordered in 10/10 seeds with exact sign-flip $P = 0.001953$.

All 220 fits were finite and stage complete, with no fallback or nonfinite values, the expected seeds and exact within-hierarchy resource equality. The second hierarchy is a replication within the same synthetic mechanism-matched task family, not an independent dataset. Model configurations, software versions, per-seed metrics and every paired contrast are in `source_data/remaining_physical_experiments/`.

5.18 Four-level hierarchy and computational reproducibility

The four-level hierarchy (H4) used the same positive-rate generator, optimizer, conductance scale, stopping rule and calibrated signal regime without retuning. Its four 16-feature signal/sensor tiers had predefined inventory counts [4, 2, 1, 1]. D1 [8], D2 [2, 3], D3 [2, 1, 2] and D4 [1, 1, 2, 2] each contained eight nonsomatic branch units per soma, 66,178 trainable parameters, 14,336 active contacts, 21,760 candidate slots and 2,944 persistent state scalars. The 360 planned fits crossed aligned and reversed placement for serial and literal grouped-point BP and for shared-soma and exact-path LocalCA; aligned serial raw-additive BP supplied the mechanism control. Ten paired seeds (10400–10409) were the inferential units.

The primary comparisons were prespecified before outcomes were analyzed: aligned serial-BP D4–D3 and D4–D1, their placement interactions, the D4 architecture-by-placement interaction, shared-soma and exact-path LocalCA D4–D3 effects and interactions, and the shunting-minus-raw-additive D4–D3 interaction. Failure of D4 to exceed D3 was defined prospectively as falsification of the simple $D_p = H_p$ trained crossover. Intervals used 50,000 paired-seed bootstrap draws, exact two-sided sign-flip tests enumerated all 2^{10} assignments, and the primary family used Benjamini–Hochberg (BH) correction, with adjusted P values denoted by q . A directional positive statement also required at least eight of ten paired differences in the predicted direction.

The initial D3 configurations did not instantiate because the automatic tier partition grouped the H4 inventory as [4], [2 + 1], [1] rather than the intended D3 row counts [4, 2, 2]. No H4 outcome was inspected before this construction error was diagnosed. After 33 targeted construction tests, the explicit D3 grouping was defined as [4], [2], [1 + 1] without examining H4 outcomes. The default automatic grouping was unchanged. Only the 90 failed D3 configurations were repeated, using the same seeds and scientific settings. The Source Data identify the software version and result origin for each row. All 360 planned seed–condition outcomes were then finite, stage complete and resource matched, with no fallback warning.

Aligned serial-BP means were 0.5963, 0.7112, 0.8671 and 0.8525 at D1–D4. D4 minus D3 was -1.46 percentage points (95% paired-seed bootstrap interval -1.74 to -1.16 ; 0/10 positive; exact $P = 0.001953$; BH $q = 0.003906$), whereas D4 minus D1 was 25.63 points (24.71–26.40; 10/10 positive; $q = 0.003906$). The D4–D3 placement interaction was -0.36 points (-0.70 to 0.04), but the D4–D1 placement interaction was 25.94 points (25.06–26.66; 10/10 positive). Thus aligned serial depth remained useful relative to D1 but saturated at D3, falsifying one-to-one tracking at H4.

Grouped-point BP stayed near 0.596 across depths. Serial minus grouped point at aligned D4 was 25.60

points (24.69–26.37; 10/10 positive), and the architecture-by-placement interaction was 25.91 points (25.06–26.62). Shared-soma LocalCA gained 5.27 points from D3 to D4 (4.84–5.72; 10/10), but exact-path LocalCA gained 0.60 points (0.04–1.25; 7/10; $P = 0.0859$) and did not meet the predefined positive-direction criterion. The aligned raw-additive D4–D3 effect was -0.55 points (-2.24 to 1.12), and its interaction with shunting was -0.90 points (-2.65 to 0.95). All null and reversed contrasts are reported in `source_data/physical_depth_h4_factorial/`, irrespective of direction.

We separately tested computational reproducibility by repeating all 220 H2/literal-point fits and 210 principal H3 fits with versioned source commit a99c3a7, using the same configurations and paired seeds. All 430 repeated fits met the source-version, completeness, finite-metric, seed and resource criteria. The median absolute original-versus-repeated difference in held-out accuracy was 0.030 percentage points, the mean difference was 0.154 points, and 357/430 pairs were within 0.1 points. The maximum difference was 15.01 points in the raw-additive forward-mechanism control. Rerun H2 and H3 aligned serial-BP depth effects were 30.84 and 30.86 points; the architecture-by-placement interactions were 32.43 and 31.27 points. Every principal shunting and local-credit comparison preserved its direction and met the corresponding predefined criterion in both analyses. This is a same-seed computational reproducibility test, not independent task or biological replication (Supplementary Fig. S26).

5.19 Fixed-depth task-family and alignment factorial

To test whether the depth benefit depended on the task’s factorization, we fixed the DendriNet architecture at D3 [2, 1, 2] and crossed three data generators, three task–sensor alignments $\alpha \in \{0, 0.5, 1\}$, serial versus literal grouped-point composition, and BP versus exact-path LocalCA. Ten paired seeds 10500–10509 gave 360 planned fits. All architectures used 66,178 trainable parameters, 14,336 active contacts, 21,760 candidate slots, 2,944 persistent state scalars and identical input tensors. Only child-to-parent composition differed between the serial and grouped-point models.

For the nested-factor family, independent, identically distributed latent log-gain variables $z_1, z_2, z_3 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ modulated the signal-bearing excitatory coordinate by

$$x_E = \max\{b_E + y\Delta + \varepsilon_E, 10^{-4}\} \exp\left(\sum_{\ell=1}^3 \sigma_{\ell} z_{\ell} - \frac{1}{2} \sum_{\ell=1}^3 \sigma_{\ell}^2\right), \quad (105)$$

where b_E is the excitatory baseline, $y \in \{0, 1\}$ is the balanced class label, $\Delta = 0.80$ is the class-signal contrast, $\sigma_{\ell} = 0.25$ is the level-wise log-gain standard deviation, and ε_E is Gaussian noise with standard deviation 0.08 before the positive-rate floor is applied. The same order—add noise, clip, then multiply by gains—is used for inhibitory sensor rates. Clipping makes the effective rate noise nonzero-mean, particularly in the class-independent middle and proximal excitatory blocks with baseline 0.02. Sensor block ℓ observed

$$s_{\ell}(\alpha) = \alpha z_{\ell} + \sqrt{1 - \alpha^2} \xi_{\ell}, \quad \xi_{\ell} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (106)$$

Here the sensor-noise variables ξ_{ℓ} are independent standard Gaussians, independent of the latent gains. The interpolation preserves unit sensor variance while changing how accurately a sensor reports the trial-specific gain that obscures the class signal.

The flat-factor control preserved the factor product and the same level-wise marginal variances. All three factors used the same deterministic eight-bin spatial partition, with independently sampled amplitudes for each factor; there was no fine-to-global nesting of supports. The local-ratio control multiplied excitation and inhibition at each level by correlated cumulative hierarchical fields. It therefore provided local divisive information rather than requiring serial cancellation of separate factors. Every family contained 10,000 balanced examples with 64 excitatory and 64 inhibitory features; the class signal remained distal-only.

The primary estimands were paired serial-minus-grouped differences and the difference of those contrasts between $\alpha = 1$ and $\alpha = 0$. A task-specific crossover was defined as the nested-family alignment interaction minus

the corresponding interaction in each mismatch family. The credit-rule endpoint was the LocalCA-minus-BP difference in that architecture-by-alignment interaction. Intervals used 50,000 paired-seed bootstrap draws. A directional claim additionally required at least 8/10 concordant seed signs. The complete $3 \times 3 \times 2 \times 2$ factorial is reported regardless of direction. Experimental conditions and primary comparisons were specified before outcomes were analyzed.

All 360 fits were finite and complete, had identical resource counts and contained no fallback. Under BP, the nested-factor serial-minus-grouped contrast changed from -0.09 percentage points at $\alpha = 0$ to 30.32 points at $\alpha = 1$, giving a 30.41 -point alignment interaction (95% paired-seed bootstrap interval 30.05 – 30.79 ; 10/10 positive; exact sign-flip $P = 0.001953$). The flat-factor interaction was also positive but smaller at 22.26 points (21.81 – 22.69 ; 10/10), whereas the local-ratio interaction was 1.30 points (1.14 – 1.45 ; 10/10) because the serial and grouped architectures both reached 0.99999 accuracy at $\alpha = 1$. The nested-minus-flat and nested-minus-local-ratio interaction contrasts were 8.16 and 29.12 points, respectively, and were positive in every paired seed.

Under exact-path LocalCA, the nested-factor serial-minus-grouped contrast changed from -0.11 points at $\alpha = 0$ to 27.59 points at $\alpha = 1$, an alignment interaction of 27.70 points (26.64 – 28.64 ; 10/10). The flat-factor interaction was 12.37 points (11.39 – 13.43 ; 10/10). The local-ratio result reversed: serial minus grouped point changed from 6.36 points at $\alpha = 0$ to -5.32 points at $\alpha = 1$, an interaction of -11.68 points (-15.29 to -8.64 ; 0/10 positive). Consequently, the nested interaction exceeded the flat and local-ratio interactions by 15.33 and 39.38 points under LocalCA. These outcomes show that the benefit of serial composition depends on task structure. The nested hierarchy was most favorable, flat multiplicative factors retained a smaller serial benefit, and locally available ratios removed or reversed that benefit. Full condition means, all paired contrasts and the two-sided sign-flip results are supplied in `source_data/task_family_alignment/`.

Identity of the historical noise tasks

The dataset key `noise_resilience` was reused for two different generators. We therefore identify the generator by cohort, rather than treating the key as a single task definition. The cohort mapping, source hashes and remaining gaps are recorded in Source Data, `release_task_identity/task_identity.json`.

Three-class noisy-line control. The tracked-clean 320-fit exact-transport/backpropagation rerun contains 160 MNIST fits and 160 noisy-line fits. Its frozen synthetic-data source, commit `4f3612a5`, implements the latter as follows. Draw a class y uniformly from $\{0, 1, 2\}$ and construct a square image L_y : class 0 is blank, class 1 has one uniformly selected horizontal row set to one, and class 2 has one uniformly selected vertical column set to one. Each input is

$$x = \text{vec}(L_y + 0.2 \epsilon), \quad \epsilon_{ab} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (107)$$

where a, b index pixels and `vec` flattens the image. The frozen dispatcher defaults to a side length of 784, producing 614,656 input coordinates; the retained checkpoint sizes independently agree with this dimension. Images are neither clipped nor normalized. The generator creates 10,000 examples and uses the run-seeded random number generators to divide them into 8,000 training, 1,000 validation and 1,000 test examples. The newer projected-noise parameters in launch configurations were not interpreted by this dispatcher. Consequently, the noise-task rows of Supplementary Table S18 tests agreement between two learning implementations on a three-class synthetic control, not robustness to corrupted MNIST.

Projected-noise MNIST protocol. The intended protocol for the prospective depth-by-feedback and subsequent routing, inhibitory-dose and fixed-budget cohorts uses flattened MNIST images $x \in [0, 1]^{784}$ with the original ten digit labels. Let $P \in \mathbb{R}^{784 \times 50}$ have independent standard-normal entries, with each column subsequently normalized to unit Euclidean length. For each example draw $z \sim \mathcal{N}(0, I_{50})$ and form

$$\tilde{x} = \text{clip}_{[0,1]}(x + 1.5Pz). \quad (108)$$

2056 The projection seed is 0; independent latent-noise seeds are 1, 2 and 3 for the training, validation and test splits.
2057 There is no additional normalization or gain. The intended split contains 48,000 training, 12,000 validation and
2058 10,000 test examples. Frozen prospective configurations specify these parameters, and the 784-input model sizes
2059 and accuracies in the depth-by-feedback cohort are consistent with this protocol. However, the executed nested
2060 generator file was not archived by hash. These records establish the intended protocol, not complete recovery of
2061 its historical execution. The retained lineage for the 320 fixed-budget noise fits in Supplementary Fig. S8 joins
2062 outcomes to the input-validity audit, but its raw resolved configurations are unavailable for rechecking and do not
2063 close this generator-provenance gap.

2064 **Inherited aggregates and release behavior.** The inherited synthetic feedback and gradient panels in Supplemen-
2065 tary Figs. S1 and S3 lack a complete join to executed configurations and generator bytes. Their task identity remains
2066 unresolved; neither generator is assigned to those aggregates retrospectively. The software release preserves the
2067 two implementations under the explicit names `legacy_noisy_lines` and `projected_noise_mnist`. It rejects
2068 the ambiguous historical key unless a generator is chosen explicitly. The projected-noise implementation is a
2069 newly frozen reference for the documented protocol, whereas the noisy-line implementation is recovered from the
2070 executed clean source commit.

2071 **5.20 Regular-tree foundation and boundary experiments**

2072 The primary feedback ladder identified neuron identity as the dominant feedback limitation in the tested standard
2073 classification tasks, with little additional benefit from exact path resolution. The regular-tree controls examined
2074 sensitivity to the local rule, inhibitory loading, physical stage count, feedback noise and dataset. Supplementary
2075 Figs. S1–S4 summarize the regular-tree mechanism, performance and feedback-control analyses. Some of these
2076 analyses were also reported in the associated preprint [1]. These figures label the normalized-additive model in
2077 Eq. 100 and the raw-additive model in Eq. 98 explicitly. Supplementary Fig. S4C contains only the shunting
2078 architecture.

2079 We assessed the provenance of historical curves by distinguishing intended configuration settings from evidence
2080 of the executed settings. The aggregate sources for Supplementary Figs. S1D,E and S3D contain no complete
2081 run-ID mapping. Associated corrective configurations specify nonnegative transfer, but do not establish which
2082 settings produced the plotted runs. The available records therefore support neither a categorical signed-input
2083 diagnosis nor a verified nonnegative-input classification. All 26 historical shunting points remain unresolved and
2084 are excluded from conductance-valid inference and from the revised figures. Historical panel identifiers refer to
2085 the archived source layouts; current S3 has panels A–C. The separate retained S8 audit joins 320 completed runs
2086 one-to-one to outcomes, recording 160 additive runs included and 160 signed-shunting runs excluded. S8 now
2087 displays only the qualified additive results. Raw resolved configurations and the documented pre-prune archive
2088 reference were unavailable for an independent execution-level recheck. Source tables, related run IDs and evidence
2089 hashes are supplied in `source_data/review_curve_lineage`.

2090 Supplementary Fig. S4 collects the nominal-depth sweep, broadcast-noise sweep, shunting-only flattened
2091 CIFAR-10 control, a separate raw-additive CIFAR-10 feedback ladder and the Fashion-MNIST replication of the
2092 feedback-resolution ladder. Panels A–C use five independent seeds per condition, panel D uses 20 paired seeds,
2093 and panel E uses ten paired seeds per architecture. Broadcast noise is zero-mean Gaussian noise added to the
2094 available teaching signal, with σ measured relative to the unperturbed signal scale. The depth sweep changes
2095 the number of dendritic stages at fixed somatic population width. CIFAR-10 images are flattened and use no
2096 convolutional front-end or competitive vision recipe. These experiments define operating regimes and feedback
2097 limits; cross-architecture differences do not isolate a backward shunting mechanism. Selected endpoints are
2098 reported in Supplementary Figs. S1, S2, S3, and S4, and Table S23.

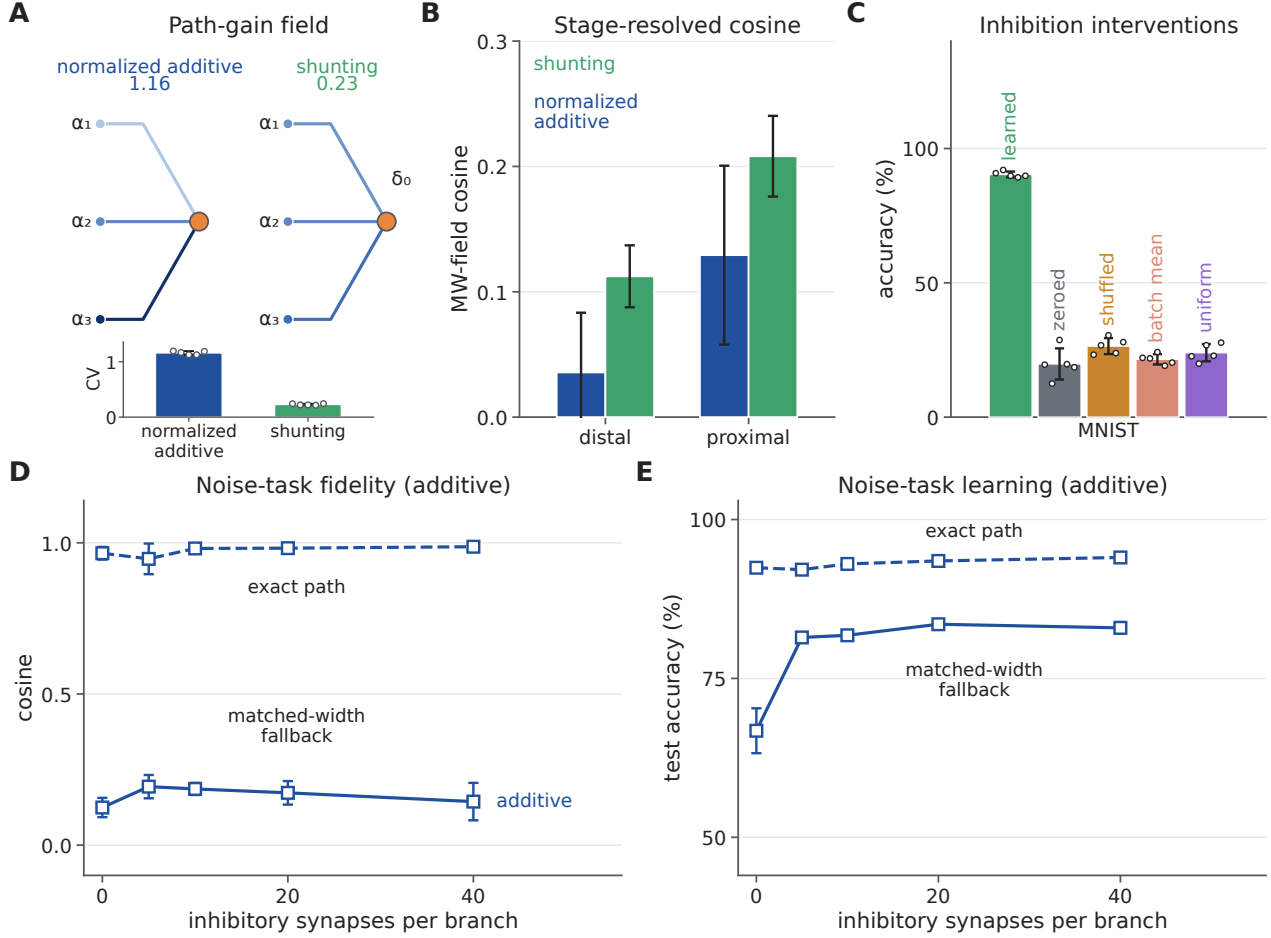


Figure S1: **Mechanistic chain from path gains to learning in regular-tree models.** MNIST panels **A–C** use two-stage [4,4] directed trees; the historical noise-resilience panels **D,E** use two-stage [3,3] additive trees. **A**, Conductance-stage path-gain field in shunting and normalized-additive models at five inhibitory synapses per branch; the inset shows paired-seed coefficients of variation (CV; standard deviation divided by mean). The displayed α values are path gains and δ_0 is the somatic error. This loading quantity omits branch derivatives and is distinct from the trained exact-error profiles in Supplementary Fig. S45D,E. **B**, Stage-resolved cosine of matched-width (MW) scalar-fallback versus exact MNIST errors; the width-matched somatic stage is excluded because its cosine is one by construction. **C**, MNIST post-training inhibition interventions: zero, sample shuffle, per-branch batch mean and a uniform matched mean. These also change forward computation and do not isolate backward shunting. **D,E**, Feedback fidelity and learning on the additive noise task; solid and dashed lines denote matched-width scalar-fallback and exact transport. Bars show standard deviations across checkpoints in **B** and retained runs in **C–E**; white points in **C** identify runs. Historical noise-task shunting aggregates with unresolved execution lineage are excluded and retained in Source Data only. The retained additive aggregates do not resolve their normalization setting. The state-matched focal analysis in main Fig. 7 provides the conductance-specific transport comparison.

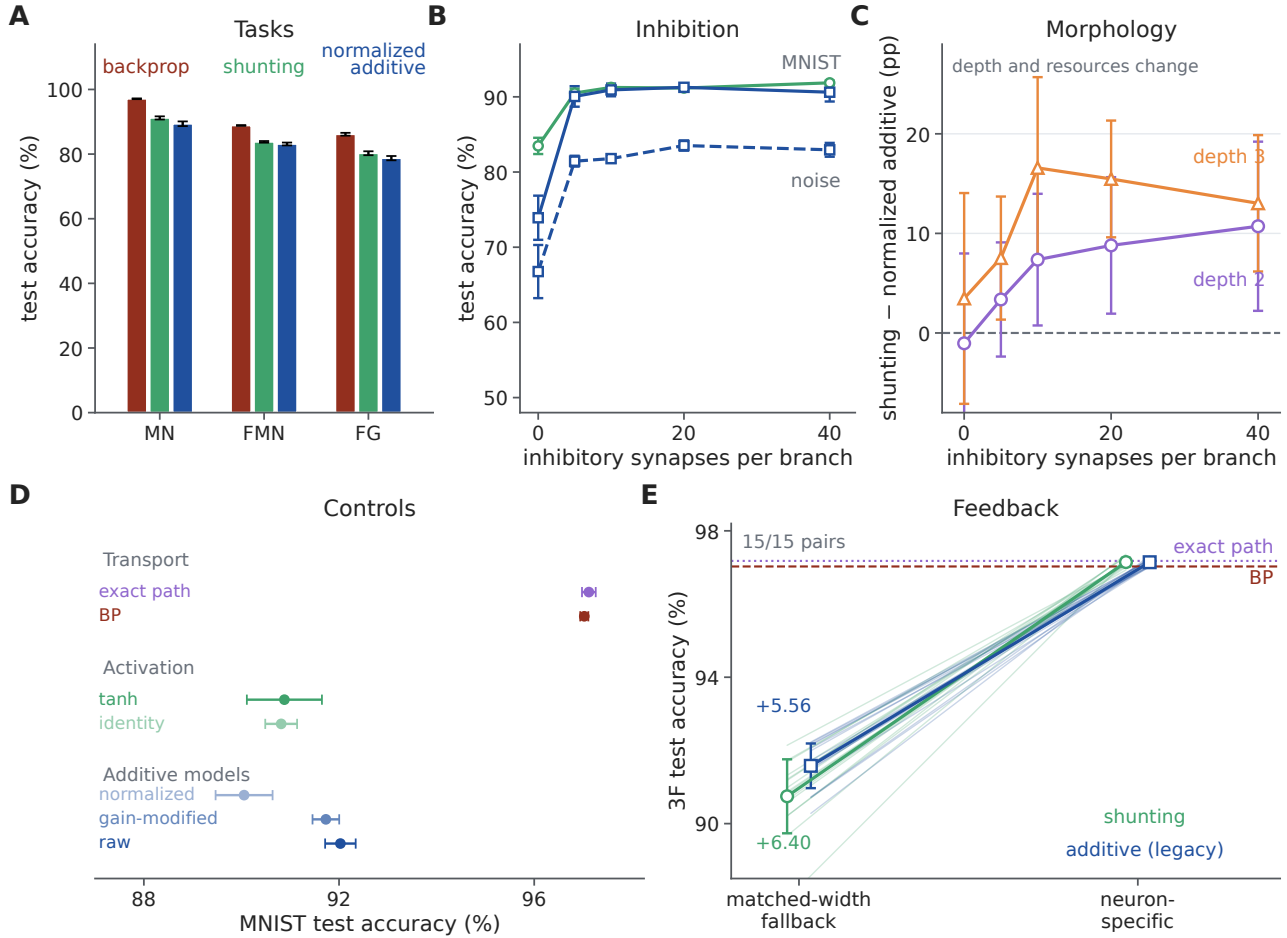


Figure S2: Matched-capacity performance, regime dependence, and feedback definition in the regular-tree models. **A**, Matched shunting backpropagation and five-factor local learning on MNIST (MN), Fashion-MNIST (FMN) and figure-ground MNIST (FG); the blue series uses the normalized-additive model. **B**, Inhibitory dose-response for both MNIST dynamics and the raw-additive noise task; noise-task shunting is excluded. **C**, Shunting-minus-normalized-additive accuracy in percentage points (pp) across two-stage [2, 2], [3, 3], [4, 4] and three-stage [2, 2, 2], [3, 3, 3] directed trees. Curves pool the listed morphologies within each stage count; standard deviations span the retained paired morphology–seed differences. This panel changes both stage count and terminal count and is an operating-regime screen rather than a resource-matched depth test. **D**, Backpropagation (BP), exact-path transport and identity-versus-tanh reactivation controls, followed by raw additive (“raw”), gain-modified raw-additive (“gain-modified”) and normalized-additive (“normalized”) controls on MNIST. **E**, Legacy fifteen-seed comparison in regular [3, 3] MNIST trees of matched-width scalar-fallback and neuron-specific feedback: the gain from preserving one soma teaching coordinate for every compartment of the corresponding modeled neuron, measured against the fallback field. Main Fig. 2 uses strict scalar feedback and provides the primary scalar-to-neuron-specific comparison. Its additive series includes retained raw-additive executions; the original five seeds lack resolved configurations, so the artwork uses “additive (legacy)”. The **E** ordinate uses the three-factor (3F) local rule; paired lines connect seeds, and horizontal BP and exact-path references come from the separate five-seed implementation check. Error bars show one standard deviation.

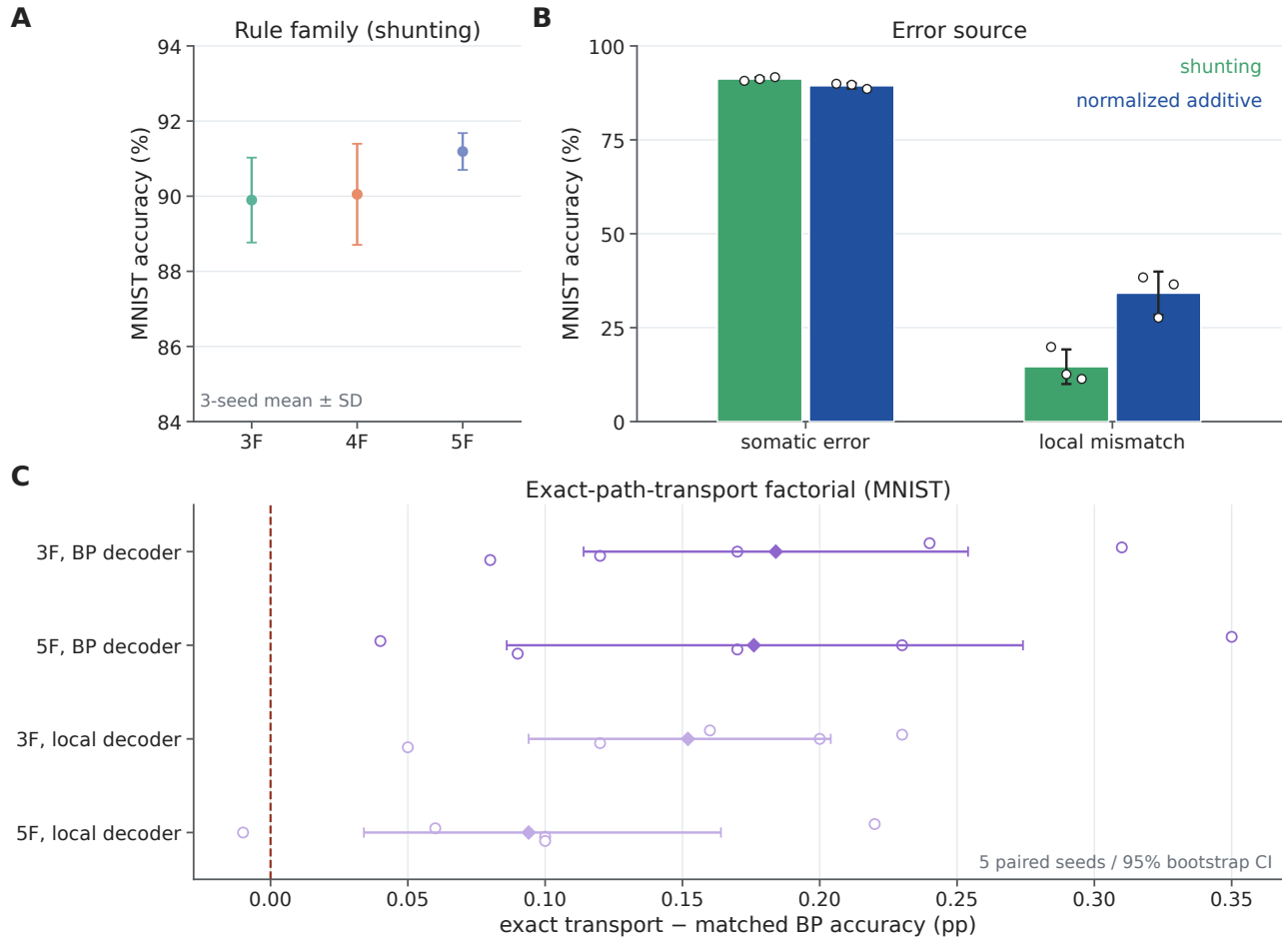


Figure S3: Local-rule and exact-transport controls quantify small remaining accuracy differences. **A**, Three-, four- and five-factor (3F, 4F, 5F) rules in the shunting model, with 4F and 5F adding bounded stage-level reliability multipliers to the 3F eligibility update: means and standard deviations from three seeds. Only aggregate means and standard deviations were retained, so individual values are not reconstructed. **B**, Somatic teaching error versus an unmatched compartment-local quantity, with all three retained seeds per condition; the blue series is the normalized-additive model. **C**, Four paired exact-transport-minus-backpropagation (BP) accuracy contrasts crossing three- or five-factor local rules with backpropagated or local decoders. Differences are in percentage points; all five paired seeds, means and 95% paired-seed bootstrap intervals (20,000 draws) are shown. No significance annotations are applied to these small-sample contrasts. Exact transport supplies the required compartment error and is an information oracle. The historical noise-task shunting ladder is excluded from the figure because its executed input convention cannot be established from its retained aggregates; those source rows remain archived.

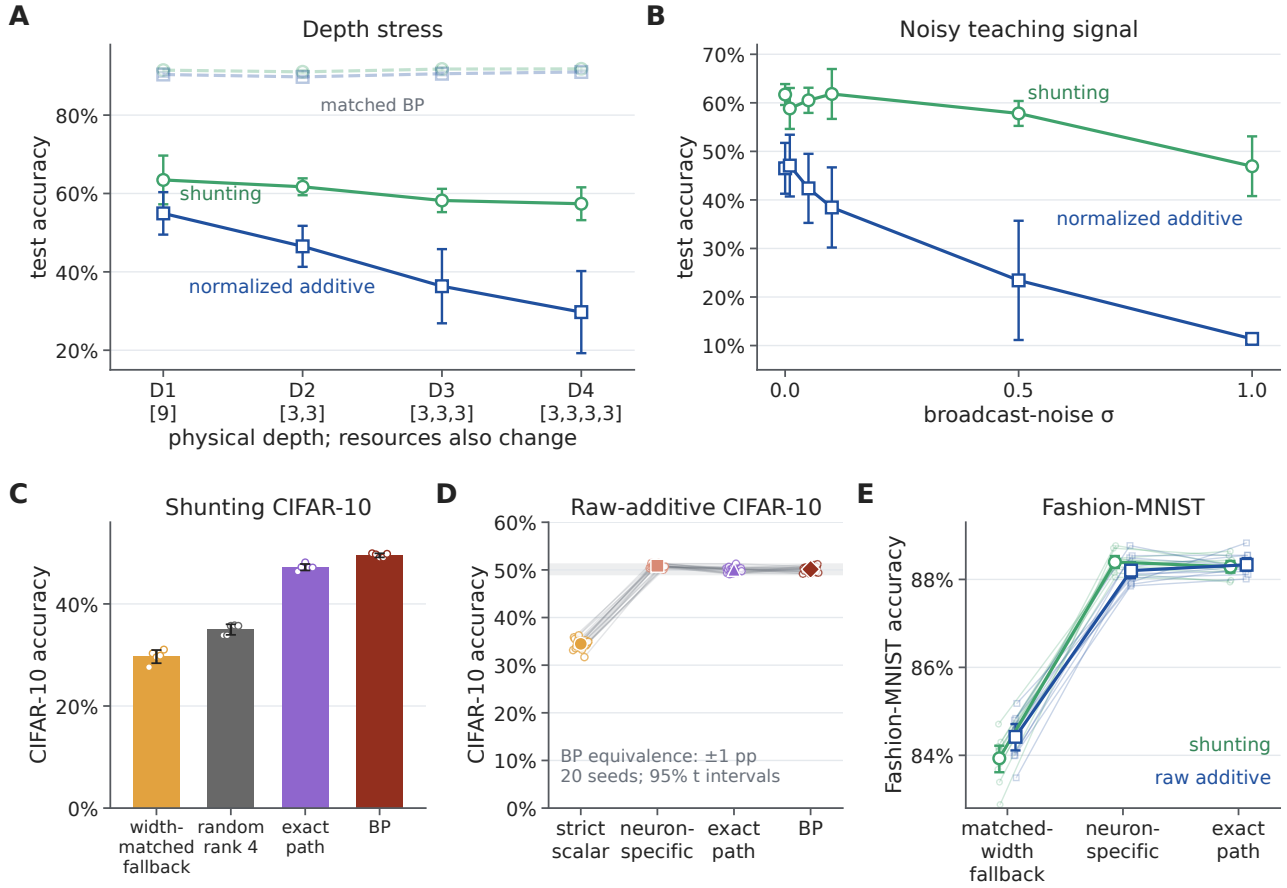


Figure S4: Additional regular-tree stress and boundary tests. **A**, Accuracy as nominal dendritic depth increases across [9], [3,3], [3,3,3] and [3,3,3,3] trees. This screen changes terminal, compartment and parameter counts with stage count and therefore does not isolate depth at fixed resources. Solid lines are local learning and faint dashed lines are matched backpropagation (BP); colour denotes shunting or normalized-additive architecture. Accuracy generally declines with depth in this screen; the resource-matched, task-aligned experiments of Supplementary Fig. S31 ask the narrower question this boundary motivates. **B**, Accuracy after adding zero-mean Gaussian noise with standard deviation σ to the broadcast teaching signal. This stress test is an empirical precursor of the admitted-noise term of the credit-operator theory, not a quantitative validation of it: it does not separately estimate the route operator, signal covariance, or forward-state effects. **C**, Flattened CIFAR-10 control in the shunting architecture, with no convolutional front end or augmentation; its four conditions use matched-width scalar-fallback feedback, a random rank-four field, exact path transport and BP. Panels **A–C** show condition means and sample standard deviations across five independent training seeds; open dots in **C** are individual runs. **D**, Raw-additive CIFAR-10 ladder in a 20-soma [3,3,3,3] tree across strict scalar, neuron-specific, exact-path and matched-backpropagation feedback (20 paired seeds). Gray lines join conditions within seed; open points are individual runs, and large symbols with bars show means and 95% Student-*t* intervals. Neuron-specific feedback improved on strict scalar feedback by 16.393 percentage points, whereas exact path transport was 0.859 points lower than neuron-specific feedback and equivalent to matched backpropagation within the prespecified ± 1 -point margin. Panels **A,B** use the normalized-additive model; **C** contains no additive condition, and **D** uses the raw-additive model. **E**, Fashion-MNIST replication of the feedback-resolution ladder in 128-soma [3,3] shunting and raw-additive trees. Thin lines connect conditions within each of ten paired seeds per architecture; large symbols and bars show means and 95% bootstrap intervals across seeds. Preserving a neuron-specific teaching coordinate improved accuracy over matched-width scalar-fallback feedback in both architectures, whereas exact path transport did not improve reliably on neuron-specific feedback. Panel **E** therefore reproduces the neuron-identity bottleneck on a second dataset without establishing a benefit from within-tree path resolution.

2099 S6 MICrONS morphology reconstruction

2100 The controlled models establish what tree-structured routes could compute. We next asked whether biological
2101 arbors contain such supports by building reciprocal cable models from reconstructed MICrONS cells and mapping
2102 their incoming synapses.

2103 6.1 Cohort and data sources

2104 The main analysis used eight layer 2–5 intratelencephalic (IT) cells from the functionally co-registered MICrONS
2105 cohort [9]. Stable nucleus identifiers were resolved to root IDs in materialization 1822 of minnie65_public.
2106 Dendritic skeletons were downloaded through the MICrONS skeleton service and incoming synapses came from
2107 synapses_pni_2. Table S24 reports the cell-level inputs to the structural analysis.

2108 Degree-two chains were collapsed into branch-to-branch cable segments. A synapse was assigned to the nearest
2109 dendritic cable node and retained if its distance was at most $5\mu\text{m}$. Across the eight cells, 28,669 classified incoming
2110 synapses passed the mapping and excitatory/inhibitory (E/I) classification criteria. The mean mapping pass fraction
2111 was 0.980. Presynaptic coarse cell type supplied E/I identity when available. Otherwise, high-confidence target
2112 structure was used as a secondary proxy from the in-development synapse_target_predictions_ssa_v2 table:
2113 spine targets were classified as excitatory and shaft or soma targets as inhibitory. This table does not currently have
2114 a stable public release identifier. Proxy and direct presynaptic labels agreed for 73.0% of 2,012 jointly labeled
2115 contacts. We therefore repeated the principal capacity comparison using only the 2,077 directly typed contacts.

2116 The independent-animal analysis used the published layer-2/3 Pinky v185 mesh, soma-valence and dense-
2117 synapse datasets [10, 11]. Before mesh topology or selected-cell synapse counts were opened, the 362 soma-table
2118 rows labelled excitatory were ordered along the second volume coordinate and divided into 12 equal-count strata.
2119 Within each stratum, the root whose first and third coordinates were nearest their global medians was selected, with
2120 root ID as a deterministic tie-break. This outcome-independent rule spans the $250 \times 140 \times 90\mu\text{m}$ volume while
2121 reducing boundary truncation; it is not a random population sample.

2122 The largest face-plus-link-edge connected component was retained from each published Hierarchical Data
2123 Format 5 (HDF5) mesh, excluding detached segmentation fragments before the soma root was chosen. Published
2124 link_edges therefore remained active in meshparty TEASAR skeletonization. We used a $7.5\mu\text{m}$ soma-collapse
2125 radius and $12\mu\text{m}$ invalidation distance, and estimated cable radius as half the opposite-surface ray distance at
2126 skeleton mesh anchors with the Embree-backed trimesh intersector (embree 2.17.7.post7). Each cell ran in an
2127 isolated process so its ray index was released before the next reconstruction. Incoming contacts were joined to
2128 E/I identity only when the presynaptic root had a corresponding label in the published soma-valence table. The
2129 reconstruction and routing settings of the minnie65 v661 reference cohort were specified before processing: a
2130 $5\mu\text{m}$ synapse-mapping criterion, identical branch-chain compression, electrical normalization, route dictionaries,
2131 route ranking, and 500 shuffle streams per cell. A cell met the predefined inclusion criteria with at least 20 mapped
2132 typed inputs, including at least three inhibitory inputs. The primary outcome was animal-level sign agreement with
2133 the minnie65 $K=4$ ancestry-minus-random-path, ancestry-minus-depth-bin and ancestry-minus-shuffled-ancestry
2134 capture contrasts. Cell-bootstrap intervals within Pinky describe volume-level stability and do not turn its 12 cells
2135 into 12 independent animals.

2136 All 12 selected Pinky cells were processed without error. Ten met the predefined direct-type criteria; the two
2137 excluded cells had 19 and 13 mapped typed inputs, respectively. At $K = 4$, mean ancestry-route capture was 0.902,
2138 compared with 0.926 for the dense principal component analysis (PCA) ceiling, 0.374 for random routes, 0.316 for
2139 depth bins and 0.245 for shuffled ancestry. Ancestry-minus-control capture advantages were 0.528 versus random
2140 routes (descriptive 95% cell-bootstrap interval 0.386–0.641; 9/10 cells; exhaustive descriptive sign-flip $P = 0.0039$),
2141 0.586 versus depth bins (0.484–0.696; 10/10; $P = 0.0020$) and 0.657 versus shuffled ancestry (0.585–0.734; 10/10;
2142 $P = 0.0020$). All three directions agree with the separate minnie65 mouse. These cell-level intervals and tests
2143 describe stability within Pinky; with one volume per animal, the biological result is directional agreement across
2144 two animals rather than a population-level test. The endpoint remains capture of a model-matched field and does
2145 not demonstrate endogenous teaching signals or learning.

2146 6.2 Normalized electrical proxy

2147 Raw axial coupling was proportional to radius squared divided by cable length; it was normalized by the median
 2148 positive value within cell and clipped to 0.05–20. Raw leak was proportional to radius times the larger of cable
 2149 length and radius; it was normalized by its median positive value and clipped to 0.1–10. Excitatory and inhibitory
 2150 synaptic areas were normalized by the median nonzero total synaptic area. The resulting quantities are passive
 2151 sensitivity weights, not fitted membrane conductances. The soma was the root of each compressed tree. Parent
 2152 traversal verified acyclicity and one path from every retained segment to the soma. The compressed cable model
 2153 used reciprocal axial coupling and a symmetric passive conductance matrix. It is therefore distinct from the directed
 2154 child-to-parent artificial network of Supplementary Note 1. The two models share a local synaptic sensitivity
 2155 multiplied by a soma-derived compartment adjoint, but only the directed model has the path product in Eq. 10.

2156 For excitatory segment i and candidate inhibitory route k , the direct model route matrix was

$$\Gamma_{ik} = A_{ik}\beta_k, \quad (109)$$

2157 where A_{ik} indicates that k lies on the path from i to the soma and β_k is its normalized inhibitory-conductance
 2158 fraction. This matrix approximates route sensitivity from normalized ancestry weights. It is distinct from the exact
 2159 Green’s-function path derivative of the reciprocal passive model used for the focal intervention. The participation
 2160 rank of the excitatory-area-weighted kernel was $(\sum_j s_j^2)^2 / \sum_j s_j^4$, where s_j are singular values.

2161 6.3 Label availability, mapping and compression sensitivity

2162 We quantified label coverage and preprocessing sensitivity using the cached raw skeletons and synapses of all eight
 2163 initial cells, without imputing new contact labels. Among 2,012 mapped contacts with both direct presynaptic and
 2164 target-structure labels, the direct/proxy contingency counts were E/E 489, E/I 84, I/E 459 and I/I 980: agreement
 2165 was 73.01%. The asymmetric errors and direct-label availability were reported by cell, soma/internal/terminal
 2166 compartment and within-cell soma-path quartile. These are observed coverage patterns, not evidence that missing
 2167 labels are random. The pooled contingency counts describe contacts nested within eight cells from one mouse
 2168 (Supplementary Fig. S33).

2169 Mapping thresholds of 2, 5 and 10 μm were crossed with hybrid labels (direct labels when available, otherwise
 2170 target-proxy probability at least 0.8) and direct-only labels. Heterogeneous radius perturbations multiplied each
 2171 compressed segment radius by $\exp(\sigma_r z - \sigma_r^2/2)$, with independent standard Gaussian z and $\sigma_r = 0, 0.25, 0.5$. Five
 2172 paired draws were used at each nonzero dose. Every condition reselected up to four inhibitory-bearing routes
 2173 by the declared anatomical leverage. Jaccard overlap compared the selected route identities with those in the
 2174 nominal hybrid-label, 5 μm model. Capture always used the same nominal model-generated kernel, excitatory
 2175 sites and weights. Thus a changed label set or radius could alter the dictionary but not its target field. This isolates
 2176 preprocessing sensitivity; it is not an endogenous task-alignment test or a calibrated uncertainty model.

2177 To assess the electrical approximation introduced by compression, we compared each compressed segment
 2178 with its constituent raw edges. Compression preserved total length to $5.7 \times 10^{-14} \mu\text{m}$ and the length–radius product
 2179 to 2.1×10^{-14} in the retained units. It did not preserve axial resistance. The mean-radius approximation $\ell/\bar{r}^2$ was
 2180 compared with the series sum $\sum_e \ell_e/r_e^2$, using mean endpoint radius on each raw edge. The median relative error
 2181 was -0.00200 , with a minimum of -0.99717 . A second electrical normalization substituted inverse exact series
 2182 resistance for inverse mean-radius resistance, preserving the original leak and synaptic terms and applying the
 2183 same within-cell median normalization and 0.05–20 clipping. Radius perturbations rescaled each segment’s series
 2184 resistance by the inverse squared radius factor. The complete factorial contained 1,056 conditions; all were retained,
 2185 and radius draws were averaged within cell.

2186 At nominal mapping and hybrid labels, mean fixed-field capture was 0.3293 under mean-radius normalization
 2187 and 0.3375 under exact series normalization. These modest mean changes do not imply equivalence of the
 2188 electrical models: the raw segment-resistance audit contains substantial local discrepancies. Historical results
 2189 remain labeled with their original normalization. Full segment audits, condition values and cache hashes are in

2190 source_data/review_morphology_uncertainty/. Figure intervals resample the eight cells 5,000 times and
2191 describe stability within this mouse.

2192 **S7 Structural credit-capacity analysis**

2193 We asked whether sparse nested supports in reconstructed arbors can efficiently approximate compartment-level
2194 credit fields. This analysis tests structural capacity using measured morphology and imposed field generators and
2195 route coefficients. Perturbation fields were generated by multiplying independent route amplitudes by the kernel
2196 in Eq. 109. For each cell and channel count, the tested dictionaries were: a dense principal-component oracle;
2197 selected real ancestry paths; random nonempty paths from the same tree; bins of cable depth; and a column-wise
2198 ancestry shuffle that preserved dictionary dimensions and sparsity. Nested-subtree routes were ranked by β_k times
2199 the fraction of total mapped excitatory synaptic area among their descendants.

2200 Both fields and dictionaries were multiplied by the square root of summed mapped excitatory synaptic area
2201 before ordinary least-squares projection, as defined in Eq. 33. Twenty independent Monte Carlo streams were used,
2202 each containing 384 perturbation fields and 64 random or shuffled dictionaries per cell and channel count. The
2203 Monte Carlo streams measure numerical sensitivity. We averaged them within each cell before inference, using the
2204 eight reconstructed cells from one animal as the units of analysis.

2205 At eight channels, morphology-selected paths captured 0.487 of the weighted energy in the model-derived
2206 route-perturbation fields (hierarchical 95% interval 0.400–0.581) at 6.92% dense-feedback wiring. This was 0.850
2207 of the dense oracle’s capture. Table S25 reports all matched controls. On the directly typed subset, weighted capture
2208 was 0.782 for the dense oracle, 0.696 for morphology, 0.347 for random paths, 0.447 for depth bins and 0.308
2209 for ancestry shuffling. Ancestry-route capture exceeded every structural control in all eight cells. The direct-type
2210 analysis tests sensitivity to E/I labels on a smaller input set; it does not estimate population capacity. Because
2211 these fields are modeled rather than observed task gradients, a descent interpretation requires treating them as exact
2212 gradients in the transformed coordinates. The alignment-controlled analysis below tests optimization under that
2213 explicit assumption.

2214 For a larger, non-overlapping-cell sensitivity analysis, we defined a 55-cell set before computing outcomes,
2215 mapped it to the public MICrONS v661 static release and excluded the initial eight-cell sample by stable nucleus
2216 identifier. Twenty independently generated perturbation streams were averaged within cell. At eight channels,
2217 morphology paths in the independent, non-overlapping sample of 47 cells captured 0.756 of modeled field energy
2218 with 8.08% of dense wiring, compared with 0.839 for the dense oracle, 0.302 for random paths, 0.441 for depth bins
2219 and 0.330 for ancestry shuffles (Supplementary Fig. S10B–D). The morphology advantage over every structural
2220 control was positive in 47/47 cells. The cohort is from the same mouse, is layer-imbalanced, descends from a
2221 convenience export, and uses an earlier reconstruction release. Direct presynaptic E/I labels were available for
2222 11,024 incoming synapses before mapping (4.71%); 10,516 mapped contacts entered the analysis. It is therefore a
2223 non-overlapping-cell robustness analysis, not an independent-animal or population replication. The candidate set
2224 and endpoint exclusions were defined before outcome inspection and accompany the source data.

2225 **7.1 Irregular-tree wavelet scale analysis**

2226 To determine the spatial scales represented by ancestry routes, we specified a multiscale analysis before inspecting
2227 its outcomes. The independent, non-overlapping public-v661 sample of 47 cells was primary and the initial eight-
2228 cell sample was secondary. We retained the existing cells, compressed segments, excitatory weights and inhibitory
2229 routes without reselection. For each cell, let w_i be mapped excitatory area and let Γ be the conductance-weighted
2230 ancestry matrix in Eq. 109. Following the multiresolution principle of tree wavelets [12], a deterministic unbalanced
2231 tree-Haar construction recursively split each current excitatory region R at the descendant subtree whose weight
2232 was closest to half the region weight. If the split produced $A \cup B = R$ with weights W_A , W_B and W_R , its normalized

2233 contrast in weighted coordinates was

$$h_{R,i} = \begin{cases} \sqrt{w_i} \sqrt{W_B / (W_A W_R)}, & i \in A, \\ -\sqrt{w_i} \sqrt{W_A / (W_B W_R)}, & i \in B, \\ 0, & i \notin R. \end{cases} \quad (110)$$

2234 Together with the weighted constant $h_{0,i} = \sqrt{w_i / \sum_j w_j}$, the recursively nested contrasts form a full orthonormal
 2235 basis. All 55 cell bases had full rank; maximum orthonormality error was 4.44×10^{-16} .

2236 Let $W = \text{diag}(w_i)$ be the excitatory-area weight matrix. To isolate within-cell address structure from scalar
 2237 broadcast, we analyzed

$$\tilde{\Gamma} = (I - h_0 h_0^T) W^{1/2} \Gamma, \quad p_r = \frac{\|h_r^T \tilde{\Gamma}\|_2^2}{\|\tilde{\Gamma}\|_F^2}. \quad (111)$$

2238 Here p_r is the fraction of centered route energy in wavelet h_r . Wavelets were called coarse when their support
 2239 exceeded one quarter of total excitatory weight, intermediate when it exceeded one sixteenth but not one quarter,
 2240 and fine otherwise. For n excitatory sites, the isotropic-noise reference for a scale containing m_s modes among $n - 1$
 2241 non-scalar dimensions was $m_s / (n - 1)$. A second null independently permuted each raw ancestry-route column
 2242 across excitatory sites before weighting and centering; 256 stable cell-specific permutations were nested within
 2243 each cell. Cell was the inferential unit and intervals bootstrapped cells.

2244 In the 47-cell cohort, coarse wavelets contained 0.4065 of non-scalar route energy despite comprising 0.1923 of
 2245 dimensions. Mean enrichment over isotropic noise was 2.351 (95% cell-bootstrap interval 1.995–2.880; 47/47 above
 2246 one; two-sided Wilcoxon $P = 1.42 \times 10^{-14}$). Fine-scale enrichment was 0.342 (0/47 above one; $P = 1.42 \times 10^{-14}$),
 2247 with intermediate enrichment 1.134 (30/47; $P = 0.0160$). The anatomy-specific null was not rejected: actual
 2248 minus permuted coarse energy was 0.0345 (−0.0025–0.0713; 31/47; $P = 0.0780$), and intermediate and fine
 2249 contrasts were also null. The initial eight-cell sample showed a coarse actual-minus-permuted excess of 0.0677
 2250 (0.0495–0.0900; 8/8; $P = 0.00781$), which therefore failed to reproduce in the larger independent, non-overlapping
 2251 sample. Thus coarse structural coordinates were enriched relative to isotropic noise, but the analysis did not
 2252 establish a replicable advantage of the observed fine ancestry layout. It also did not measure alignment with task
 2253 credit or use of these coordinates during learning.

2254 The preceding fields were generated from the ancestry kernel being tested. To remove this direct correspondence,
 2255 we formed an independent response operator on the trees from the initial eight-cell sample. A focal shunt of 10^{-3}
 2256 times local leak-plus-synaptic conductance was added at every inhibitory-bearing segment of the reciprocal passive
 2257 model. After an independently solved state-matching compensatory current restored the baseline somatic voltage,
 2258 we finite-differenced the log magnitude of the exact excitatory-conductance gradient at every excitatory-bearing
 2259 segment. The resulting response operator was full rank in all eight cells and placed 17.6% of its weighted energy,
 2260 on average, outside direct ancestry support. For a dictionary Φ , expected capture was evaluated exactly as the
 2261 squared Frobenius norm after projection of this operator onto $W^{1/2} \Phi$, divided by its total weighted Frobenius norm.

2262 To distinguish fine ancestry from coarse depth and degree, we added 200 surrogate trees per cell. At each
 2263 topological depth, child nodes were randomly assigned among the original parent slots, preserving every node's
 2264 depth and every parent's out-degree. All edge and node attributes remained attached to their original child nodes.
 2265 The same leverage rule was then applied to the surrogate dictionary. Row-shuffled selected paths preserved each
 2266 selected column's values and nonzero count. At eight channels, morphology exceeded random real paths and
 2267 row-shuffled paths in every cell, but did not reliably exceed depth bins or degree- and depth-matched surrogate
 2268 trees (Table S26). These controls support the value of sparse ancestry structure relative to random assignment,
 2269 while limiting the evidence for a further contribution from fine topology beyond depth and degree.

2270 7.2 Anatomical capacity after equalizing the common broadcast

2271 The original eight-cell analysis exposed a common-mode mismatch: depth bins included a constant field, whereas
 2272 selected subtree profiles did not. We therefore gave every family an identical budgeted broadcast column. The

eight cells were development data for this correction. The protocol and input hashes were frozen before computing the new independent-generator outcomes in 47 disjoint v661 cells and the Pinky cohort. Both follow-up cohorts had previously been used for route-generated positive controls; this is a frozen follow-up on existing anatomy, not external preregistration or new animal sampling. Stable nucleus identifiers verify that v661 excludes the original eight cells; the original eight cells and the 47-cell v661 extension belong to one mouse, while Pinky supplies a second animal.

Every dictionary has one all-site constant and $K - 1$ spatial columns, for $K = 1, 2, 4, 8, 16$ when site counts permit. The focus budget is eight. Ancestry columns retain the existing ordering by local inhibitory fraction times excitatory-area-weighted descendant fraction, without inspecting the response. Random routes sample inhibitory sites without replacement. Site shuffling independently permutes each ancestry column, preserving its values and nonzero count while leaving the constant untouched. Surrogates preserve node depths and parent out-degrees while reassigning parent slots, then recompute ancestry and its ranking. The depth dictionary contains a constant and the first $K - 1$ inherited depth bins, giving exactly the original K -bin span. Its additional nonzero count is retained. Equal column budgets need not give equal numerical rank: random and overlapping routes can be redundant.

The response matrix H contains finite-difference changes in the logarithm of absolute excitatory synaptic gradients after focal shunts, with the soma re-clamped. It uses the inherited dimensionless cable model, excitatory and inhibitory scales 0.35, reversal potentials 1 and -0.2 , and derivative dose 0.001. Let $Y = W^{1/2}H$, with W proportional to excitatory contact area, and let q be the normalized weighted constant. The residual field is $R = (I - qq^T)Y$. Total capture is $\|P_D Y\|_F^2 / \|Y\|_F^2$; residual capture is $\|P_D R\|_F^2 / \|R\|_F^2$, where P_D projects onto the weighted dictionary. Since every dictionary contains q , total capture equals broadcast capture plus $(1 - \text{broadcast capture})$ times residual capture, cell by cell. The common-constrained oracle adds the leading $K - 1$ left singular vectors of R .

We average 200 randomized controls within each cell before inference. Paired cell means and 95% bootstrap intervals use 20,000 draws; Holm adjustments cover four spatial controls separately for each endpoint and cohort at $K = 8$. Other budgets are descriptive. All cohorts remain separate. Pinky retains ten of twelve originally selected cells under the inherited typed-contact criterion; eight qualify at $K = 8$, and none at $K = 16$. The other two qualifying cells remain in lower-budget outcomes. Their small input-bearing domains explain near-saturation at eight columns.

The analysis retains 65 response operators and 149,331 dictionary evaluations, including 148,200 randomized realizations. Validation checks frozen inputs, original ancestry-capture reproduction, weighted energy identities, oracle ceilings, depth-span equivalence, numerical ranks, common-mode coverage and exact ancestry-shuffle nonzero matching. Five independent numerical tests pass. All per-draw results, operator matrices, eligibility records and bootstrap summaries are available in `source_data/anatomy_commonmode/`. Squared-energy capture here differs from the one-minus-residual-norm score used in the older route-generated replication, whose endpoint is labeled separately.

S8 Synapse-resolved inhibitory route analysis

We next asked whether observed inhibitory contacts are distributed across descendant domains in a way that could support spatially restricted route control. The analysis measures anatomical association; it does not measure inhibitory activity or learning signals. The inhibitory-census endpoints, controls, and matching criteria were specified before the data join. The cohort was the complete stable-nucleus-ID intersection between the 47-cell v661 sensitivity cohort and the v795 inhibitory census: 20 targets, with no selection by an outcome. Because the two releases use different coordinate representations, the raw-to-slanted affine map was estimated from paired published inhibitory-synapse locations and required to reproduce all paired locations to less than $10^{-6} \mu\text{m}$ before any target was analyzed. Five targets had different root IDs across releases and were joined by stable nucleus identity. The source tables contained 116,401 input locations and 2,789 known contacts from 116 inhibitory cells before spatial mapping; 114,775 inputs and 2,637 known contacts from 114 inhibitory cells mapped within $5 \mu\text{m}$. The packaged census table has no per-target proofreading-status field, so no such status is inferred.

2320 The connection analysis retained one largest synapse per published axonal clump. This avoided counting imme-
 2321 diately adjacent contacts from the same presynaptic axonal event as independent control sites. A matched candidate
 2322 pool was constructed separately for every retained contact. Matching preserved target cell and compartment and
 2323 minimized difference in log distance to the soma. All connection outcomes were averaged within target before
 2324 testing. There were 402 inhibitory cell–target connections with at least two retained clumps, nested within 20
 2325 targets.

2326 To measure descendant-domain overlap, we began with a binary indicator of mapped input locations below
 2327 candidate site k . Multiplying each entry by the square root of its synapse size and normalizing to unit length gave
 2328 the vector r_k . Two control sites therefore have overlap $r_k^\top r_l$. This value is one for identical weighted descendant
 2329 domains and zero for disjoint domains. It is an anatomical co-control statistic, not a measured correlation of
 2330 inhibitory activity. Table S31 reports the predefined target-level endpoints.

2331 We then tested whether the primary target-level associations persisted under two more stringent controls. First,
 2332 connection contrasts were averaged within each of 92 presynaptic inhibitory axons before testing, which accounts
 2333 for one axon contacting multiple targets. Second, candidate sites were ranked jointly by proximity in log path
 2334 distance and three-dimensional Euclidean distance, and 500 matched assignments were repeated within target. The
 2335 coarse tree-distance contrast survived presynaptic-axon aggregation. Shared-path and descendant-domain overlap
 2336 did not; no endpoint survived joint 3D and path-distance matching (Table S32). Thus the primary result establishes
 2337 spatial co-targeting, not geometry-independent organization over credit domains.

2338 The route-capacity matrix used all actual inhibitory-bearing segments as columns, rather than one column per
 2339 presynaptic inhibitory cell. Row weights were square roots of mapped input size and column weights were square
 2340 roots of known inhibitory synapse size. Sites were selected in descending order of column norm. Random-site and
 2341 shuffled controls were independently generated 200 times per cell and budget. This diagnostic asks whether the set
 2342 of observed locations contains an efficient ancestry basis; it does not show that the highest-leverage subset was
 2343 selected or used in vivo. The target is the actual-route matrix from which the selected columns are drawn. This
 2344 analysis therefore measures internal compressibility, rather than independent routing performance.

2345 S9 Focal shunting perturbations

2346 Route capacity does not show that conductance can selectively alter transport. We therefore compared a focal shunt
 2347 with a baseline-current-matched injection at the same site of each reciprocal reconstructed cable and quantified the
 2348 resulting change in the exact modeled gradient.

2349 9.1 Adjoint formulation and baseline-current-matched current injection

2350 For passive conductance matrix G and current-source vector $\mathbf{b}$, the voltage vector was $\mathbf{V} = G^{-1}\mathbf{b}$. Let $\mathbf{e}_s$ select the
 2351 somatic coordinate and let w_i denote the excitatory conductance at site i . For a loss depending on somatic voltage,
 2352 the adjoint error and conductance gradient were

$$\mathbf{q} = G^{-1}\mathbf{e}_s \frac{\partial \mathcal{L}}{\partial V_s}, \quad \frac{\partial \mathcal{L}}{\partial w_i} = q_i(E_E - V_i). \quad (112)$$

2353 Adding focal inhibitory conductance κ_k at node k , selected by the coordinate vector $\mathbf{e}_k$, gives

$$G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top, \quad \mathbf{b}' = \mathbf{b} + \kappa_k E_1 \mathbf{e}_k.$$

2354 The shunt therefore changes both conductance loading and the forward source. Holding the somatic loss derivative
 2355 fixed, the Sherman–Morrison identity gives the transport change

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (113)$$

2356 The reversal E_1 does not appear in Eq. 113 because the adjoint transport depends on G' , not on the forward source
 2357 $\mathbf{b}'$. It remains in the solved voltage and in every inhibitory driving force. Corollary 2 further identifies its spatial

form: all descendants of k share one adjoint attenuation factor, and each off-path sister block along the ancestral path shares another. This is a gain on the baseline-weighted ancestry partition, with additional site-specific variation entering the full gradient through the changed driving force. Let $r_k = (G^{-1})_{kk} > 0$. Then

$$\Delta \mathbf{q} = -a_k G^{-1} \mathbf{e}_k, \quad a_k = \frac{\kappa_k q_k}{1 + \kappa_k r_k}, \quad \frac{\partial |a_k|}{\partial \kappa_k} = \frac{|q_k|}{(1 + \kappa_k r_k)^2} \geq 0. \quad (114)$$

The derivative is strictly positive when $q_k \neq 0$; if $q_k = 0$, the perturbation vanishes. Its amplitude saturates at $|q_k|/r_k$. For a nonzero perturbation, the normalized spatial profile is fixed by the Green's-function column $G^{-1} \mathbf{e}_k$ in this linear passive state. For a descendant set $D(k)$ and a prespecified matched comparison set $C(k)$, define

$$S_k(D, C) = \frac{\text{median}_{i \in D(k)} |(G^{-1})_{ik}|}{\text{median}_{j \in C(k)} |(G^{-1})_{jk}|}. \quad (115)$$

For $\kappa_k > 0$, $q_k \neq 0$ and a positive comparison median, the common amplitude $|a_k|$ cancels. Under these conditions, $S_k > 1$ is necessary and sufficient for a positive descendant-minus-comparison contrast in absolute adjoint change at fixed k . It is only a transport-level criterion. Relative or logarithmic changes additionally divide by the baseline q_i , and the full synaptic gradient also contains the post-perturbation driving force. This is why axial coupling, leak, background conductance and local input resistance must be varied explicitly rather than inferred from topology alone. The baseline-current-matched current injection applied $\kappa_k(E_I - V_k)$ at the same node without changing G . This equals the current carried by the added shunt at the unperturbed voltage; its local voltage effect matches the shunt only to first order in dose. An independently solved state-matching compensatory current injection restored baseline somatic voltage after both interventions. This is not a continuously adjusted voltage clamp. The somatic scalar error was therefore fixed. Shunting changed adjoint transport, whereas the current-injection control left it unchanged. Dendritic voltages were not clamped, so both interventions could also change local driving forces.

9.2 Prespecified endpoint and numerical validation

Before running the analysis, we fixed the site-selection rule, primary dose, comparison groups, and localization endpoint. Focal sites were selected by a depth-stratified rule without access to gradient outcomes. A site required at least three excitatory descendants and three comparison sites. At most 16 eligible sites were sampled per cell across quartiles of topological depth. The primary model used excitatory and inhibitory synaptic scales of 0.35 and normalized reversal potentials $E_E = 1$ and $E_I = -0.2$. Synapses were classified as descendants, ancestors, sister-subtree sites or unrelated sites. Unrelated synapses within each template were matched to descendant cable depth. For baseline gradient g_i and perturbed gradient g'_i , we defined $m_i = |\log[(|g'_i| + \varepsilon)/(|g_i| + \varepsilon)]|$, where $\varepsilon = 10^{-15} \max(1, \max_i |g_i|)$. For each site, the localization index was the median m_i among descendants minus the median among depth-matched unrelated synapses. Site effects were first averaged within cell.

The somatic loss was one-half the squared difference between somatic voltage and a target one normalized unit below its baseline value. Conductance doses were 0.25, 0.5, 1, and 2 times the local leak-plus-synaptic conductance. The state-matching current was solved after each perturbation and then treated as fixed when evaluating the post-intervention conductance gradient. The topology control sampled 200 relation templates from other focal sites, stratified by focal-site topological-depth quartile when another site was available; otherwise it sampled another site in the same cell.

The primary unit-dose test included 101 sites in eight cells. Mean localization was 0.104 under shunting and 0.035 under the baseline-current-matched current injection control. The paired cell-level difference was 0.069 log units (95% bootstrap interval 0.053–0.086), positive in all eight cells. The relation control used 200 templates sampled by the stratified procedure with the fallback described above. True focal relations exceeded this control by 0.080 (0.059–0.098), again in all eight cells. The dose contrast increased from 0.018 at 0.25 local-conductance units to 0.128 at two units.

We separated the two factors in the exact gradient algebraically. Writing $d_i = E_E - V_i$ and $d'_i = E_E - V'_i$, full shunting uses $q'_i d'_i$, adjoint-only substitution uses $q'_i d_i$, and driving-force-only substitution uses $q_i d'_i$. At unit

dose, their mean localization indices were 0.104, 0.130 and 0.026, respectively, compared with 0.035 for the current-injection control. Adjoint-only minus driving-force-only localization was 0.104 (95% interval 0.080–0.125), positive in all eight cells.

Full shunting and the driving-force-only state provide a direct comparison of adjoint transport: they share the same post-shunt voltage field and differ only in whether the adjoint field is updated. Their localization difference was 0.079 and was positive in all eight cells. This is the exact adjoint-replacement contrast plotted in main Fig. 8D.

The gradient factors and localization statistic are nonlinear, so these four outcomes do not form an additive decomposition. To allocate the full-shunt-minus-current-injection contrast, we used two-factor Shapley values. For v_{00} equal to current-injection localization, v_{10} adjoint-only, v_{01} driving-force-only and v_{11} full-shunt localization,

$$\phi_q = \frac{1}{2}[(v_{10} - v_{00}) + (v_{11} - v_{01})], \quad (116)$$

$$\phi_d = \frac{1}{2}[(v_{01} - v_{00}) + (v_{11} - v_{10})]. \quad (117)$$

The identity $\phi_q + \phi_d = v_{11} - v_{00}$ held at every site to a maximum absolute residual of 5.6×10^{-17} . Averaging sites within cells, the adjoint allocation was 0.087 (hierarchical 95% interval 0.063–0.111), while the driving-force allocation was -0.017 (-0.029 to -0.009). The former was positive and the latter negative in all eight cells. Within this algebraic allocation, the adjoint change accounted for more than the net positive contrast, while the driving-force change partially cancelled it. The baseline-current-matched current injection is the relevant control but is not a literal no-factor corner of a physical 2×2 intervention. Using the unperturbed field as that corner assigned 0.104 (0.074–0.136) to the adjoint and approximately zero to the driving force. Both analyses allocate a nonlinear model outcome. The factor substitutions are algebraic comparisons, not independent biological interventions.

All conductance matrices in the primary analysis were positive definite. The somatic state-matching residual was numerically zero. Analytic adjoint gradients agreed with finite differences to a maximum relative error of 1.03×10^{-6} . Table S27 reports prespecified conductance-scale and reversal-potential sensitivity runs. The direction of the shunting-minus-current-injection effect was positive in every condition, with seven or eight positive cells. These results establish the mechanism in a passive model on measured trees; they do not establish endogenous inhibitory teaching signals in vivo.

With proxy-classified contacts removed, the primary shunt-minus-current-injection contrast was 0.101 (0.075–0.129) across 55 sites and was positive in all eight cells. In the independent, non-overlapping v661 sample, 45 cells supplied 235 eligible sites. The same contrast was 0.138 (0.116–0.162), positive in 45/45 cells. Forty cells with at least two eligible sites supplied the reassigned-relation control; the true relation exceeded this control by 0.118 (0.094–0.144), with 39 positive cells and one numerical tie (Supplementary Fig. S10F,G). Cells, rather than sites or relation draws, were the units of inference.

To test sensitivity to electrical calibration, we also computed axial and leak conductances in physical units from the same compressed geometry. For segment radius r , length ℓ , axial resistivity R_a and specific membrane resistance R_m , a cylindrical non-root segment had axial conductance $\pi r^2 / (R_a \ell)$ and membrane leak $2\pi r \ell / R_m$; the root was assigned spherical area $4\pi r^2$. All conductances were converted to nanosiemens (nS). Because MICrONS synapse size is not an electrical conductance, synaptic sizes retained the prespecified relative scaling with the median physical leak as reference. We varied R_a and R_m while retaining the site-selection rule, relative dose, gradient definition and current-injection control. For each condition and intervention, we re-solved the compensating somatic current to restore that condition's baseline somatic voltage. Table S28 shows the resulting electrotonic dependence. The large contrast observed under the normalized proxy was recovered only in low- R_m , high-conductance settings; at $R_a = 150 \Omega \text{cm}$ and $R_m = 15,000 \Omega \text{cm}^2$ it was negligible or small. These are sensitivity settings, not fits to the recorded cells, and the model remains passive and steady-state.

A broader parameter matrix, specified before numerical validation, tested membrane resistance, background conductance and dose definition jointly. It used $R_a = 150 \Omega \text{cm}$ and crossed $R_m \in \{300, 1000, 15,000\} \Omega \text{cm}^2$, distributed leak-like background conductance of 0, 1 or 4 times baseline leak, and three dose definitions. Relative-local doses were 0.25, 1 and 2 times membrane-plus-synaptic conductance; fixed doses were 0.05, 0.5 and 5 nS; input-normalized doses were 0.25, 1 and 4 times $[(G^{-1})_{kk}]^{-1}$. The same 101 sites in eight cells supplied 16,362

2441 site-condition-intervention rows. Every conductance matrix was positive definite (minimum eigenvalue 0.107 nS)
2442 and the maximum somatic-state residual was 3.33×10^{-16} .

2443 At fixed absolute dose, the conductance-specific contrast did not follow the raw focal-shunt localization alone.
2444 At 0.5 nS, shunt-minus-current-injection localization was 0.0104 (0.0080–0.0126), 0.0073 (0.0044–0.0104), and
2445 -0.0039 (-0.0075 to -0.0004) at $R_m = 300, 1,000$ and $15,000$, respectively. At 5 nS the corresponding values
2446 were 0.0974, 0.0741 and 0.0061; the last interval included zero. Distributed background conductance restored
2447 the input-normalized effect in the high-resistance condition: at unit input-conductance dose, the contrast changed
2448 from -0.0004 with no background to 0.1210 and 0.2587 at background multipliers one and four. Across all
2449 8,181 focal-shunt rows, every descendant gradient magnitude was attenuated, none was enhanced and no sign flip
2450 occurred. The passive mechanism therefore selectively attenuated gradients; sign reversal would require dynamics
2451 or active conductances absent from this phase (Supplementary Fig. S11).

2452 9.3 Weak-channel steady-state linearization check

2453 We checked the local-Jacobian calculation after adding weak voltage-dependent currents to the high-conductance
2454 cable. The physical reconstructed-tree system used $R_a = 150 \Omega \text{cm}$, $R_m = 1,000 \Omega \text{cm}^2$ and distributed background
2455 conductance equal to baseline leak. Steady-state sodium (Na), potassium (K), calcium (Ca), hyperpolarization-
2456 activated cyclic nucleotide-gated (HCN) and N-methyl-D-aspartate receptor (NMDA) current families had fixed
2457 reversal potentials, half-activation voltages, slopes and gate powers specified in the predefined parameter file. Global
2458 and segment-level maximal conductances were drawn independently from prespecified log-normal distributions.
2459 Median maximal Na, K, Ca and HCN conductances were 0.06, 0.10, 0.025 and 0.05 times baseline leak, respectively;
2460 NMDA conductance was 0.15 times the excitatory conductance. Gate activation further scales these currents. These
2461 prespecified weak-channel draws are not cell-specific density estimates or a test of strongly regenerative operation.

2462 For each draw, a residual-reducing Newton solve found the nonlinear steady state. The predefined acceptance
2463 criteria required all voltages to lie in the specified range, maximum absolute state residual below 10^{-10} , and
2464 minimum eigenvalue of the symmetrized local Jacobian above 10^{-8} . The implicit-adjoint calculation then used
2465 the exact inverse of this baseline Jacobian. For channel family a , let $\bar{g}_a$ be maximal conductance, $h_a(V)$ the
2466 steady-state gating factor and E_a the reversal potential. The current $I_a(V) = \bar{g}_a h_a(V)(V - E_a)$ contributed the
2467 diagonal term $\bar{g}_a[h_a(V) + h'_a(V)(V - E_a)]$; thus the baseline Jacobian includes voltage-dependent gate derivatives.
2468 Focal shunting and a baseline-current-matched current injection were evaluated at 0.25, 1 and 4 times local input
2469 conductance within this fixed local linearization. Sherman–Morrison updated the Jacobian inverse for the added
2470 focal conductance, and a compensating somatic current restored baseline somatic voltage in that linearized system.
2471 The active gates and their derivatives were not re-evaluated at a new nonlinear equilibrium after each finite dose.
2472 Up to 256 attempts were permitted to obtain 64 accepted draws per cell. All first 64 attempts were accepted in
2473 every cell, giving 512 equilibria, 101 focal sites and 38,784 site-draw-condition rows. All values were finite, the
2474 largest equilibrium residual was 9.51×10^{-11} and the smallest Jacobian eigenvalue was 1.084.

2475 Channel draws and focal sites were averaged within each of the eight cells before inference. At doses 0.25, 1
2476 and 4, mean shunt localization was 0.159, 0.529 and 1.339, respectively (Table S7). At unit dose, shunting exceeded
2477 the current-injection control by 0.382 (95% cell-bootstrap interval 0.356–0.409), positive in 8/8 cells (two-sided
2478 Wilcoxon $P = 0.0078$). Every modeled descendant gradient was attenuated, none was enhanced or changed sign,
2479 and descendant gradient energy fell to 0.330 of baseline. The dose response closely followed the passive response
2480 at the same electrical calibration. This agreement checks the weak-channel linearization; it does not establish
2481 robustness to strongly active dendrites, channel kinetics or spiking.

2482 S10 Functional topology and task-derived credit

2483 Anatomical routes can be available without aligning with the credit required by a measured task. We therefore used
2484 recorded visual responses to define a prediction objective on reconstructed target cells. Within each model, we
2485 changed the backward field while holding the forward morphology fixed.

10.1 Direct anatomy–function join

Functionally imaged excitatory partners were linked to anatomical contacts through MICrONS/CAVE and DANDI mappings. The analysis contained 69 connected presynaptic partners across seven postsynaptic target cells. Automatic matches came from `coregistration_auto_phase3_fwd_apl_vess_combined_v2` in materialization 1822. We retained residual at most 10 and score at least zero, then deduplicated each presynaptic root by ascending residual and descending score. All mapped contacts from a retained presynaptic root contributed to its partner-level contact summary. For the one-site-per-partner model, contacts were first grouped by postsynaptic segment. The partner was assigned to the segment with greatest summed contact size, breaking ties by synapse count, and its major branch was the first segment below the soma on that segment’s ancestry chain. Thus a partner with contacts on several segments supplied one activity feature and one modeled learning site; its total contact size and synapse count still included all retained contacts.

The target universe was the initial eight-cell morphology convenience sample. The analysis plan, specified before outcome inspection, included every target in that sample with a DANDI scan and directly connected, functionally imaged excitatory partners; seven targets qualified. Predefined inclusion criteria required exactly one postsynaptic target in an extract, agreement between partner and contact counts, at least five mapped partners for the topology screen, at least two sites after any requested task filter, and at least one nonempty inhibitory nested-subtree route. The primary task analysis applied neither a manual-match restriction nor a repeat-reliability threshold. The exact target/session/scan pairs are listed in Table S29. The original analysis plan did not record a deterministic rule for choosing among multiple eligible scans of the same target. We therefore treat these scan choices as part of the initial selected-scan sample, not as the output of a coverage-ranking procedure. The omitted structural target, root 864691136043380566 (nucleus 292648), had one source-data row, for session 8/scan 4, but has `dandi_asset=False` and no DANDI asset identifier or path. No additional exclusion reason is inferred.

To assess sensitivity to the initial scan selection, a scan-complete analysis applied one deterministic rule to the complete extracted scan universe: retain every automatic-conservative target–session–scan cohort with at least five unique connected imaged presynaptic roots. Thirteen scans across the same seven targets qualified. Pairwise statistics were computed within scan, scan values were averaged within target, and the seven targets remained the units of inference. Mean shared-path Spearman correlation was -0.0245 (95% target-bootstrap interval -0.214 to 0.147); the partial shared-path correlation after geometry and depth control was -0.0485 (-0.224 to 0.118), positive in three of seven targets. Negative tree-distance correlation and the same-major-branch contrast were also centered near zero (Table S8). Thus the deterministic scan-complete cohort preserves the null result from the initial selected-scan analysis.

Each target had 464 trials and 280 unique condition hashes, of which 136 were repeated. Trial families comprised 384 Clip, 40 Monet2 and 40 Trippy presentations. For each trial and region of interest, the response was the mean raw `RoiResponseSeries` fluorescence over samples whose timestamps fell from trial start through trial stop, including both endpoints. Start and stop offsets were zero. No trials were dropped; every response used here was finite. Table S29 gives target-level asset identifiers and sample counts.

Functional similarity between two partners was the Pearson correlation of their vectors of mean raw fluorescence over the 136 condition identities with at least two trials. Repeat reliability was computed separately for each region of interest: successive occurrences of every repeated condition were assigned alternately to two groups, responses were averaged within each group, and the two vectors of condition means were compared by Spearman correlation. No reliability correction was applied, and the primary cohort retained all partners regardless of this diagnostic; thresholds of 0.1 and 0.2 were used only in sensitivity analyses.

The controlled shared-ancestry statistic was computed separately within each target. Functional similarity and shared-path fraction were rank transformed. The two controls were $\log(1 + d_{\text{Euclidean}})$ and $\log(1 + |d_{\text{soma},i} - d_{\text{soma},j}|)$, where the second term is the absolute difference in soma path length; each control was also rank transformed. We regressed each ranked variable on an intercept and the two ranked controls by ordinary least squares, then took the Pearson correlation between the two residual vectors. The mean target-level partial-rank effect was -0.069 with a target-bootstrap 95% interval of -0.255 to 0.108 ; three of seven targets had positive effects. Pairwise partner comparisons were not treated as independent replicates.

Several features limit the interpretation of this cohort. MICrONS calcium imaging measured excitatory neurons, the observed partners cover only a small fraction of each target’s complete inputs, and response reliability is modest. The volume comes from one animal, and the recordings were not made during learning.

10.2 Task-derived branch credit

A branch model predicted the standardized postsynaptic response from the standardized responses of its connected presynaptic partners. Let x_i be the response of partner/site i , w_i its input weight, and $b(i)$ the dominant major branch receiving its contacts. Branch b had summed input d_b , activity h_b and readout weight v_b ; c was an output bias. The fitted prediction $\hat{y}$ was

$$d_b = \sum_{i:b(i)=b} x_i w_i, \quad h_b = \tanh(d_b), \quad \hat{y} = \sum_b v_b h_b + c. \quad (118)$$

For N training trials with target responses y_t , the objective was

$$\frac{1}{2N} \sum_{t=1}^N (\hat{y}_t - y_t)^2 + \frac{\lambda}{2} (\|\mathbf{w}\|_2^2 + \|\mathbf{v}\|_2^2), \quad \lambda = 10^{-3}, \quad (119)$$

with no penalty on c . The exact voltage-space error at site i was $(\hat{y} - y) v_{b(i)} [1 - \tanh^2(d_{b(i)})]$.

Whole condition-hash stimulus identities, not trials of the same identity, were split: 20% of the 280 identities and all of their trials were held out. Input columns and the target were standardized separately using training-set means and sample standard deviations (ddof = 1); invalid or zero scales were replaced by one and remaining nonfinite standardized values by zero. A stable SeedSequence combined base seed 20260721, target root, stream and replicate. There were ten fixed split and initialization replicates per target.

With n_{site} input sites and n_{branch} occupied branches, input weights were initialized as $\mathcal{N}(0, 0.12^2/n_{\text{site}})$, branch readout weights as $\mathcal{N}(0, 0.25^2/n_{\text{branch}})$, and $c = 0$. Each method used 1,200 full-batch Adam steps with learning rate 0.015, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. The exact model was trained first. Its exact site-error fields on training trials defined the dictionaries and its held-out exact field defined reported capture. All projected-learning methods then restarted from the identical initial state. Only the input-weight error was projected; $\mathbf{v}$ and c always received their exact gradients.

The primary budget requested four channels, reduced only when a target had fewer sites or nonempty candidate routes. Ancestry candidates were columns of ancestry multiplied by local inhibitory fraction g_i/g_{tot} ; routes were ranked by this fraction times their descendant-site fraction. Random paths sampled nonempty candidates uniformly without replacement. Each ancestry-shuffle column was permuted independently over sites. Depth controls were path-length quantile bins, scalar feedback was an all-ones column, and the dense oracle used the leading right singular vectors of the exact training error. Projectors used relative singular-value tolerance 10^{-10} . No held-out response was used to fit a dictionary.

Held-out normalized MSE divided a method’s held-out MSE by the MSE of predicting the training mean response. Replicates were averaged within target before inference; the seven target cells were the units. Intervals used 20,000 target bootstrap draws. Paired method comparisons used two-sided Wilcoxon tests, and the capture-utility analysis used method-label permutations stratified within target.

Table S30 reports the four-channel primary analysis. The dense four-dimensional oracle nearly matched exact learning, establishing that feedback dimension alone was not limiting in this surrogate. Ancestry routes did not reliably outperform ancestry shuffling in capture or learning. The within-target association between capture and normalized error was $\rho = -0.150$ with a target-stratified method-label permutation $P = 0.419$. The target-bootstrap 95% interval for ρ was -0.449 to 0.144 . The coefficients are least-squares oracles and therefore upper bounds on the capacity of each feedback family, not biologically implemented encoders.

Sensitivity analyses at one, two and eight channels, with direct manual matches, with repeat-reliability thresholds, and with a linear branch model did not establish a reliable morphology-specific learning benefit. Some subsets improved morphology-specific capture without improving learning. The separation between capture and learning motivates the distinction between anatomical route capacity and task alignment.

10.3 Complete reconstructed-tree learning across all eligible scans

To retain the complete spatial state, we replaced the major-branch reduction with a segment-resolved model. Each complete compressed reconstruction was instantiated as a reciprocal passive conductance system. Each imaged partner supplied a trainable non-negative excitatory conductance at its dominant mapped segment. For trial-specific soma prediction error, the exact input gradient factored into the local term $x_i(E_E - V_i)$ and the exact soma-to-segment adjoint from the inverse conductance operator. Readout and bias parameters received exact gradients under every method.

Restricted rules received scalar output error, local eligibility and one fixed site profile lying in a four-route span. That profile was calibrated before training from a conductance-weighted topology dictionary, an equal-rank random selection of nonempty anatomical routes, or a column-value-preserving site shuffle. No target response entered this calibration. Each trial scaled the same profile by the scalar error; the rule did not infer four independent trial-dependent coefficients. The comparison therefore tests this frozen-profile rule within each dictionary, rather than all estimators its four-route span could support. Ten seed-fixed splits per scan held out complete stimulus identities. Runs were averaged within scan and then within target. The resulting design contained four methods, 13 scans and ten splits, for 520 fits. The exclusion criteria were missing routes, rank mismatch, nonfinite values, incomplete scan coverage or failed finite-difference validation. None of the runs met an exclusion criterion. One exact conductance coordinate per run agreed with finite differences to maximum relative error 3.05×10^{-7} .

Mean normalized held-out MSE was 0.788 for exact compartment error, 0.832 for topology, 0.833 for random routes and 0.839 for site shuffling (Table S9). The topology MSE improvement over shuffling was 0.0070 (95% target-bootstrap interval -0.0207 to 0.0321) and over random routes 0.0013 (-0.0090 to 0.0113). At the common exact checkpoint, topology achieved update match 0.232 in eligibility-weighted coordinates, compared with 0.358 for random routes and 0.439 for site shuffling. Here the reported metric is normalized update match, $1 - \sum_i \|\mathbf{u}_i - \hat{\mathbf{u}}_i\|^2 / \sum_i \|\mathbf{u}_i\|^2$, for exact and prescribed trial updates $\mathbf{u}_i$ and $\hat{\mathbf{u}}_i$. It is not a projection-capacity bound because the frozen-profile coefficients are prescribed.

A retrospective geometry check separated this coefficient restriction from the capacity of the fixed route spans. Full checkpoint states were not archived, so we reexecuted the 130 exact-learning fits using their resolved settings, stimulus splits and recorded random-route identities. At each resulting checkpoint we compared the frozen profile, a trialwise exact-field projection, and an eligibility-weighted update oracle. For fixed dictionary Φ and trial eligibility vector ℓ_i , define $E_i = \text{diag}(\ell_i)$. The update oracle fitted

$$\mathbf{c}_i^* = \arg \min_{\mathbf{c}} \|\mathbf{u}_i - E_i \Phi \mathbf{c}\|_2^2, \quad \hat{\mathbf{u}}_i = E_i \Phi \mathbf{c}_i^*.$$

This is the best attainable per-trial update inside that dictionary's eligibility-weighted span. Its normalized update match equals projected energy capture. Split-level values were averaged within scan and then within target, with 20,000 bootstrap draws over the seven targets.

The within-span update oracle increased mean match from 0.2322 to 0.2436 for ancestry, from 0.3584 to 0.3847 for random anatomical routes, and from 0.4392 to 0.4677 for shuffled routes. Ancestry-minus-random oracle capture was -0.1411 (95% target-bootstrap interval -0.2125 to -0.0388); ancestry-minus-shuffle was -0.2241 (-0.3927 to -0.0605). The unprojected baseline transfer profile reached update match 0.9756 (0.9641–0.9858), so baseline transfer alone was substantially closer to the exact field than its restriction to these four ancestry routes. Reexecution changed normalized exact MSE by more than 10^{-5} in three of 130 fits (maximum 5.74×10^{-4}); maximum frozen-profile match discrepancy was 2.57×10^{-4} . No reexecuted fit was excluded. The analysis therefore uses reexecuted checkpoints and cannot certify exact reproduction of every archived state. The tables and replay discrepancies are supplied in `source_data/fulltree_within_span_oracle/`. This analysis measures common-state geometry; it does not train an oracle feedback rule or establish its eventual accuracy. Even with the complete segment tree and trial-dependent oracle coefficients, ancestry alignment was not recovered at the evaluated states. The result remains conditional on the objective, response window, observed inputs and biophysical calibration.

10.4 Nested linear baselines and observed-input sensitivity

To establish the predictive difficulty of the measured-response task, we fitted an intercept-only training-mean predictor, ordinary least squares and ridge regression on the same 130 outer stimulus-identity splits as the complete-tree analysis. All 13 scans and seven targets were retained. Three inner folds partitioned training stimulus identities; no outer test identity entered preprocessing, reliability filtering or penalty selection. The primary linear models used the training-quantile input transform from the complete-tree analysis (10th and 90th percentiles, clipped at three), followed by column centering and standardization using training data. These quantities were refitted inside each inner fold. Raw standardized inputs supplied a secondary ridge reference.

Ridge regression minimized mean squared training error plus $\lambda \|\mathbf{w}\|^2$, with an unpenalized intercept. The penalty grid was $\{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$; the minimum mean inner-fold validation error selected the penalty, with ties resolved toward the earlier, smaller penalty. An eigendecomposition solved each fit. The selected primary penalties were 0.1 in 90 splits, 1 in 36 and zero in four. Held-out normalized MSE used the same training-mean denominator as the nonlinear models. Archived nonlinear outcomes were read unchanged and paired to the new linear fits; they were not refitted under the sensitivity masks.

Observed-input sensitivity used three explicit restrictions. First, the manual-only mask retained partners with archived manual-match provenance. Second, repeat reliability was the Spearman correlation between condition means formed from alternating repeats; it was recomputed using only inner training identities during tuning and only outer training identities for the final fit. Partners passing thresholds 0, 0.1 or 0.2 were retained. Third, five independent partner orderings per split defined nested masks retaining 25%, 50% or 75% of observed inputs, with at least one input. Empty manual or reliability masks used the intercept-only model and were retained. A separate measurement-noise series added Gaussian predictor noise with SD 0.25, 0.5 or 1 in units of the training-standardized input, in both training and test examples. The same underlying noise draws were paired across doses. Noise was applied after the quantile transform and training standardization; the intercept was refitted after perturbation.

We fixed this secondary protocol before inspecting its baseline outcomes; it was not publicly preregistered. Its 3,380 new evaluations are archived in `source_data/review_response_baselines/`. Mask draws were averaged within split, then ten splits within scan, and scans within target. All intervals used 20,000 bootstrap resamples of the seven target means. Input hashes and outer split counts were verified against the retained complete-tree records. A separate audit hashed all 130 reconstructed split masks and verified stimulus-identity disjointness. In 338 condition-level checks across one split per scan, replacing every held-out predictor and response by extreme values left selected penalties, masks and training reliability exactly unchanged. Historical trial-mask arrays were not separately archived; split reconstruction uses the unchanged helper and seed rule, checked against all retained train/test counts. Sensitivity contrasts are descriptive; the seven targets from one mouse remain the inferential units.

Mean normalized MSE was 1 for the training mean, 0.807 for least squares and 0.803 for nested ridge (Supplementary Fig. S34). Exact nonlinear learning reached 0.788, a paired nonlinear-minus-ridge difference of -0.0152 (95% target-bootstrap interval -0.0271 to -0.0042 ; six of seven targets lower). The nonlinear model therefore provided a small predictive improvement over this linear baseline on the measured task. The corresponding subtree-minus-ridge difference was 0.0293 (-0.0130 to 0.0736), which does not establish a benefit for the frozen subtree rule. Manual-only inputs raised ridge MSE to 0.912, with 3.18 retained partners on average compared with 8.98 for all inputs. Retaining only 25% raised MSE to 0.917; adding unit-SD predictor noise raised it to 0.869. The strictest reliability mask retained 5.63 partners and reached MSE 0.837. These manipulations measure sensitivity to degradation of the observed inputs. They neither recover unobserved partners nor estimate the power of a complete-population experiment, and do not establish that missing inputs explain the absence of preferential ancestry alignment.

S11 Alignment-controlled optimization

The measured-response results leave open whether poor task alignment limits the usefulness of ancestry routes. To isolate alignment, we imposed a known fraction of field energy in the selected ancestry span and tested whether capture predicted optimization. For each reconstructed cell, we rebuilt the conductance-weighted ancestry kernel

and selected eight nested-subtree routes by inhibitory fraction times descendant excitatory synaptic-area fraction before drawing any gradient fields. Coordinates were weighted by the square root of summed mapped excitatory synaptic area. Let Q be an orthonormal basis for those selected routes. Paired unit vectors $\mathbf{u}_{\parallel} \in \text{span}(Q)$ and $\mathbf{u}_{\perp} \perp \text{span}(Q)$ generated

$$\mathbf{g}(a) = \sqrt{a}\mathbf{u}_{\parallel} + \sqrt{1-a}\mathbf{u}_{\perp}, \quad a \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}. \quad (120)$$

Every field therefore had unit energy and exactly fraction a in the morphology-selected span. The parallel and orthogonal directions were reused across alignment levels.

Controls were eight random paths from the same tree, eight path-distance quantile bins, and an independent row permutation of each selected morphology route. The last control preserved each column’s values and nonzero count while breaking ancestry. Random and shuffled dictionaries were redrawn within each nested stream and held fixed across alignment levels in that stream. Each of the eight cells contributed 40 streams with 256 fields per stream.

Optimization used a positive diagonal quadratic whose curvature spectrum ran from 0.5 to 1.5. The spectrum was independently permuted within each stream and then fixed across methods and alignment levels. One-step comparisons matched update norm at 0.1 times exact-gradient norm. Iterative projected descent used learning rate 0.25 for 20 steps. Progress was normalized by the corresponding full-gradient progress. Supplementary Fig. S5 and Table S35 report the principal values. Across the three control dictionaries and alignment levels, the median within-cell Spearman association between initial capture and 20-step progress was $\rho = 0.990$ (cell range 0.969–0.996). This analysis establishes conditional sufficiency of aligned routes by linking capture to loss reduction under a controlled objective. Because alignment is imposed and coefficients are fitted from the exact gradient, it is neither an independent biological result nor a model of how a teaching circuit learns its coefficients.

S12 External signed-credit analysis and task-interference identity

At the neuronal level, the framework gives a simpler prediction: when a causal task assigns opposite learning signs to two neurons, their measured plasticity-related contrasts should separate along the corresponding signed coordinate. The following reanalysis tests that prediction but does not address dendritic morphology. The Francioni et al. workbook was downloaded from the source-data link of the published article; file identity was verified by SHA-256 be1b87481d0f1d1bffc85d8a4cabfd37b7055d640306afab5b3622de0ced8264. The animal-level contrasts in sheet “Ext 13E” were used directly. The endpoint was the difference in soma–dendrite residual, in z-score units, between error-reduction and error-increase epochs. P+ and P- denote populations assigned opposite causal signs in the source brain–computer-interface mapping. For the source-defined P+ and P- groups, we required the table’s signed-separation column to equal P+ minus P- to numerical precision. The orthonormal common and signed projections sum exactly to total two-coordinate energy. Sheet “5E” supplied the four neuron-level distributions in the display but was not used to increase the sample size of the animal-level test.

For $n = 6$ animals, the one-sided exact random-sign test enumerated all 2^6 sign assignments. Bootstrap intervals sampled animals with replacement. The directions were fixed by the experimental mapping: P+ positive, P- negative, and P+ minus P- positive. This is a retrospective test because the prediction was formulated after publication of the experiment, although it was evaluated from the numerical source table rather than the displayed figure.

12.1 Route overlap and task interference

We finally examined how overlap between addressed regions and leakage outside an intended route determine interference with an already learned task. Branch-specific plasticity during motor learning suggests that different tasks can recruit different dendritic regions [13]. Let $\mathbf{w}_A$ be an optimum of a quadratic Task-A objective with Hessian H_A , and let an approximate Task-B teaching field produce update $-\eta M_B \mathbf{g}_B$, where η is the step size, $\mathbf{g}_B$ is the Task-B gradient and M_B is its routing operator. Expansion about the Task-A optimum removes the first-order term, leaving the exact loss increase

$$\mathcal{L}_A(\mathbf{w}_A - \eta M_B \mathbf{g}_B) - \mathcal{L}_A(\mathbf{w}_A) = \frac{\eta^2}{2} \mathbf{g}_B^T M_B^T H_A M_B \mathbf{g}_B. \quad (121)$$

2707 In the equal-route example, opposing updates on the shared fraction ω have magnitude two, whereas leakage onto
2708 the Task-A-only fraction has magnitude λ . Dividing by route size gives $\eta^2[4\omega + \lambda^2(1 - \omega)]/2$. Numerical vector
2709 updates matched this identity to 4.2×10^{-17} absolute error (Supplementary Fig. S6). Separated routes therefore
2710 protect a learned task only when both anatomical overlap and teaching-field leakage are small. The published
2711 somatostatin (SST) perturbation is consistent with the resulting prediction that loss of branch-selective inhibition
2712 increases effective leakage and cross-task interference [13], but it does not uniquely identify the credit-routing
2713 mechanism. This static identity does not model dendritic calcium, spikes, eligibility timing, or the SST manipulation
2714 in the source study.

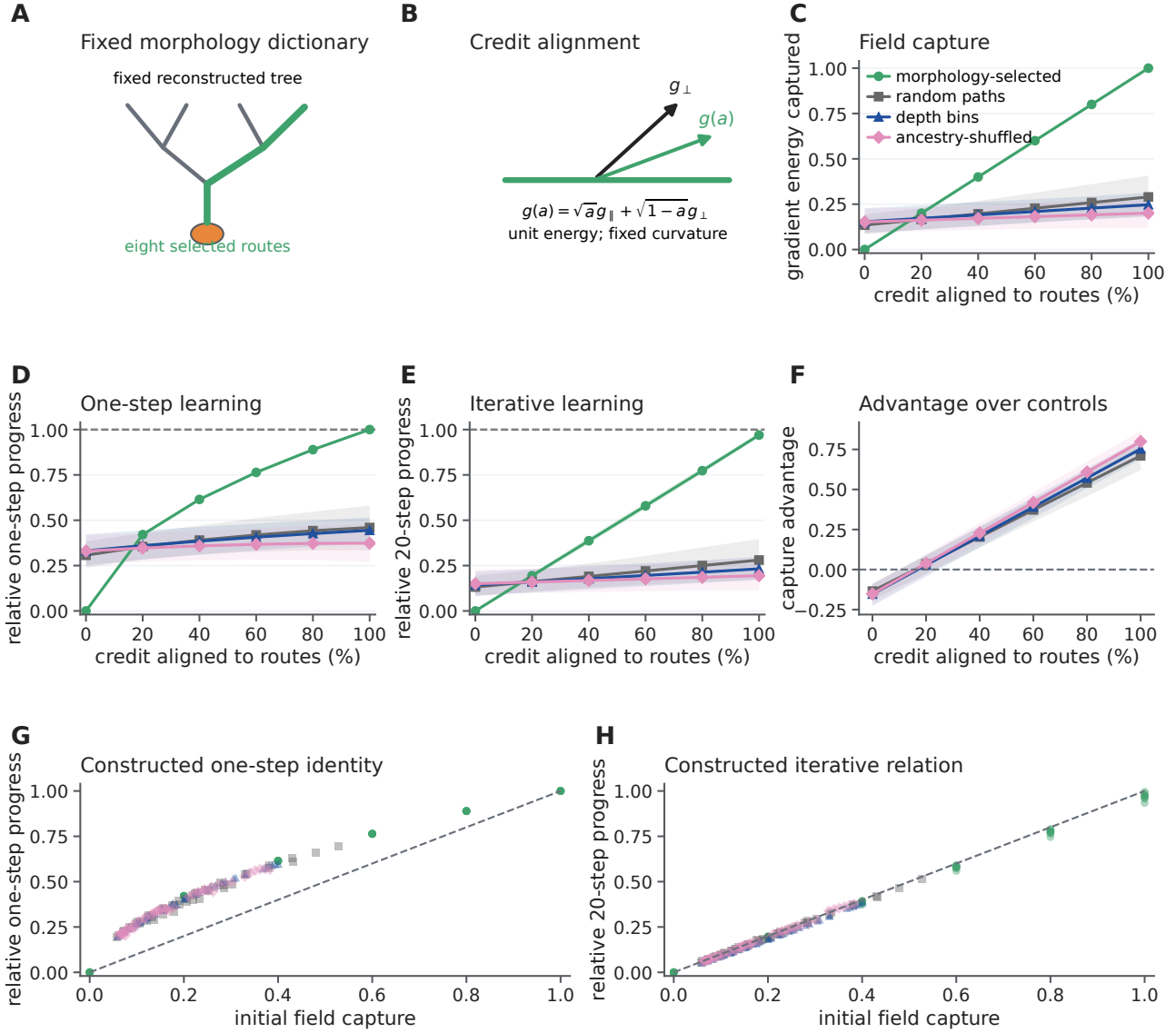


Figure S5: **Constructed route alignment controls projected learning.** All panels use the initial eight reconstructed MICrONS arbors from one mouse. The anatomy supplies a fixed route dictionary, whereas the credit fields, projection coefficients and quadratic objective are imposed. **A**, A fixed eight-route dictionary on a reconstructed tree. **B**, Unit-energy fields $g(a) = \sqrt{a}g_{\parallel} + \sqrt{1-a}g_{\perp}$ interpolate between its span and orthogonal complement; a is aligned energy. **C**, Capture across imposed alignment levels. **D**, Loss reduction from one norm-matched projected-gradient step, relative to the full-gradient step. **E**, Loss reduction after 20 projected steps, relative to the corresponding full-gradient sequence. **F**, Ancestry-route capture relative to random paths, depth bins and ancestry shuffling. **G,H**, Initial capture versus one- and 20-step progress; across the control dictionaries the median within-cell rank correlation between initial capture and 20-step progress is $\rho_s = 0.99$. Means in **C–F** are across eight cells with hierarchical 95% bootstrap intervals over cells and nested Monte Carlo streams. Ancestry-route capture equals imposed alignment by construction; coefficients are oracle projections. This is a conditional positive control, not independent evidence about the recorded neurons.

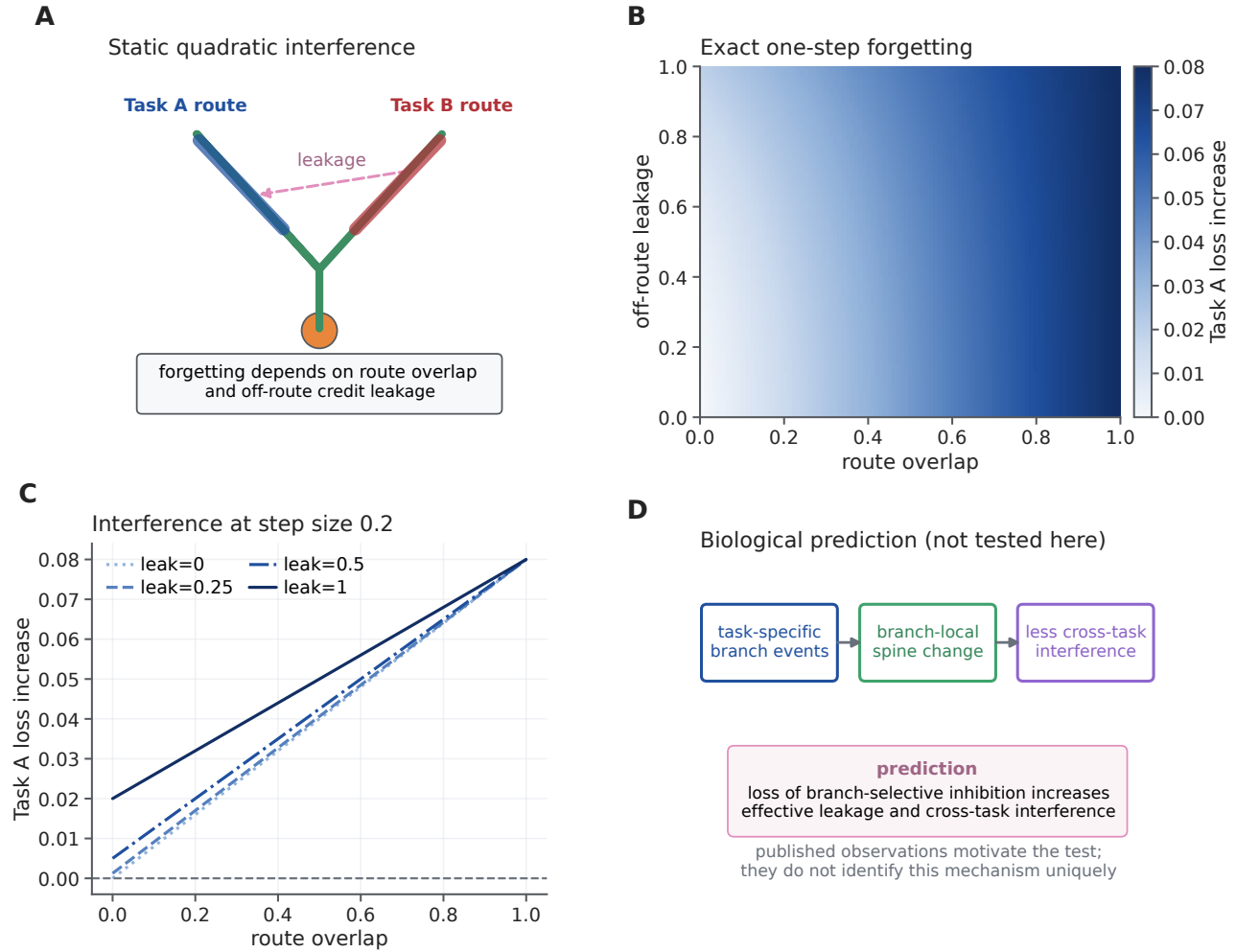


Figure S6: **Route overlap and off-route leakage determine one-step task interference.** The drawn branches are abstract route supports in an exact quadratic calculation; no conductance morphology is simulated and no network is trained. **A**, Two tasks address partly separated dendritic routes; an approximate teaching field can also leak onto the other route. **B**, Exact Task-A loss increase after one Task-B update across route overlap and off-route leakage. **C**, Slices at step size 0.2 for leakage 0, 0.25, 0.5 and 1; colors identify leakage levels. **D**, Published branch-specific events and spine changes motivate the prediction that loss of selective inhibition increases leakage and interference. Panels **A–C** are an exact static quadratic consequence and numerical verification, not a temporal neural model. Panel **D** is a cross-study interpretation rather than a reanalysis of the original motor-learning data.

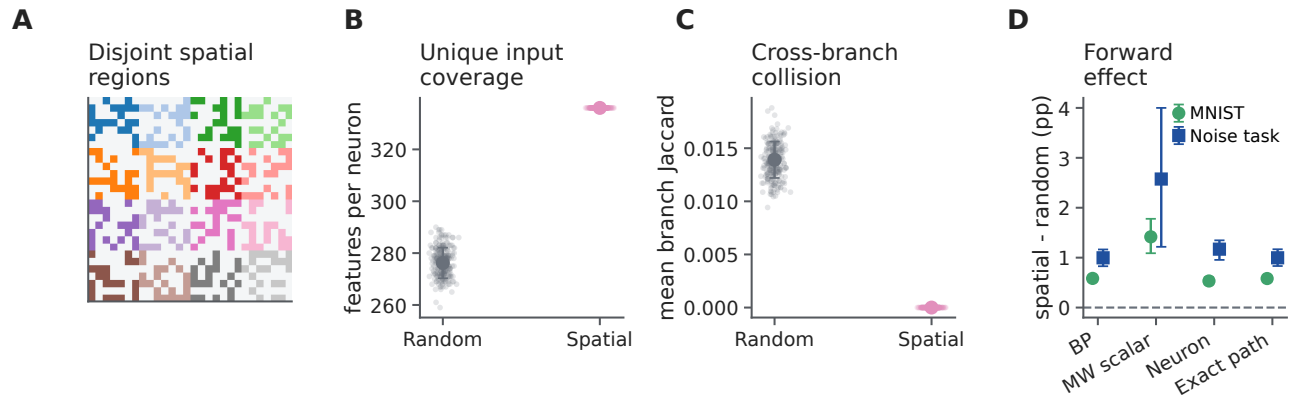


Figure S7: **The fixed spatial map changes forward input coverage.** All panels use depth-four regular directed $[2, 2, 2, 2]$ trees with 16 terminal branches. The spatial and random conditions differ only in which input coordinates reach those terminal branches; if an effect persists under backpropagation, it is a forward sparsity prior rather than a local-credit effect. **A**, Example contacts from one neuron. Colors distinguish the 16 distal branches, which occupy disjoint image regions. **B**, Unique input coordinates reached by each neuron under independently sampled random branches and the spatial map. **C**, Mean pairwise Jaccard overlap between branch input sets (intersection divided by union). **D**, Spatial-minus-random accuracy under backpropagation (BP), matched-width scalar-fallback feedback (MW scalar), neuron-specific sharing (Neuron) and exact path transport. MNIST averages shunting and raw-additive models within seed; the randomly projected noise task uses the conductance-valid raw-additive model only. Panels **B,C** show means $\pm$ s.d. across neuron-level map summaries; panel **D** shows means and 95% paired-seed bootstrap intervals. Each learning condition contains ten paired seeds. The map has the same number of contacts in both conditions but greater unique coverage and no cross-branch collisions in the spatial condition; its performance effect is therefore not specific to local credit assignment. Table S21 gives the map coverage and overlap values.

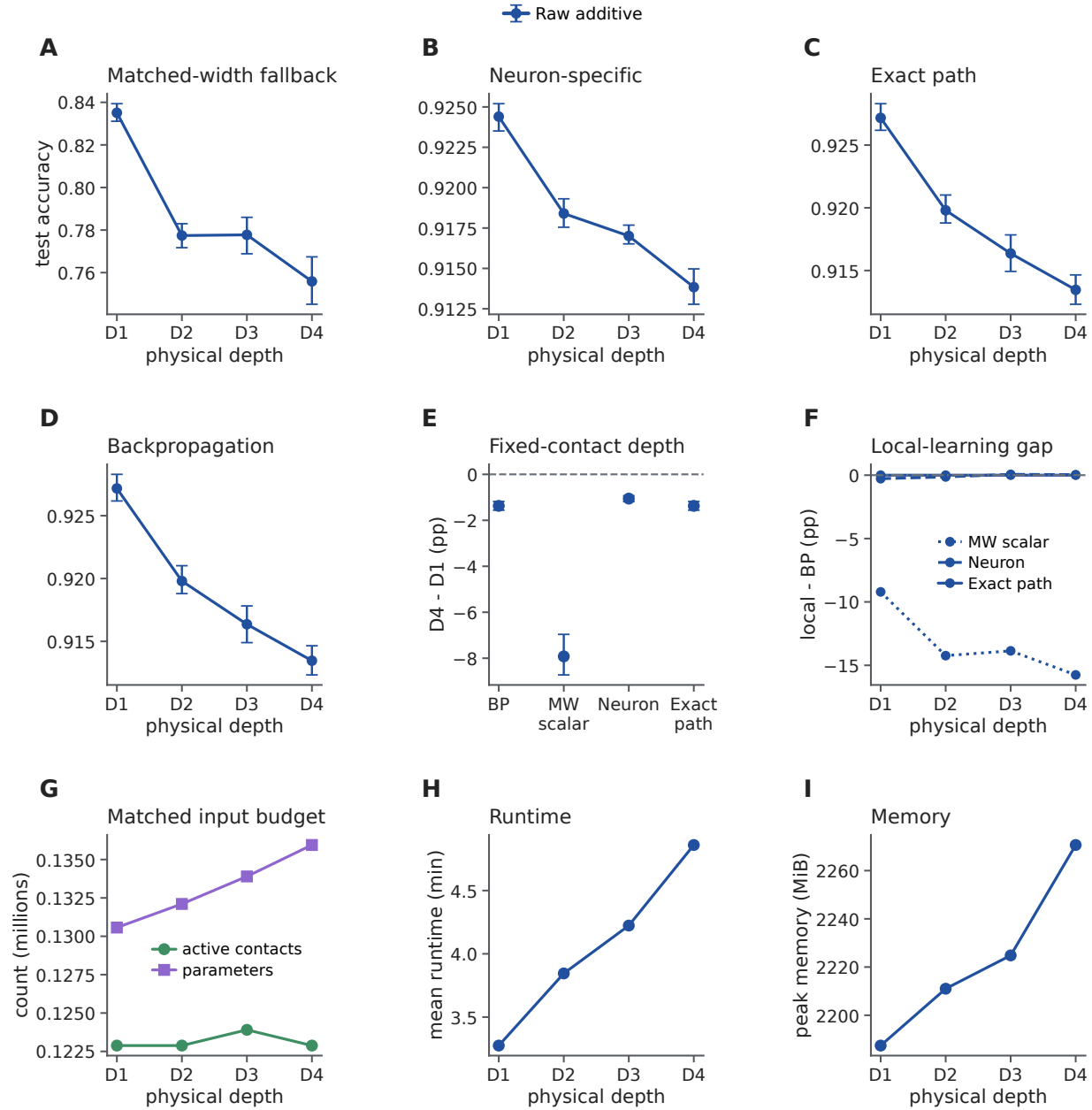


Figure S8: Fixed terminal-branch and input-contact budgets do not make greater dendritic depth intrinsically beneficial. The historical noise-resilience cohort uses conductance-valid raw-additive trees with 16 terminal branches: D1 [16], D2 [4, 4], D3 [2, 2, 4] and D4 [2, 2, 2, 2]. These cohort-specific labels are not the morphologies in Supplementary Fig. S31. The experiment asks whether accuracy rises when fixed terminal and contact budgets are distributed across more serial stages. **A–D**, Held-out accuracy under matched-width scalar-fallback, neuron-specific, exact-path and backpropagated learning, respectively. **E**, Prespecified D4-minus-D1 accuracy contrasts, in percentage points (pp). **F**, Local-minus-backpropagated (BP) accuracy in pp; line styles identify the three local fields. **G**, Active contacts and trainable parameters. **H, I**, Runtime in minutes and peak memory in mebibytes (MiB). Each condition has ten paired seeds; symbols and intervals are means and 95% paired-seed bootstrap intervals. All four retained D4-minus-D1 contrasts were negative in 10/10 pairs (Wilcoxon $P = 0.001953$). The design fixes 16 terminals and 960–968 active contacts per soma, but not compartment or parameter count. Shunting series fail the nonnegative-conductance criterion on this task and are excluded from inference. Its exact executed noise generator remains unresolved (SI S5).

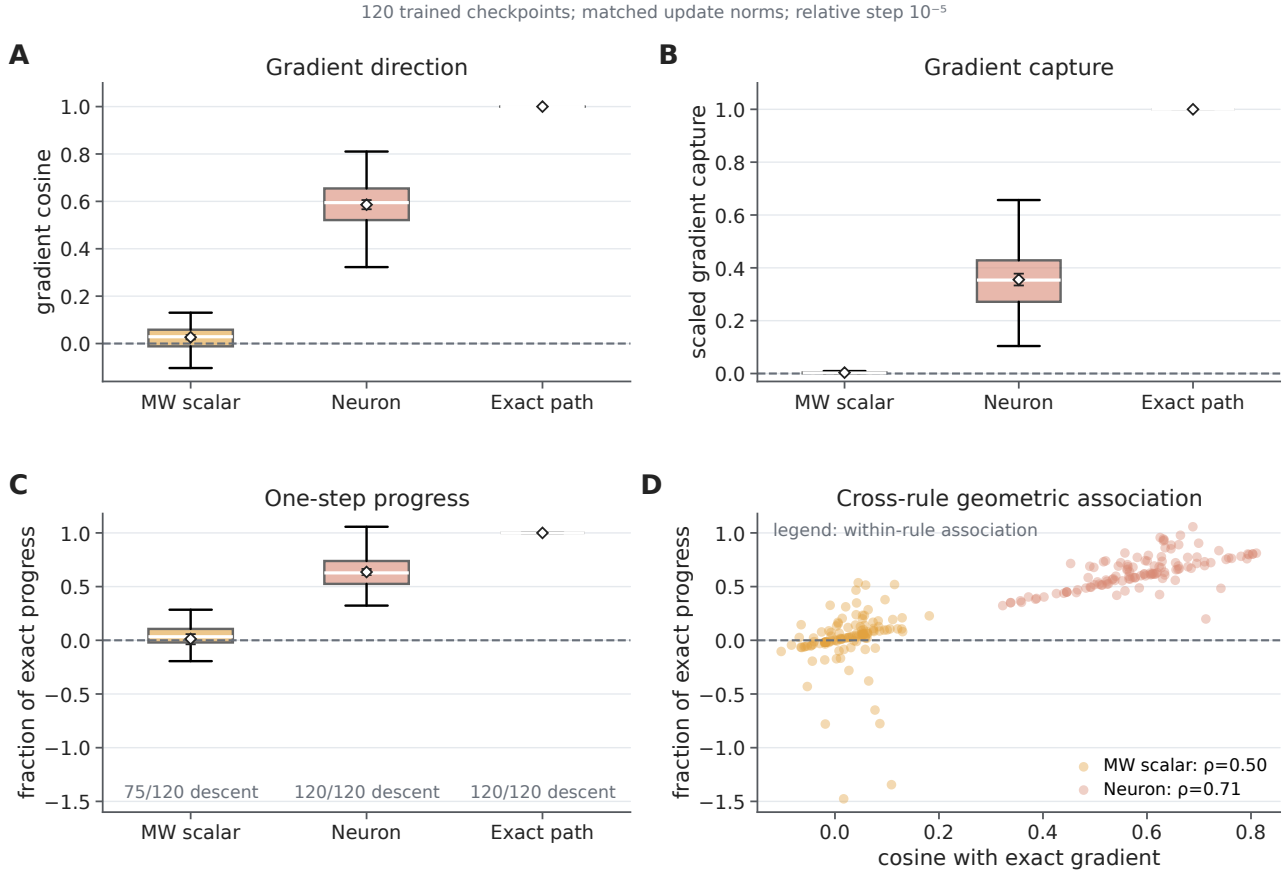


Figure S9: Gradient geometry describes a norm-matched step at trained checkpoints. No new training is introduced. The diagnostic uses 120 conductance-valid backpropagation checkpoints from the MNIST and historical noise-resilience depth cohorts with directed morphologies [2], [2, 2], [2, 2, 2] and [2, 2, 2, 2]. **A**, Candidate-versus-exact dendritic gradient cosine for matched-width scalar-fallback (MW scalar), neuron-specific (Neuron) and exact-path feedback. **B**, Eligibility-weighted gradient energy retained after optimally rescaling the single candidate update direction; this squared-cosine quantity does not measure full-span capacity. **C**, Loss decrease after matching each candidate update to the exact-gradient norm; labels count checkpoints with decreased loss. **D**, Cosine versus relative one-step progress for matched-width scalar-fallback and neuron-specific feedback. The legend reports within-rule rank correlations separately; pooling the separated rule clusters would not establish predictive performance within a rule. Boxes in **A–C** show the interquartile range, whiskers the 5th–95th percentiles, and diamonds means with 95% checkpoint-bootstrap intervals. Every panel uses one held-out batch per trained checkpoint and relative step 10^{-5} . The relationship is therefore local and explicitly norm matched; it does not validate initialization-based long-horizon model selection.

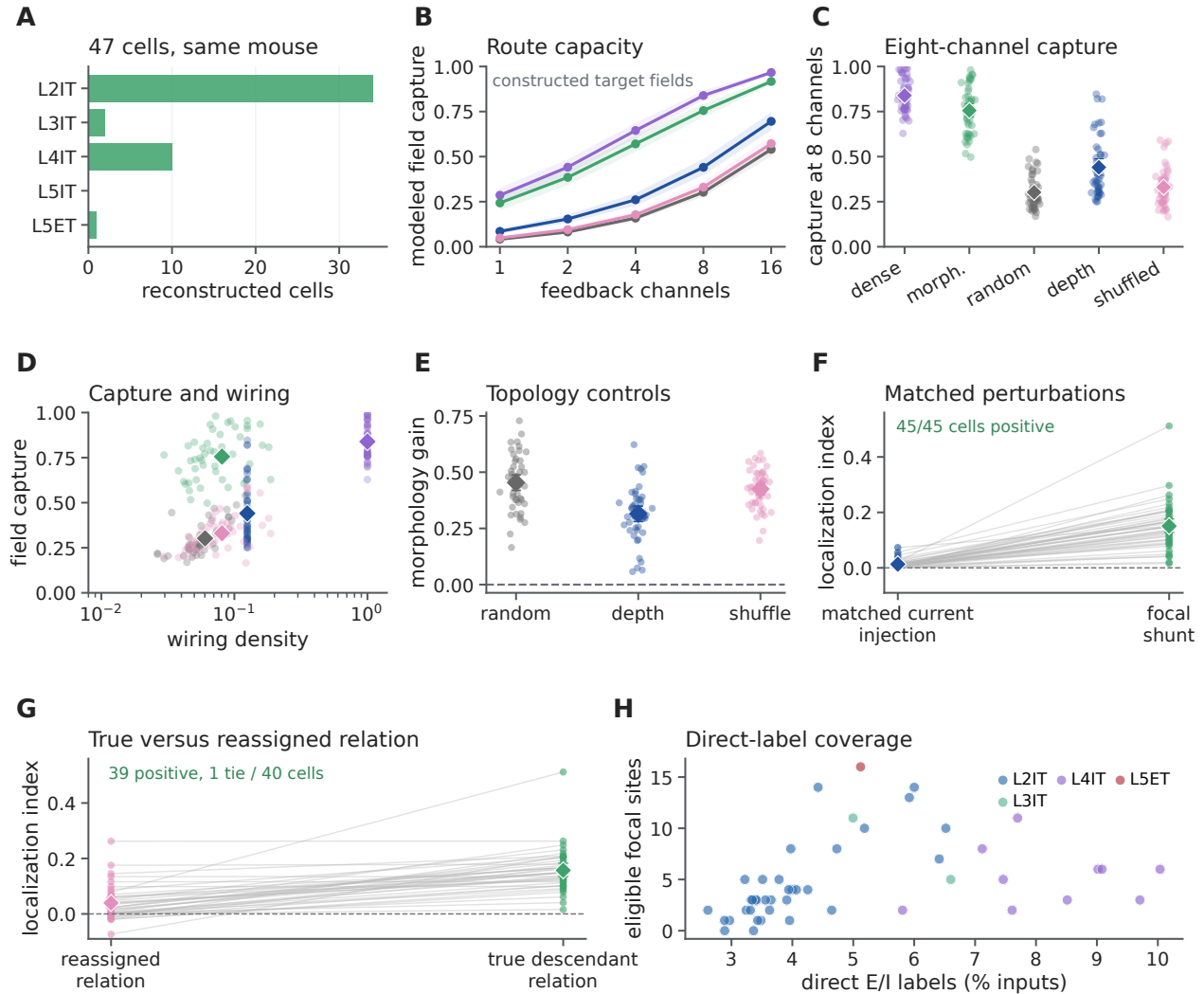


Figure S10: **Structural routing and focal shunting remain robust in a non-overlapping public-v661 cell sample.** **A**, Composition of the 47-cell sample by excitatory cell class (L, cortical layer; IT, intratelencephalic; ET, extratelencephalic). **B**, Cell-mean weighted modeled field-energy capture across feedback-channel budgets for a dense principal-component-analysis (PCA) oracle, morphology-defined paths, random paths, depth bins, and ancestry shuffles ($n = 47$ cells through eight channels; $n = 46$ at 16 channels). Twenty perturbation streams were averaged within cell; bands are 95% cell-bootstrap intervals. **C**, Cell-level capture at eight channels, using the dictionaries in **B**. **D**, Eight-channel wiring-capture trade-off; morphology paths used 8.1% of dense-feedback wiring. **E**, Within-cell morphology-minus-control capture for random paths, depth bins, and ancestry shuffles; all three contrasts were positive in 47/47 cells. **F**, Localization of exact-gradient changes after a unit-dose focal shunt and baseline-current-matched current injection ($n = 45$ cells, 235 sites). **G**, Localization for true descendant relations and relation templates reassigned to other focal sites ($n = 40$ cells, 230 sites). **H**, Direct presynaptic excitatory/inhibitory (E/I) label coverage and eligible focal sites per cell. Diamonds and intervals in **C**, **E**–**G** show means and 95% cell-bootstrap intervals. Cells, not sites or streams, are the inferential units. The sample excludes the initial eight nuclei but comes from the same mouse. It uses v661 reconstructions and direct E/I calls only (4.71% of incoming synapses). These are structural and passive-model tests, not measurements of learning in vivo.

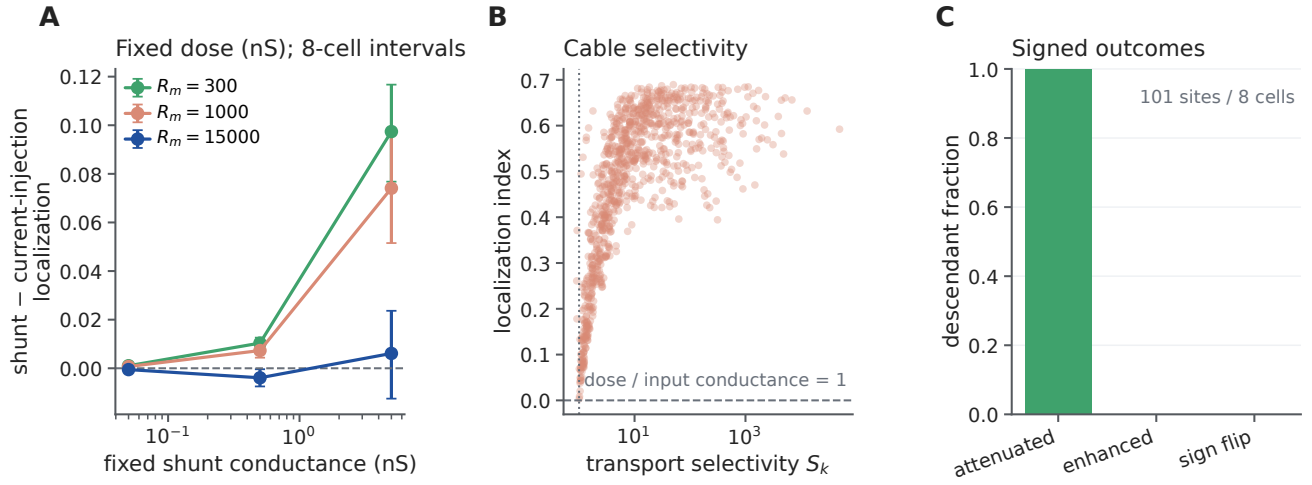


Figure S11: Fixed-absolute dose and background conductance bound passive focal-shunt selectivity. All panels use passive reciprocal-cable models of the initial eight reconstructed MICrONS cells; membrane resistance and background conductance are sensitivity parameters rather than fitted cell properties. **A**, Shunt-minus-current-injection localization across fixed absolute conductance doses at three membrane-resistance settings with no added background conductance. Each curve averages focal sites within each of eight cells before the cell mean; error bars are 95% cell-bootstrap intervals. **B**, Transport selectivity S_k , the descendant-to-control ratio of median absolute Green-function entries for focal site k , versus full synaptic-gradient localization at unit input-conductance-normalized dose across 101 sites, three membrane resistances and three background levels. Points are site-regime values and are descriptive because sites and regimes are repeated within cell. The dotted line marks $S_k = 1$. **C**, Fraction of descendant gradients attenuated, enhanced or sign reversed at $R_m = 1,000 \Omega \text{cm}^2$, background multiplier one and unit input-conductance dose. All tested passive focal-shunt conditions had the same qualitative signed outcome: attenuation without enhancement or sign reversal. R_m denotes specific membrane resistance; these settings are sensitivity parameters, not cell-specific fits.

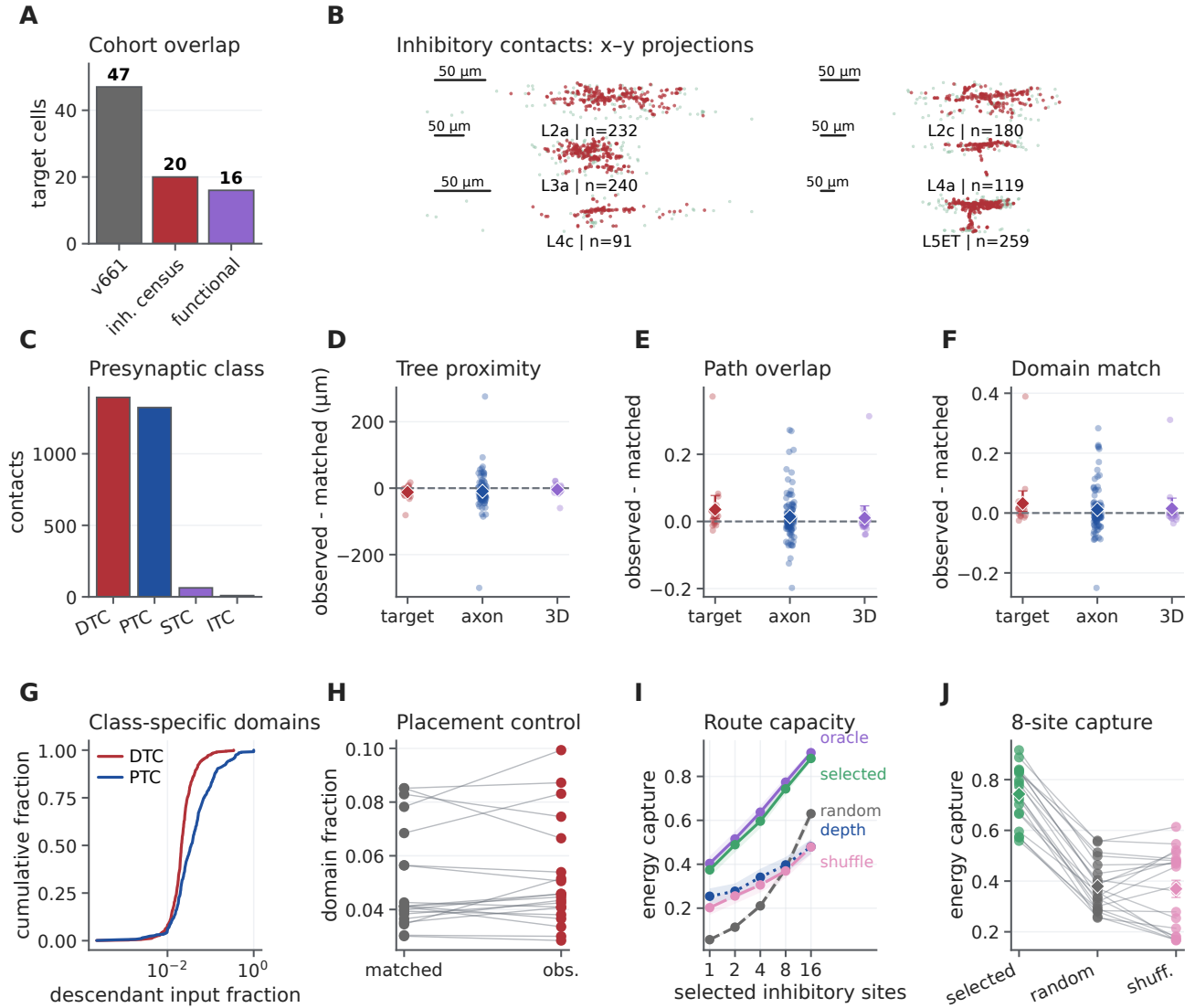


Figure S12: **Synapse-resolved inhibition reveals class-dependent dendritic scales.** **A**, Overlap between the 47-cell v661 sensitivity cohort, the v795 inhibitory census and available functional summaries. **B**, Six layer-diverse target reconstructions in x-y projection; red points are mapped known inhibitory contacts. Each tree is independently scaled and has a physical 20 or 50 μ m scale bar; n counts mapped inhibitory contacts, and L/ET denote cortical layer/extratelencephalic class. **C**, Contacts in the overlap by the published inhibitory targeting class. DTC, distal dendrite-targeting; PTC, perisomatic-targeting; STC, sparsely targeting; ITC, inhibitory-targeting. **D–F**, Observed-minus-matched tree distance, shared-path fraction and weighted descendant-domain overlap, respectively. Target denotes the primary 20-target analysis; axon first averages within each of 92 presynaptic axons; 3D jointly matches path distance and Euclidean proximity within target. Diamonds and intervals show unit means and bootstrap 95% intervals. Only tree distance persisted after axon aggregation, and no endpoint persisted after joint 3D matching. **G**, Cumulative descendant-input fractions for DTC and PTC contacts, retaining one contact per axonal clump. **H**, Overall inhibitory-contact domain size versus matched locations; no reliable global localization advantage was detected. **I**, Weighted modeled-field capture versus site budget for the dense oracle, anatomically selected inhibitory routes, random sites, depth bins and shuffled routes. **J**, Eight-site capture within each of 20 targets. Sites in **I,J** are selected by anatomical leverage; shading in **I** and bars in **J** show means $\pm$ standard errors across the 20 targets. Panel **H** connects paired target values without an interval; panels **D–F** use 10,000 bootstrap draws over the displayed target or axon units. This is a structural capacity analysis, not evidence that the routes carried teaching signals in vivo. Target cells are nested within one mouse.

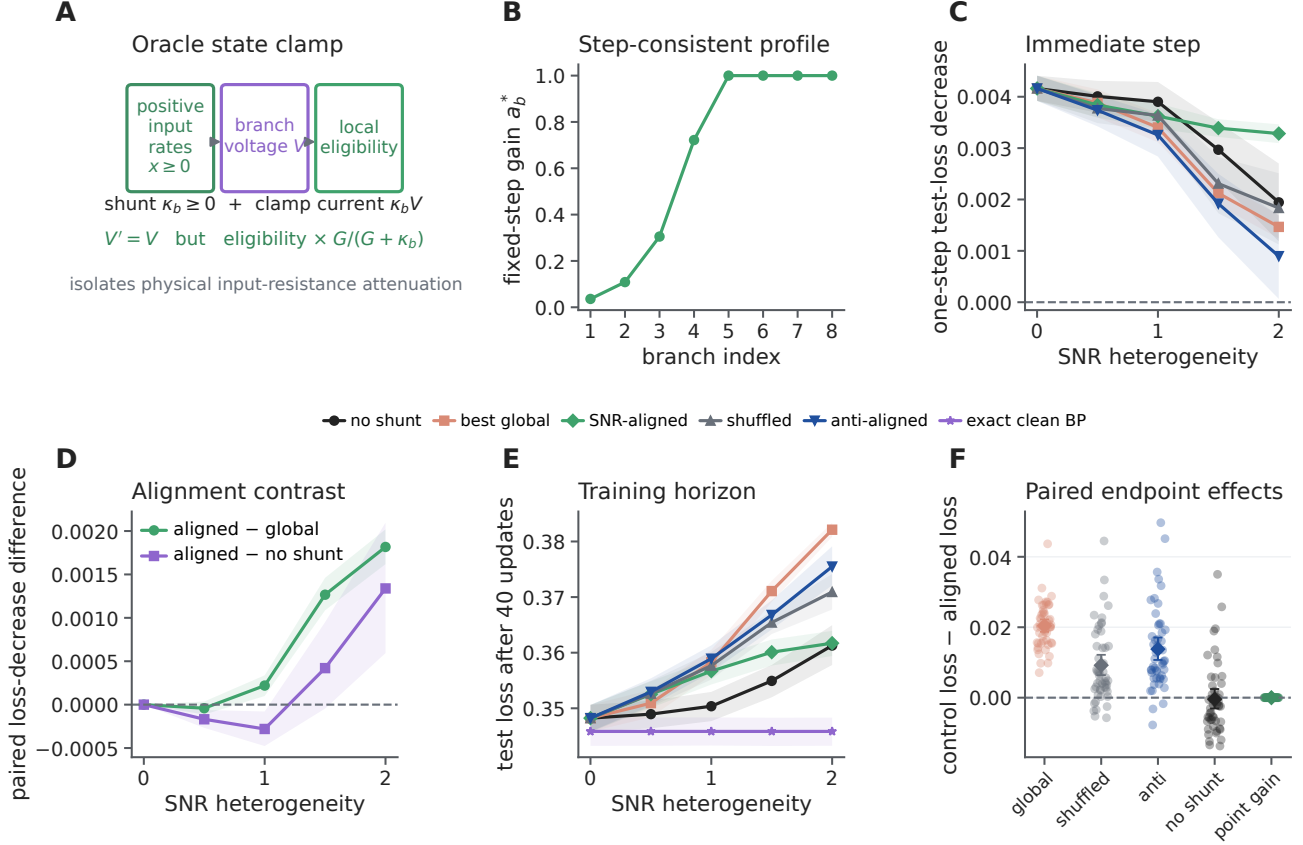


Figure S13: A state-matched positive conductance approximates the predicted fixed-step branch-reliability factor, with a bounded training benefit. This synthetic positive-rate task has eight parallel, nonserial branch blocks with heterogeneous signal and noise. It tests reliability-dependent gain and is not a reconstructed-arbor or physical-depth experiment. **A**, A nonnegative shunt κ_b is paired with compensating current $\kappa_b V_b$, preserving branch voltage while reducing physical input resistance and local eligibility. **B**, Prescribed reliability profile $\min\{1, S_b/[c(S_b + N_b)]\}$ at maximum heterogeneity with $c = 0.5$; S_b and N_b are branch signal and noise energies. SNR denotes signal-to-noise ratio. **C**, Mean one-step held-out loss decrease across 50 paired seeds for aligned, global, shuffled, anti-aligned, unshunted and exact clean-backpropagation (BP) rules. Lines and shading show seed means and 95% paired-seed bootstrap intervals. **D**, Aligned-minus-global and aligned-minus-no-shunt one-step contrasts. At heterogeneity two, the former is positive in 49/50 seeds; the latter has a positive mean interval, is positive in 33/50 seeds and has two-sided Wilcoxon $P = 0.0030$. **E**, Held-out loss after 40 updates. Exact clean backpropagation is the unattenuated optimization reference. **F**, Paired final-loss contrasts at maximum heterogeneity, displayed as control minus aligned so positive values favor aligned shunting. Aligned shunting beats the global, shuffled and anti-aligned shunts but has no reliable final difference from the unshunted noisy rule. The independently computed explicit point-gain control receiving the same sample-dependent factors agrees to numerical precision. All input rates and excitatory and shunt conductances are nonnegative. The oracle state clamp and fixed reliabilities isolate mechanism; they do not constitute autonomous shunt learning.

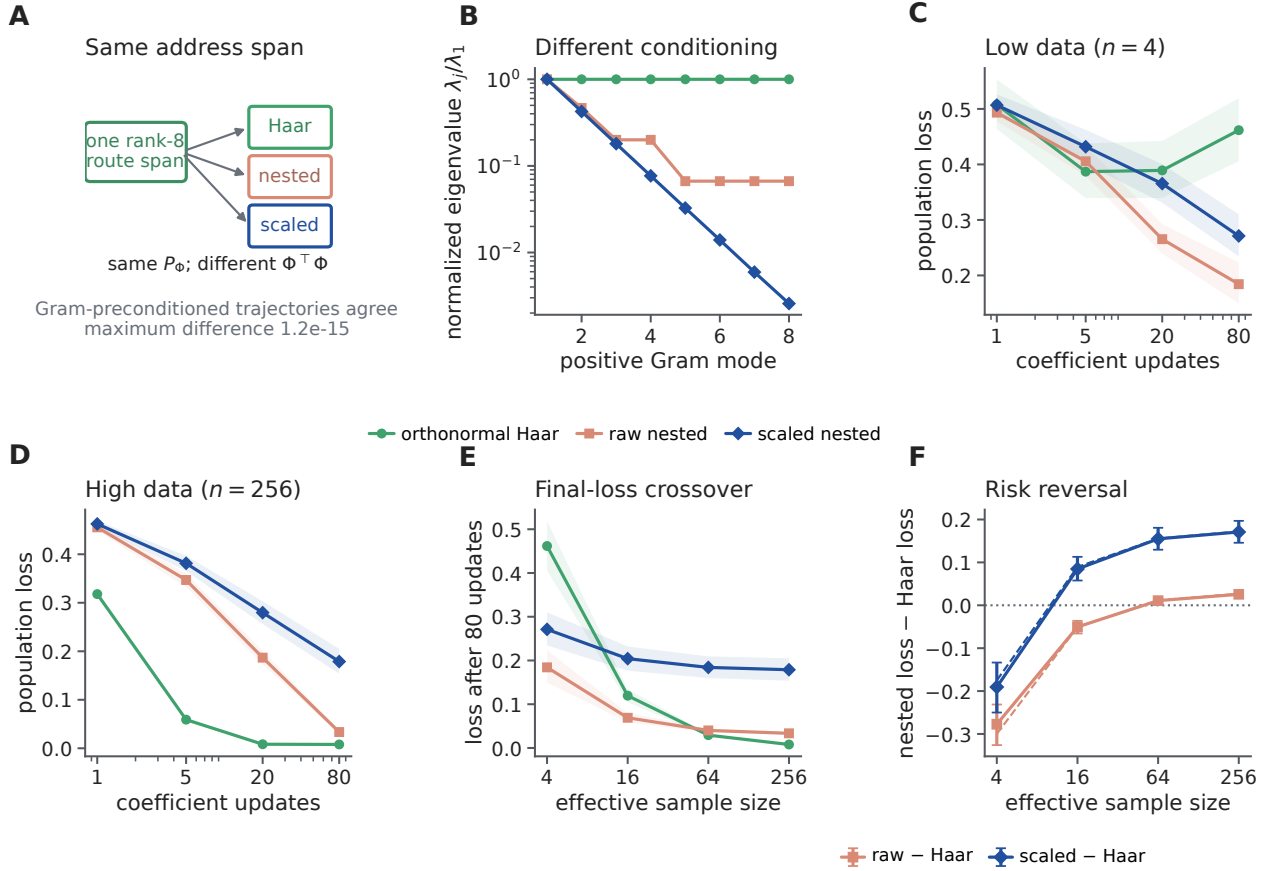


Figure S14: **Same address span produces a finite-data coefficient-learning crossover.** A balanced tree indexes 16 abstract coefficient coordinates; no forward dendritic neuron is simulated. **A**, Orthonormal Haar, redundant nested and statically scaled nested dictionaries parameterize the identical rank-eight route span. Φ is the dictionary and P_Φ its projector; $\Phi^T \Phi$ is the Gram matrix. **B**, Positive Gram eigenvalues normalized to the largest, λ_j/λ_1 ; condition numbers are 1, 15 and 388.52. **C,D**, Mean population-loss trajectories under common noisy local target estimates at effective sample sizes 4 and 256. Shading shows 95% paired-seed bootstrap intervals across 50 seeds. At low data, slow nested modes filter variance; at high data, their retained bias dominates. **E**, Final loss across the sample-size series. **F**, Paired raw-nested-minus-Haar and scaled-nested-minus-Haar loss contrasts; negative values favor the nested coordinates. Solid points and bars are observed means and intervals; dashed curves are the exact finite-time risk from Eq. 57. The reversal is a bias–variance effect within one span. The equality note in **A** reports that Gram-preconditioned controls make all parameterizations agree to 1.15×10^{-15} and do not assert a biological implementation of the preconditioner.

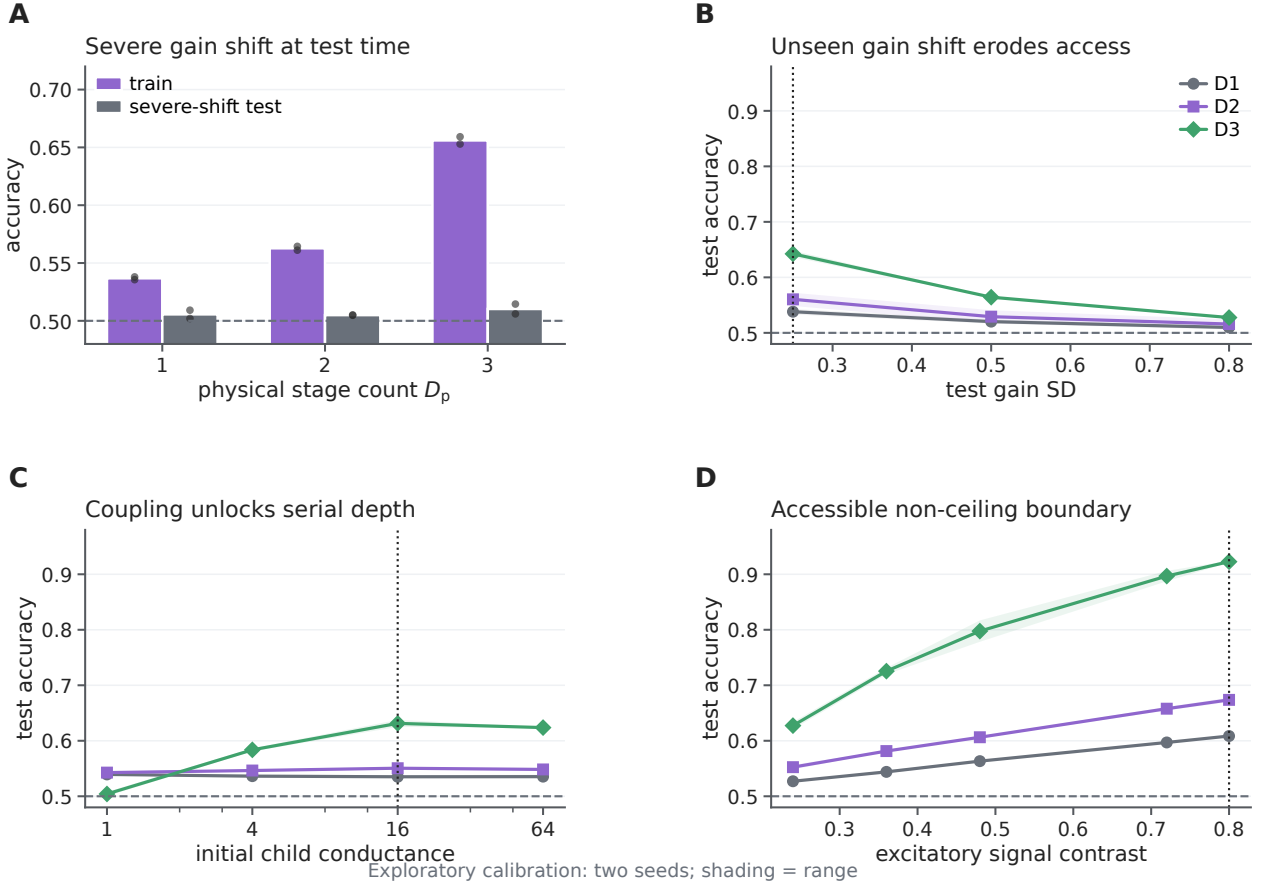


Figure S15: Transparent calibration of the nonlinear physical-depth operating point. All panels use the three-level hierarchical gain-load task and serial shunting morphologies D1 [8], D2 [2, 3] and D3 [2, 1, 2], each with eight nonsomatic compartments per soma. They test the prerequisite for the main hypothesis: whether the mechanism-matched divisive task is accessible at every depth before comparing depth benefits. **A**, The original aligned shunting-backpropagation (BP) pilot at signal contrast 0.24 fit the training distribution increasingly with depth but remained at chance under the prespecified severe gain shift (test gain standard deviation (SD) 1.2). Bars are two-seed means and dots are the individual exploratory seeds. **B**, With training gain SD fixed at 0.25, increasing unseen test gain variation eroded accessibility at every depth. The dotted line marks the matched train/test condition. **C**, At matched gain and signal contrast 0.24, stronger child coupling selectively made serial D3 computation accessible; the dotted line marks child conductance 16 used subsequently. **D**, At matched gain and child conductance 16, increasing excitatory signal contrast moved D1–D3 away from chance without saturating D1 or D2. The separately run 0.80 boundary (dotted line) met the predefined accessibility criterion and selected the operating point used for the analysis seeds. Lines show two-seed means; shading spans the two exploratory values and is not a confidence interval. All pilot, ladder and boundary seeds were excluded from the confirmatory contrasts.

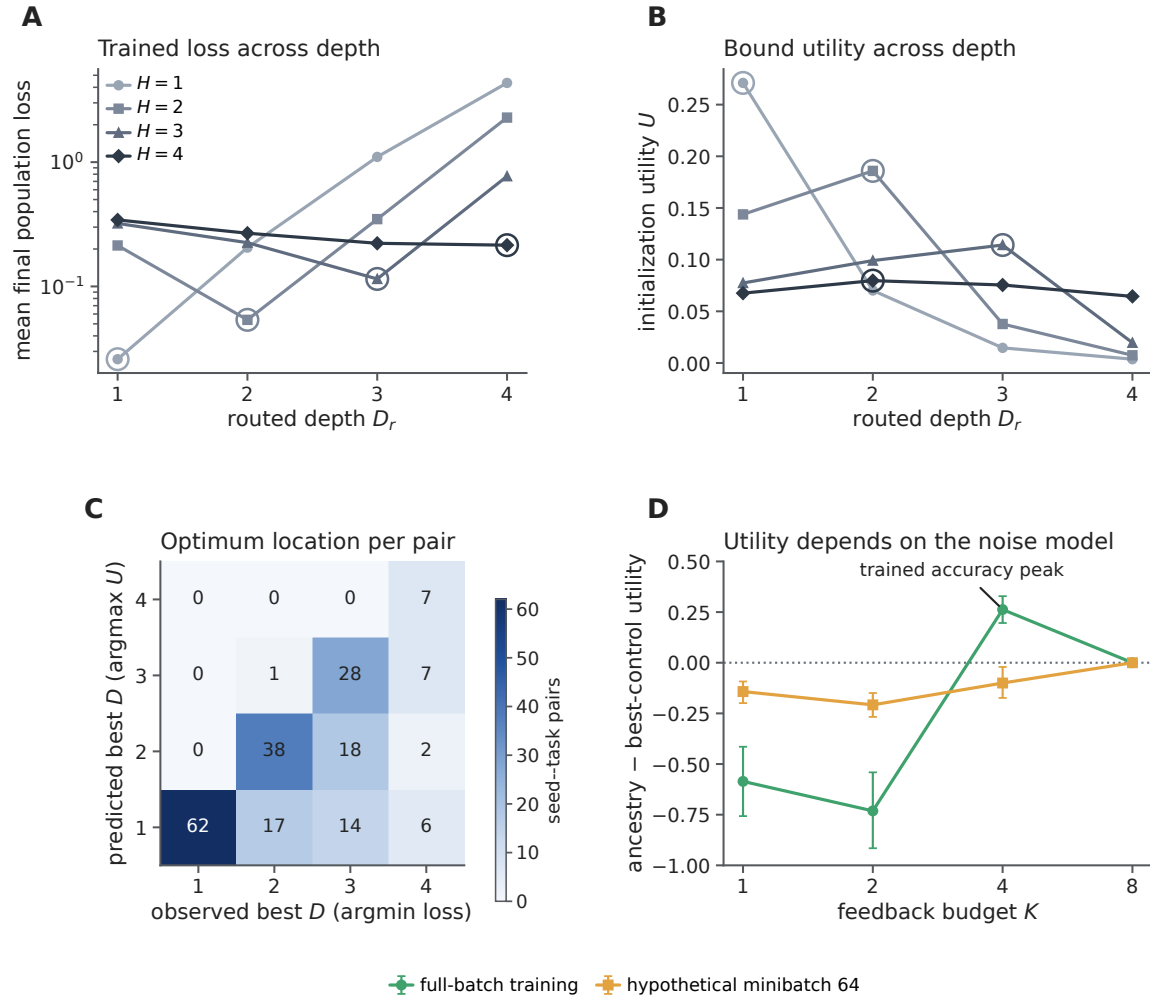


Figure S16: **Initialization scores locate conditional route optima.** Panels **A–C** use the abstract 16-coordinate balanced-tree credit-operator experiment; the tree indexes Haar credit modes and is not a forward dendritic morphology. Panel **D** is a separate reanalysis of the trained eight-stream subtree-address factorial from main Fig. 3. **A**, Mean final population loss across routed depth D_r for task depths $H_c = 1–4$ in the hierarchy-depth routing experiment (aligned tree, $n = 50$ seeds); open circles mark the loss minimum. **B**, Mean initialization utility (Supplementary Eq. 48) for the same conditions; open circles mark the utility maximum. The group-level argmax matches the trained optimum exactly at $H_c = 1, 2, 3$; at the full-rank boundary $H_c = 4$ the one-step bound is conservative and favours $D_r = 2$. **C**, Predicted (argmax utility) versus observed (argmin loss) best depth; heatmap entries count all 200 seed–task pairs, 135 pairs lie on the diagonal. **D**, Ancestry-minus-best-control initialization utility across K in the dendritic-tree subset of the factorial, using the global logistic curvature bound. The comparator is the maximum over deranged, depth-interleaved, random-sparse and random-rank routes. The deterministic full-batch score recovers the mean $K = 4$ ancestry-accuracy peak shown in main Fig. 3; adding hypothetical minibatch-64 variance does not. Points and bars are means and 95% seed-bootstrap intervals ($n = 20$). At the individual-seed level, the two scores select the accuracy-contrast optimum in 13/20 and 8/20 cases, respectively. Raw within-family accuracy instead peaks at $K = 8$ in the group means. These are retrospective, distinct optimum criteria. Analyses: `scripts/analyze_interior_optimum.py` and `scripts/analyze_review_evidence.py`.

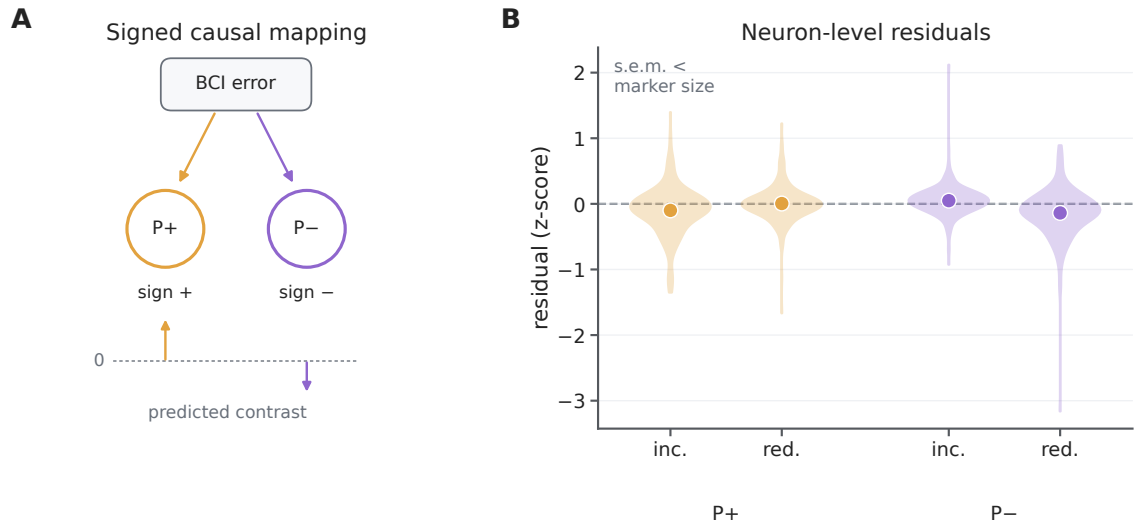


Figure S17: Published animal data are consistent with signed neuron-specific teaching coordinates. This reanalysis contains no reconstructed dendritic morphology and measures no within-tree learning signal; it tests only the sign of two neuron-level coordinates. **A**, In the source brain–computer-interface (BCI) experiment, P+ and P- populations had positive and negative causal decoder weights, predicting opposite dendritic residual contrasts. **B**, Descriptive neuron-level residual distributions from published source sheet “5E” for error-increasing (inc.) and error-reducing (red.) conditions, in the source z-score normalization; dots show means and bars show neuron-level standard errors. These bars are descriptive and are not animal-bootstrap intervals. The paired animal contrasts and mode-energy decomposition use $n = 6$ animals and are reported in Table S33; animals are the inferential units for those tests. This retrospective external comparison is consistent with signed neuron-specific information but does not resolve within-tree transport or uniquely identify a conductance mechanism.

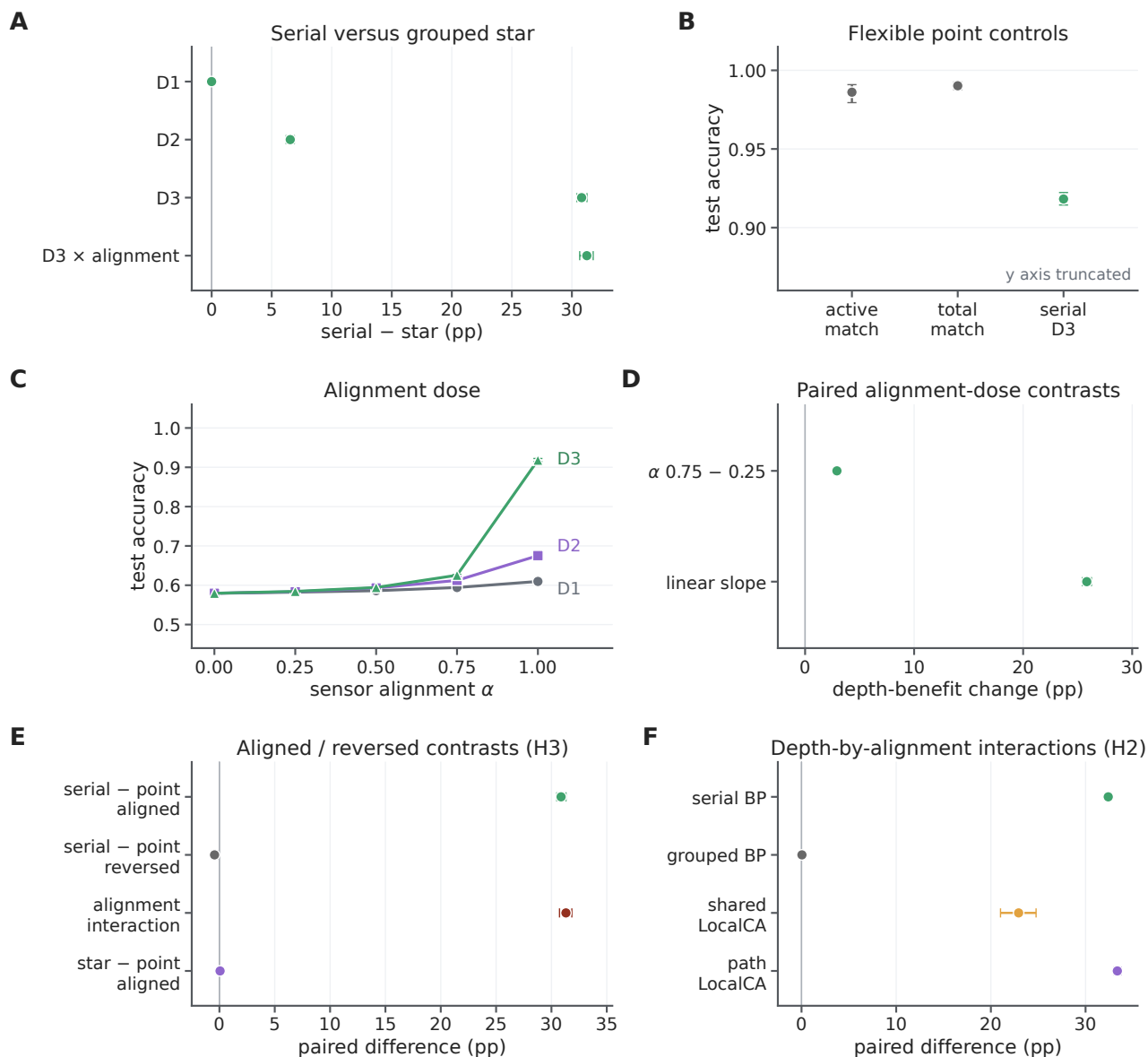


Figure S18: Matched point controls and paired alignment contrasts qualify the physical-depth benefit. Panels A–E use the three-level gain-load task with D1 [8], D2 [2, 3] and D3 [2, 1, 2]; F uses an independently seeded two-level task with D1 [8] and D2 [4, 1]. Vectors are proximal-to-distal and all have eight nonsomatic compartments per soma. **A**, Serial-minus-grouped-star accuracy effects and the aligned-minus-reversed change in that D3 effect, in percentage points (pp). **B**, Flexible point networks matched to active or total parameter count versus serial D3. **C**, Accuracy across sensor–task alignment α . **D**, Change in the D3-minus-D1 advantage from $\alpha = 0.25$ to 0.75 and its fitted slope across doses. Intermediate doses were specified after the endpoint results and constitute interpolation. **E**, D3 serial-minus-point effects for aligned and reversed placement, their interaction, and the aligned star-minus-point effect. **F**, Two-level depth-by-alignment interactions: the D2-minus-D1 accuracy benefit under aligned minus reversed placement, separately for serial and grouped-point backpropagation (BP) and shared-soma or exact-path local credit assignment (LocalCA). Points and curves are means with 95% paired-seed bootstrap intervals, using ten seeds per cohort. Complete depth curves appear once in Supplementary Fig. S31. These are calibrated task-family controls, not independent demonstrations of a universal depth advantage.

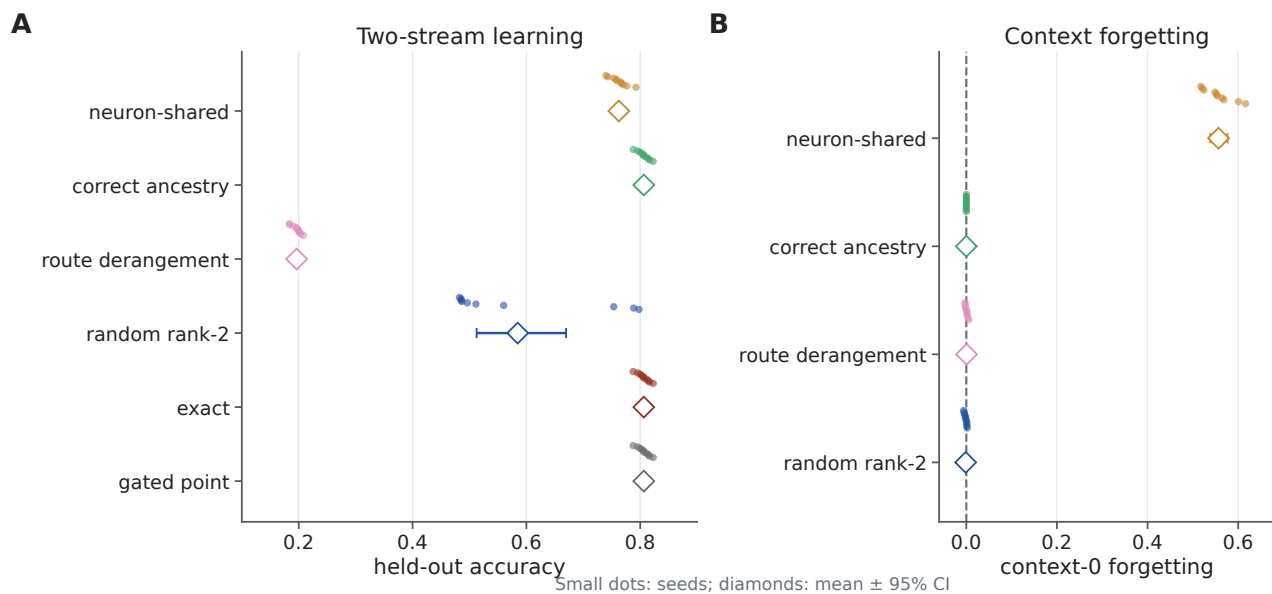


Figure S19: Two-branch routing supports learning and limits interference. This minimal task uses two sibling branches without serial depth. **A**, Held-out accuracy for neuron-shared, correct-ancestry, deranged, random rank-two, exact and gated-point rules in the two-stream experiment (Table S1). **B**, Forgetting, defined as the decrease in first-context accuracy after adaptation to the second. Small forgetting for deranged feedback reflects its poor initial solution. Small points are ten paired seeds; larger symbols and bars show means and 95% paired-seed bootstrap intervals. Exact path transport supplies the full error information; gated-point learning supplies equivalent branch selection locally. The separate 120-checkpoint norm-matched diagnostic appears in Supplementary Fig. S9.

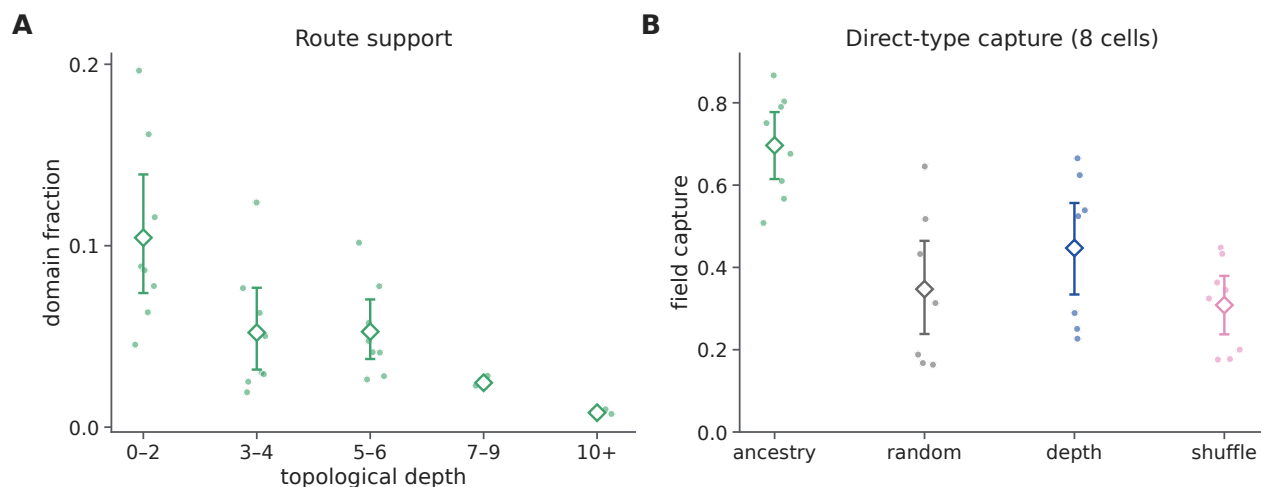


Figure S20: Morphology-route support and direct-type capacity controls. Both panels use the initial eight reconstructed MICrONS arbors from one mouse and modeled route-generated fields. The experiment tests structural capacity, not task learning or an observed biological credit field. **A**, Mean ancestry-domain support fraction within topological-depth bins; a domain comprises the input-bearing descendants of a route origin. **B**, Eight-channel model-field capture for ancestry, random, depth-bin and shuffled dictionaries, restricting inputs to directly typed presynaptic contacts. Points are reconstructed-cell summaries; intervals are 95% cell-bootstrap intervals. The original route-generated capacity values are retained in Tables S25 and S26.

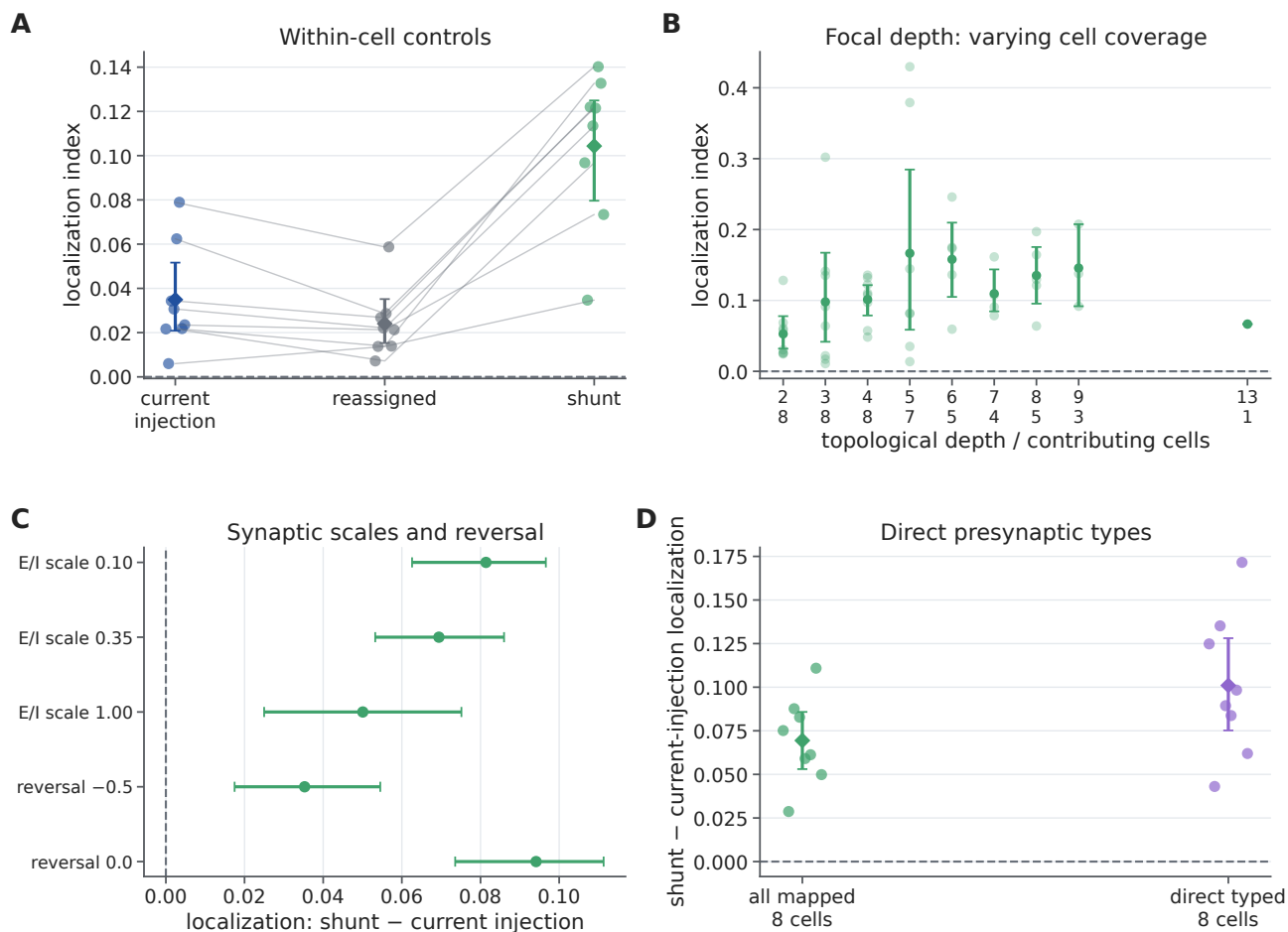


Figure S21: **Within-cell focal-shunting controls and sensitivity.** All panels use passive reciprocal-cable models of the initial eight reconstructed cells and ask whether a focal shunt changes descendant gradient transport more selectively than a baseline-current-matched injection. **A**, Within-cell controls. Shunt and baseline-current-matched current injection values are localization indices (descendant minus depth-matched unrelated median absolute log-gradient change); the relation-reassignment value compares the true relation with a matched foreign relation template. Positive values favor descendant localization for the physical shunt or true relation. **B**, Cell means and 95% cell-bootstrap intervals within focal-site depths 2/3/4/5/6/7/8/9/13, with 8/8/8/7/5/4/5/3/1 eligible cells. No curve connects means from changing cell sets. **C**, Dimensionless excitatory/inhibitory (E/I) conductance-scale and normalized inhibitory-reversal sensitivity. **D**, All mapped contacts versus directly typed presynaptic contacts, summarized by shunt-minus-current localization. Sites are nested within eight reconstructed cells. Panels **A–D** show 95% cell-bootstrap intervals; **A,B,D** use 10,000 draws and **C** uses the retained source-summary intervals. The intervention schematic, relation selectivity, dose response, adjoint-replacement contrast and electrotonic calibration appear as main Fig. 7.

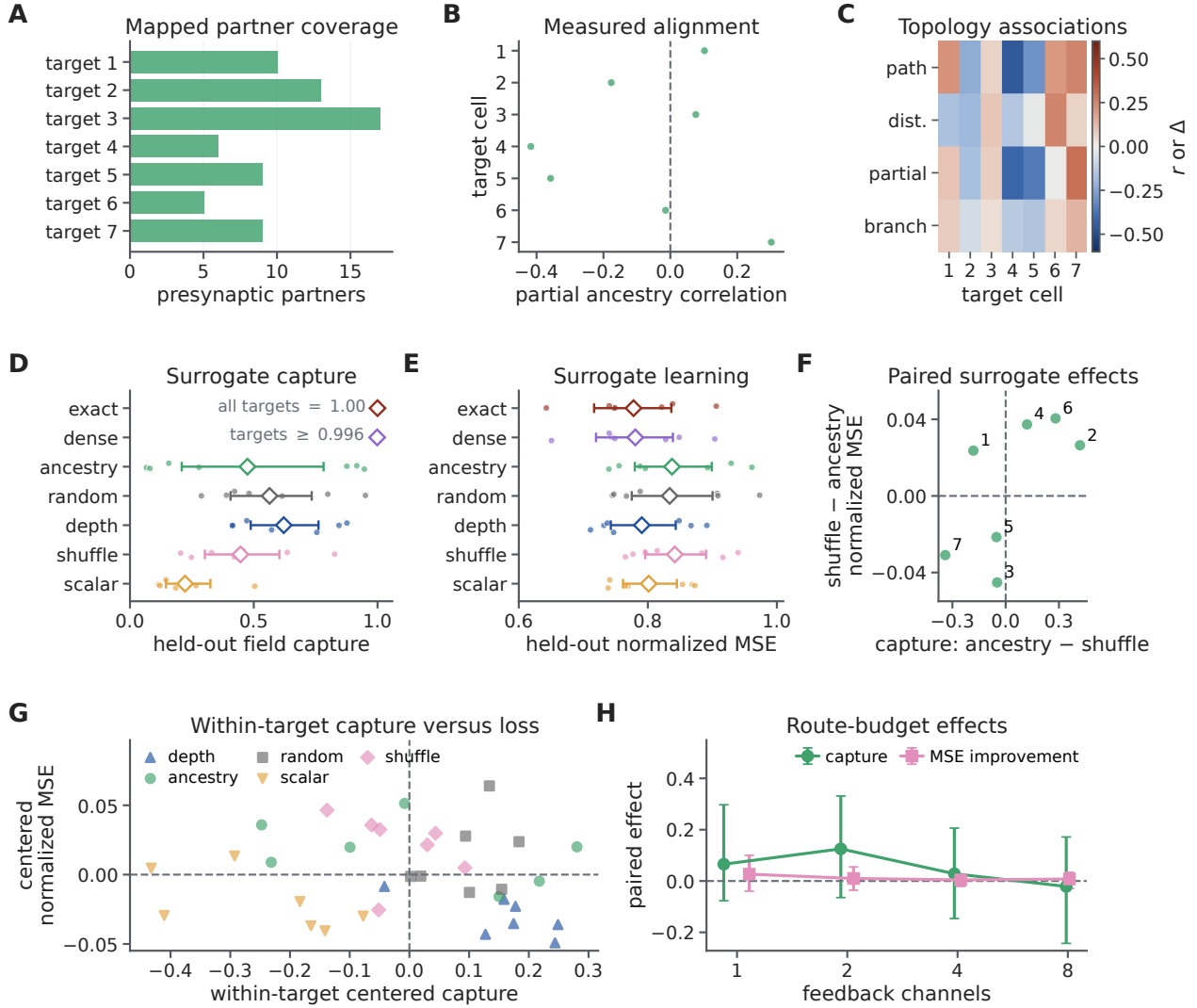


Figure S22: Measured-response topology and major-branch learning diagnostics. Reconstructions and measured responses come from seven targets and their mapped presynaptic partners in one mouse. Learning panels use the major-branch surrogate, not the complete segment-resolved model of main Fig. 8. Only the exact or trialwise projected input-error field changes. **A**, Number of mapped functional presynaptic partners for each of seven targets. **B**, Shared-ancestry/response rank correlation after controlling Euclidean separation and soma-to-contact depth differences. **C**, Target-specific associations: shared-path correlation (path), negative-tree-distance correlation (dist.), the partial correlation in **B** (partial), and the same-major-branch response-similarity contrast (branch). Colors encode signed correlation r or contrast Δ . **D,E**, Four-channel exact-field capture and held-out normalized MSE for exact errors, the dense principal-component oracle, ancestry routes, random routes, depth bins, shuffled ancestry and scalar broadcast. Projection coefficients are per-trial oracles; exact and dense capture approach the ceiling. **F**, Target-level contrasts: ancestry-minus-shuffle capture and shuffle-minus-ancestry normalized MSE, so positive values on either axis favor ancestry. **G**, Capture versus normalized mean squared error (MSE), each centered by its target-specific mean across the five restricted dictionaries. **H**, Ancestry-minus-shuffle capture (green) and shuffle-minus-ancestry normalized MSE (pink) across bandwidth; both are positive when ancestry improves the endpoint. Points in **B,F,G** describe target relations; bars in **D,E,H** are means and 95% target-bootstrap intervals (20,000 draws). Ten splits are averaged within each target before inference. These projected-field surrogate results differ from the frozen profile rule in the complete-tree experiment.

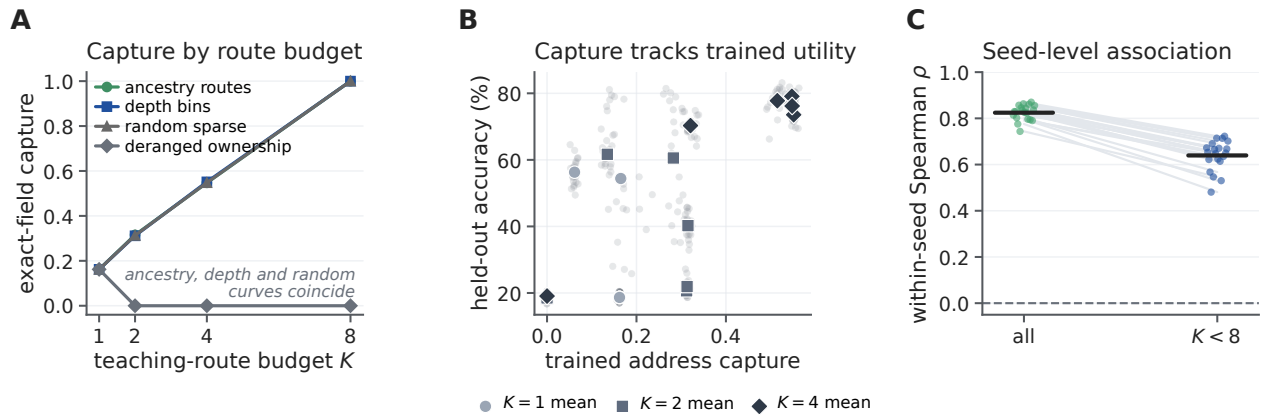
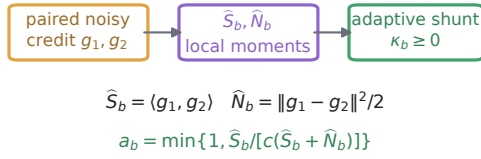


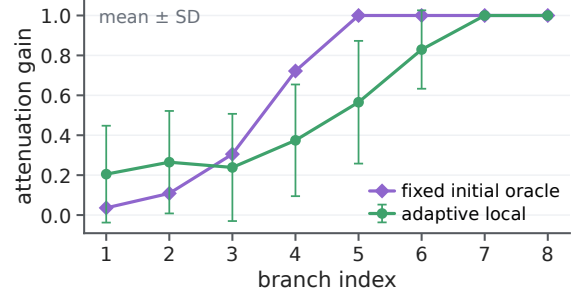
Figure S23: The trained partition residual explains bandwidth and assignment failures but not fine support-specific differences. All panels reanalyze the eight-context task with eight terminal parameter blocks arranged in a balanced routing hierarchy. Dendritic, flat, grouped-point and gated-point implementations received identical route matrices; internal routing-tree nodes did not perform serial forward computation. **A**, Capture of the trained exact per-example coefficient field by the best coefficient on each available route; K is the feedback-route budget. Curves and intervals are means and 95% seed-bootstrap intervals over 20 independent seeds. Coordinate-to-arbor derangement excludes the target leaf and has zero capture for $K > 1$; three partitions that retain the target have similar capacity at matched K . **B**, Trained address capture—checkpoint capture of the exact per-example coefficient field, in the vocabulary of the main text—versus held-out accuracy for the six restricted route families at $K < 8$. Small grey points are seed-condition means across the four implementation-equivalent representations; large symbols are condition means. **C**, Spearman correlations computed separately within each independent seed for all route conditions and after excluding $K = 8$; lines pair the same seed and horizontal bars show means. This deterministic reconstruction is a post-hoc diagnostic, not an independent prediction test.

A

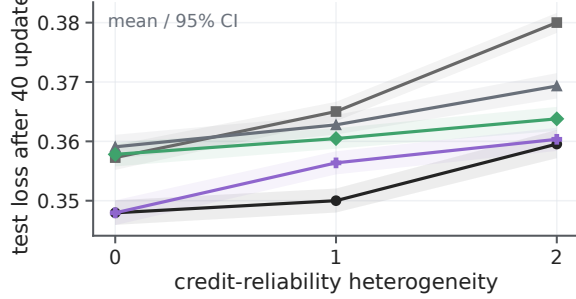
Local reliability estimate

**B**

Estimated branch ordering

**C**

No endpoint gain over no shunt

**D**

Final-loss boundary at high heterogeneity

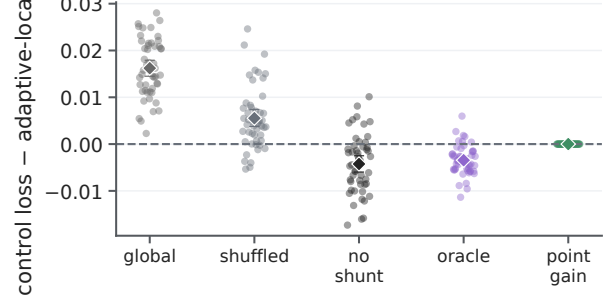


Figure S24: Locally estimated reliability recovers placement ordering but not an absolute shunting advantage. This synthetic positive-rate binary task uses eight parallel, nonserial branch blocks with heterogeneous signal and noise; it is not a reconstructed arbor or a physical-depth experiment. **A**, Two independent noisy local gradient observations estimate branch signal and noise moments, $\hat{S}_b$ and $\hat{N}_b$, which set gain a_b and nonnegative shunt κ_b for branch b ; c is the fixed step fraction. **B**, Fixed initial-oracle and final adaptive attenuation profiles at maximum heterogeneity; adaptive points show mean $\pm$ s.d. over 50 independent seeds. **C**, Final test loss after 40 updates across reliability heterogeneity, comparing adaptive local, global and shuffled gains with no shunt and the fixed initial oracle. Curves and shading are means and 95% seed-bootstrap intervals. **D**, Control minus adaptive-local final loss at maximum heterogeneity; positive values favor adaptive local. Small points are paired seeds and large symbols are means with 95% paired-seed bootstrap intervals. Adaptive local beats global and shuffled attenuation, loses to no shunt and the fixed initial oracle, and is numerically identical to the independently computed point gain.

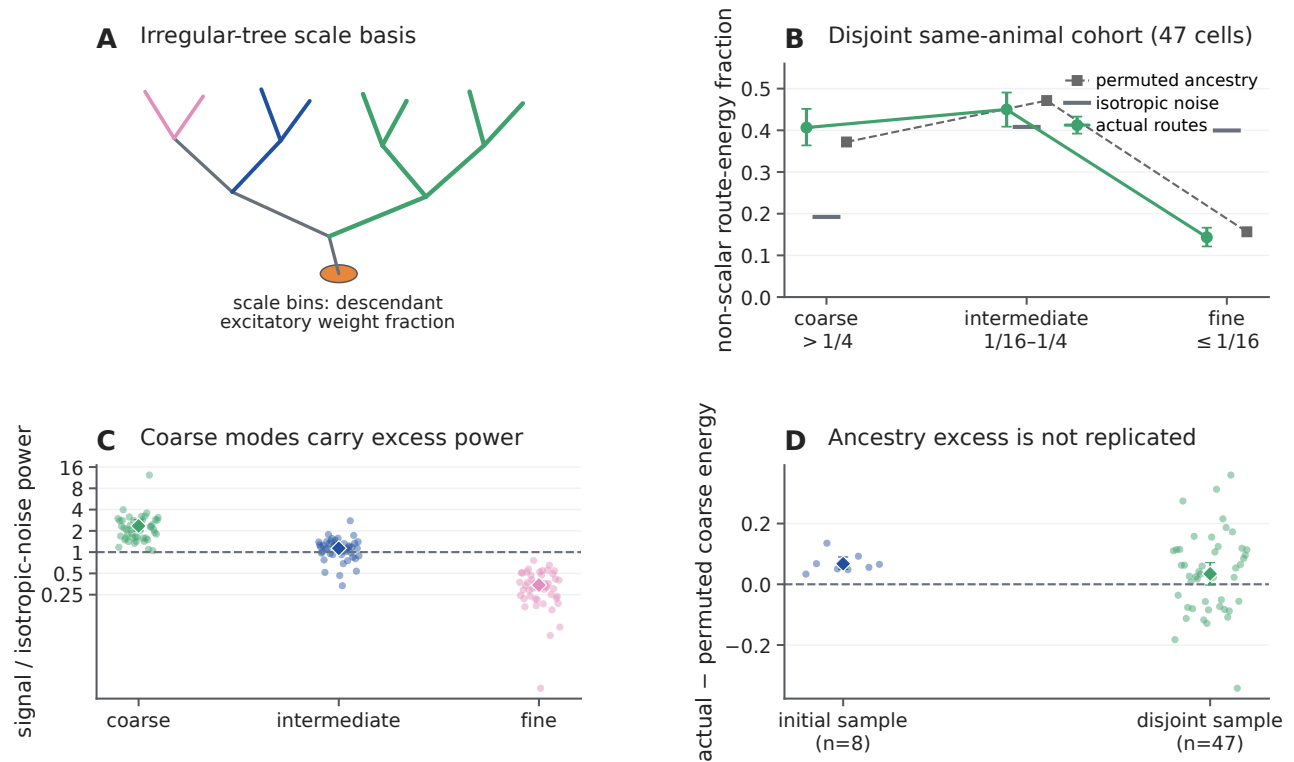


Figure S25: **Irregular reconstructed trees concentrate modeled route-field energy coarsely, but the anatomy-specific excess does not replicate.** **A**, Weighted unbalanced tree-Haar contrasts recursively partition each reconstructed arbor into coarse, intermediate and fine supports, defined by descendant excitatory-weight fractions $> 1/4$, $1/16-1/4$ and $\leq 1/16$, respectively. **B**, Non-scalar conductance-weighted ancestry-route energy fraction in the disjoint 47-cell cohort, compared with column-permuted ancestry and dimension-matched isotropic noise. Actual-route points and intervals are cell means and 95% cell-bootstrap intervals; null markers are cell means. **C**, Signal-to-isotropic-noise power by scale for all 47 cells; diamonds show means and 95% cell-bootstrap intervals. **D**, Actual-minus-permuted coarse energy in the original and disjoint cohorts. The positive original-cohort contrast is not reliable in the 47-cell cohort. Cells are the inferential units; all reconstructions derive from one mouse. Modeled route fields are not measured teaching signals.

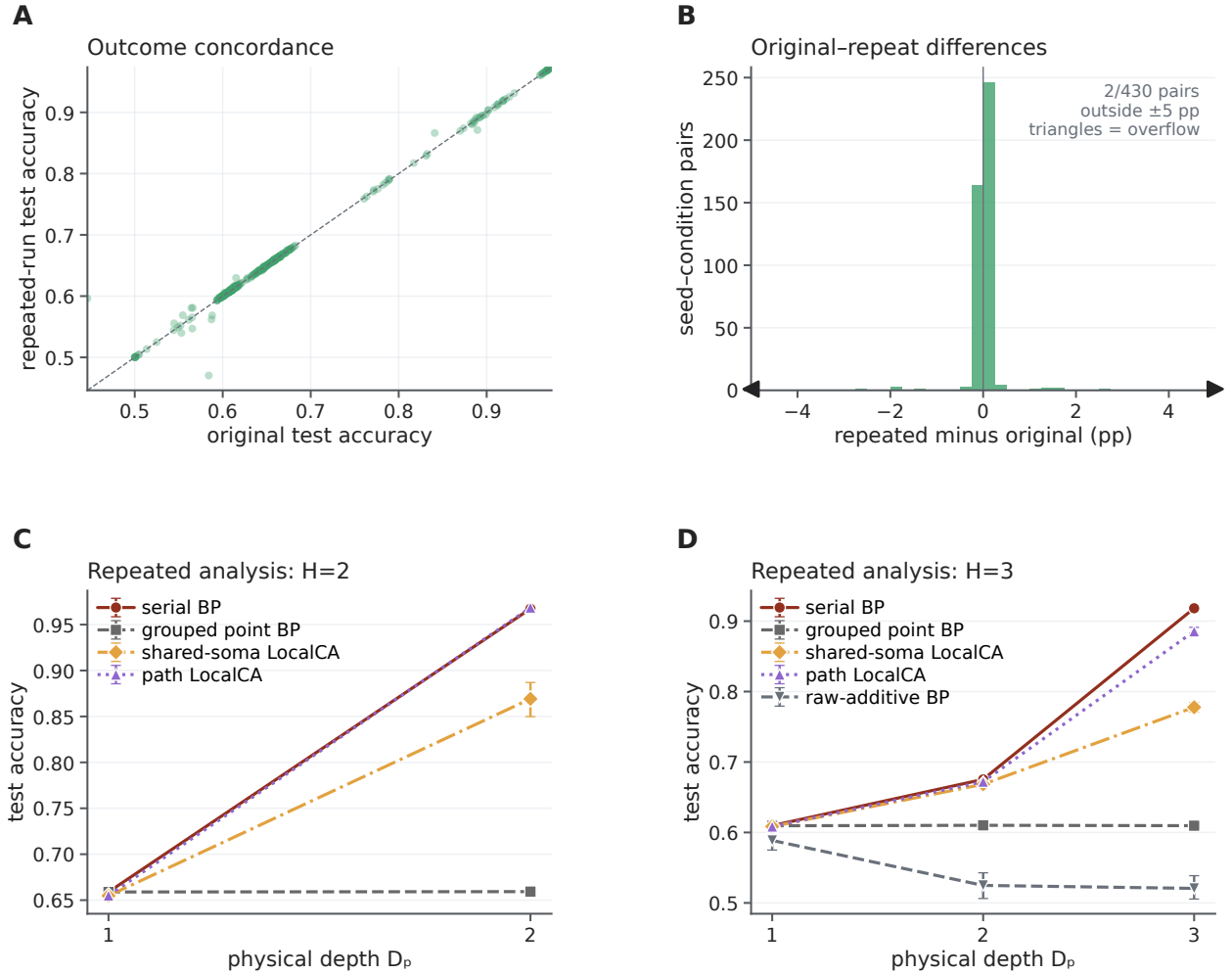


Figure S26: **Independent computational reruns reproduce the H2/H3 physical-depth results.** H2 is the two-level gain-load task with D1 [8] and D2 [4, 1]; H3 is the three-level task with D1 [8], D2 [2, 3] and D3 [2, 1, 2]. Morphology vectors are proximal-to-distal, and every architecture has eight nonsomatic compartments per soma. **A**, Held-out accuracy in the repeated analysis versus the original analysis for each of 430 same-seed fit pairs. The dashed line is identity. **B**, Distribution of paired repeated-minus-original accuracy differences in percentage points (pp); overflow triangles retain two of 430 pairs outside the displayed ± 5 -point range. **C,D**, H2 and H3 accuracy in the repeated analysis for serial backpropagation (BP), literal grouped-point BP, shared-soma local credit assignment (LocalCA), exact-path LocalCA and the H3 raw-additive control. Lines and intervals are paired-seed means and 95% seed-bootstrap intervals ($n = 10$). The H2/H3 depth, placement, point-control and local-credit conclusions were unchanged within the original training budget. These same-seed reruns test computational reproducibility, not independent scientific replication.

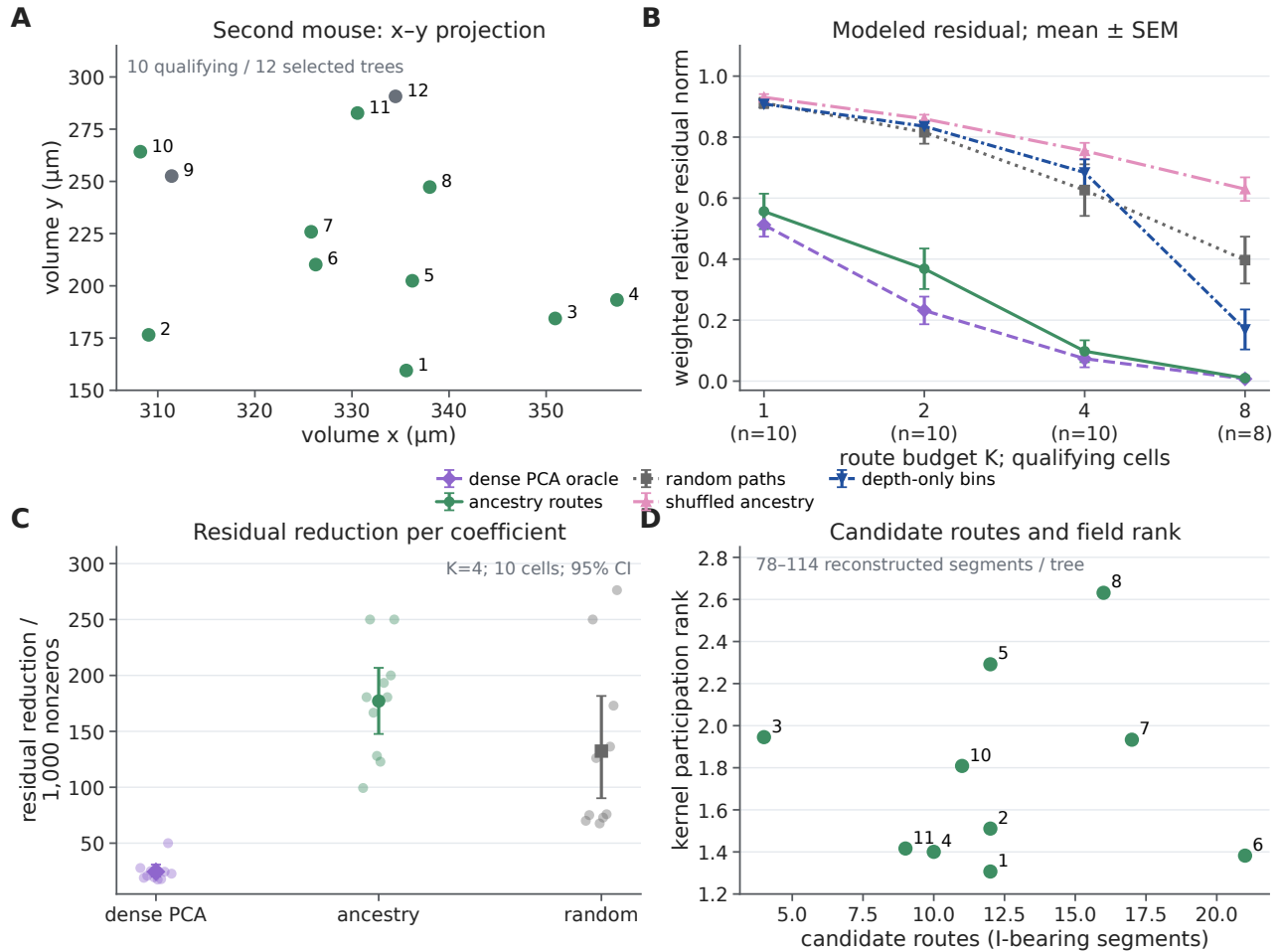


Figure S27: **A second animal supplies structural replication with small-tree limits.** **A**, Outcome-independent 12-cell Pinky v185 sample across the public volume; green cells meet and gray cells fail the predefined criteria. Numbers give selection order. **B**, Weighted residual across feedback budgets for the dense principal-component-analysis (PCA) ceiling, subtree routes and controls; lower is better. Curves are means $\pm$ standard errors across eligible cells. The eligible count is ten at $K = 1, 2, 4$ and eight at $K = 8$, under the inherited site-count and candidate-budget eligibility rule. **C**, Four-channel one-minus-weighted-residual-norm score per 1,000 nonzero delivery coefficients; this is not squared-energy capture. Points are cells; larger symbols and bars are means and 95% cell-bootstrap intervals (50,000 draws). **D**, Inhibitory-bearing candidate segments versus kernel participation rank (effective rank), with reconstruction-size ranges; I-bearing denotes inhibitory-contact-bearing segments. This exposes why small trees can approach the rank ceiling with few routes. Cells are nested within one volume and are not independent animals. Cross-cohort residual comparisons appear in the route-generated comparisons in Supplementary Fig. S10. The target fields are model matched and do not measure biological teaching signals.

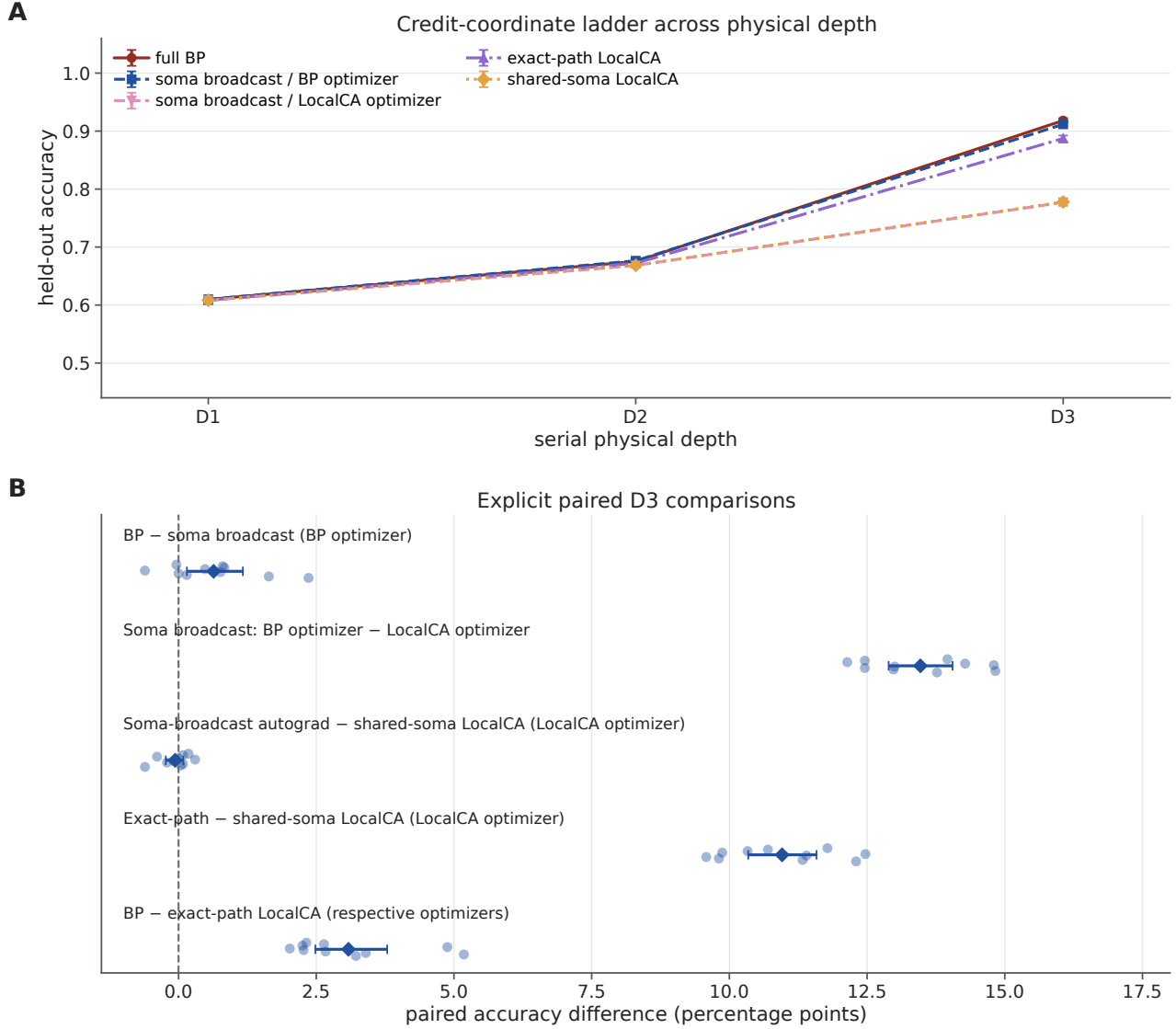


Figure S28: **Teaching-coordinate and optimizer diagnostics for the physical-depth experiment.** Both panels use the three-level hierarchical gain-load task. Panel **A** compares D1–D3; panel **B** focuses on the D3 serial conductance morphology [2, 1, 2]: two proximal, two middle and four distal compartments per soma. They dissect the backward learning coordinate while holding each forward architecture fixed within its feedback comparison. **A**, Credit-coordinate ladder from full backpropagation (BP) through soma-broadcast automatic differentiation (autograd) to shared-soma and exact-path local credit assignment (LocalCA). **B**, Five separately labeled pairwise D3 contrasts under the stated optimizer recipes; signs follow the displayed left-minus-right interventions. At D3, full BP exceeds soma-broadcast autograd by 0.64 percentage points under the standard BP optimizer; exact-path LocalCA exceeds shared-soma LocalCA by 10.96 points under the split LocalCA optimizer. Changing the broadcast optimizer costs 13.47 points. Path specificity therefore interacts with optimization in this task. These secondary diagnostic analyses are not independent trained experiments. Panel **A** shows means; small points in **B** are ten paired seeds, and larger symbols are means. Intervals are 95% paired-seed bootstrap intervals.

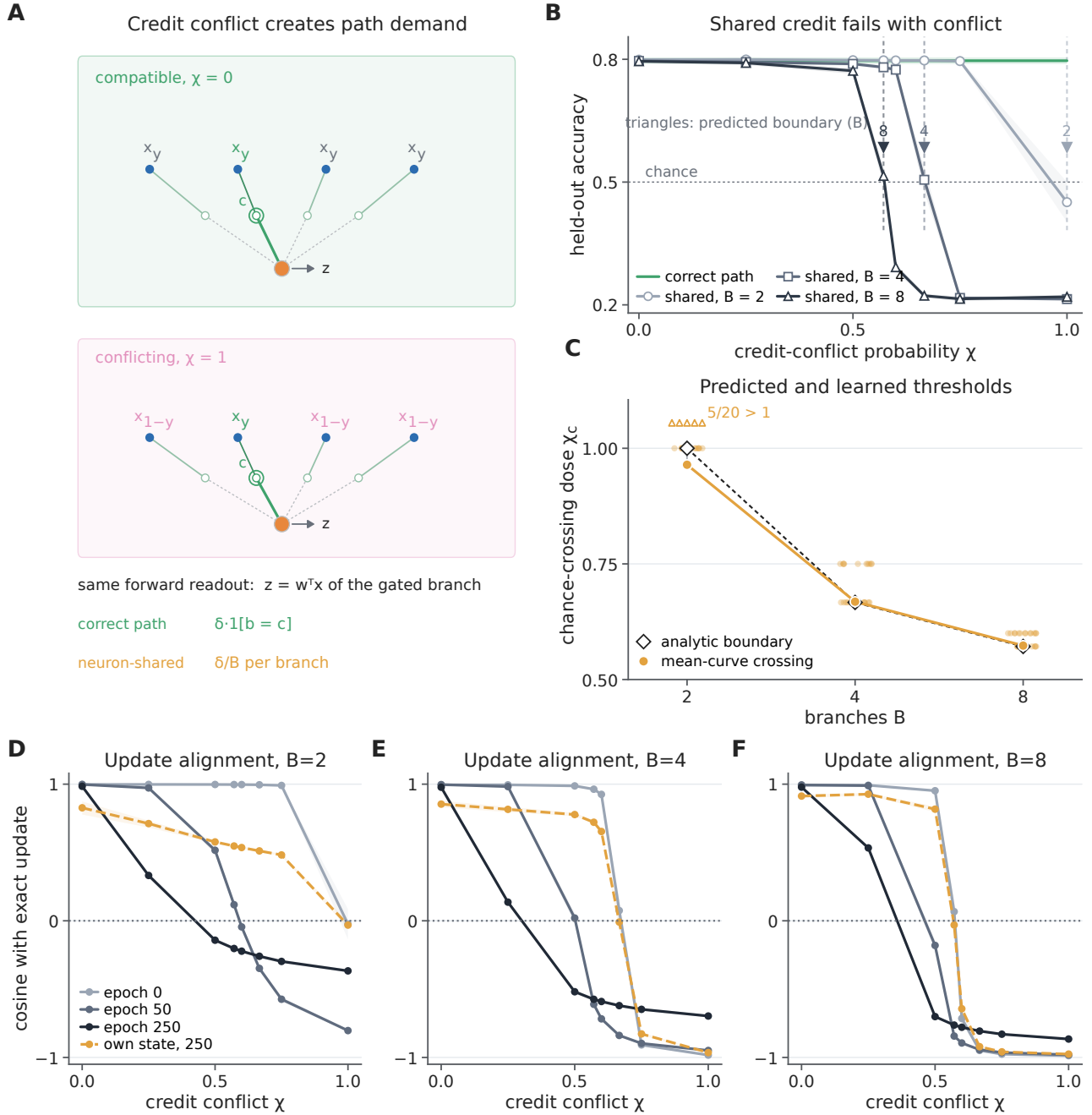


Figure S29: Branch selection and trained-state geometry constrain shared feedback. One logistic output combines B parallel branch weight vectors without serial or conductance nonlinearities. **A**, Context selects one branch's contribution to the logit, while all branches form active eligibilities. At conflict dose $\chi = 0$, nonselected images copy the selected view; at $\chi = 1$, they come from the opposite class. Exact feedback is $\delta \mathbf{1}[b = c]$; shared feedback is δ/B , where δ is output error, b branch and c selected context. **B**, Held-out accuracy across conflict dose. The green band encloses nearly coincident exact-route curves for $B = 2, 4, 8$; colored curves use shared feedback. Triangles mark the centered mean-field boundary $\chi_c = B/[2(B-1)]$. **C**, Analytic boundaries versus crossings interpolated from mean trained accuracy. Interpolation is descriptive. Open overflow triangles retain five of twenty $B = 2$ seeds without a crossing in the tested range. Analytic backpropagation (BP) and the equivalent gated-point implementation coincide with exact routing (main Fig. 1D,E). **D–F**, Shared-versus-exact gradient cosine for two, four and eight branches. Solid curves use common exact-rule states at epochs 0, 50 and 250; dashed curves use each shared rule's own state at epoch 250. Curves and shading show 20 paired-seed means and 95% seed-bootstrap intervals (5,000 draws). All six checkpoints and both state comparisons remain in Source Data. Missing gates admit distractor eligibilities with zero exact derivatives; trained-state geometry can depart from the initialization approximation.

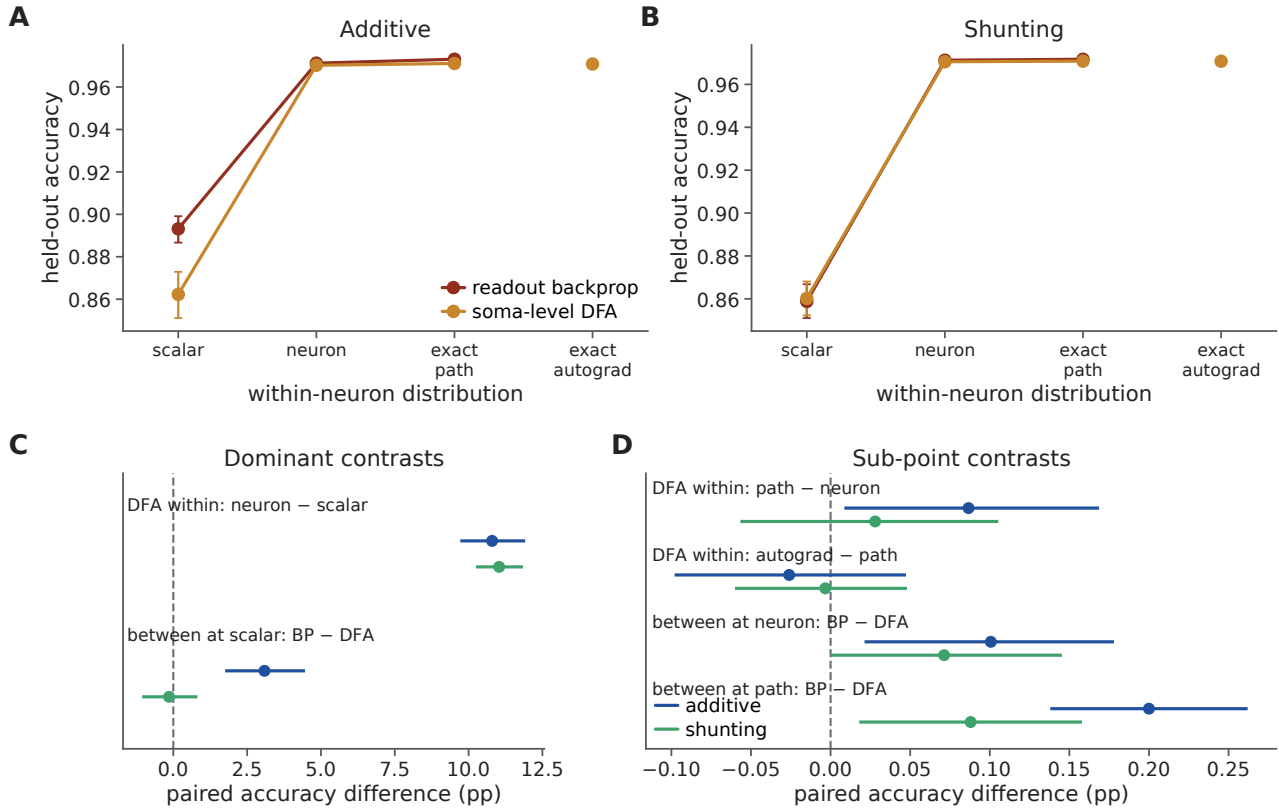


Figure S30: The within-neuron ladder survives an approximate between-neuron source. The between-by-within factorial crosses the MNIST feedback ladder of main Fig. 1 with the production of the per-neuron signal: exact readout backpropagation (the paper’s default map F) versus a fixed random soma-level feedback matrix (direct feedback alignment, DFA), seeded per run, inside the same local credit-assignment (LocalCA) rule; a soma-DFA trainer with exact autograd inside each tree anchors the DFA-by-exact cell. All new cells reuse the ladder’s architectures, contacts and training protocol, and the exact-readout row reuses the frozen ladder results. **A,B**, Held-out accuracy under both signal sources for raw-additive and shunting trees, respectively. The within-neuron conditions are strict scalar, neuron-specific, exact-path and exact automatic-differentiation (autograd) feedback. **C**, Dominant paired contrasts: under DFA, neuron-specific feedback beats the scalar rung by +10.8 (additive) and +11.0 (shunting) points, 15/15 seeds each (exact sign-flip $P = 6.1 \times 10^{-5}$); the between-source gap at the scalar rung is +3.1 points for additive trees and null for shunting. **D**, Sub-point contrasts: exact path transport adds at most +0.09 points over neuron-specific feedback under DFA, the transport and autograd implementations coincide, and the readout-minus-DFA gap at the neuron and path rungs is no more than about 0.2 points. Blue is additive and green shunting in **C,D**. Symbols and whiskers show means and 95% seed-bootstrap intervals over 15 paired seeds.

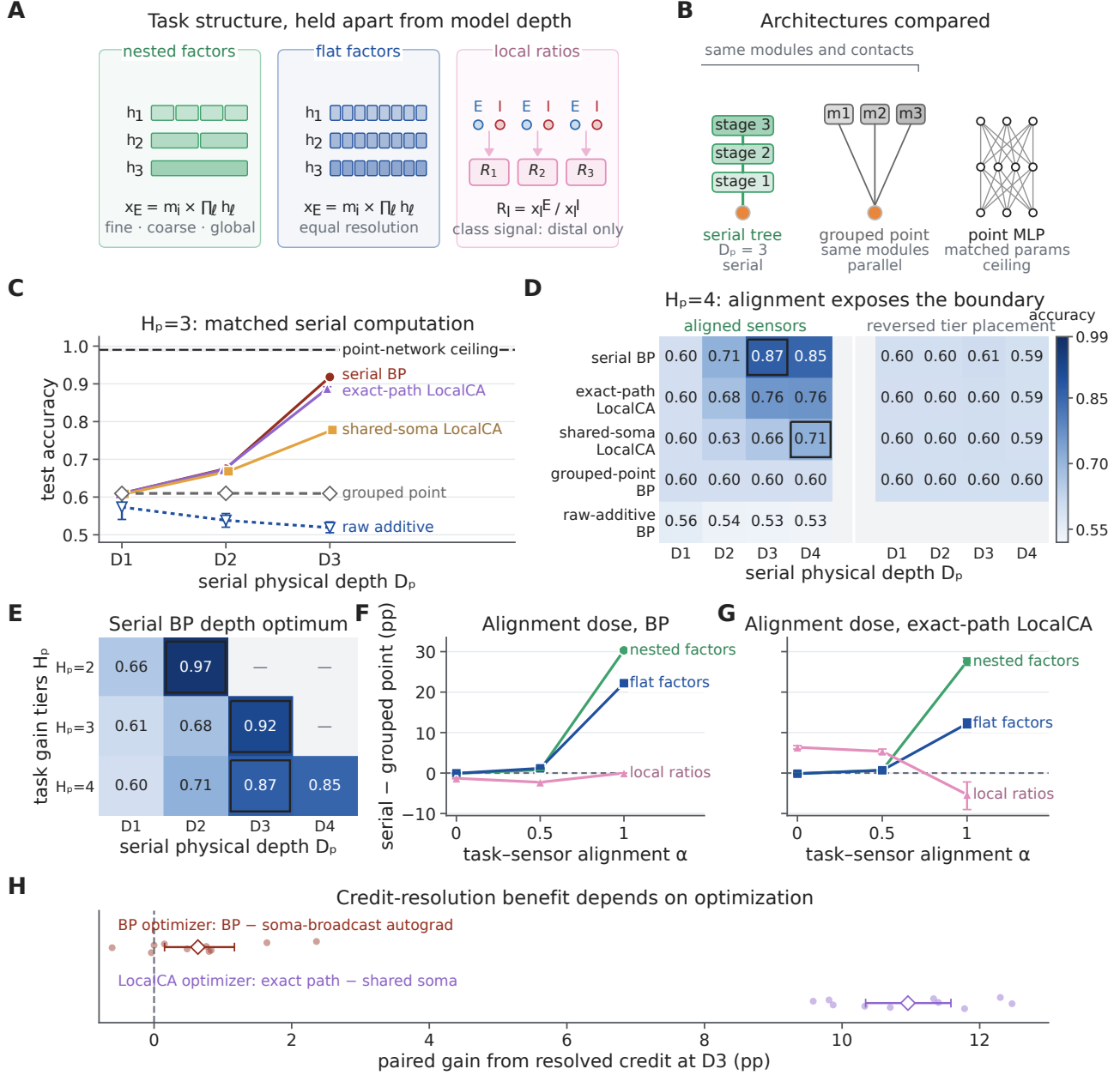


Figure S31: Serial computation and feedback depend on task and optimization. **A**, Nested factors use 4/2/1 spatial groups; independent flat factors share an eight-group partition; h_l are gain factors and m_i the class-dependent signal; local excitatory/inhibitory (E/I) ratios $R_l = x_l^E / x_l^I$ allow cancellation within modules. α is factor-sensor correlation. **B**, Serial D3, resource-identical grouped-point and approximately parameter-matched dense multilayer-perceptron (MLP) controls. Only the first two share modules and contacts. **C**, Three-level aligned task across D1–D3 and feedback conditions. **D**, Four-level task, including D4, with aligned or reversed placement. **E**, Descriptive depth optima across independent two-, three- and four-level cohorts; blank cells were untested. **F,G**, Serial-minus-grouped accuracy at fixed D3 across task family and alignment under backpropagation (BP) and exact-path local credit assignment (LocalCA), respectively. **H**, Credit-resolution effects: BP minus soma-broadcast under the BP optimizer, 0.637 points (95% interval 0.152–1.168); exact-path minus shared-soma under LocalCA optimization, 10.957 points (10.342–11.582). These are recipe-specific contrasts, not a complete interaction test. Points, paired seeds; curves/bars, means and 95% paired-seed bootstrap intervals; ten seeds per cohort. H_p counts task gain tiers and D_p physical stages; D1–D4 denote those stage counts within each cohort. Branching vectors and heatmap-outline criteria are defined in Supplementary Section S5.

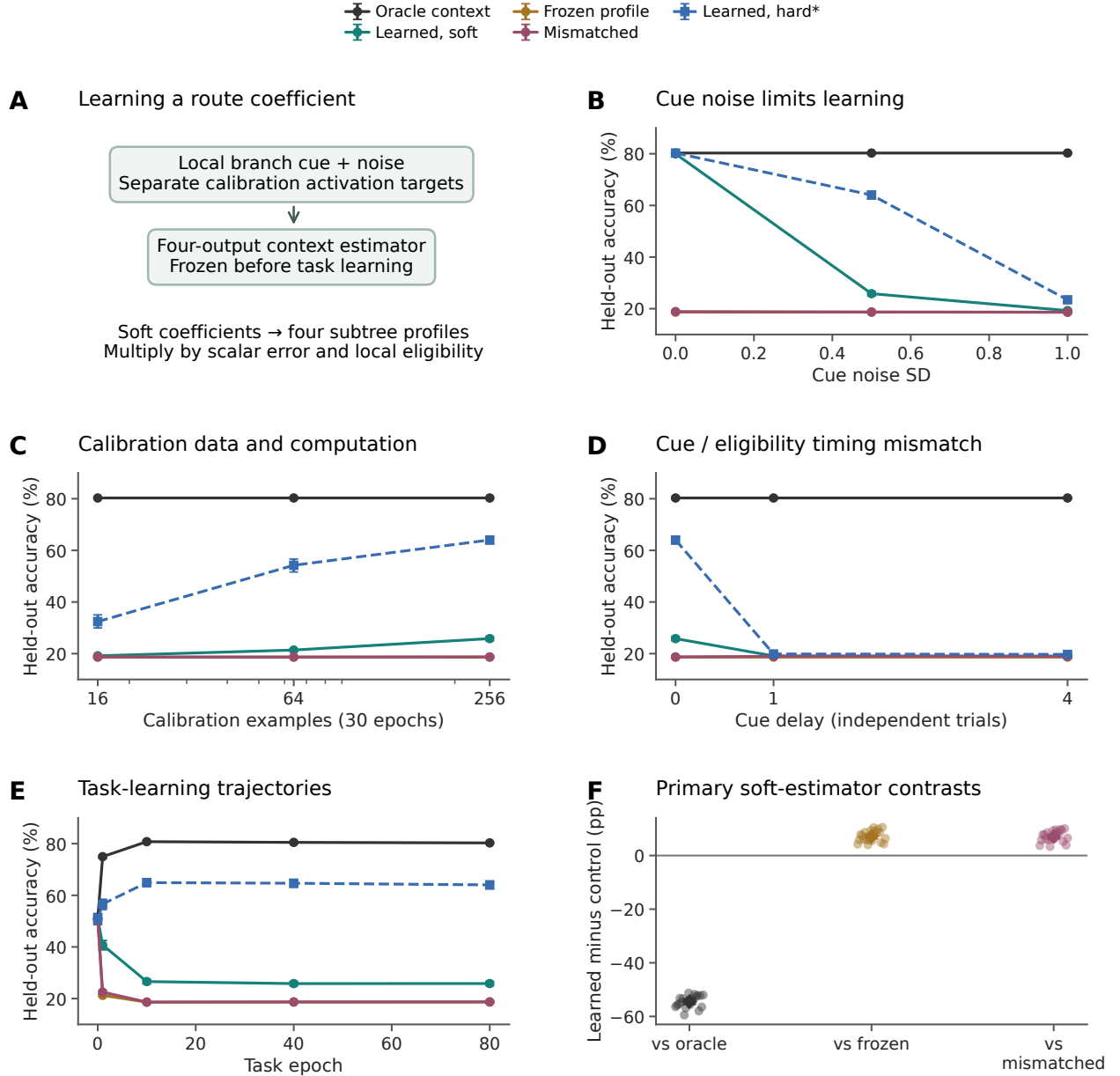


Figure S32: An explicit context cue supports learned coefficients, with strong dependence on cue quality and readout. The eight-context task uses four fixed ancestry routes and one scalar output error; there is no serial forward dendritic computation. **A**, Separate calibration cues and local route-activation targets train a four-output softmax encoder, which is then frozen during task learning. **B**, Held-out accuracy versus cue-noise standard deviation (SD) at 256 calibration trials and zero delay. **C**, Calibration set size at noise SD 0.5 and zero delay; 30 epochs are fixed, so both data and computation increase. **D**, Cue delay at 256 calibration trials and noise SD 0.5. Delays mismatch independent contexts and are measured in trials, not physiological time. **E**, Task-learning trajectories in the primary condition. **F**, Primary soft-estimator-minus-control accuracy differences in percentage points, relative to oracle, frozen and mismatched coefficients; all 20 paired seeds are shown. In **B–E**, dashed blue curves marked with an asterisk use maximum-probability route selection by the same encoder. This hard readout was specified after the soft outcomes, with no encoder refit, and is an exploratory paired sensitivity. The primary soft rule remains far below the oracle at noise SD 0.5; hard selection narrows but does not eliminate this gap. Curves, diamonds and bars show means and 95% paired-seed bootstrap intervals (5,000 draws for this figure; 20,000 for the primary numerical contrasts), with 20 fresh analysis seeds. The supplied cue and activation targets are explicit information resources, not demonstrated biological teaching signals.

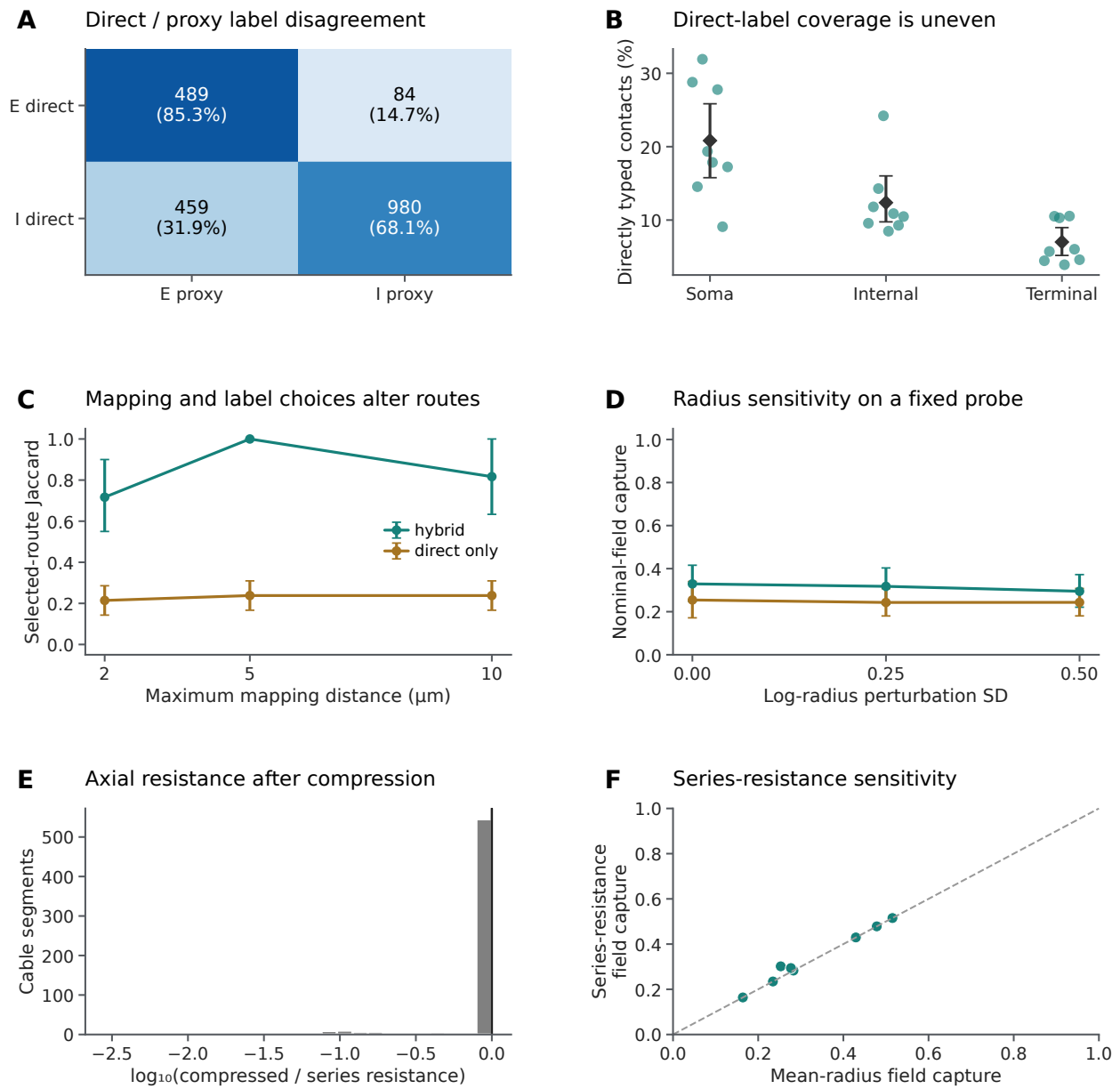
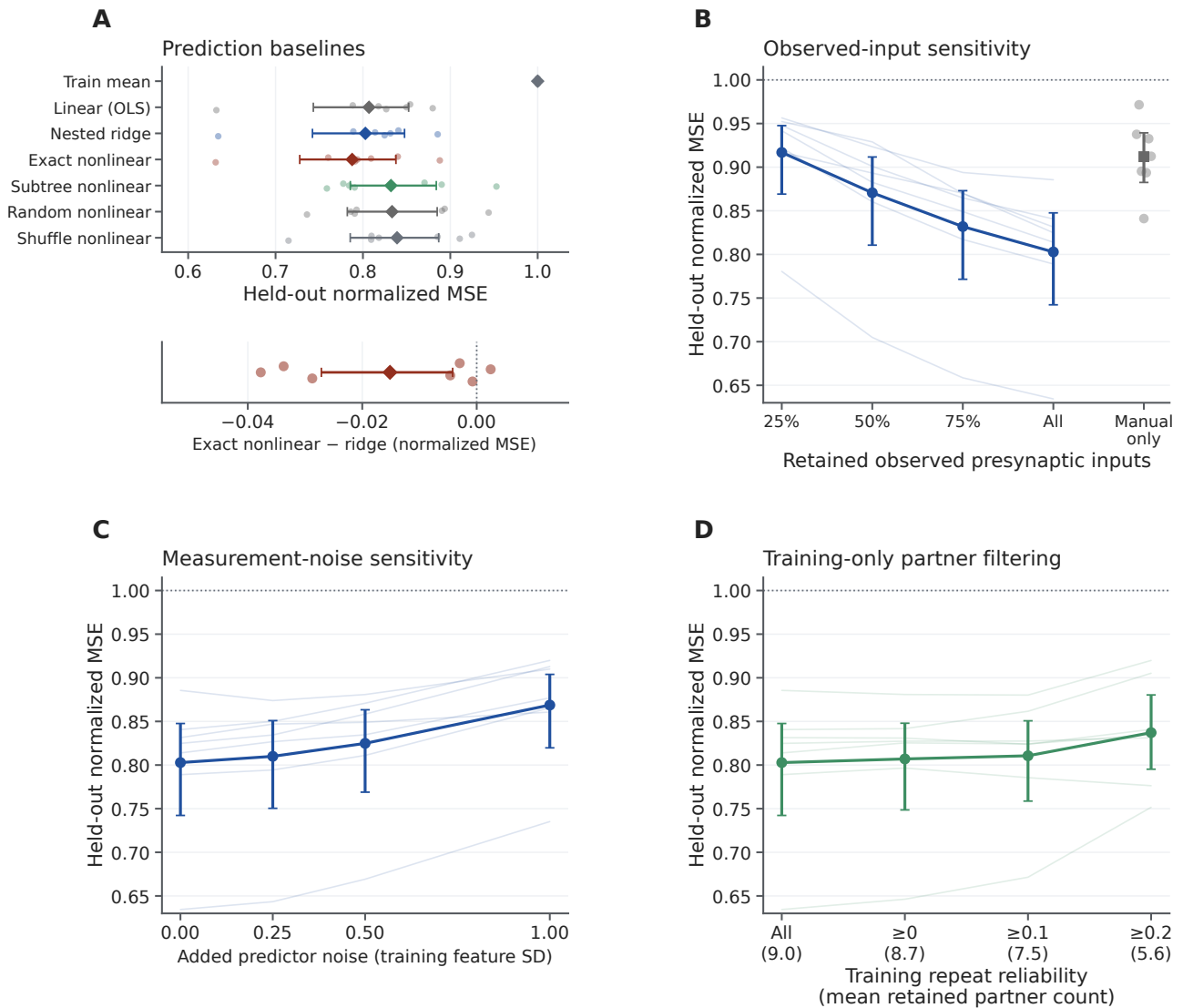


Figure S33: Label coverage and geometric preprocessing affect anatomical route dictionaries. All analyses use eight initial reconstructed cells from one mouse. **A**, Direct-presynaptic versus target-proxy excitatory/inhibitory (E/I) labels for 2,012 jointly labeled mapped contacts; counts and row percentages expose the asymmetric disagreement. Contacts are nested observations. **B**, Direct-label fraction by compartment: individual cells and cell means with 95% cell-bootstrap intervals. Label availability is not assumed random. **C**, Jaccard overlap of the four selected routes with the nominal hybrid-label, 5 μm dictionary across mapping thresholds and label choices. **D**, Capture of the fixed nominal model-generated field under heterogeneous log-radius perturbations, indexed by their standard deviation (SD); five draws per nonzero dose are averaged within cell. Curves and bars in **C,D** show means and 95% cell-bootstrap intervals (5,000 draws). **E**, Distribution of the mean-radius compressed axial resistance divided by the exact series sum over raw edges. Length preservation does not imply axial-resistance preservation. **F**, Paired cell capture under mean-radius and exact series-resistance normalizations, with leak terms held fixed. The diagonal marks equality. Across 1,056 retained sensitivity conditions, target fields, probe sites and weights remain those of the nominal model. These are preprocessing sensitivities, not calibrated geometric uncertainties or tests of endogenous task alignment.



Seven targets; thin traces/dots: target means; diamonds/lines: means and 95% target-bootstrap intervals.

Figure S34: Linear baselines explain most measured-response prediction, and degrading observed inputs worsens performance. Thirteen scans retain the complete-tree analysis's 130 outer stimulus-identity splits. All preprocessing, reliability masks and ridge tuning use training identities only. **A**, Training-mean, ordinary least-squares (OLS) and nested ridge-regression baselines against the unchanged archived nonlinear fits. Lower normalized MSE is better. Below, paired exact-nonlinear-minus-ridge differences show the small additional nonlinear benefit: mean -0.0152 (95% interval -0.0271 to -0.0042). **B**, Ridge prediction when retaining nested random fractions of observed presynaptic inputs; five masks per split are averaged before inference. Manual-only partners are a separate provenance restriction, not a random retention dose. Empty masks use the training mean and remain included. **C**, Added Gaussian predictor noise in both training and test data, in units of training input standard deviation (SD) after the quantile transform. **D**, Reliability filtering based only on training repeats; parentheses give mean retained partner counts. Thin traces or dots show seven target means after averaging splits within scan and scans within target. Larger symbols and bars are means and 95% target-bootstrap intervals (20,000 draws). All targets come from one mouse. Only ridge is refitted under the sensitivity conditions. Removing observed inputs does not recover unknown partners, establish population power or explain away the absence of preferential ancestry alignment.

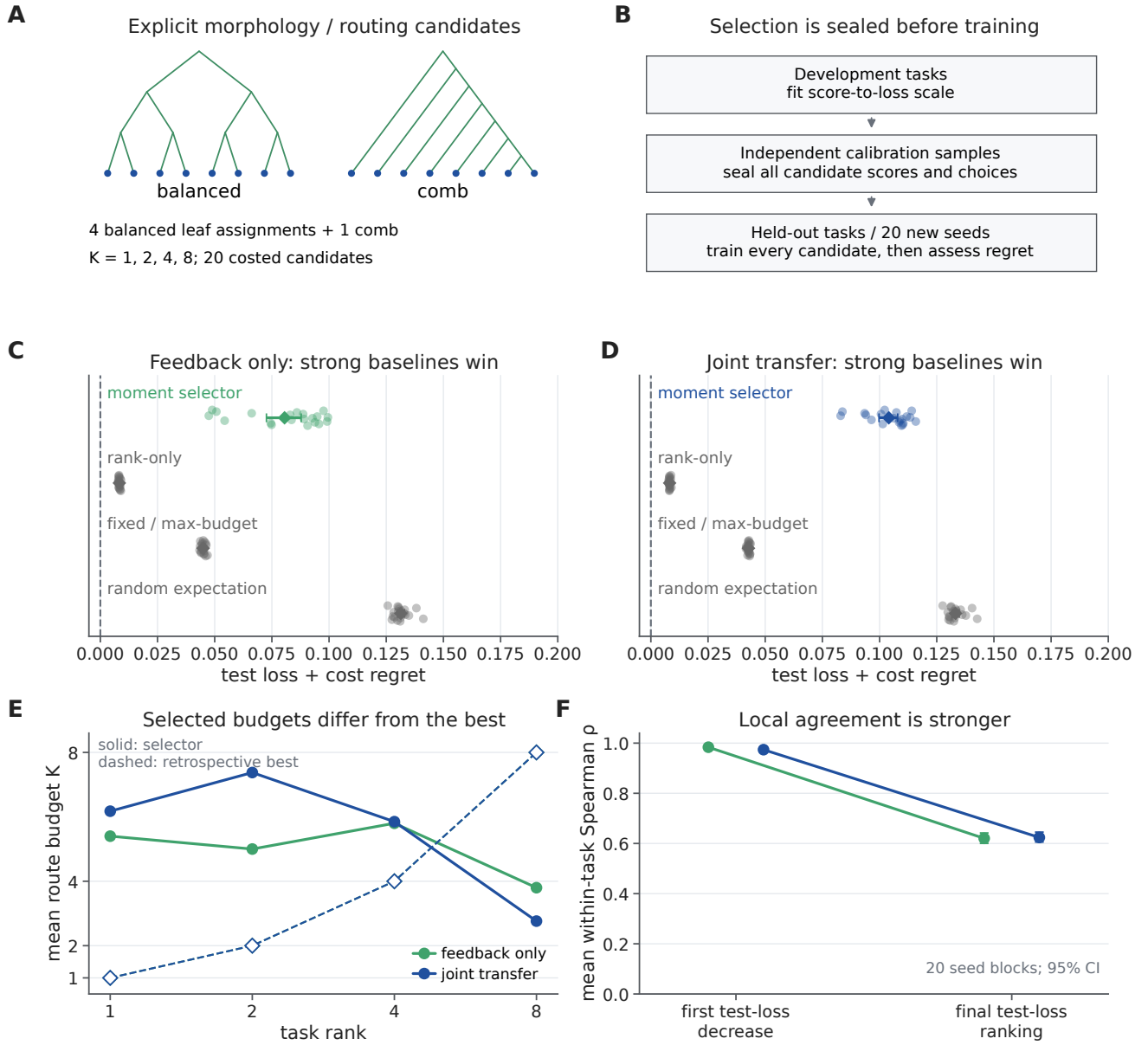


Figure S35: The original scalar initialization score predicts a first step but fails at prospective endpoint selection. This is the unchanged original 20-seed confirmation experiment, retained after replacing its main-text figure. **A**, Four balanced leaf assignments and one comb supply route budgets $K = 1, 2, 4, 8$: twenty candidates with prescribed cost; K counts feedback channels. **B**, Development, independent calibration, sealed choices and confirmation training. **C,D**, Final test half-mean-squared-error (MSE) plus cost, expressed as excess over the best trained candidate, for feedback-only and joint passive-transfer arms, respectively. Small points are twenty whole-seed means; diamonds and whiskers are means and 95% seed-bootstrap intervals (10,000 draws). The development-fixed/max-budget and privileged generating-rank baselines outperform the moment selector; random expectation averages all candidates uniformly. **E**, Selected mean budget versus imposed rank; dashed curves show the retrospectively best candidate's budget. **F**, Mean within-task Spearman correlation of initialization utility with actual first-step test-loss decrease and final test-loss ranking. Intervals resample whole seeds. All 12,800 confirmation candidate fits and 38,400 checkpoints remain included. Every model has a freely trained dense weight matrix; passive transfer changes conditioning and cost but does not necessarily restrict representation. This experiment is distinct from the locally connected multi-affine construction and the separately frozen finite-horizon follow-up.

Exact finite-domain diagnostic: 140 tasks $\times$ 12 candidates; all 256 inputs

A Fixed spectrum; different interactions

All 105 perfect matchings

$$f_M = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j$$

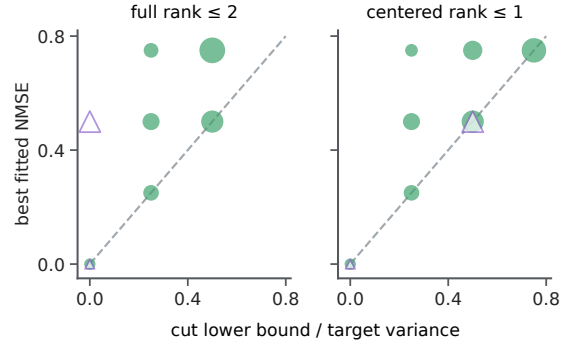
All 35 four-plus-four partitions

$$f_A = \frac{1}{2} \left(\prod_{i \in A} x_i + \prod_{i \notin A} x_i \right)$$

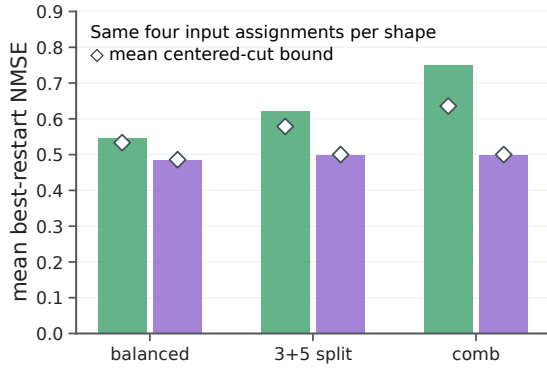
$$\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4; \text{rank} = 8$$

28 coefficients $\cdot$ 14 edges $\cdot$ root-only output

B The centered constraint is stronger



C Shape effects at matched assignments



D Selection within twelve candidates

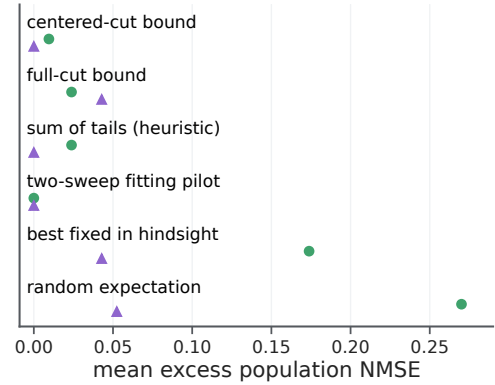


Figure S36: **Interaction constraints distinguish scalar trees at an identical input-sensitivity spectrum.** **A**, All 105 perfect matchings and 35 four-plus-four partitions of eight independent inputs $x_i \in \{-1, 1\}$, with target f and common input-gradient second moment $I_8/4$. This moment is not a model's changing parameter-gradient covariance. Every tree has one named input per leaf, seven scalar multi-affine nodes, 28 coefficients and fourteen edges, with only the root read out. **B**, Maximum cut-tail bounds from the full rank-two matrix (left) and centered rank-one matrix (right), versus the smallest achieved population normalized mean squared error (NMSE) over four alternating-least-squares (ALS) restarts. Points aggregate coincident task–candidate cases; size indicates multiplicity. Dashed lines show equality. **C**, Family means across the same four leaf assignments for each shape. White diamonds mark mean centered-cut bounds. **D**, Mean excess NMSE above the best fitted candidate in the fixed pool of twelve. Score ties use tolerance 10^{-10} and lexicographic candidate ID. The sum of tails is a heuristic, and the two-sweep pilot already fits each candidate. Best-fixed selection uses family outcomes in hindsight; random expectation averages uniformly over candidates. All 140 targets, twelve candidates, four restarts and 32-sweep fits are retained. The task sets are exhaustive finite families, so no sampling confidence intervals are drawn. A positive fit-to-bound gap may reflect optimization error, bound looseness or both. The bound and exact representation theorem apply specifically to nodes affine in each child.

Full-target Fourier construction; specific to nodes affine in each child

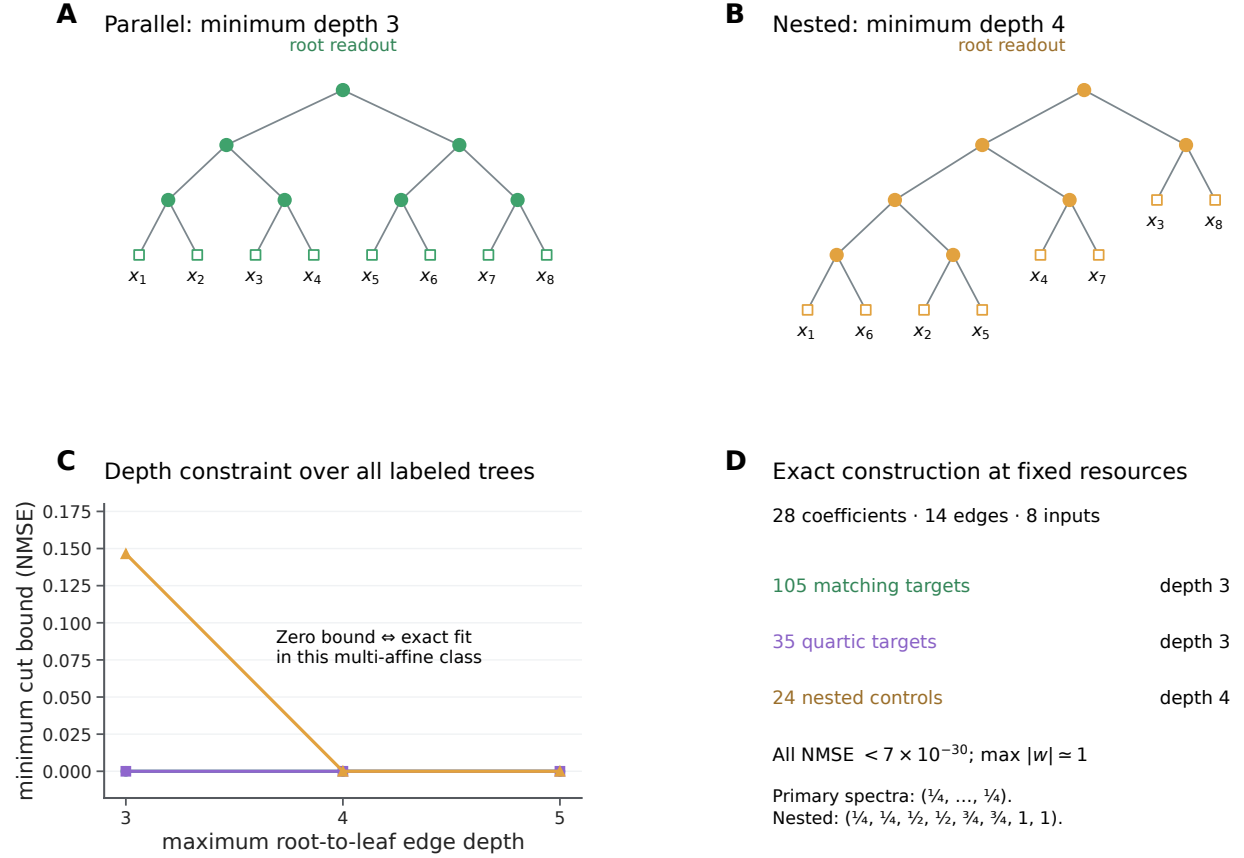


Figure S37: **Exact-target construction identifies compatible grouping and minimum depth.** **A,B**, Actual trees returned for the first enumerated matching and first seeded nested-prefix control, selected by enumeration order. Leaves access the indicated input only; internal nodes implement Eq. 65. **C**, Minimum possible maximum centered-cut bound over all labeled binary trees at each depth limit, calculated separately for every target. Curves coincide within family. Every depth-three tree incurs normalized mean squared error (NMSE) at least 0.146446609407 on each nested control; a zero-bound tree exists at depth four. Both primary families admit zero-bound trees at depth three. Proposition 9 establishes the equivalence between zero centered tails and exact representation in this class. **D**, Counts, minimum exact depths, reconstruction error and coefficient magnitude. All 164 targets reconstruct with NMSE below 7×10^{-30} and maximum absolute coefficient one, up to numerical rounding, in these constructions. Every construction has 28 coefficients and fourteen edges. The 24 unique nested controls have rank eight but an anisotropic spectrum different from the primary families; they are a separate shape/depth positive control. The corrected two-pass dynamic program uses the complete target Fourier tensor from all 256 inputs. This is an exact-target capacity certificate, not a prediction from limited calibration data or a theorem for general nonlinear or conductance dendrites.

Fresh seeds: 20 independent blocks; fixed-cache endpoint; 95% seed-bootstrap intervals

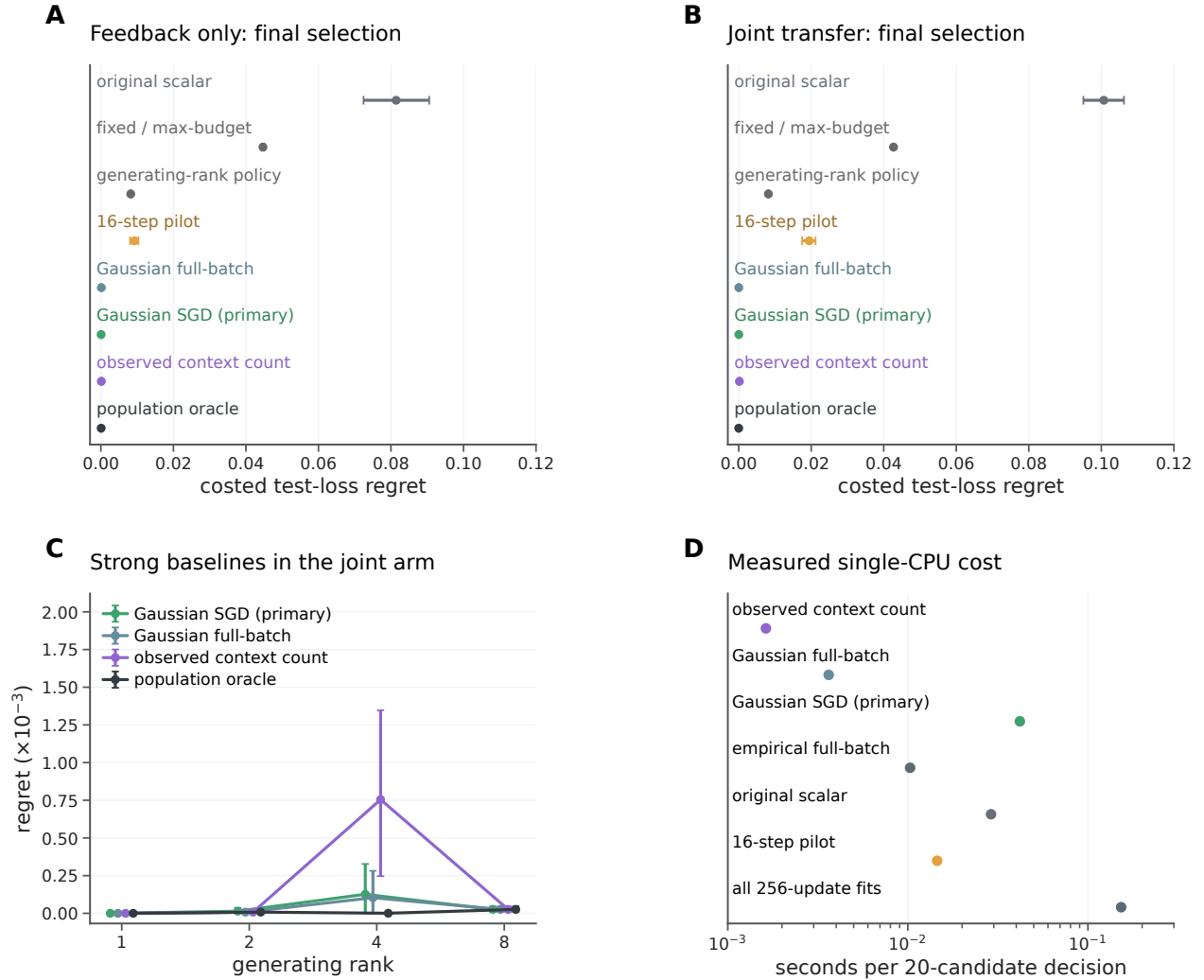


Figure S38: **Finite-horizon forecasts repair scalar extrapolation, with a strong inexpensive baseline.** The primary Gaussian stochastic-gradient-descent (SGD) forecast was frozen before twenty new seed blocks supplied 320 tasks per arm and training regime. **A,B**, Final costed-test-loss regret for the feedback-only and joint-transfer fixed-cache learners, respectively; regret is excess above the best trained candidate. All candidate outcomes are retained; generating-rank and population-oracle policies receive explicitly privileged information. The primary forecast estimates its model from independent calibration observations. The observed-context-count baseline and its conditioning variant were added after development inspection and separately frozen before fresh outcomes. **C**, Differences among strong selectors at each rank in the joint arm. The full-batch and count-only baselines remain competitive. Curves and whiskers in **A–C** are means and pointwise 95% intervals from 10,000 whole-seed bootstrap draws, after averaging nested task conditions within each of twenty seeds. **D**, Median measured elapsed time on one CPU per twenty-candidate decision. Plug-in timings include calibration-model preparation; count-only time excludes shared calibration generation; the pilot includes sixteen candidate updates and calibration evaluation. Times measure these implementations, not hardware-independent algorithmic complexity. All finite-horizon methods choose feedback budget $K = r$, the generating task rank, on all fixed-cache tasks. The large improvement over the original scalar score therefore does not establish a large practical advantage for fine morphology selection in this linear family.

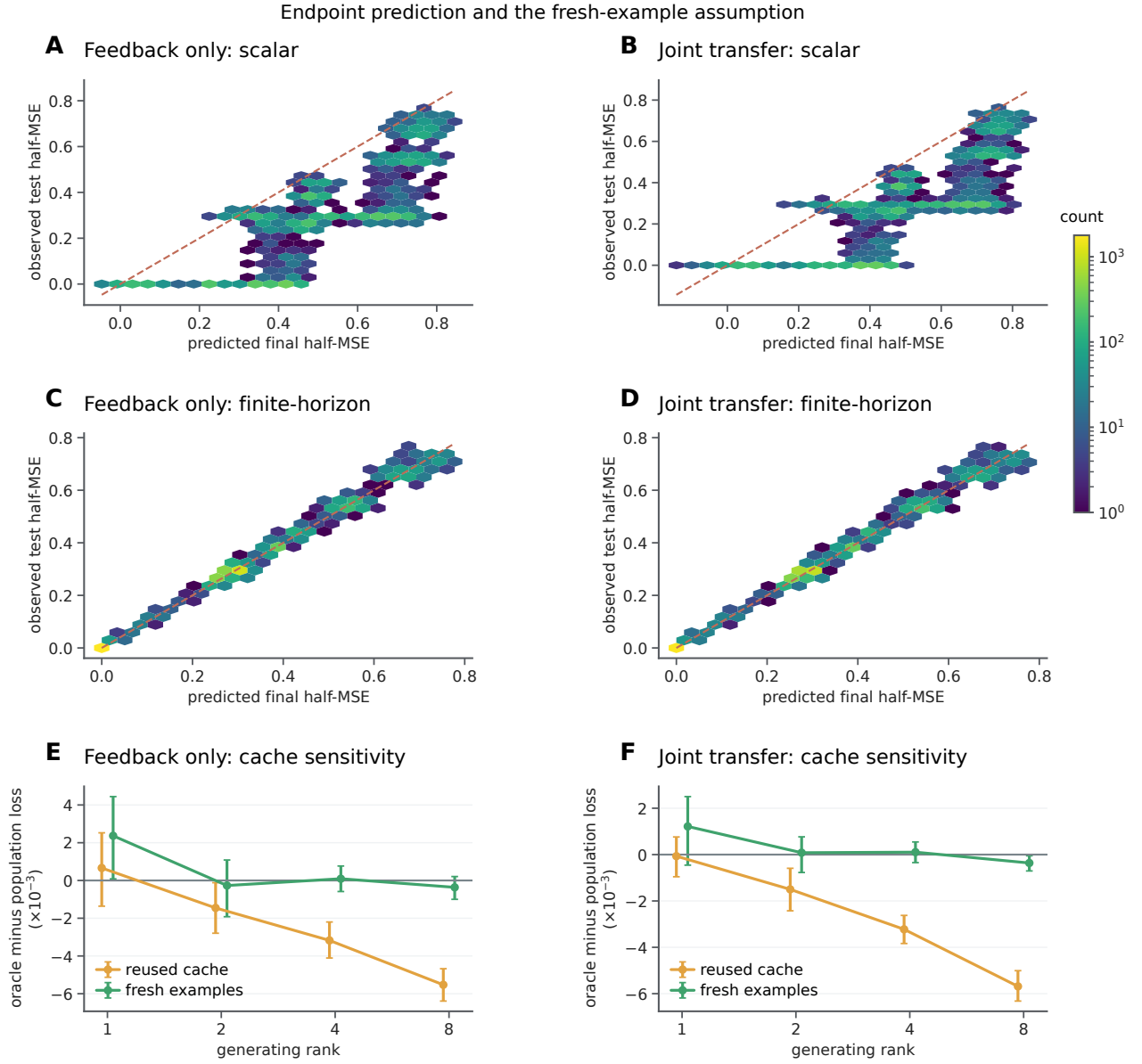


Figure S39: Endpoint forecasting improves while reused data violate the fresh-example model. **A,B**, Original scalar forecasts versus final test half-mean-squared-error (MSE) in the feedback-only and joint arms of the new fixed-cache cohort. **C,D**, The calibration-derived Gaussian stochastic-gradient-descent (SGD) forecast on the same 6,400 candidate-task cases per arm. Hexagons encode counts on a common logarithmic color scale; dashed lines mark equality. **E,F**, Population-oracle predicted loss minus the exact population loss of the realized trained weights, at label-noise standard deviation (SD) 0.75 and update 256. Fresh independent examples and reused finite-cache minibatches are paired by task seed. Means and pointwise 95% intervals use 10,000 whole-seed bootstrap resamples of twenty seed averages. The fresh-example recurrence is exact under its Gaussian-input, conditional-linear and independence assumptions; applying it to a reused cache is a stated misspecification. All 25,600 fresh-cohort candidate fits and 76,800 checkpoints across both regimes are retained. Prediction correlations that are undefined for constant numerical outcomes remain missing with explicit defined-task counts. These panels test a linear learning-dynamics model, not a nonlinear dendritic morphology law.

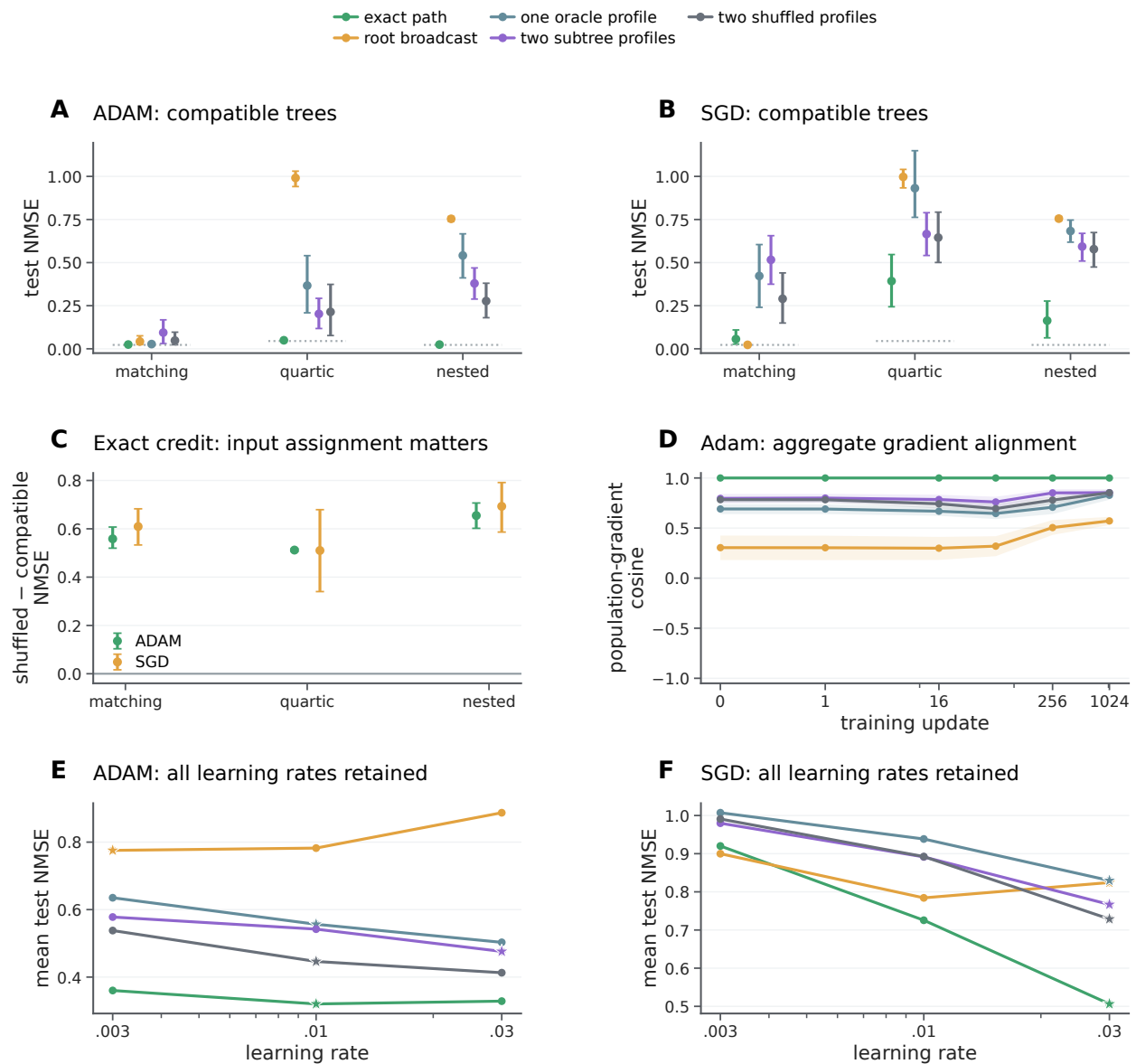


Figure S40: **Task-dependent credit requirements in bounded multi-affine trees.** **A,B**, Final test normalized mean squared error (NMSE) on oracle-compatible trees under Adam and stochastic gradient descent (SGD), respectively. Exact path, root broadcast, one oracle profile, two subtree profiles and two shuffled profiles share the forward model, initialization, examples and minibatches. Dotted segments mark known label-noise floors: 0.0225 for matching/nested tasks and 0.045 for quartic tasks. Projection coefficients access the exact current path field and are oracle diagnostics; root broadcast supplies the same scalar error to each internal node. **C**, Paired exact-credit NMSE increases after shuffling leaf input assignments while preserving tree shape and parameter count. Chance-compatible shuffles are retained. **D**, Adam gradient cosine between the exact and delivered aggregate clean-population updates, evaluated at each rule's own trained state. Each seed averages the three task families before aggregation; these are not comparisons at a common model state or mean per-example cosines. Points and whiskers in **A–C** and bands in **D** are means and pointwise 95% intervals from 10,000 whole-seed bootstrap draws, with twenty independent fresh seeds per family. **E,F**, Final NMSE at all three learning rates, averaged over both compatible and shuffled assignments and all three families. Stars mark rates selected separately for each rule and optimizer using only development seeds. These means therefore differ from compatible-only panels **A,B**. Every SGD choice was the upper tested rate; the result is a comparison of the stated training recipes, not a fully optimized SGD limit. Matching and quartic tasks share the input-gradient second moment $I_8/4$; nested tasks have a different spectrum. The oracle-compatible architectures here are independent of the noisy morphology-selection experiment. All 3,600 fresh fits and all tested rates are retained.

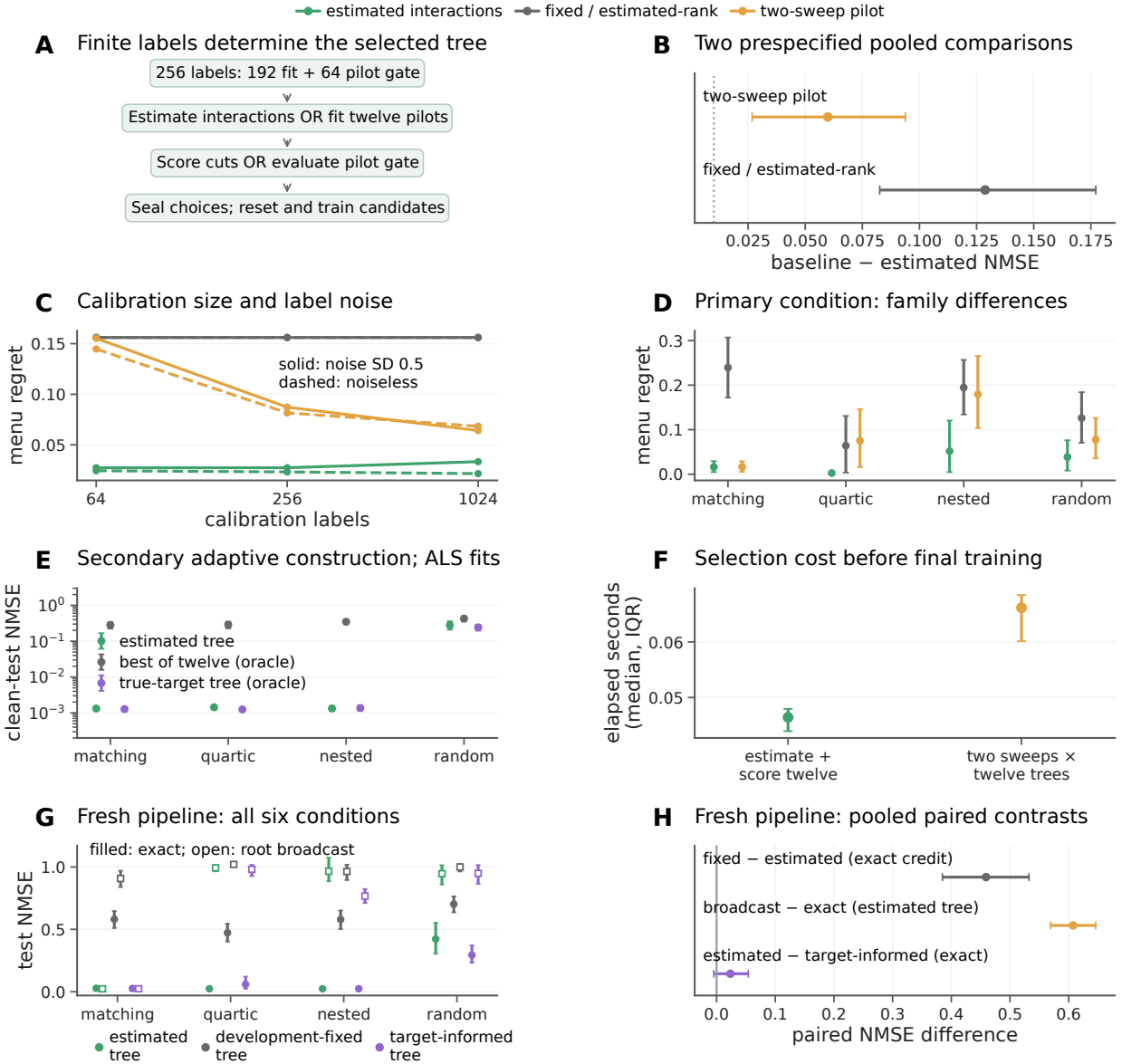


Figure S41: Finite calibration, adaptive construction and direct gradient learning. **A**, Primary calibration: 256 labels, noise standard deviation (SD) 0.5; 192 observations for estimation/pilot fitting and 64 for pilot gating. The 255-term Walsh estimator receives no family labels, true support or coefficients. The pilot fits all twelve candidates for two sweeps. Choices are sealed; training restarts independently. **B**, Prespecified pooled fixed-baseline-minus-estimated and pilot-minus-estimated normalized mean squared error (NMSE). Whiskers are Bonferroni-adjusted 97.5% paired bootstrap intervals; the dotted line marks the 0.01-NMSE meaningful-improvement margin. **C**, Excess clean-test NMSE above the best trained candidate across calibration sizes; solid curves use noise SD 0.5 and dashed curves no noise. **D**, Family-specific primary regret for estimated interactions, the development-fixed/estimated-rank baseline and the fitting pilot; pooled superiority is not uniform across families. **E**, Secondary adaptive trees trained with alternating least squares (ALS), versus the best trained twelve-candidate menu and a full-target construction oracle. Random-interaction errors remain. **F**, Elapsed time (median and interquartile range) on one CPU for eighty tasks: estimation/scoring of twelve trees versus pilot fitting/gating, excluding final training. All 2,240 ALS fits are retained. **G,H**, Independent twenty-seed Adam cohort. **G**, Three structures and two credit rules in four non-isospectral families: filled symbols, exact credit (Adam 0.01); open, root broadcast (0.003). Label-noise floor: 0.0225; **E** uses clean targets. **H**, Fixed-minus-estimated error under exact credit and broadcast-minus-exact error on estimated trees use Bonferroni-adjusted 97.5% intervals; estimated-minus-target-informed error under exact credit uses a pointwise 95% interval. Target-informed construction is not a performance ceiling. All 480 fits are retained. Other whiskers: pointwise 95% intervals, 10,000 whole-seed bootstrap draws ($n = 20$); pooled statistics average families within seed. Both cohorts are independent of Supplementary Fig. S40.

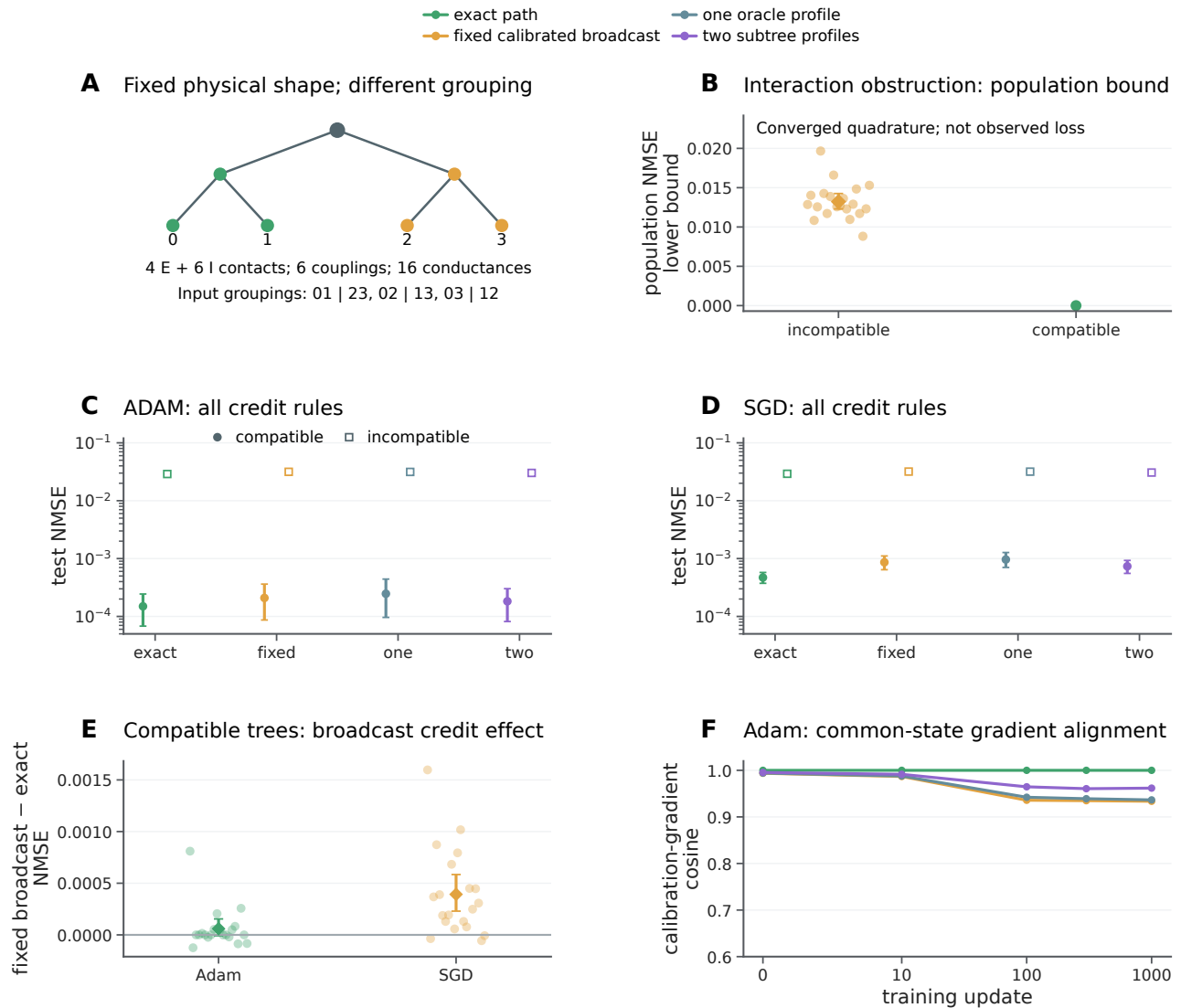


Figure S42: Input grouping and restricted feedback in positive-conductance trees. **A**, Seven-compartment directed shunting model with the same physical shape, sixteen positive conductances (four excitatory and six inhibitory contacts plus six couplings) and soma-only readout under all three input groupings. Teacher-task permutations preserve the input-gradient spectrum. These are planted mechanistic controls within the stated model class. **B**, Analytic population lower bound from interactions crossing the student's two proximal input blocks, evaluated by converged quadrature. Dots average the six incompatible task/grouping pairs within each of twenty seeds; the mean bound is 0.013222 population normalized mean squared error (NMSE). This is neither an observed loss nor a certified numerical integration bound. **C,D**, Final held-out sample NMSE for exact path, fixed calibrated broadcast, one oracle profile and two subtree-profile credit rules under Adam and stochastic gradient descent (SGD), respectively. Circles average three compatible task permutations within seed; open squares also average the two incompatible groupings per task. Rates were selected on development seeds and frozen before fresh outcomes. The fixed broadcast profile is calibrated from initial student path sensitivities; projection coefficients access the current exact field and are oracle controls. **E**, Paired compatible-tree fixed-broadcast-minus-exact NMSE differences. The Adam interval includes zero; the small SGD effect does not establish a universal need for precise credit. Dots are the twenty independent seed means. Means and whiskers in **B–E** use 10,000 whole-seed bootstrap draws and pointwise 95% intervals. Population-normalized bounds in **B** and split-variance-normalized test errors in **C–E** have distinct denominators. **F**, Mean aggregate gradient cosine on 256 calibration examples for compatible Adam models, evaluated at common exact-learning checkpoints; no uncertainty band is shown. All 1,440 fresh fits and 7,200 checkpoint records are retained. The interaction bound relies on independent input blocks, identity somatic activation and no direct somatic synapses; it is separate from the multi-affine cut-rank theorem.

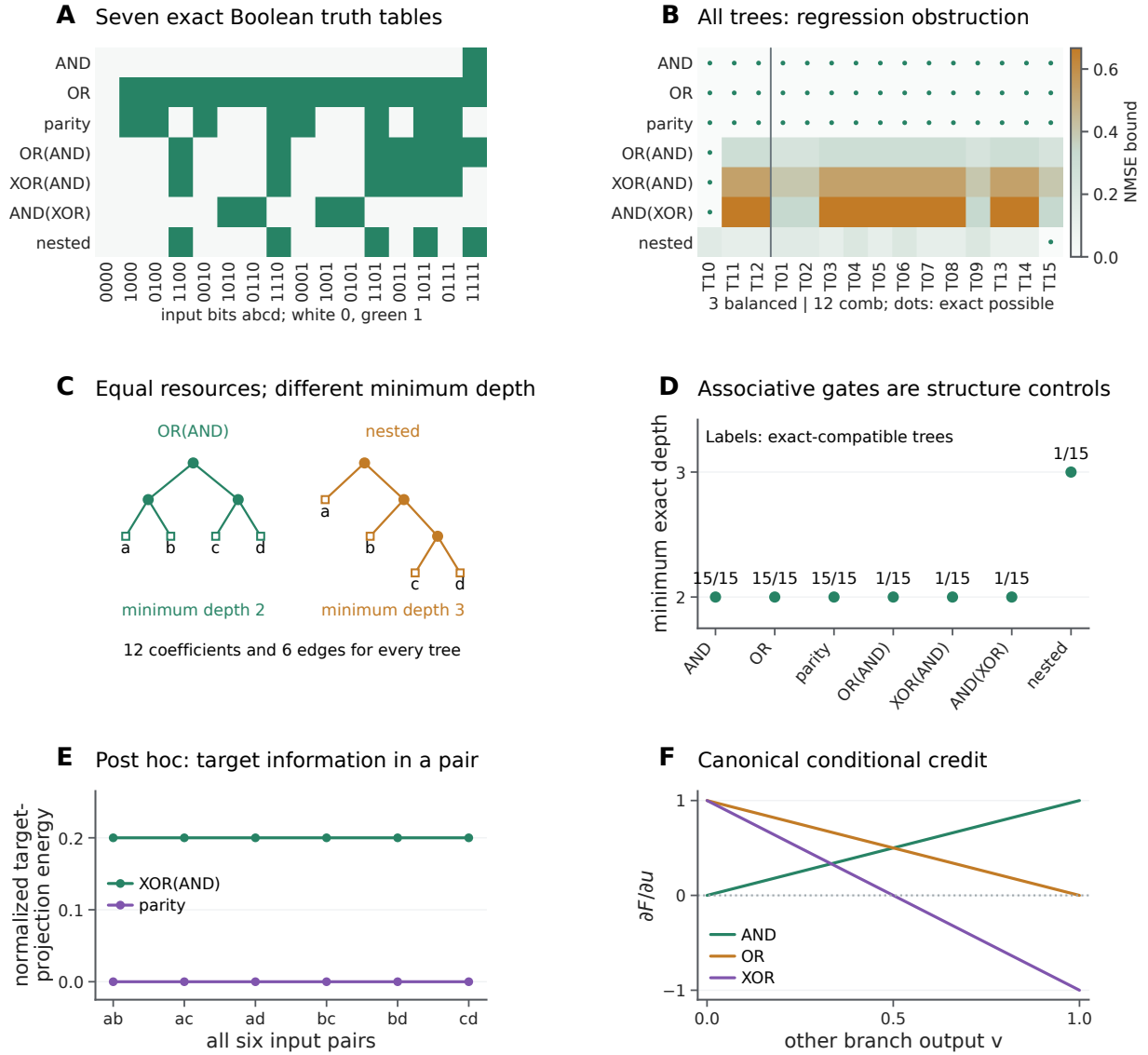


Figure S43: Boolean interactions distinguish structure, depth and canonical credit. **A**, Complete truth tables of seven four-input templates, with sixteen equally weighted patterns (white, 0; green, 1). Mixed targets are $(a \wedge b) \vee (c \wedge d)$, $(a \wedge b) \text{ XOR } (c \wedge d)$, $(a \text{ XOR } b) \wedge (c \text{ XOR } d)$, and nested $a \wedge [b \vee (c \wedge d)]$. **B**, Centered-cut normalized-mean-squared-error (NMSE) regression lower bounds for all fifteen labeled binary trees. Green dots mark 49 exactly representable family-tree pairs, independently certified with coefficients in $[-2, 2]$. The largest coefficient magnitude in these constructions is $4/\sqrt{15}$. All trees have twelve coefficients and six edges; T identifiers and cut sets are supplied in Source Data. A positive bound concerns truth-value regression, not threshold classification. **C**, OR-of-AND and the nested function each use two AND gates and one OR gate, but require minimum exact depths two and three. **D**, Minimum exact depth and compatible-tree counts. AND, OR and parity admit all fifteen trees; each mixed-gate target admits one. All seven input-sensitivity matrices have rank four, but their spectra generally differ. **E**, Explicitly post hoc target-projection energy $\|\mathbb{E}[z | x_S]\|^2$ for all six two-input subsets, with normalized target z . Parity has zero energy on every proper subset; XOR-of-AND has energy $1/5$ on every pair, including crossed pairs. This describes local target information and does not predict grouping compatibility or prove broadcast-learning failure. **F**, Canonical derivatives $\partial F / \partial u$ with respect to left-branch output u , versus right-branch output v : AND, v ; OR, $1 - v$; XOR, $1 - 2v$. These raw Boolean coordinates are illustrative; training can change every coefficient and need not preserve them. No learning outcome is inferred from a derivative sign alone. Panels are exhaustive analytic diagnostics, with no sampling error bars; the multi-affine and independent-input assumptions are explicit in Supplementary Note 4.21.

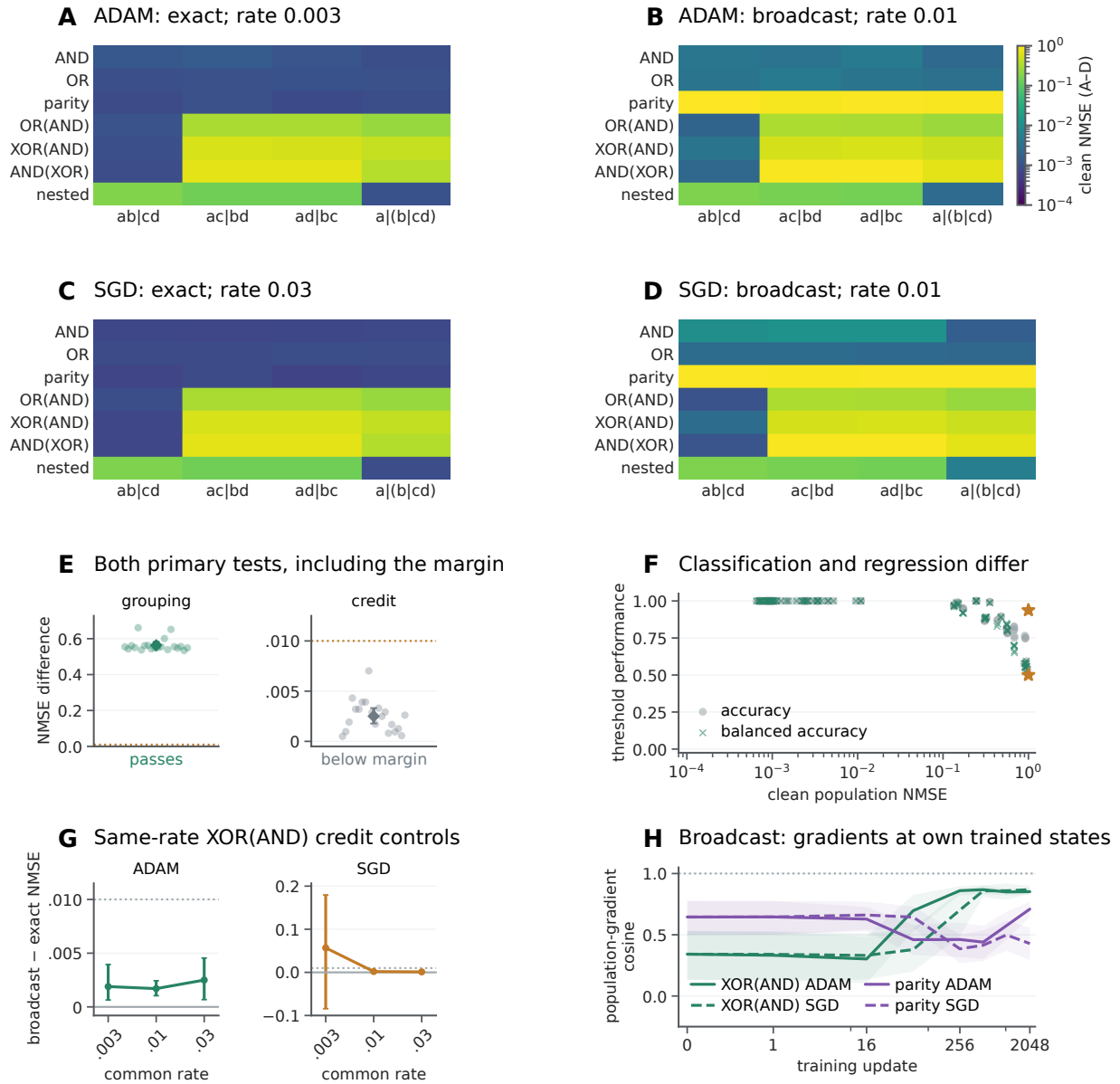


Figure S44: **Boolean grouping benefit and a small primary credit effect under frozen recipes.** **A–D**, Clean population normalized mean squared error (NMSE), averaged over twenty fresh seeds: Adam exact (**A**), Adam broadcast (**B**), stochastic gradient descent (SGD) exact (**C**) and SGD broadcast (**D**). Heatmaps share a logarithmic scale; columns show three balanced groupings and one nested tree. Rates in **A–D** are 0.003, 0.01, 0.03 and 0.01, respectively, frozen using five development seeds. Seeds vary permutations, initialization and noisy observations of fixed templates. **E**, Prespecified Adam XOR-of-AND contrasts: mean crossed-minus-aligned exact-credit error and aligned broadcast-minus-exact error. Dots, twenty paired seed differences; diamonds/whiskers, means and Bonferroni-adjusted 97.5% intervals (10,000 seed-bootstrap draws). Dotted lines: 0.01-NMSE mean-effect margin. Grouping passes both criteria; the positive credit difference is below the margin. Both compatible rules classify all sixteen patterns perfectly in every seed. The two axes have different ranges. **F**, Accuracy (gray) and balanced accuracy (green) versus NMSE for 112 condition means. Raw-output classification threshold: 0.5. Orange stars show the AND/OR constant-majority references: NMSE one, accuracy 15/16 and balanced accuracy 1/2. **G**, Same-rate credit contrasts on aligned XOR-of-AND trees, with descriptive 95% intervals; Adam and SGD axes have different ranges. The uncertain low-rate SGD result is retained. **H**, Broadcast gradient cosine on the aligned balanced tree for XOR-of-AND (green) and parity (purple), with solid Adam and dashed SGD curves, evaluated on all sixteen clean patterns at each learner's own state, before clipping or optimizer transformation. Bands are pointwise 95% seed-bootstrap intervals. The large parity credit effect is descriptive. All rates, 6,720 fresh fits and 53,760 checkpoints are retained; statistical scopes were fixed before training.

2715 **S13 Statistical analysis, reproducibility and interpretation**

2716 Central comparisons report paired effect sizes, the number of positive cell or seed pairs, and 95% intervals unless
2717 a different level is specified for that analysis. Tests were two sided unless a direction had been specified before
2718 analysis. Exact Wilcoxon signed-rank tests were used for small paired cell cohorts. Reconstructed-cell intervals
2719 resampled cells first and, where relevant, one Monte Carlo stream within cell. Measured-response intervals
2720 resampled target cells. Artificial-network summaries used independently initialized training runs. Inhibitory-census
2721 outcomes were first averaged within postsynaptic target. Signed-credit intervals and exact sign tests used animals;
2722 neuron-level source rows were descriptive only.

2723 Bootstrap intervals used 20,000 draws unless an experiment's protocol states otherwise; the Fashion-MNIST
2724 ladder, the physical-depth, H4 and task-family cohorts and the external animal reanalysis used 50,000 paired draws.

2725 Benjamini–Hochberg correction was applied within each prespecified comparison family—the archived four-
2726 budget subtree scan across $K = 1, 2, 4, 8$, the six bandwidth-matched routing contrasts and the H4 primary family.
2727 The additional four-budget ancestry-minus-best-control reanalysis enumerated all 2^{20} paired sign assignments
2728 for a two-sided mean-difference test and applied Holm correction across the four budgets. Accuracy differences
2729 were reconstructed on the exact $1/2048$ test-count grid before testing. Initialization-score correlation intervals
2730 used 5,000 whole-seed bootstrap draws. The local-context soft-encoder analysis used Holm correction across its
2731 three primary contrasts; the hard-readout follow-up remains exploratory. The prospective selection experiment
2732 resampled 20 whole seed blocks 10,000 times. The finite-horizon, morphology-calibration, credit and conductance
2733 studies also used 10,000 whole-seed draws, retaining task, candidate and optimizer conditions together within each
2734 seed. The two primary comparisons in each of the finite-calibration and separate end-to-end experiments used
2735 Bonferroni-adjusted 97.5% intervals and a prespecified 0.01 improvement margin in normalized MSE (NMSE).
2736 Other bridge intervals are pointwise and descriptive. Exhaustively enumerated structural targets have no biological
2737 sampling interpretation. Response-baseline sensitivity retained seven targets and 20,000 draws. Other parameter
2738 sweeps and sensitivity analyses received no multiplicity correction, are labeled secondary, and are not interpreted
2739 as independent confirmation.

2740 For the cross-animal structural check, the mouse is the replication unit. Contrasts were first summarized
2741 within each volume; no cell-level test was used to claim an animal-level P value from two mice. The 12-cell
2742 Pinky cohort was sized to span 12 equal-count spatial strata while keeping complete mesh skeletonization and
2743 500-shuffle processing feasible. The task-family factorial used ten paired seeds, matching the sample size used for
2744 the physical-depth experiments; positive directional language requires a paired 95% interval excluding zero and at
2745 least 8/10 seed signs.

2746 Cross-architecture conclusions exclude pooled error-field metrics that mix dendritic stages with a somatic stage
2747 whose reconstruction is exact by construction. They also exclude unweighted path-gain dispersion as evidence of
2748 gradient direction and post hoc comparisons between unmatched training cohorts. Table S34 records the supported
2749 interpretation and boundary of each evidence class.

2750 Every empirical main panel is accompanied by a machine-readable source table containing the plotted values,
2751 biological or training unit, nesting variables, seed, condition names, and analysis version. Automated reproducibility
2752 checks verify that figure scripts read these tables rather than duplicating values in plotting code. Authentication
2753 tokens, restricted caches, and raw data already hosted by MICrONS or DANDI are not redistributed.

2754 The release also includes the three regular-tree sweep files, representative resolved configurations, the versioned
2755 run manifest, the seed-level result tables, and an executable NumPy reference for the raw additive forward model
2756 and derivatives. Exact source copies of the morphology, routing-capacity, focal-perturbation and projection-bound
2757 analyses are provided with SHA-256 provenance. The measured-response directory contains the CAVE join,
2758 extraction of Neurodata Without Borders (NWB) files from DANDI, and task-derived learning scripts together
2759 with a machine-readable four-channel specification and target manifest. The expanded release also contains the
2760 stable-ID inhibitory-census join, coordinate-transform checks, target-level topology tables, the verified Francioni
2761 source checksum and extracted contrasts, and the static interference verification grid. Private filesystem paths in
2762 portable configurations were replaced by paths relative to the article directory; numerical settings were not changed.
2763 The provenance manifest records each origin and every portability edit.

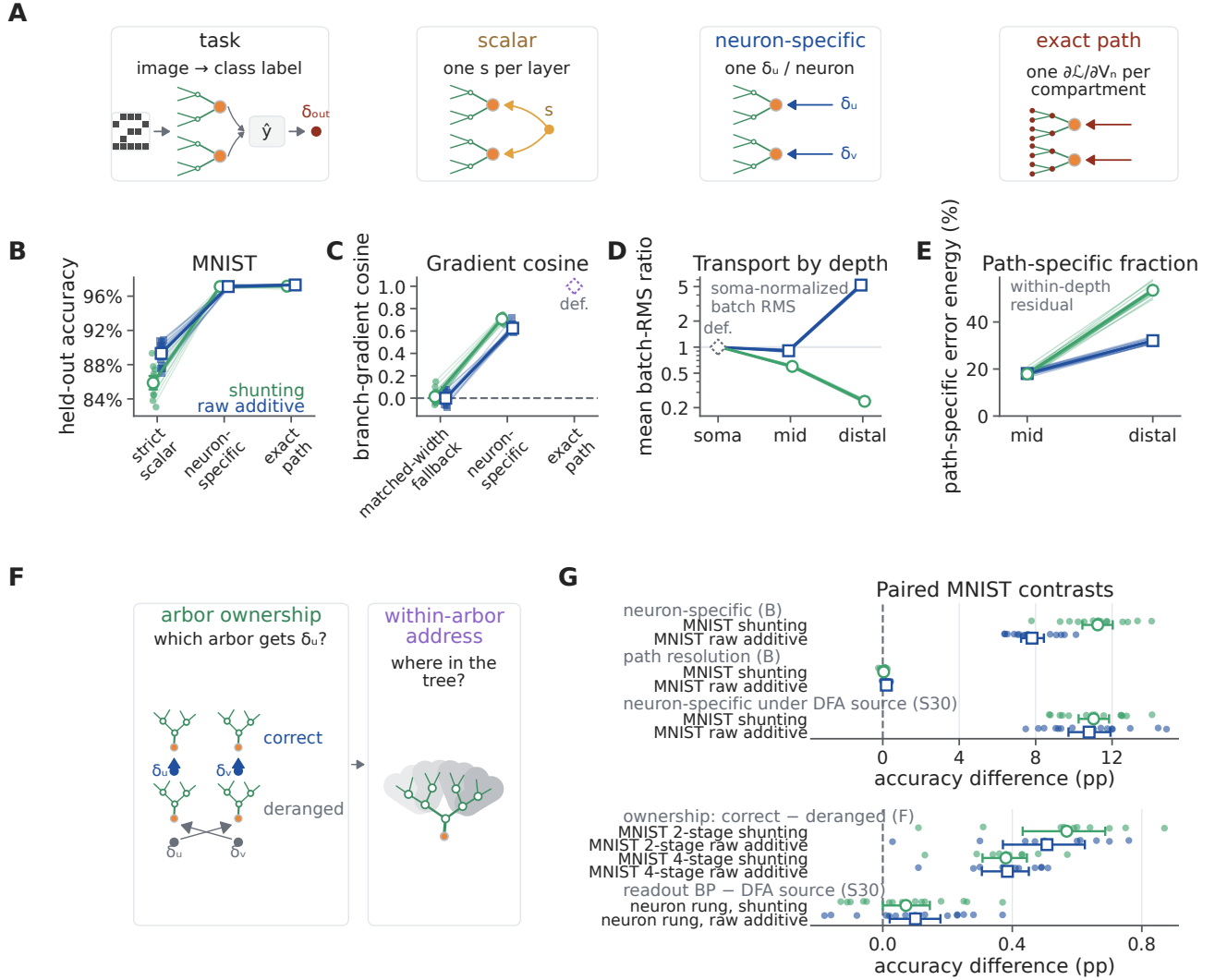


Figure S45: Neuron-specific feedback accounts for most of the MNIST accuracy gain. Panels A–E and upper G use flattened MNIST and 128 directed [3,3] trees. Green circles denote shunting and blue squares raw-additive. The purple diamond in C marks the exact-path cosine of one by definition, not a measured cohort. Card colors in A,F distinguish concepts. A, The feedback ladder supplies one strict scalar per layer, one coordinate per neuron repeated over its arbor ($K = 1$), or exact compartment errors. The strict scalar s follows the population-error reduction defined in Methods. B, Held-out accuracy. C, Branch-gradient cosine with the exact gradient at fixed checkpoints trained with neuron-specific feedback. Matched-width scalar-fallback and neuron-specific fields are evaluated at identical weights and states. D, Exact voltage-error magnitude by depth at exact-path checkpoints from B: the batch root-mean-square (RMS) ratio $\text{RMS}(\delta_{n,u}^V) / \text{RMS}(\delta_{0,u}^V)$, averaged over neurons and paths. Indices $n, 0, u$ denote compartment, soma and neuron. The log ordinate has somatic value one by normalization. E, Residual error-energy fraction after subtracting the mean across paths within depth d : $\|\delta_d^V - \langle \delta_d^V \rangle_{\text{path}}\|_2^2 / \|\delta_d^V\|_2^2$. The soma has zero residual and is omitted. Distal shunting attenuates magnitude but retains a larger path-specific fraction; no additional models are trained. F, Ownership assigns a neuronal coordinate to its arbor; four gray shades distinguish within-arbor addresses. G, Upper: neuron-specific-minus-strict-scalar and exact-path-minus-neuron-specific effects, followed by the neuron-identity contrast under fixed random direct feedback alignment (DFA; Supplementary Fig. S30). Lower: correct-minus-deranged ownership in separate [2,2] and [2,2,2,2] cohorts (six and 30 nonsomatic compartments), and readout-backpropagation-minus-DFA effects at the neuron rung (path-rung gap ≤ 0.2 percentage points). The strips have different percentage-point (pp) scales. Small dots are paired differences; symbols and bars are means and 95% seed-bootstrap intervals. There are 15 paired seeds per architecture in B–E, upper G and S30, and ten in the ownership experiments.

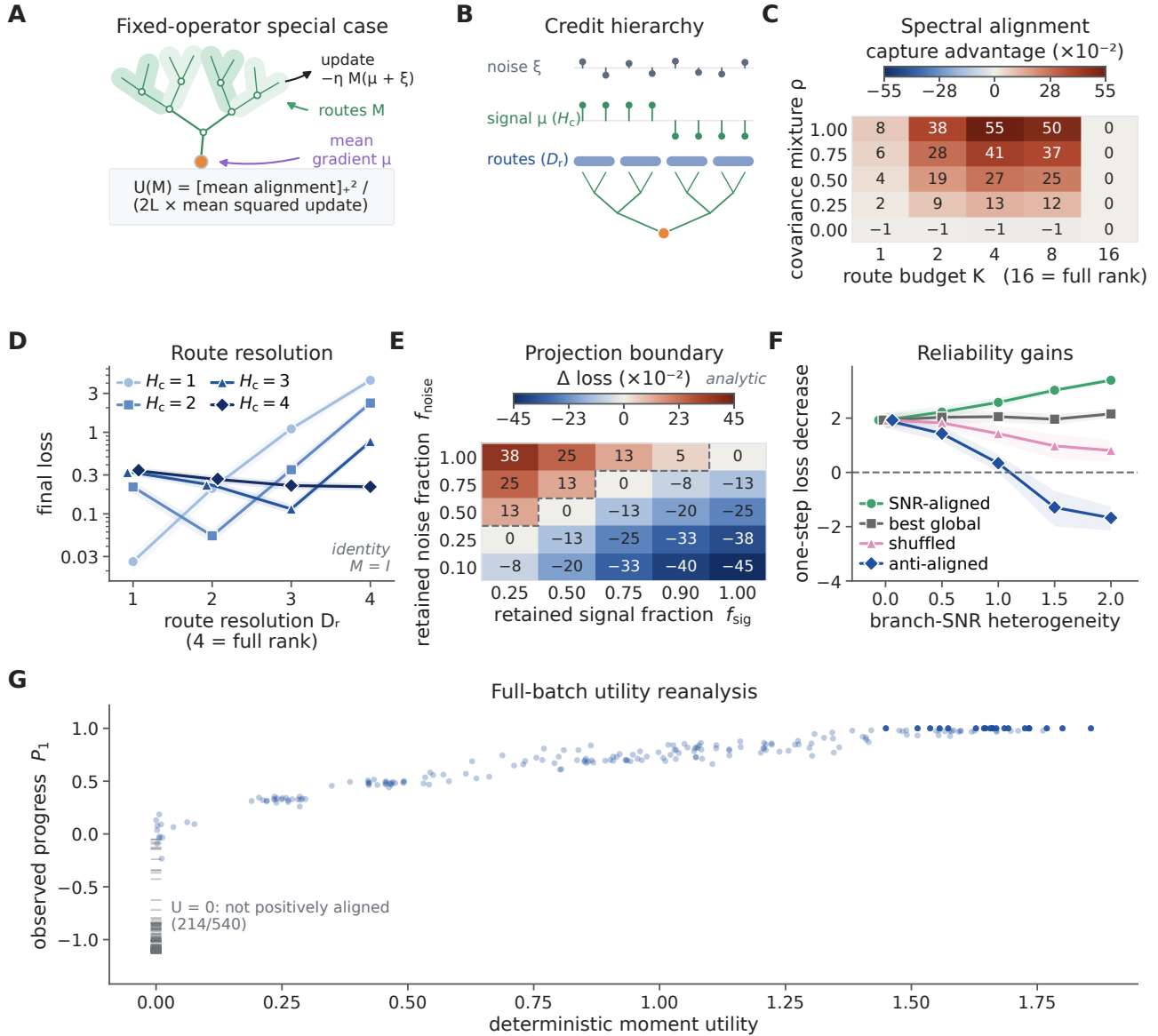


Figure S46: Update utility relates routing to signal, noise and training. **A**, Fixed-operator special case: M maps $\mu + \xi$ to a delivered update; μ is the mean gradient, ξ zero-mean noise with covariance Σ , L the curvature bound and η the step size. In the schematic, mean alignment is $\mu^T M \mu$, and mean squared update is $\mathbb{E} \|M(\mu + \xi)\|^2$. The positive part $[z]_+ = \max(z, 0)$ excludes ascent directions. The general bound instead uses the delivered update's mean and covariance (Supplementary Section S3); utility is zero for nonpositive descent alignment. **B**, Eight-terminal illustration of signal hierarchy H_c and feedback resolution D_r . Quantitative screens use 16 Haar coordinates and no forward neuron. **C**, Subtree-minus-random capture over budget and covariance-mixture coefficient ρ . Endpoints are isospectral; intermediate mixtures also change concentration. The full-rank contrast is zero. **D**, Final quadratic loss; amplified fine-scale noise makes matched signal and route resolution the designed optimum. **E**, Analytic projected-minus-unprojected one-step loss, $(f_{\text{noise}} - f_{\text{sig}})/2$, where f_{sig} and f_{noise} are retained signal and noise fractions under the stated normalization. Negative values favor projection. **F**, Gains matched to signal-to-noise ratio (SNR), a common global gain, shuffled gains and reversed gains over eight blocks; a point implementation supplied with identical gains matches. **G**, Retrospective full-batch analysis of 540 fits (27 conditions, 20 paired seeds; main Fig. 3). Deterministic moment utility with global logistic curvature correlates with norm-matched one-step progress (P_1) at Spearman $\rho = 0.951$, 95% seed-block interval 0.949–0.954. Nonpositive alignment gives zero utility in 214 fits. Panels **C,D,F**: 50 paired seeds; lines/ribbons in **D,F** follow the specified analytic construction. Panel **G**: 5,000 bootstrap draws over 20 seed blocks.

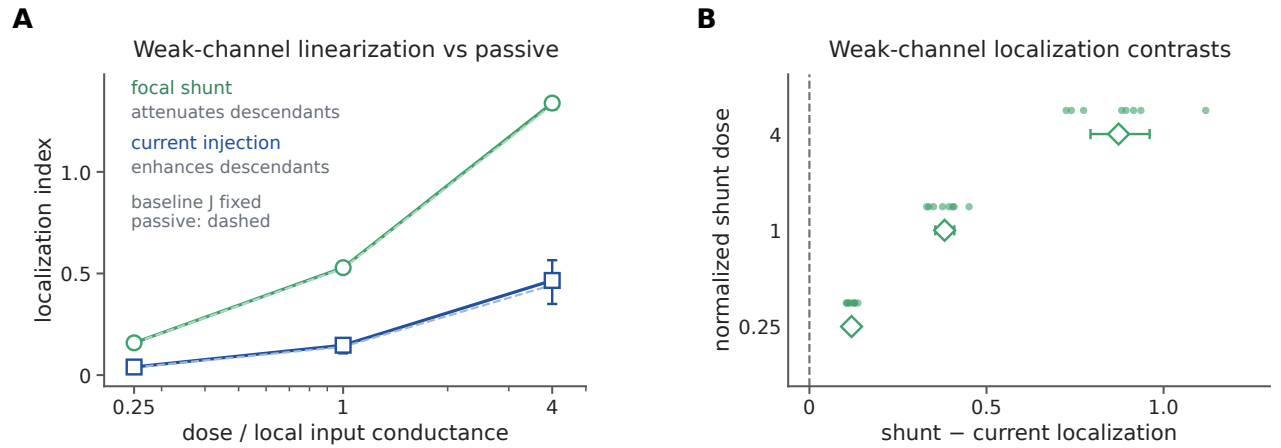


Figure S47: Weak voltage-dependent currents provide a local-linearization check near the passive response. **A**, Focal-shunt and baseline-current-matched injection localization across doses relative to local input conductance. Solid curves use a fixed Jacobian around each solved weak-channel steady state; dashed curves show the passive model at the same electrical calibration. The Jacobian J includes baseline voltage-gate derivatives. A separately solved somatic current restores baseline somatic voltage within each linearized intervention. Shunting attenuates descendant gradients; the matched injection enhances them, although the unsigned localization index is positive for both. **B**, Paired shunt-minus-current localization contrasts. Small points show eight cell means; larger diamonds and bars show cohort means and 95% cell-bootstrap intervals. The three doses all have positive contrasts in eight of eight cells (two-sided Wilcoxon $P = 0.0078$). The 64 channel draws and focal sites are averaged within each cell and are not independent replicates. Both panels use $R_a = 150 \Omega \text{cm}$, $R_m = 1,000 \Omega \text{cm}^2$ and background conductance equal to baseline leak. Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times excitatory conductance. This weak-channel ensemble tests the local calculation, not strong regenerative responses, channel kinetics or a newly solved nonlinear equilibrium at each dose.

2764 An exact package lock was not archived for the June exact-transport or backpropagation jobs, nor for the
2765 measured-response analysis. These versions are therefore not guessed from the current environment. The complete
2766 July feedback analysis records Python 3.10.13, its Git identity and selected source hashes. Its archived source
2767 state nevertheless includes tracked modifications. Table S12 and the accompanying reproducibility README
2768 distinguish exact, partial and unverifiable provenance fields.

Table S2: Confirmatory subtree-address factorial. K is feedback-channel budget; entries are mean accuracies or paired mean accuracy differences across 20 seeds. All accuracies and contrasts are fractions (0.01 is one percentage point). The best matched route control is the post-training maximum within seed over route derangement, depth-interleaved, random-sparse and random-rank fields. Positive contrasts favor correct ancestry.

K	Correct accuracy	Versus deranged	Versus best control	Versus rewired tree
1	0.1866	0.0000	−0.3583	0.0000
2	0.4486	0.2639	−0.1540	0.2315
4	0.8010	0.6106	0.0127	0.0502
8	0.8100	0.5587	0.0000	0.0000

For the best-control contrast, 0/20, 5/20, 15/20 and 0/20 paired seeds favored ancestry at $K = 1, 2, 4, 8$, respectively; the full-rank conditions are exactly tied by construction. At $K = 4$, the paired bootstrap 95% interval was 0.00593–0.01946. The main display uses the paired Wilcoxon signed-rank test with Holm-adjusted $P = 0.0101$ across four budgets. For sensitivity, an exact two-sided paired mean-sign-flip test gave $P = 0.00259$ (Holm-adjusted $P = 0.00519$ across $K = 1, 2, 4, 8$). The archived two-sided Wilcoxon test gave $P = 0.0048$ (Benjamini–Hochberg-adjusted $P = 0.0064$); the new sign-flip test uses the paired mean and a different multiplicity correction. Integer accuracy counts on the 2,048-example test set were restored before testing to preserve ties. Correct-versus-rewired intervals were 0.210–0.254 at $K = 2$ and 0.042–0.060 at $K = 4$, with 20/20 seeds positive in both comparisons.

Table S3: Credit-operator signal–noise experiment and factorial reanalysis. The signal, depth and reliability contrasts use 50 independent paired seeds. The depth contrast combines four task depths within seed; its positive reported magnitude favors matched depth. K denotes channel budget, ρ covariance-mixture coefficient and V delivered-update noise covariance. Correlation intervals use 5,000 whole-seed bootstrap draws over 20 seeds in the 540-fit dendritic-tree subset. The factorial was trained with full batches; the minibatch score is a hypothetical diagnostic.

Endpoint	Effect	95% interval	Positive	Control / interpretation
Ancestry–random capture fraction, $\rho = 1, K = 4$	0.5520	0.5353–0.5686	50/50	Dense rank- K upper bound; full rank tied
Matched-depth benefit in final quadratic loss	0.0957	0.0782–0.1134	48/50 seeds	Best mismatched minus matched loss; four task-depth contrasts averaged within seed
Aligned–global one-step loss decrease	1.2415	1.1160–1.3728	50/50	Quadratic-loss units; explicit point gain matched exactly
Full-batch utility versus one-step progress, Spearman ρ	0.9509	0.9486–0.9542	—	Global logistic curvature bound; $V = 0$
Full-batch utility versus final accuracy, Spearman ρ	0.9242	0.9108–0.9401	—	Same initialization score
Hypothetical minibatch utility versus one-step progress, Spearman ρ	0.9373	0.9337–0.9412	—	Batch size 64; estimated V
Hypothetical minibatch utility versus final accuracy, Spearman ρ	0.9157	0.9049–0.9289	—	Global logistic curvature bound

Two-sided Wilcoxon values for the spectral, depth and reliability contrasts were 1.78×10^{-15} , 3.00×10^{-13} and 1.78×10^{-15} , respectively. All 25 signal–noise condition means followed the analytical projection boundary. Static route-wise gains preserved projectors to 1.33×10^{-15} . The signal–noise experiment is a fixed-state quadratic validation, not a trained positive-conductance network.

Table S4: Step-consistent state-matched positive-conductance reliability experiment at maximum branch signal-to-noise-ratio (SNR) heterogeneity. Effects are paired mean differences across 50 confirmatory seeds; intervals are paired-seed bootstrap intervals. Positive one-step effects favor aligned shunting; negative final-loss effects favor the left-hand aligned condition. The count column reports positive signed differences, not the number of seeds favoring aligned shunting.

Endpoint	Effect	95% interval	Positive signed differences	Two-sided Wilcoxon P
Aligned–global, one-step loss decrease	0.001817	0.001618–0.002020	49/50	3.55×10^{-15}
Aligned–no shunt, one-step loss decrease	0.001339	0.000594–0.002096	33/50	0.0030
Aligned–global, final test loss	−0.020404	−0.022220–−0.018705	0/50	1.78×10^{-15}
Aligned–shuffled, final test loss	−0.009204	−0.012171–−0.006387	7/50	6.92×10^{-9}
Aligned–anti-aligned, final test loss	−0.013781	−0.017077–−0.010708	3/50	9.52×10^{-13}
Aligned–no shunt, final test loss	0.000425	−0.002456–0.003071	33/50	0.154
Aligned–explicit point gain, final test loss	0	0–0	0/50; 50 ties	1

The independently computed state-matched unattenuated current control agreed with the unshunted noisy rule to numerical precision. Exact clean backpropagation had lower mean final loss than every attenuated condition. Shunts were fixed from initial oracle reliabilities and forward voltage was clamped exactly; excitatory conductance parameters, not shunts, were trained.

Table S5: Adaptive local conductance-reliability experiment at maximum heterogeneity. Effects are paired mean differences across 50 analysis seeds, with paired-seed bootstrap intervals. Positive one-step values favor adaptive local; negative final-loss values favor adaptive local over the named control. The ordering column counts seeds favoring adaptive local, using the sign appropriate to each endpoint.

Endpoint	Effect	95% interval	Local favored	Two-sided Wilcoxon P
Local-global, one-step loss decrease	0.001420	0.001158–0.001684	45/50	4.49×10^{-13}
Local-global, final test loss	–0.016217	–0.017913–0.014501	50/50	1.78×10^{-15}
Local-shuffled, final test loss	–0.005528	–0.007418–0.003766	41/50	7.80×10^{-8}
Local-no shunt, final test loss	0.004250	0.002499–0.005975	11/50	1.78×10^{-5}
Local-initial oracle, final test loss	0.003449	0.002577–0.004305	7/50	4.44×10^{-9}
Local-adaptive point gain, final test loss	0	0–0	0/50; 50 ties	1

The estimator used paired noisy local gradient observations, not oracle signal/noise energies. It retained paired probes, a known step fraction and an exact state-matching clamp. Adaptive local recovered relative placement value but did not beat the unshunted rule.

Table S6: Same-span noisy coefficient-learning contrasts after 80 updates. Effects are paired population-loss differences across 50 confirmatory target-and-noise seeds, with paired-seed bootstrap intervals. n_{eff} controls observation-noise variance as defined below the table. Positive values mean greater loss than the orthonormal Haar coordinates.

Coordinate contrast	n_{eff}	Effect	95% interval	Positive	Two-sided Wilcoxon P
Raw nested-Haar	4	–0.2774	–0.3260–0.2313	0/50	1.78×10^{-15}
Scaled nested-Haar	4	–0.1906	–0.2500–0.1335	10/50	1.05×10^{-7}
Raw nested-Haar	256	0.02555	0.02171–0.02952	50/50	1.78×10^{-15}
Scaled nested-Haar	256	0.17091	0.14576–0.19686	50/50	1.78×10^{-15}

All dictionaries had the same rank-eight projector. Gram condition numbers were 1 for Haar, 15 for raw nested and 388.52 for scaled nested coordinates. Here n_{eff} sets observation-noise variance proportional to $1/n_{\text{eff}}$; it is an effective sample-size parameter. Median exact-risk crossover sample sizes were 38.53 for raw nested and 8.25 for scaled nested coordinates. Oracle Gram-preconditioned field trajectories agreed to 1.15×10^{-15} .

Table S7: Weak-channel steady-state linearization check. Reported cell means first average 64 accepted channel draws and focal sites within each of eight reconstructed cells. Localization is descendant-minus-control median absolute natural-log gradient change, not a fraction. Energy ratios divide post-shunt descendant squared-gradient energy by its baseline value. Interventions use each accepted state’s fixed local Jacobian, including baseline gate derivatives. The small prespecified channel conductances test the local calculation around a high-conductance baseline; they do not test strongly regenerative dynamics.

Dose	Shunt localization	Current-injection localization	Shunt minus current injection	Shunt energy ratio
0.25	0.1587	0.0398	0.1189	0.7154
1	0.5287	0.1470	0.3817	0.3302
4	1.3387	0.4657	0.8730	0.0625

Dose is relative to local input conductance. Shunt-localization cell-bootstrap intervals were 0.1538–0.1628, 0.5130–0.5414 and 1.3066–1.3642. At unit dose, the shunt-minus-current localization contrast was 0.3817 with a cell-bootstrap interval of 0.356–0.409. At every dose and in every cell, all modeled descendant gradients were attenuated without enhancement or sign reversal. Draws are nested sensitivity samples, not independent biological replicates.

Table S8: Functional topology across all eligible scans. Associations between anatomical proximity and response similarity were computed within 13 scans, averaged within each of seven target cells, then summarized across targets. Positive values indicate stronger response similarity for greater shared ancestry or proximity; the partial correlation controls Euclidean separation and soma-to-contact depth differences.

Endpoint	Mean	95% target-bootstrap interval	Positive targets
Shared-path Spearman ρ	-0.0245	[-0.2142, 0.1466]	4/7
Negative tree-distance Spearman ρ	-0.0175	[-0.1297, 0.1016]	3/7
Partial shared-path ρ	-0.0485	[-0.2237, 0.1183]	3/7
Same-major-branch contrast	-0.0027	[-0.0833, 0.0699]	4/7

Two-sided exact Wilcoxon values were 1.0, 0.8125, 0.8125 and 1.0, respectively. Pairwise partner observations and multiple scans are nested within targets; all targets are from one mouse.

Table S9: Complete reconstructed-tree learning across all eligible scans. Means average ten held-out-stimulus splits within each of 13 scans, then scans within each of seven target cells. Mean squared error (MSE) is normalized to the training-mean predictor. Each restricted learning rule scales one fixed site profile in a four-route span by scalar prediction error. Common-checkpoint update match is one minus normalized squared update error for that prescribed profile, rather than the best attainable projection into the full span.

Teaching field	Normalized held-out MSE	Common-checkpoint update match	Fits
Exact compartment error	0.7877	1.0000	130
Topology routes	0.8321	0.2322	130
Random routes	0.8333	0.3584	130
Site-shuffled routes	0.8391	0.4392	130

Positive MSE improvement favors topology. Topology improved over site shuffling by 0.0070 (95% interval -0.0207 to 0.0321) and over random routes by 0.0013 (-0.0090 to 0.0113). All restricted dictionaries had identical effective rank within run. None of 520 fits was excluded. A separate common-state diagnostic at 130 reexecuted exact-learning checkpoints gave per-trial eligibility-weighted within-span oracle captures of 0.2436, 0.3847 and 0.4677 for ancestry, random and shuffled routes, respectively. Those oracle values are geometric upper bounds at the evaluated states; no oracle-feedback learning accuracy was measured.

Table S10: Architecture and data specifications for the regular-tree MNIST feedback experiments. The branching vector is ordered proximal to distal. E/I denotes excitatory/inhibitory; TopK selects the active contacts from separate candidate arrays for each sign. Active contacts and total trainable candidate parameters are reported separately.

Component	Specification
Data	MNIST tensors; 784 flattened pixels; no normalization or augmentation; seed-controlled 48,000/12,000 train/validation split; official 10,000-image test set.
Modeled population	One layer of 128 excitatory modeled neurons; no separate inhibitory-neuron population.
Tree	[3,3]: nine distal, three proximal and one somatic compartment per neuron; no somatic input synapses.
Input contacts	Per dendritic compartment, separate TopK arrays selected 40 excitatory and 20 inhibitory inputs from 784 candidates per sign; 92,160 active contacts among 2,408,448 candidate synaptic parameters.
Parameterization	Nonnegative synaptic and child-coupling conductances obtained by softplus transformation; inactive local weights were not updated.
Reactivation	Learned $[\tanh(m(V - b)) + 1]/2$ at distal, proximal and somatic stages. Shunting used analytical initialization; raw-additive used 0.1/0.9 occupancy-quantile calibration.
Readout	Linear map from 128 soma activities to 10 class scores.
Model size	2,414,602 trainable raw parameters in the feedback experiments, of which 2,408,448 were candidate synaptic parameters.

Table S11: Training protocols for the regular-tree results. Each protocol used minibatches of 256, categorical negative log likelihood, no weight decay, no mixed precision and restoration of the best validation checkpoint. 3F and 5F denote three- and five-factor local updates; ReLU denotes rectified linear unit. TopK rates apply to the candidate synaptic parameters, whereas child-coupling, reactivation and decoder rates apply to their separately listed parameter groups.

Protocol	Seeds; epochs	Core output	Rule and feedback	Adam learning rates
Main feedback ladder	42–56; 180	Identity	3F, local decoder; strict scalar feedback, neuron-specific sharing or exact path transport	TopK 1.5×10^{-3} ; child coupling 7×10^{-4} ; reactivation 5×10^{-4} ; decoder 1.5×10^{-3}
Matched-width scalar-fallback comparison	42–56; 180	Identity	3F, local decoder; matched-width scalar-fallback feedback	Same as the main feedback ladder
Exact path transport factorial	42–46; 200	ReLU	3F or 5F crossed with local or backpropagated decoder; exact path-transported compartment error	TopK 1.5×10^{-3} ; child coupling 7×10^{-4} ; reactivation 5×10^{-4} ; decoder 1.5×10^{-3}
Backpropagation reference	42–46; 200	ReLU	Standard backpropagation through the matched shunting tree and decoder	TopK and decoder 10^{-3} ; child coupling and reactivation 5×10^{-4}

Adam used $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The 3F local-gradient elementwise clip was 5. The exact path transport and backpropagation protocols shared architecture, seeds, epoch budget, and checkpoint-selection procedure, but used the method-specific learning rates listed above. Both were separate from the 180-epoch feedback-definition protocol.

Table S12: Computational environments and retained records for the regular-tree analyses. “Not recorded” means that the value was unavailable and was not inferred from the current environment.

Analysis	Retained outputs	Hardware	Software environment
Matched feedback experiment, 120 runs (90 plotted)	All resolved configurations, checkpoints, logs, final metrics, versioned sweep manifests and SHA-256 hashes	Heterogeneous NVIDIA accelerators; scheduler placement did not change any scientific configuration	Python 3.10.13; Git commits 6c1aaa25 and bd5f905e; executable source hashes and scientific signatures identical across feedback fields within each architecture
Coordinate and gradient-diagnostic experiment, 60 runs	All resolved configurations, checkpoints, logs, final metrics, versioned sweep manifests and SHA-256 hashes	NVIDIA H200; 4.006 summed GPU-hours across both arrays	Python 3.10.13; Git commit 4ba38bde; archived source state includes tracked modifications; generator and model-source hashes are recorded
Depth-by-feedback experiment, 640 runs (480 included)	All resolved configurations, checkpoints, learning curves, held-out metrics, resource records, versioned manifests, source hashes and the outcome-independent 1,840-run validity table	73.29 summed GPU-hours over heterogeneous NVIDIA accelerators	Versioned source identity plus numerical-equivalence checks for two later changes that are inactive in every resolved configuration; signed-noise shunting conditions are excluded from inference
Fixed-checkpoint diagnostic, 3,200 rows (2,400 included)	Five feedback-field labels and four step sizes at each of 160 checkpoints; the reported analysis uses 120 conductance-valid independently trained checkpoints	Fixed-checkpoint analysis; no additional model training	Complete balance and finite-value checks; checkpoint-clustered bootstrap analysis and exclusion record supplied with the source tables
Repeated analysis of exact path transport and back-propagation, 320 runs	Resolved configurations, seed records, checkpoints, learning summaries, final metrics, scheduler and training logs, versioned sweep manifests and SHA-256 hashes	NVIDIA H200 for all included runs; incomplete runs are excluded	Versioned, unmodified root commit 74792ca; nested synthetic-data commit 4f3612a5; signed transfer retained for raw-additive noise, explicit ReLU transfer used for positive-conductance shunting noise
Exact path transport factorial, 20 runs	Resolved configurations, checkpoints, logs and grouped result table	NVIDIA A100-SXM4-40GB; 2.953 GPU-hours total	Exact commit, Python and package environment not recorded
Backpropagation reference, 5 runs	Resolved configurations, checkpoints, logs and grouped result table	NVIDIA A100-SXM4-40GB; 0.474 GPU-hours total	Exact commit, Python and package environment not recorded

Portable sweep files, all resolved configurations, source-data tables, source hashes and validation records for the current 15-seed feedback ladder, the fixed-checkpoint diagnostic and the 320-run same-seed rerun are supplied with the article. Hardware times are sums of Slurm elapsed time over GPU tasks, not wall-clock latency.

Table S13: Notation used in the theory and analyses. Symbols with more than one context-specific role are distinguished in their definitions; feedback-channel count, algebraic rank, routed depth and physical depth are separate quantities.

Symbol	Definition
u, n, i	Indices for a neuron, compartment and synapse, respectively.
V_n, a_n	Conductance-stage voltage and transmitted activity of compartment n ; $a_n = f_n(V_n)$.
$g_{c \rightarrow n}^{\text{den}}$	Nonnegative child-to-parent coupling conductance.
$\mathcal{S}_n^{\text{E}}, \mathcal{S}_n^{\text{I}}$	Excitatory and inhibitory contact sets on compartment n ; contact class is fixed.
$g_i, x_i, g_i x_i$	Nonnegative maximal synaptic conductance, presynaptic activity and resulting open conductance. These are distinct from bold gradient symbols.
$E_i^{\text{rev}}, E_{\text{E}}, E_{\text{I}}, E_{\text{L}}$	Contact-specific, excitatory, inhibitory and leak reversal potentials. E/I polarity is set by reversal, not by a signed conductance.
$G_n^{\text{E}}, G_n^{\text{I}}$	Aggregate activity-weighted excitatory and inhibitory open conductances.
$g_n^{\text{tot}}, R_n^{\text{tot}}$	Total compartment conductance and its reciprocal input resistance.
δ_u	Exact neuron-level learning signal for point neuron u .
N_u	Number of routed nonsomatic compartments in neuron u .
$A_u, \mathbf{c}_u$	Fixed $N_u \times K$ compartmental address matrix and example-dependent vector of K signed channel values; $\hat{\delta}_u^V = A_u \mathbf{c}_u$.
$\delta_{0,u}^V, \delta_{n,u}^V$	Exact somatic and compartment voltage-space errors for neuron u .
$\alpha_n^{\text{cond}}, \bar{\alpha}_n$	Conductance-only and effective path gains; the latter includes reactivation derivatives.
$\bar{\alpha}_w, \text{CV}_w$	Eligibility-energy-weighted mean and coefficient of variation of the effective path gain.
α, ρ, a	Sensor alignment in the physical-depth task, second-moment mixture coefficient in the spectral experiment, and imposed route-span energy fraction.
χ, χ_c	Credit-conflict probability in the branch-conflict task and its centered mean-field boundary. The archived run configurations denote the same dose by α .
$\mathbf{w}, \mathbf{g}, \hat{\mathbf{g}}$	Parameter vector, exact population gradient, and its unbiased stochastic estimate.
$\mathbf{q}$	In reciprocal cables, loss sensitivity to injected current. In the algebraic credit comparison, $q_n = \partial u_0 / \partial u_n$ is a root-output path derivative; multiplying by scalar output error gives credit.
$\mathbf{d}, \mathbf{m}, V, U$	Routed-update estimator, its conditional mean, its centered covariance, and optimized nonnegative-step bound in Eq. 43.
$M, \Sigma, U(M)$	Fixed parameter-gradient operator, exact-estimator noise covariance, and the moment score in the special case $\mathbf{d} = M(\mathbf{g} + \boldsymbol{\xi})$.
$K, \text{rank}(\Phi)$	Number of fixed feedback profiles and their effective algebraic rank; redundant columns need not increase rank.
$d_{\varepsilon,u}, r_{\text{eff}}$	Smallest tested dictionary budget attaining aggregate capture $1 - \varepsilon$, and participation rank $(\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ of the stated second moment.
H_c, D_{T}	Task-credit hierarchy and routed address depth in the fixed-state signal-noise experiment.
$H_p, D_p; \text{D1--D4}$	Physical task hierarchy and physical dendritic stage count. D1–D4 mean $D_p = 1, \dots, 4$ only within the stated cohort; the associated branching vectors can differ between cohorts. In $[b_1, \dots, b_{D_p}]$, factors are ordered soma-to-distal.
A_T	Binary synapse-by-route ancestry matrix of tree T .
Γ_{ik}, Γ_u	Rectangular anatomy-derived route profile from route k to site i , and diagonal channel-gain matrix in the main delivery model, respectively. A passive focal shunt has a diagonal gain form in its baseline-weighted ancestry partition, not in every selected route dictionary.
$\Phi_{T,K}$	K -channel feedback dictionary implemented by topology, route gains, channel allocation and weights.
D_e, U	Diagonal eligibility/anatomical-weight matrix in the route dictionary and orthonormal tree-Haar basis, respectively.
W	Positive diagonal coordinate-weight matrix; summed excitatory synaptic area in the structural MICrONS analysis.
$P_{\Phi,W}, \bar{P}_{\Phi}$	Weighted projector onto $\text{col}(\Phi)$ and its ordinary Euclidean representation in coordinates transformed by $W^{1/2}$.

Supplementary Table S13 continued

Symbol	Definition
$\mathcal{C}_K$	Per-field capture or aggregate energy capture at budget K , as specified; neither is the directional $\cos^2$ diagnostic.
$\mathbf{t}_u, \bar{\Sigma}_t$	Exact target credit field over the compartments of neuron u and its uncentered second moment in the $W^{1/2}$ -transformed coordinates.
$G, \mathcal{R}, R_a, R_m$	Passive conductance matrix, inverse response operator $\mathcal{R} = G^{-1}$, axial resistivity, and specific membrane resistance.
κ_k, κ_b	Focal shunt conductance at compartment k and branch-specific reliability shunt, respectively.
B	Number of branches in the branch-conflict task; mini-batch size only when used as the subscript in $\langle \cdot \rangle_B$.
$S_K, \bar{\Sigma}_g$	Feedback-channel assignment matrix in Eq. 29 and transformed uncentered gradient second moment, respectively.

Table S14: Dictionary constructions, coefficient sources and forward models. The five constructions represent a delivered field as a linear combination of profiles but differ in coordinates, admissible profiles, metric and delivered information. A dictionary may be redundant; its rank need not equal its column count.

Object	Coordinates, profiles and rank	Coefficients, metric, state dependence and forward model
Compartment addresses A_u	N_u nonsomatic compartments by K delivery profiles; rank at most K . Shared and subtree indicators are nonnegative; the general linear model allows signed profiles.	Signed example-dependent coefficients; Euclidean field capture. A layer scalar shares a coefficient across neurons; neuron-specific feedback retains ownership. Fixed profiles differ from exact state-dependent path transport of one somatic error. Directed conductance and raw-additive forward trees.
Haar basis / projector P_K	Sixteen abstract gradient coordinates; selected orthonormal columns have rank K . The global mode is nonnegative and contrast modes are signed.	Exact or noisy projected gradient coefficients in Euclidean coordinates; optional fixed reliability gains. Profiles stay fixed within a condition. Quadratic objective with no simulated forward neuron.
Context field $\Phi^{(K)}$	Eight contexts by eight terminal blocks, rank K , unit row norm. Ancestry fields are nonnegative; dense controls may be signed. Nonzero support and assignment vary by control.	Each trial delivers $\delta_i \Phi_{c_i}^{(K)}$; local input eligibility produces the parameter update. Euclidean directional and update-moment diagnostics differ from span capture. Context-gated logistic block model; internal feedback-tree nodes have no forward state.
Anatomical dictionary A_K	Input-bearing segments by K nonnegative profiles. Historical analyses selected K descendant supports; the common-mode comparison uses one constant plus $K - 1$ supports. Realized rank can be below K .	Free fitted coefficients measure representational capacity under mapped excitatory-area weighting W . No training or biological coefficient estimator. Targets are independently modeled reciprocal-cable response fields or explicitly route-generated positive controls.
Functional dictionary D	Mapped partner-input coordinates by four selected weighted routes, with effective rank matched to controls.	Historical training projects baseline transfer once, $v_D = P_D v_0$, then delivers $\delta_i^{\text{out}} v_D$: one trial-varying scalar and one fixed profile. New trialwise field and eligibility-weighted update oracles assess capacity only. Complete reciprocal passive tree with measured visual-response inputs.

Table S15: Principal regular-tree learning and gradient results. Accuracy is reported as a percentage and gradient cosine is dimensionless; values are means $\pm$ standard deviations over the stated independent seeds. 3F denotes the three-factor local rule. Strict scalar feedback, neuron-specific sharing and exact path transport form the matched 90-fit accuracy ladder; the matched-width scalar-fallback rows provide a separate comparison with the strict-scalar condition. Gradient-cosine entries are a separate fixed-checkpoint diagnostic evaluated at checkpoints trained with neuron-specific sharing; it compared matched-width scalar-fallback feedback with neuron-specific sharing, and the exact-path cosine is one by construction. The final two rows are an independent rule-and-decoder implementation check. All rows labeled raw-additive use Eq. 98, not the normalized-additive model used in Supplementary Figs. S1–S4.

Architecture	Feedback or learning condition	n	Test accuracy (%)	Dendritic-gradient cosine
Shunting	Strict scalar feedback, 3F	15	85.88 ± 1.62	—
Shunting	Matched-width scalar-fallback feedback, 3F	15	90.74 ± 0.54	0.012 ± 0.059
Shunting	Neuron-specific sharing, 3F	15	97.13 ± 0.07	0.708 ± 0.027
Shunting	Exact path transport, 3F	15	97.17 ± 0.13	1 by definition
Raw-additive	Strict scalar feedback, 3F	15	89.31 ± 1.27	—
Raw-additive	Matched-width scalar-fallback feedback, 3F	15	91.59 ± 0.67	-0.001 ± 0.047
Raw-additive	Neuron-specific sharing, 3F	15	97.12 ± 0.13	0.625 ± 0.035
Raw-additive	Exact path transport, 3F	15	97.31 ± 0.12	1 by definition
Shunting implementation check	Exact path transport, 3F, local decoder	5	97.18 ± 0.12	1 by definition
Matched reference	Backpropagation	5	97.03 ± 0.08	1 by definition

Table S16: Strict scalar feedback and matched-width scalar-fallback comparison. The strict-minus-fallback accuracy differences are paired over 15 seeds per architecture and reported in percentage points with 95% paired-seed bootstrap intervals. The practical-equivalence criterion was specified before outcome inspection and required the entire interval to lie within ± 0.5 points.

Architecture	Strict scalar feedback (%)	Matched-width scalar-fallback (%)	Strict minus fallback (points)	Seed ordering
Shunting	85.88	90.74	$-4.859 [-5.669, -3.994]$	0/15 strict higher
Raw-additive	89.31	91.59	$-2.274 [-2.699, -1.796]$	0 higher, 1 tie

Table S17: Conductance-valid depth-by-feedback results. Accuracy contrasts are ranges of condition means over one to four dendritic stages. MNIST retains shunting and raw-additive models; the noise task retains the raw-additive model. The exact-transport comparison reports the largest absolute difference between the three-factor local rule and matched backpropagation among the 12 retained task–model–depth conditions.

Task	Contrast	Accuracy difference	Paired-seed ordering
MNIST	Neuron-specific sharing minus matched-width scalar-fallback feedback	4.92–6.99 percentage points	Positive in 80/80 pairs
Noise resilience, raw-additive	Neuron-specific sharing minus matched-width scalar-fallback feedback	9.76–14.16 percentage points	Positive in 40/40 pairs
Retained conditions	Exact 3F minus matched backpropagation	At most 0.14 percentage points in absolute mean difference	12 matched task–model–depth conditions

Each condition contains ten independent paired seeds. Exact transport is an information oracle rather than a proposed biological signal. Synthetic-noise shunting rows with a signed transfer output are excluded by an outcome-independent conductance-validity rule.

Table S18: Repeated exact-transport/backpropagation analysis. Within each task and neuron model, accuracy was first averaged over depths one through four for each paired seed. Accuracies are percentages and exact-minus-BP differences are percentage points (BP, backpropagation); intervals are 20,000-draw paired-seed bootstrap intervals and P values are descriptive two-sided Wilcoxon tests across ten paired seeds per row. ReLU denotes rectified linear unit.

Task	Core	Exact accuracy	BP accuracy	Exact – BP (95% interval)	Exact > BP	P
MNIST	Raw-additive	97.013	97.008	0.005 (–0.028–0.041)	4/10	1.000
MNIST	Shunting	96.960	96.973	–0.013 (–0.047–0.022)	4/10	0.492
Noisy lines	Raw-additive	33.378	33.490	–0.112 (–0.465–0.247)	4/10	0.625
Noisy lines	Shunting (ReLU)	36.690	36.788	–0.098 (–0.653–0.390)	7/10	1.000

The 320 runs comprise 160 seed-, task-, core- and depth-matched method pairs. One MNIST raw-additive depth-averaged seed was tied. Exact path transport is an information-oracle implementation control; intervals containing zero do not constitute a formal equivalence test.

Table S19: Bandwidth-matched neuron-to-tree routing control. Values are correct minus fixed-deranged held-out accuracy in percentage points. Each row uses ten paired seeds; intervals are 95% paired-seed bootstrap intervals; P values are exact two-sided Wilcoxon tests.

Task	Neuron model	Depth	Difference (95% interval)	Positive pairs	P
MNIST	Raw-additive	2	0.51 (0.37–0.62)	10/10	0.0020
MNIST	Raw-additive	4	0.39 (0.31–0.45)	10/10	0.0020
MNIST	Shunting	2	0.57 (0.43–0.69)	10/10	0.0020
MNIST	Shunting	4	0.38 (0.31–0.44)	10/10	0.0020
Noise task	Raw-additive	2	0.70 (0.56–0.84)	10/10	0.0020
Noise task	Raw-additive	4	0.15 (0.02–0.28)	8/10	0.1055

Five contrasts remained significant after Benjamini–Hochberg correction across the six tests; the raw-additive depth-four noise contrast did not. Across conditions, 58/60 paired seeds favored correct assignment. The modest effect relative to the much larger contrast between matched-width scalar-fallback feedback and neuron-specific sharing separates a dominant coordinate-bandwidth contribution from a smaller assignment-specific routing contribution.

Table S20: Outcome-independent conductance-validity accounting for the artificial-tree experiments. “Included” counts retained training runs, or checkpoints in the diagnostic row; these unit types are not added together. The entire inhibitory-dose family is excluded because its prespecified endpoint compares against invalid signed-input shunting cells.

Experiment	Planned units	Invalid shunting	Unmatched controls	Included units
Depth by feedback	640 runs	160	0	480
Matched routing	160 runs	40	0	120
Inhibitory dose	400 runs	200	200	0
Spatial topology	320 runs	80	0	240
Fixed-budget depth	320 runs	160	0	160
Checkpoint diagnostic	160 checkpoints	40	0	120

The exclusion rule is task plus core plus resolved transfer activation; no accuracy or gradient outcome enters it. The complete 1,840-run validity table and all included row tables are supplied as Source Data.

Table S21: Properties of fixed random and spatial sparse input maps. Each row summarizes 1,280 neuron-level maps: 128 neurons for each of ten topology seeds. Both maps contain 336 active contacts per neuron. Jaccard overlap is intersection divided by union of branch input sets; the feature-degree coefficient of variation is the standard deviation divided by the mean degree. Neurons are nested within topology seeds.

Connectivity property	Random map	Spatial map
Unique input coordinates, mean $\pm$ s.d.	276.2 $\pm$ 5.9	336.0 $\pm$ 0.0
Input coverage fraction	0.352	0.429
Repeated-contact fraction	0.178	0.000
Mean pairwise branch Jaccard overlap	0.0139	0.0000
Feature-degree coefficient of variation	1.507	1.155

Table S22: Fixed-checkpoint credit-to-step analysis at relative step 10^{-5} . Values are means over 120 conductance-valid independently trained backpropagation checkpoints. Gradient capture is the squared cosine after optimally rescaling one update direction, not full dictionary-span capacity. One-step progress is the loss decrease from a norm-matched candidate update divided by that from the exact update.

Feedback field	Gradient cosine	Gradient capture	Relative progress	Descent checkpoints
Matched-width scalar-fallback feedback	0.027	0.003	0.013	75/120
Neuron-specific sharing	0.586	0.355	0.636	120/120
Exact path transport	1.000	1.000	1.000	120/120

Across matched-width scalar-fallback feedback and neuron-specific sharing, gradient cosine predicts relative one-step progress with Spearman $\rho = 0.893$ (checkpoint-clustered 95% bootstrap interval 0.860–0.923); gradient capture gives $\rho = 0.859$ (0.825–0.890). These are fixed-state optimization diagnostics, not convergence or generalization comparisons.

Table S23: Selected endpoints from the expanded regular-tree regime analyses. BP denotes backpropagation and σ the standard deviation of added teaching-signal noise. Values are test-accuracy percentages (means $\pm$ s.d.); the first three retained rows use five independent seeds per condition and the raw-additive CIFAR-10 comparison uses 20 paired seeds.

Analysis	Contrast	Endpoint 1	Endpoint 2
Depth stress	One to four dendritic layers	Shunting: $63.5 \pm 6.2\%$ to $57.4 \pm 4.2\%$	Normalized-additive: $54.9 \pm 5.4\%$ to $29.7 \pm 10.5\%$
Broadcast-noise stress	$\sigma = 0$ to $\sigma = 1$	Shunting: $61.7 \pm 2.2\%$ to $46.9 \pm 6.2\%$	Normalized-additive: $46.5 \pm 5.2\%$ to $11.4 \pm 0.9\%$
Flattened CIFAR-10	Matched-width scalar-fallback versus random rank four	Matched-width scalar-fallback: $29.7 \pm 1.3\%$; random rank four: $35.0 \pm 1.0\%$	Exact path transport: $47.2 \pm 0.6\%$; BP: $49.5 \pm 0.4\%$
Raw-additive CIFAR-10	Strict scalar feedback versus neuron-specific sharing	Strict scalar feed-back: $34.47 \pm 1.06\%$; neuron-specific sharing: $50.87 \pm 0.38\%$	Exact path transport: $50.01 \pm 0.54\%$; BP: $50.12 \pm 0.56\%$

The historical noise-task feedback ladder is excluded here because its shunting execution remains unresolved (Source Data, `review_curve_lineage`); its archived aggregates are retained. The first two rows compare complete forward architectures and therefore establish robustness regimes rather than a causal backward effect of shunting; their additive entries use the normalized-additive model. Exact path transport is an information oracle. The first flattened-CIFAR-10 analysis uses the shunting model only. The final row uses the raw-additive model and 20 paired seeds. Both are harder-data mechanism controls, not state-of-the-art image classification comparisons.

Table S24: Reconstructed-cell inputs to the structural analysis. Synapse counts are mapped excitatory (E) or inhibitory (I) contacts, using direct presynaptic types where available and the target-structure proxy otherwise. Type identifies cortical layer (L) and intratelencephalic class (IT). Depth counts topological stages; effective rank is kernel participation rank. Map pass is the fraction of incoming contacts meeting the 5 μ m dendritic-distance criterion. The final row sums E/I contacts and averages depth, segments, effective rank and map pass.

Root ID	Type	Depth	Segments	E synapses	I synapses	Effective rank	Map pass
864691135060651419	L2IT	11	146	4661	1378	38.36	0.983
864691135277186789	L3IT	10	104	3780	983	14.79	0.981
864691135324623260	L4IT	6	40	1534	506	16.66	0.967
864691135196576298	L5IT	5	22	1254	394	11.00	0.974
864691135409937097	L2IT	9	78	2870	744	21.15	0.984
864691135739661169	L5IT	6	34	970	446	11.96	0.982
864691135810666525	L2IT	14	128	4725	1934	31.29	0.986
864691136043380566	L2IT	9	64	1987	503	21.25	0.986
Total or mean		8.75	77.0	21,781	6,888	20.81	0.980

Table S25: Eight-channel routing capacity on fields generated from the ancestry-route kernel in eight reconstructed cells. Capture is the fraction of weighted field energy in the dictionary span; wiring density is the fraction of nonzero delivery coefficients relative to dense feedback. Ancestry advantage is ancestry-minus-control capture. Intervals resample cells and nested Monte Carlo streams hierarchically; PCA denotes principal component analysis.

Dictionary	Mean capture	95% interval	Wiring density	Ancestry advantage
Dense PCA oracle	0.569	0.481–0.665	1.000	–
Ancestry-selected paths	0.487	0.400–0.581	0.069	–
Random nonempty paths	0.285	0.191–0.398	0.081	0.202 (0.161–0.242)
Depth-only bins	0.222	0.157–0.292	0.125	0.265 (0.229–0.302)
Ancestry shuffle	0.198	0.132–0.272	0.069	0.289 (0.253–0.324)

The morphology advantage was positive in 8/8 cell means for each structural control; the two-sided exact paired Wilcoxon value was $P = 0.0078125$ in each exploratory comparison. The cohort was selected before this journal synthesis but was not a population-random sample.

Table S26: Eight-channel ancestry dictionaries evaluated on an independently generated reciprocal-cable shunt-response operator. Capture is the expected weighted response-energy fraction in the stated subspace, averaged over eight cells. Ancestry difference is ancestry-minus-control capture; ordered cells count positive paired differences. The dense oracle uses singular value decomposition (SVD).

Dictionary	Mean capture	Ancestry difference	Ordered cells
Dense SVD oracle	0.694	–	–
Ancestry-selected paths	0.458	–	–
Random real paths	0.363	0.095	8/8
Depth bins	0.445	0.014	5/8
Row-shuffled selected paths	0.249	0.209	8/8
Degree- and depth-matched surrogate tree	0.423	0.035	6/8

The target was the finite-difference response of the exact soma-state-matched excitatory log-gradient field to every focal shunt, rather than a field sampled from the tested ancestry kernel. Ancestry-minus-control cell-bootstrap intervals were 0.064–0.124 for random paths, –0.037–0.068 for depth bins, 0.179–0.244 for row shuffling, and 0.0069–0.0658 for matched surrogate trees. The paired two-sided Wilcoxon values were 0.0078, 0.844, 0.0078, and 0.078, respectively. The cell-bootstrap interval for the matched surrogate excluded zero whereas the prespecified exact paired test did not; we therefore treat that comparison as inconclusive. This remains a passive-model response on one mouse’s reconstructed anatomy.

Table S27: Focal-perturbation sensitivity at unit dose in the normalized conductance proxy. The first row is the reference for this proxy, not the standard physical-cable calibration of Table S28. Difference is shunt-minus-current-injection localization, where localization is the descendant-minus-depth-control median absolute log-gradient change. Scale multiplies normalized excitatory and inhibitory synaptic conductance; E_I is the inhibitory reversal potential in normalized units.

Scale	E_I	Shunt localization	Current-injection localization	Difference	Positive cells	P
0.35	-0.2	0.104	0.035	0.069	8/8	0.0078
0.10	-0.2	0.101	0.019	0.081	8/8	0.0078
1.00	-0.2	0.114	0.064	0.050	7/8	0.0156
0.35	0.0	0.115	0.021	0.094	8/8	0.0078
0.35	-0.5	0.089	0.054	0.035	7/8	0.0156

Values are means of site-averaged cell effects. P is the exact two-sided paired Wilcoxon value over reconstructed cells. Sensitivity rows reuse the same eight anatomical cells and are not independent replications.

Table S28: Physical passive-cable sensitivity of the focal shunt contrast. Difference is the mean shunt-minus-current-injection localization index; positive values indicate greater descendant localization under shunting. Intervals bootstrap cells after averaging sites within cell. Median axial/leak is the cohort-median conductance ratio. R_a is axial resistivity and R_m is specific membrane resistance.

Cohort	(R_a, R_m)	Median axial/leak	Difference	Positive cells	95% interval
Original eight	(150, 300)	1.36	0.0674	8/8	0.0502–0.0898
Original eight	(150, 1000)	4.53	0.0164	8/8	0.0068–0.0290
Original eight	(150, 3000)	13.60	0.0010	5/8	-0.0043–0.0068
Original eight	(150, 15000)	68.02	-0.0019	2/8	$[-0.00459, -3.47 \times 10^{-5}]$
v661 sensitivity	(150, 300)	2.74	0.0808	45/45	0.0687–0.0937
v661 sensitivity	(150, 3000)	27.41	0.0113	45/45	0.0090–0.0138
v661 sensitivity	(150, 15000)	137.07	0.0024	45/45	0.0018–0.0030

R_a is in Ωcm and R_m in Ωcm^2 . Synaptic area was calibrated relative to the median physical leak and is not a measured conductance. Specific resistances are sensitivity settings rather than fitted cell-specific values. Complete R_a/R_m combinations are supplied as source data.

Table S34: Supported conclusions and unresolved claims for each evidence class. Scope is limited to the stated models, tasks, information sources and sampling units; a positive result in one class does not establish the stronger claim in the final column.

Evidence	Supported conclusion	Not established
Exact factorization	Conductance-tree gradients separate local eligibility from transported compartment error.	A biological circuit computes exact gradients or has exact path transport.
Regular-tree feedback	Preserving neuron-specific information closes most of the scalar-feedback gap in both tested architectures; exact path transport makes 3F sufficient; feedback quality orders norm-matched one-step progress.	Universal advantage of shunting, or superior accuracy, memory or latency relative to backpropagation.
Branch-conflict path demand	Branch-selective information becomes useful when off-route eligibilities conflict; the shared-credit failure boundary moves with branch count as predicted.	A dendrite-exclusive mechanism, a need for multiple feedback errors rather than an equivalent eligibility gate, or prevalence of this regime in natural tasks.
Subtree-address factorial	Task-aligned nested ancestry has a small advantage over matched non-anatomical routes at intermediate bandwidth and over a rewired tree at intermediate budgets.	A uniquely dendritic advantage; matched gated point and flat implementations reproduce the same result, and unrestricted low-rank routes win at the smallest budgets.
Moment-bound theory	Update moments bound one-step loss change. The original scalar endpoint selector loses to rank-only and fixed baselines; a separately frozen finite-horizon forecast improves on fresh seeds but competes with a cheap observed-count baseline.	General-purpose fine-morphology selection, universal superiority to exact gradients or a finite-training optimality guarantee.

Table S34 continued

Evidence	Supported conclusion	Not established
Multi-affine structure	Identical target input-sensitivity spectra can coexist with different tree compatibility; cut matrices with their constant inside row removed exactly characterize representation in the affine-in-child scalar model.	A theorem for arbitrary nonlinear or conductance trees, or equality of model parameter-gradient moments.
Finite noisy morphology selection	A generic sparse interaction estimator improves two prespecified pooled comparisons within the declared eight-variable task mixture; a separate cohort confirms estimated-tree gradient learning.	Arbitrary-function or high-dimensional guarantees, an isospectral claim for the random-coefficient mixture, or uniform family superiority.
Actual credit learning	Frozen Adam recipes interact with task family on estimated trees; additional oracle-compatible controls retain all rules and rates.	An identical-rate or optimizer-independent credit intervention, universal superiority of subtree profiles, or a biological coefficient estimator.
Matched credit geometry	Pairwise and quartic targets share input spectra and a compatible tree but develop different credit spectra. Exact credit substantially improves quartic learning over tested fixed profiles in fresh paired fits, including a common-rate comparison.	A task-only initialization spectrum, proof of minimum external feedback rank, or optimizer-independent asymptotic necessity.
Conductance grouping	Fixed physical shape with mismatched input access creates omitted interactions and a population obstruction; exact-gradient grouping effects persist on fresh seeds.	A universal morphology theorem, learned physical-depth optimality, or necessary precise credit under Adam.
Exact-target construction	Complete Fourier access gives bounded-coefficient exact trees and certifies minimum depth within the stated class.	A selector inferred from limited calibration data, a universal task-complexity law, or biological discovery of optimal morphology.
Physical-depth controls	Serial divisive composition helps when task factors match physical order. H4 saturates at D3 under BP; flexible point MLPs beat serial trees. Extending training preserves the depth benefit but reverses the LocalCA accuracy advantage of exact credit, which retains lower cross-entropy.	A dendrite-exclusive expressivity advantage, natural-task prevalence, biological availability of exact path transport or optimizer-independent equivalence of BP and LocalCA.
State-matched reliability conductance	Fixed positive shunts approximate the step-consistent oracle reliability ordering and outperform global, shuffled and anti-aligned shunts under heterogeneous branch noise.	Exact block-scalar attenuation, autonomous shunt learning, a dendrite-exclusive operation or sustained final-loss superiority over unshunted noisy learning.
Same-span coefficient learning	Coordinate conditioning creates the finite-data bias–variance crossover predicted by exact linear risk, despite identical route capacity.	A biological Gram preconditioner, unconditional benefit of orthogonal or ill-conditioned coordinates, or new address span.
Conductance-valid artificial-tree analyses	Neuron-specific, assignment, spatial and fixed-budget conclusions persist in the 1,000 included artificial-tree runs; the inhibitory-dose family is excluded.	Any claim derived from signed synthetic inputs driving positive-conductance shunting cells.
Fixed spatial topology	Spatial partitioning improves forward learning and input coverage at fixed contact count.	Task-aligned dendritic credit routing; the effect persists under backpropagation and a randomly projected task.
MICrONS capacity	With a budgeted common broadcast, ancestry captures additional independent modeled response variation; the disjoint 47-cell cohort favors ancestry over depth bins and degree–depth surrogates. Cohort-specific rank and cost differences are retained.	Endogenous task-gradient alignment or learning in the recorded mice; a universal fine-topology advantage independent of field, rank and wiring cost.
Inhibitory census	Inhibitory classes occupy different dendritic scales and repeated contacts are spatially co-targeted; actual-route dictionaries are internally compressible.	Geometry-independent descendant-route organization, inhibitory activity, a teaching signal, or biological selection of leverage-ranked routes; all targets come from one mouse.
Focal perturbation	In the normalized proxy, an adjoint-replacement control attributes descendant localization to changed adjoint transport; physical calibration specifies where the effect is large or small.	That the effect is universal across electrotonic regimes or that inhibitory synapses conveyed teaching signals during the animal’s experience.

Table S34 continued

Evidence		Supported conclusion	Not established
Weak-channel check	focal	Descendant-enriched attenuation persists in the fixed local Jacobian of 512 accepted weak-channel steady states near the passive baseline.	Strongly regenerative operation, channel kinetics, spikes, cell-specific channel fits or independent biological replication by the nested draws.
Measured responses		Ancestry-response similarity shows no preferential alignment in seven targets across thirteen scans. The separate 520-fit learning comparison diagnoses restriction of a nearly fixed transfer profile to incompletely covering routes.	Absence of topology-function organization in cortex, failure of every state-dependent coefficient rule, or generalization to other response windows, coverage and biophysical calibration.
Projection coefficients		Oracle projections measure capacity; a separately calibrated local-context encoder improves learning over a frozen profile when an explicit cue and route-activation targets are supplied.	Autonomous discovery of the cue or target coordinates, or a demonstrated biological circuit implementing oracle coefficients.
Alignment-controlled optimization		Rotating exact credit into a fixed reconstructed-tree basis changes both capture and loss reduction under matched quadratic objectives.	That the reconstructed cells acquired or used the imposed gradients in vivo.
Signed animal contrast		Known P+/P- causal sign predicts the dominant dendritic population contrast across six animals.	Within-tree routing or a unique conductance/shunting mechanism.
Interference identity		Route overlap and off-route leakage are sufficient to produce exact one-step forgetting in a static quadratic task.	Temporal neural dynamics or a causal explanation of the published motor-learning experiment.

Table S29: Target-level inputs for the measured-response branch-model analysis. Every target contributed 464 trials, 280 unique stimulus identities and 136 repeated identities; all extracted response values were finite. L/IT denote cortical layer/intratelencephalic class; identifiers link anatomical reconstructions, imaging scans and archive assets. Sites and occupied branches are defined below the table.

Nucleus ID	Root ID	Type	Session	Scan	DANDI asset prefix	Sites	Branches
256456	864691135810666525	L2IT	6	6	06c05eec	10	5
256550	864691135060651419	L2IT	9	4	ad27b3c6	13	5
258209	864691135277186789	L3IT	8	5	436f762f	17	8
264871	864691135739661169	L5IT	7	3	67384ebb	6	4
264888	864691135196576298	L5IT	7	3	67384ebb	9	5
264889	864691135324623260	L4IT	6	6	06c05eec	5	3
291111	864691135409937097	L2IT	6	4	dc68c26d	9	6

Full DANDI asset identifiers and NWB paths are included in the machine-readable target manifest. “Sites” are connected, functionally imaged presynaptic partners after automatic conservative matching; “Branches” are occupied first-level dendritic branches in the fitted model.

Table S30: Four-channel measured-response task analysis of the initial selected-scan cohort. Lower normalized mean squared error (MSE) is better; PCA denotes principal component analysis. Field capture is a fraction of held-out coefficient-field energy under best-fitting projection coefficients. Values average ten fixed stimulus splits within each of seven target cells before averaging targets. Learning projects the trial-dependent exact site-error field; this oracle coefficient rule differs from the frozen-profile rule in the complete-tree model.

Feedback field	Held-out field capture	Held-out normalized MSE
Exact backpropagation	1.000	0.778
Dense PCA oracle	0.999	0.781
Ancestry-selected paths	0.475	0.837
Random nonempty paths	0.564	0.834
Depth-only bins	0.621	0.791
Ancestry shuffle	0.446	0.842
Scalar broadcast	0.222	0.801

Nested-subtree routes minus subtree-assignment-shuffle capture was 0.028 (target bootstrap 95% interval -0.146 to 0.206 ; 3/7 targets positive). The reduction in error from morphology relative to ancestry shuffle was 0.004 (interval -0.021 to 0.027 ; 4/7 targets improved). Neither contrast was reliable.

Table S31: Synapse-resolved inhibitory-census endpoints. DTC and PTC denote distal dendrite-targeting and perisomatic-targeting presynaptic classes, respectively. Domain and capture differences are fractions; tree distance is in micrometres. Connection-geometry contrasts subtract matched locations from observed contacts; class and capacity contrasts follow the row labels. Differences are first averaged within each of 20 postsynaptic targets. Confidence intervals bootstrap target cells. Directional one-sided Wilcoxon tests were specified before the join for the first four endpoints and the three selected-site capacity comparisons; overall placement is two sided.

Endpoint	Mean difference	95% interval	Ordered targets	<i>P</i>
Same-axon tree distance (μm)	-12.003	$[-21.460, -4.151]$	15/20 lower	0.0032
Same-axon shared path	0.0357	$[0.0091, 0.0766]$	15/20 higher	0.0028
Same-axon domain overlap	0.0322	$[0.0060, 0.0743]$	14/20 higher	0.0053
DTC minus PTC median domain fraction	-0.0162	$[-0.0231, -0.0105]$	20/20 lower	9.5×10^{-7}
All contacts minus matched locations	0.0016	$[-0.0022, 0.0056]$	10/20 higher	0.546
Eight selected sites minus random sites	0.3638	$[0.3326, 0.3948]$	20/20 higher	9.5×10^{-7}
Eight selected sites minus depth bins	0.3462	$[0.3126, 0.3793]$	20/20 higher	9.5×10^{-7}
Eight selected sites minus ancestry shuffle	0.3738	$[0.3437, 0.4037]$	20/20 higher	9.5×10^{-7}

Table S32: Dependence and spatial sensitivity of the repeated inhibitory-contact analysis. Contrasts subtract matched from observed contacts, as in Table S31. Ordered units count axons or targets with the stated sign. Shared path and domain overlap are fractions; tree distance is in micrometres. Joint 3D matching controls three-dimensional Euclidean proximity as well as path distance.

Control	Endpoint	Mean difference	Ordered units	<i>P</i>
Presynaptic-axon aggregation	Tree distance (μm)	−9.368	62/92 lower	0.0008
Presynaptic-axon aggregation	Shared path	0.0143	45/92 higher	0.104
Presynaptic-axon aggregation	Domain overlap	0.0117	41/92 higher	0.393
Joint 3D and path matching	Tree distance (μm)	−4.272	13/20 lower	0.108
Joint 3D and path matching	Shared path	0.0107	8/20 higher	0.826
Joint 3D and path matching	Domain overlap	0.0151	9/20 higher	0.54

The first control averages connection contrasts within each of 92 presynaptic inhibitory axons before testing. The second retains the target as the unit and draws matched sites by joint percentile rank of log path distance and three-dimensional Euclidean proximity. All units remain nested within one mouse. Only the coarse tree-distance contrast persisted after presynaptic-axon aggregation; no endpoint persisted after joint spatial matching.

Table S33: Retrospective six-animal signed-credit reanalysis of Francioni et al. source sheet “Ext 13E”. The contrast is error-reduction minus error-increase soma–dendrite residual, in the source z-score units. P+ and P- denote populations with positive and negative causal decoder weights. Directions follow this published mapping; intervals bootstrap paired animal-level contrasts, retaining animals as the inferential unit.

Contrast	Mean	95% interval	Ordered animals
P+	0.0753	[0.0295, 0.1214]	5/6 positive
P-	−0.1498	[−0.2054, −0.0866]	6/6 negative
P+ minus P-	0.2251	[0.1416, 0.3108]	6/6 positive

Exact one-sided random-sign permutation *P* values were 0.0313, 0.0156, and 0.0156. The signed and common modes contained 83.7% and 16.3% of pooled squared projection energy across the six animal contrast vectors, respectively. The pooled fraction is a ratio of summed energies, not the mean of animal-level fractions. The signed fraction was 45.8–97.1% across individual animals, with an animal-bootstrap 95% interval of 60.6–95.7% for the pooled fraction and a leave-one-animal-out range of 76.5–90.2%. Animal-level values are provided in Source Data.

Table S35: Alignment-controlled capture and one-step optimization. Values are means across eight reconstructed cells after averaging 40 nested Monte Carlo streams per cell. One-step progress is normalized by a norm-matched full-gradient update and therefore differs from the squared-energy capture ratio; the selected rows show imposed aligned-energy fractions 0, 0.4 and 1 from the full sweep. All quantities are dimensionless; lower field capture means less representable energy, whereas higher normalized progress means greater loss reduction.

Aligned energy	Dictionary	Field capture	One-step progress
0.0	Ancestry-selected paths	0.000	0.000
0.0	Random paths	0.135	0.307
0.0	Depth bins	0.153	0.331
0.0	Ancestry shuffle	0.152	0.331
0.4	Ancestry-selected paths	0.400	0.615
0.4	Random paths	0.197	0.390
0.4	Depth bins	0.191	0.385
0.4	Ancestry shuffle	0.172	0.359
1.0	Ancestry-selected paths	1.000	1.000
1.0	Random paths	0.290	0.460
1.0	Depth bins	0.247	0.444
1.0	Ancestry shuffle	0.201	0.373

At 40% alignment, morphology capture exceeded the three controls by 0.203–0.228 and the difference was positive in 8/8 cell means. At zero alignment morphology lost to all controls in 8/8 cells; at complete alignment it exceeded all controls in 8/8 cells. Monte Carlo streams and gradient fields are nested sensitivity samples, not biological replicates.

Table S36: Cohort and reproducibility index. Counts refer to distinct runs within each experiment. Reanalyses of the same runs do not add independent observations. The cited Supplementary Information (SI) and corresponding Source Data directories provide complete condition matrices, source identifiers and exclusion records. BP denotes backpropagation; H2–H4 denote two- to four-level latent-factor hierarchies, distinct from physical stage count D. Counts such as 480/640 give retained/planned fits, not a ratio of independent cohorts.

Experiment	Paired unit and design	Protocol and audit record
Image feedback ladders	MNIST: 15 seeds, 90 fits; Fashion-MNIST: 10 seeds, 60 fits; CIFAR-10: 20 seeds, 80 fits	SI S5; Tables S10–S12, S15–S16. Checkpoints and resolved configurations retained.
Random between-neuron feedback	15 seeds per dynamics; 120 new fits, with frozen exact-readout references	SI Fig. S30; feedback matrix fixed within run.
Depth, ownership and input topology	480/640 depth fits; 120/160 ownership; 240/320 spatial; 160/320 fixed-budget fits retained	SI S5; Table S20 records conductance-valid exclusions. Separate 320-fit exact/BP rerun.
Branch-conflict selection	20 seeds, three branch counts, eight nested conflict doses	SI S5 and Fig. S29; paired inputs and masks; six-checkpoint gradient audit on original seeds.
Two-branch / eight-context routing	10 seeds for two branches; 20 seeds and 2,700 fits for full factorial	SI S5; Tables S1–S2. Equivalent implementations are controls, not replication.
Local context-coefficient encoder	20 fresh seeds; 2,160 primary fits; paired exploratory hard-readout sensitivity	SI S5, Fig. S32; separate calibration cues; encoder frozen before task learning.
Moment, reliability and coefficient screens	50 seeds per screen; physical fixed reliability: 2,250 outcomes; adaptive: 1,050; same-span: 4,800 checkpoint rows	SI S4; Tables S3–S6. Seed is the unit; checkpoints remain nested.
Prospective linear selection	Original: 5 development / 20 held-out seeds, 12,800 confirmatory fits; new dynamics: 20 fresh seeds, 25,600 fits	SI S4, Figs. S35, S38–S39; independent calibration and sealed choices.
Matched fixed-profile credit	Three development and twenty fresh paired seeds; 432 development and 1,200 fresh fits	SI S4; saved checkpoints, matched input spectra, calibrated/sign controls and common-rate sensitivity.
Interaction structure	Census of 140 primary and 24 nested targets; 7,872 fits and 164 exact constructions	SI S4, Figs. S36–S37; full-domain capacity analysis, not sampled generalization.
Structure and learning studies	Twenty paired seed blocks each: 2,240 calibration, 3,600 credit and 1,440 conductance fits; 480 further end-to-end fits	SI S4, Figs. S40–S42; frozen choices, full controls and independent test outcomes.
Boolean structure and learning	Seven fixed truth tables; all 15 trees; 20 new seed blocks and 6,720 fits	SI S4, Figs. S43–S44; exact capacity, frozen optimizer recipes and separate classification metrics.
Physical depth and point controls	Ten seeds per cohort: H3 270 fits; point/optimizer controls 200; alignment dose 90; grouped H3/H2 220; H4 360; task-family 360	SI S5, Figs. S15, S28, S31. A 430-fit same-seed rerun tests implementation stability; sixty further fits extend the budget to 600 epochs (main Fig. 5).
Anatomical dictionaries	Eight initial and 47 disjoint cells, one mouse; 12 selected/10 eligible Pinky cells, a second mouse	SI S6–S7; Tables S24–S28. Cohorts analyzed separately.
Anatomical common broadcast	65 modeled operators, 149,331 dictionary evaluations	SI S7; frozen follow-up on disjoint 47 cells and eligible Pinky cells; 200 randomized controls per cell.
Inhibitory census	20 postsynaptic targets; 92-axon sensitivity aggregation, one mouse	SI S8; Tables S31–S32. Synapses and connections nested.
Passive / weak-channel perturbations	Eight initial cells; 45/47 disjoint cells eligible; weak channels: 64 draws per initial cell	SI S9; Tables S7, S27–S28. Sites and channel draws averaged within cell.
Measured responses	Seven targets, 13 scans, ten splits; complete tree: 520 fits; oracle geometry: 130 exact reexecutions	SI S10; Tables S8–S9, S29–S30. Reexecution discrepancies retained.
Measured-response baselines	Same 130 splits, 13 scans, seven targets; 3,380 new evaluations	SI S10, Fig. S34; nested ridge tuning and training-only masks; draws averaged before targets.

Experiment	Paired unit and design	Protocol and audit record
Imposed alignment / external data	Eight cells with 40 nested streams; six animals in external signed-credit reanalysis	SI S11–S12; Tables S33, S35. External neurons are descriptive.

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